

Erdős Problem 817

Collected Research Notes

Corrected reader edition through finite-period control

Corrected edition: 16 September 2026

This edition collects all 29 unique mathematical notes in the supplied source collection. The original Markdown files, implementations, proof data, audits and patches remain in the dated original source distribution. Byte-identical duplicate notes are printed once; all original paths are indexed. This reader repairs the conversion of inline mathematical powers and repeated-block notation, and incorporates the published separation-bound and equation-tag corrections in Chapter 24.

The original anonymous/deleted contributor retains credit for the four-term construction and signed-block proof. The contributor ~~sneed-and-feed~~ retains separate credit for the finite-upper-bound Lean extension. Each collected note retains its own attribution, original date, assumptions and evidence boundary.

The finite-period theorems in Chapter 28 have been reviewed as ordinary mathematical proofs; their producer and independently implemented auditor passed fresh bounded replays in normal and optimized Python. Earlier chapters retain their stated proof and computation scopes. The distinct original Lean results keep their existing attribution and scope. Historical publication-status statements describe the source snapshot; the current repository status is recorded separately.

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An unweighted ternary-variance refinement of the $k = 4$ lower bound

Source: `history/EP817_PR2_variance_contribution_20260914/notes/ternary-variance-lower-bound.md`

Identity: `81a329e64b8f54838228faf23ef8f9f60c2f11c9ac0d4a4067bc89b55122471b` Date: 14 September 2026.

Base: `23c0110c95b5a2036bdc04a1f352b6e5e27742c8`.

Status: ordinary mathematical derivation, with exact bounded computational checks; submitted for independent review. **Not Lean-checked.** No existing Lean source, receipt, or accepted claim status is changed by this contribution.

The dependency is the minimal signed-block/short-kernel classification proved in `paper/ep817_k4_rate.tex`, especially `lem:short-kernel` and `eq:ternary-count`. That construction and proof remain attributed to the anonymous/deleted Reddit contributor. The finite upper-bound Lean extension remains separately credited to **sneed-and-feed**. The calculations below are an additional workbench contribution, not a priority claim for either earlier result. No identification of the anonymous contributor is attempted.

1. Statement

Let $A = \{a_1, \dots, a_n\}$ be a set of distinct positive integers, $n \geq 1$, with no nonconstant four-term arithmetic progression in its set $H(A)$ of subset sums. Write $N = \max A$, $S = \sum_i a_i$, $Q = \sum_i a_i^2$, and

$$T(A) = \left\{ \sum_i x_i a_i : x_i \in \{0, 1, 2\} \right\}, \quad M = |T(A)|.$$

Crucially, every element of $T(A)$ is counted **once**, regardless of the number of ternary representations. Put $n = 3q + r$, $0 \leq r < 3$, and $M_n = 19^n q 3^r$.

Theorem. Under these hypotheses,

$$M \geq M_n, \quad 19(M^2 - 1) \leq 192Q \leq 192nN^2.$$

Consequently the extremal function in Problem 817 satisfies

$$g_4(n) \geq \left\lceil \sqrt{\frac{19}{192n}(M_n^2 - 1)} \right\rceil.$$

In particular,

$$g_4(n) \geq \sqrt{\frac{19}{192n}(19^{2n/3} - 1)} = \left(\sqrt{\frac{19}{192}} + o(1) \right) \frac{19^{n/3}}{\sqrt{n}}$$

as a lower estimate. This improves the existing interval-counting lower estimate by an $\text{order-}\sqrt{n}$ factor. It does not assert that the $\sqrt{n}$ denominator is optimal, and does not change the established exponential rate. The remaining gap to the upper construction is polynomial, not exponential.

2. Exact generating polynomial for distinct ternary sums

Use the source theorem to choose minimal signed relations $r^{(j)}$, with disjoint supports B_j of sizes $m_j \geq 3$, one representative from each opposite pair. Let F be the uncovered coordinates. The complete short kernel is precisely

$$\left\{ \sum_j t_j r^{(j)} : t_j \in \{-2, -1, 0, 1, 2\} \right\}.$$

For a block B , write $S_B = \sum_{i \in B} a_i$ and $Q_B = \sum_{i \in B} a_i^2$. Since its signed relation has coefficients ± 1 on the block, its positive and negative parts have the same sum $S_B/2$; in particular S_B is even. Define

$$P_B(z) = \prod_{i \in B} (1 + z^{a_i} + z^{2a_i}) - z^{S_B/2} \prod_{i \in B} (1 + z^{a_i}).$$

Proposition (unweighted polynomial factorization).

$$\sum_{u \in T(A)} z^u = \prod_{i \in F} (1 + z^{a_i} + z^{2a_i}) \prod_j P_{B_j}(z).$$

Each factor P_B has coefficients zero or one. The full product also has coefficients zero or one; this is an identity of unweighted image polynomials, not just an equality of supports or an equality after evaluation at $z = 1$.

Proof. Reflect a coordinate x_i to $2-x_i$ where $r_i = -1$, leaving positive coordinates unchanged. On the reflected cube the equivalence relation is translation by a constant vector. Each class has exactly one representative with minimum coordinate zero, obtained by subtracting the minimum.

The vectors excluded from that representative set have all reflected coordinates in $\{1, 2\}$. Before reflection, a positive coordinate therefore lies in $\{1, 2\}$, and a negative coordinate in $\{0, 1\}$. Their generating polynomial is

$$z^{\sum_{r_i=1} a_i} \prod_{i \in B} (1 + z^{a_i}) = z^{S_B/2} \prod_{i \in B} (1 + z^{a_i}).$$

Subtracting it from the full ternary polynomial selects one representative per class, and gives P_B . The short-kernel theorem ensures that two local representatives have the same value exactly when they are the same class. It also ensures that an equality of global sums cannot change a free coordinate or the class of any block. Thus the Cartesian product of all local representative sets maps bijectively onto $T(A)$. Multiplying their polynomials proves the proposition. QED.

3. Exact variance of the uniform distinct-image distribution

Let U be uniformly distributed on the finite set $T(A)$, not the ternary cube.

Proposition (exact second moment).

$$\mathbb{E}U = S, \quad \text{Var}(U) = \frac{2}{3} \sum_{i \in F} a_i^2 + \sum_j \kappa_{m_j} Q_{B_j},$$

where

$$\kappa_m = \frac{(2/3)3^m - (1/4)2^m}{3^m - 2^m} = \frac{8 \cdot 3^m - 3 \cdot 2^m}{12(3^m - 2^m)}.$$

Proof. In a block of size m , the 3^m ternary assignments, counted with multiplicity, have mean S_B and variance $(2/3)Q_B$. The 2^m assignments subtracted in the preceding proposition have the distribution of $S_B/2$ plus a sum of independent fair binary choices of the weights a_i . They have the same mean S_B and variance $Q_B/4$. Subtract the unnormalized zeroth, first, and second moments of these two multisets. The

resulting polynomial has $3^m - 2^m$ coefficients equal to one, so division by that number gives mean S_B and variance $\kappa_m Q_B$ for the uniform distinct block image.

The global representative map is a bijection. Uniform measure on its domain is therefore the product of the uniform local measures, and pushes forward to uniform measure on $T(A)$. These independent contributions have additive means and variances. A free coordinate has variance $2a_i^2/3$. QED.

For every $m \geq 3$,

$$\frac{16}{19} - \kappa_m = \frac{5(8 \cdot 3^m - 27 \cdot 2^m)}{228(3^m - 2^m)} \geq 0,$$

because $(3/2)^m \geq (3/2)^3$. Equality holds at $m = 3$. Also $2/3 < 16/19$. Consequently

$$\text{Var}(U) \leq \frac{16}{19}Q.$$

The distinction between the two probability spaces matters. For $A = \{1, 4, 5\}$, the ternary cube has 27 assignments but only 19 distinct sums. The cube variance is 28, while the uniform distinct-image variance is $672/19$. Using 28 for the latter distribution would invalidate the argument.

4. A discrete variance packing inequality

Lemma. If V is uniform on M distinct integers, then

$$\text{Var}(V) \geq \frac{M^2 - 1}{12}.$$

Proof. List the integers as $v_1 < \dots < v_M$. Then $v_j - v_i \geq j - i$ for $j > i$, and

$$M^2 \text{Var}(V) = \sum_{i < j} (v_j - v_i)^2 \geq \sum_{d=1}^{M-1} (M - d)d^2 = \frac{M^2(M^2 - 1)}{12}.$$

The last equality follows by the finite formulas for sums of squares and cubes, or directly by induction on M . The $M = 1$ case is also included. QED.

Applying the lemma to the preceding exact variance yields

$$\frac{M^2 - 1}{12} \leq \text{Var}(U) \leq \frac{16}{19}Q,$$

which proves $19(M^2 - 1) \leq 192Q$. No density heuristic, concentration approximation, or assumption of distinct ternary representations is used.

5. Exact residue-sensitive optimization of the class count

The source quotient gives $M = 3^{|F|} \prod_j (3^{m_j} - 2^{m_j})$. For $m \geq 3$,

$$3^m - 2^m \geq 19 \cdot 3^{m-3},$$

since this is equivalent to $8 \cdot 3^{\{m-3\}} \geq 2^m$. The inequality is strict for $m > 3$. If h is the number of blocks, $h \leq q = \text{floor}(n/3)$, so

$$M \geq 19^h 3^{n-3h} \geq 19^q 3^r = M_n,$$

where the second inequality uses $19 < 27$. Equality $M = M_n$ holds precisely when there are q blocks, all of size three, and r free coordinates. This characterizes equality in the class-count bound, not in the variance bound or in the extremal problem for N . Combining this with Section 4 proves the theorem in Section 1.

6. Distinctness-aware, integer-only finite bound

Distinct positive generators bounded by N satisfy

$$S \leq nN - \frac{n(n-1)}{2}, \quad Q \leq Q_{\max}(n, N) := nN^2 - n(n-1)N + \frac{n(n-1)(2n-1)}{6}.$$

Thus every admissible N satisfies all three integer constraints

$$N \geq n, \quad 2nN \geq M_n + n(n-1) - 1, \quad 192Q_{\max}(n, N) \geq 19(M_n^2 - 1).$$

The companion script defines `finite_lower(n)` as the least integer satisfying them. The quadratic expression is increasing for $N \geq n$; binary search therefore computes the exact threshold without floating-point square roots. For reference, completing the square gives the quadratic threshold

$$N \geq \frac{n-1}{2} + \sqrt{\max \left\{ 0, \frac{19(M_n^2 - 1)}{192n} - \frac{n^2 - 1}{12} \right\}},$$

used together with $N \geq n$ and the interval constraint. When the radicand before taking its maximum is negative, the quadratic adds no restriction.

n	Combined finite lower bound
1	1
2	3
3	5
4	11
5	27
6	49
9	724
12	11,841
30	352,128,817,514

The first three bounds are attained by $\{1\}$, $\{1,3\}$, and $\{1,4,5\}$, respectively; their binary subset-sum sets can be inspected directly. No claim of exactness is made for the other entries.

7. Reproducibility and formalization boundary

Run from the repository root:

```
python certificates/verify_variance_lower_bound.py \
  --output certificates/ternary_variance_receipt.json
```

The default exhaustive domain is all nonempty subsets of $\{1, \dots, 24\}$ with size at most five: 55,454 candidate sets, of which 2,551 are admissible. For each admissible set the script independently enumerates the entire short kernel, reconstructs the minimal blocks, compares the full polynomial coefficient map against the directly computed unweighted ternary image, checks its exact rational mean and variance, and checks the finite bounds and the class-count equality criterion. This includes 823 cases with a size-three block and 179 with a size-four block; 1,549 have no signed block.

Seven separate targeted cases cover a pair of size-three blocks, a size-three block with two free coordinates, a mixed size-three/size-four pair, and single blocks of sizes three through six. Further bounded checks cover canonical cube moments through dimension eight, the partition optimizer through $n=200$, and the coefficient bound through $m=256$. Six regression checks distinguish weighted from unweighted variance, detect an incorrect polynomial shift, and reject inadmissible or invalid inputs. All calculations use integers or `fractions.Fraction`. The receipt hashes the executable and the ordered exhaustive transcript. Finite checks are corroboration, not the general proof.

No Lean executable was available in the session environment. A new local Lean build was therefore not run, and no uncompiled Lean file is presented as a formal certificate. The existing PR #1 receipt was inspected, not regenerated.

The next formalization units are precise:

1. Prove the minimal-block and complete short-kernel classification already stated in the source manuscript, including disjoint magnitude layers.
2. Construct a finite equivalence from the product of canonical local representatives to the distinct ternary image; this is the essential cardinality and probability-measure bridge.
3. Prove the polynomial identity and rational centered-second-moment identity.
4. Prove the integer spacing/variance inequality and combine it with the residue-sensitive product bound to obtain the finite lower theorem.
5. Add the real nth-root squeeze only after those finite claims are checked.

These are targets, not assertions of completed formalization. The accepted Lean upper-bound theorem, Core hashes, and existing verification statuses remain untouched.

References and dependencies

Anonymous/deleted Reddit contributor, & The Clankers. (2026, September 13). *The exponential rate for four-term-progression-free subset-sum sets* [Workbench manuscript]. `paper/ep817_k4_rate.tex` at the base commit above. The original proof route belongs to the anonymous contributor; reconstruction and verification are attributed separately in that manuscript.

sneed-and-feed. (2026, September 13). *Formalize finite upper bound theorem and core helper lemmas in Extended.lean* [Pull request #1]. <https://github.com/KokunoYumeto/erdos-problem-817-workbench/pull/1>.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* [Preprint]. arXiv:2606.24139. Cited for the surrounding problem and published abstract bounds, not as a source of the new variance formula.

General-k exponential capacity, exact carry certificates, and improved k=5,6 constructions

Source: `history/EP817_CUMULATIVE_PR2_PR3_20260914/PR3_general_k/notes/general-k-capacity.md` **Identity:** 45c4e09b089b65afe65ae661494c4b82bf41e8f95b76971421f8243eec6f3894 14 September 2026. Research continuation in the Erdős Problem 817 workbench. Base revision: 23c0110c95b5a2036bdc04a1f352b6e5e27742c8.

Evidence status. The general statements below have ordinary mathematical proofs. The finite blocks have independently replayed integer certificates. These claims are submitted for independent review; no new Lean elaboration is claimed. Existing Core/Extended sources, their receipts, PR #1, and the separate variance PR #2 remain unchanged.

1. Problem, attribution, and results

For a finite set A of distinct positive integers, write

$$H(A) = \left\{ \sum_{a \in U} a : U \subseteq A \right\}, \quad S(A) = \sum_{a \in A} a.$$

The empty sum belongs to $H(A)$. For $k \geq 3$, call A k -admissible when $H(A)$ contains no tuple $(x, x+d, \dots, x+(k-1)d)$ with d a nonzero integer. Define

$$g_k(n) = \min\{\max |A| : |A| = n, A \text{ is } k\text{-admissible}\} \quad (n \geq 1).$$

These are the conventions of Erdős–Sárközy and Korsky. The anonymous/deleted Reddit contributor supplied the workbench's original $k=4$ construction and signed-block proof. The Lean finite-upper-bound extension is separately credited to sneed-and-feed. The present general- k arguments do not assume the proposed variance refinement in PR #2; the original $k=4$ block is used as a control example. No claim of literature priority is made for the arguments here.

The continuation establishes the following.

Theorem 1 (all- k capacity). For every fixed integer $k \geq 3$, the limit

$$\Lambda_k = \lim_{n \rightarrow \infty} g_k(n)^{1/n}$$

exists. It has the exact finite-certificate characterization

$$\Lambda_k = \inf_{\substack{B \neq \emptyset \text{ finite, distinct, positive} \\ q > \sum B; H(B) \bmod q \text{ has no nonzero-step } k\text{-AP}}} q^{1/|B|}.$$

The modulus q ranges over all integers, including composite integers. A modular progression here is an ordered tuple with nonzero step in $\mathbb{Z}/q\mathbb{Z}$; repetition of residues is allowed. Every test in the displayed infimum is finite and decidable.

Theorem 2 (all-length decision). For any finite set D of nonnegative integers containing 0, any $q \geq 2$, and any $k \geq 3$, a finite directed graph decides whether

$$L_t(D, q) = \left\{ \sum_{j=0}^{t-1} d_j q^j : d_j \in D \right\}$$

is k -progression-free for every t . Digits $\geq q$ and signed carries are retained. An unsuccessful certificate returns an explicit integer progression and its digit representations. When D is contained in $[0, q-1]$, the graph has at most $2k!$ states.

Theorem 3 (explicit improved bounds). Put $W = (1, 4, 5, 17, 21, 22)$. For $n \geq 1$ write

$$j = \lfloor (n-1)/6 \rfloor, \quad s = 1 + ((n-1) \bmod 6).$$

With w_s the s -th entry of W ,

$$\boxed{g_5(n) \leq w_s 97^j, \quad g_6(n) \leq w_s 93^j.}$$

In particular,

$$\Lambda_5 \leq 97^{1/6}, \quad \Lambda_6 \leq 93^{1/6}.$$

These improve the corresponding upper rates in Korsky's inspected v1, namely $\sqrt{5}$ and $7^{2/5}$. The exact comparisons are $97 < 5^3$ and $93^5 < 7^{12}$. No floating-point comparison is needed.

Theorem 1 identifies the full fixed- k exponential-rate problem with a uniform bound on finite certificate capacities. Determining that infimum numerically for every k , or establishing the sharp large- k law, remains a separate mathematical task. Section 10 gives the exact outstanding inequality rather than an assumed premise for any theorem above.

2. The four-term obstruction and its exact witness map

For each $k \geq 4$ there is an inclusion of sets of admissible generator sets

$$\mathcal{A}_{4,n} \hookrightarrow \mathcal{A}_{k,n}, \quad A \mapsto A.$$

A nonconstant k -AP would contain a nonconstant 4-AP in its first four positions. This proves the inclusion and transfers upper constructions in that direction.

The inclusion is strict already for $k=5$ and $A=\{1,2\}$. Its subset-sum image is $\{0,1,2,3\}$, so it is 5-admissible. The short relation $2*1-2=0$ has nonzero magnitude-one and magnitude-two layers. Feeding its literal coefficient classes into the four-term witness construction gives

$$P_1 = 0, \quad P_2 = 1, \quad N_1 = 2, \quad N_2 = 0,$$

and the tuple

$$(P_1 + N_2, P_1 + P_2, N_1 + N_2, N_1 + P_2) = (0, 1, 2, 3).$$

Thus the map from this coefficient relation to a four-term progression works exactly as before, but its codomain is the allowed four-term witness set for a 5-admissible object. The general- k development below retains simultaneous progression equations instead of asserting that these coefficient layers vanish.

3. An exact composition of admissible generator sets

Let A and B be finite sets of distinct positive integers, and put

$$R = 2S(A) + 1, \quad C = A \cup RB.$$

All members of RB exceed every member of A, so $|C|=|A|+|B|$. At the level of chosen subsets, the map

$$\mathcal{P}(A) \times \mathcal{P}(B) \longrightarrow \mathcal{P}(C), \quad (U, V) \longmapsto U \cup RV$$

is a bijection. Its inverse intersects with A and RB and divides the latter elements by R. It induces the following value-space bijection:

$$E_{A,B} : H(A) \times H(B) \longrightarrow H(C), \quad (x, y) \longmapsto x + Ry.$$

Indeed, $0 \leq x \leq S(A) < R$, so the inverse on $H(C)$ is

$$z \longmapsto (z \bmod R, \lfloor z/R \rfloor).$$

Collisions among different subsets within $H(A)$ or $H(B)$ remain present; this statement is about the distinct numerical image sets.

Lemma 3.1 (progression decomposition). Every k -term progression in $H(C)$ pulls back under E to a k -term progression in each factor.

Proof. Write its terms $u_i = x_i + Ry_i$. For $0 \leq i \leq k-3$,

$$0 = u_i - 2u_{i+1} + u_{i+2} = (x_i - 2x_{i+1} + x_{i+2}) + R(y_i - 2y_{i+1} + y_{i+2}).$$

The first parenthesis is an integer in $[-2S(A), 2S(A)] = [-(R-1), R-1]$. It is divisible by R and hence equals zero. The second parenthesis therefore also equals zero. These consecutive second-difference identities prove that both coordinate sequences are arithmetic progressions. QED.

The corresponding constructive contrapositive is a map

$$\text{AP}_k^*(H(C)) \longrightarrow \text{AP}_k^*(H(A)) \sqcup \text{AP}_k^*(H(B)).$$

Pull back the tuple; send it to the first component when its first common difference is nonzero, otherwise to the second. The second difference is then nonzero because the original difference is $d_x + Rd_y$. Thus every failure of the composition has an explicit failure witness in a factor.

Corollary 3.2. If A and B are k -admissible, then C is k -admissible. Furthermore,

$$2S(C) + 1 = (2S(A) + 1)(2S(B) + 1).$$

The equality follows by substituting $S(C) = S(A) + (2S(A) + 1)S(B)$; every term and constant is retained.

4. Proof that every fixed- k exponential rate exists

Define

$$F_k(n) = \min_{A \in \mathcal{A}_{k,n}} (2S(A) + 1), \quad F_k(0) = 1.$$

Admissible sets exist for every n : powers of 3 have 3-progression-free subset sums, as consecutive binary base-3 digit equations show, and hence are k -admissible. The minima are minima of nonempty sets of positive integers.

Corollary 3.2 proves

$$F_k(m+n) \leq F_k(m)F_k(n).$$

Let $\alpha = \inf_{m \geq 1} \log(F_k(m))/m$. Fix m and write $n = tm + r$ with $0 \leq r < m$. Repeated composition gives

$$\frac{\log F_k(n)}{n} \leq \frac{t \log F_k(m) + \log F_k(r)}{n}.$$

The finitely many remainder terms are bounded with m fixed. Hence the limit superior is at most $\log(F_k(m))/m$, and then at most α upon taking the infimum. Every term is at least α , so

$$\lim_{n \rightarrow \infty} F_k(n)^{1/n} = \inf_{m \geq 1} F_k(m)^{1/m}.$$

This also supplies a direct proof of the subadditivity step without importing a limit theorem as a black box.

The cost and maximum are related by the exact inequalities

$$2g_k(n) + 1 \leq F_k(n) \leq 2ng_k(n) + 1.$$

For the left inequality use $\max(A) \leq S(A)$ on a minimizer for F . For the right inequality use $S(A) \leq n \max(A)$ on a minimizer for g . Since $g_k(n) \geq 1$,

$$\frac{F_k(n)}{2n+1} \leq g_k(n) \leq \frac{F_k(n)}{2}.$$

Taking n th roots proves existence of Λ_k and

$$\Lambda_k = \inf_{n \geq 1} F_k(n)^{1/n}.$$

For orientation, the elementary one-shift argument gives $\Lambda_k > 1$. If B and $B+a$ belong to a k -AP-free final cube, each maximal a -chain in B has length at most $k-2$, because adding a extends its end by one. Thus

$$|B \cup (B+a)| \geq \frac{k-1}{k-2}|B|.$$

Iterating and using $|H(A)| \leq n \max(A) + 1$ proves

$$\Lambda_k \geq \frac{k-1}{k-2}.$$

The sharper polynomial factor from Korsky's adaptive recurrence is independent of the existence proof. Also $\Lambda_{k+1} \leq \Lambda_k$ follows from the admissibility inclusion.

5. Finite cyclic certificates are cofinal for the exact rate

Let M_k be the class of pairs (q, B) satisfying:

- $q \geq 2$ is an integer;
- B is a nonempty finite set of distinct positive integers, with $S(B) < q$;
- $D = H(B)$, reduced in $\mathbb{Z}/q\mathbb{Z}$, has no ordered k -term progression with nonzero step.

Reduction

$$\rho_q : D \hookrightarrow \mathbb{Z}/q\mathbb{Z}$$

is injective because $0 \leq d \leq S(B) < q$. It preserves addition modulo q , not unrestricted integer addition. The equations needed for progression lifting are accounted for next.

Lemma 5.1 (all-length cyclic lift). For (q, B) in M_k , define

$$A_t = \bigcup_{j=0}^{t-1} q^j B.$$

Then $|A_t| = t|B|$, $\max(A_t) = q^{t-1} \max(B)$, and A_t is k -admissible.

Proof. Since $1 < b < q$, equality $q^i b = q^j b'$ with $i < j$, a contradiction. The equal-level case gives $b = b'$. Each subset of A_t chooses a subset in every block, so

$$H(A_t) = L_t(D, q).$$

The map D^t to L_t sending $(d_0, \dots, d_{t-1})$ to $\sum_j d_j q^j$ is a bijection, with inverse the actual base- q digits. Suppose y_i is a k -AP in this language. Modulo q , the digits form a k -AP; their common difference must be zero. All low digits therefore equal one actual digit c . The map

$$\{y \in L_t : y \equiv c \pmod{q}\} \longrightarrow L_{t-1}, \quad y \longmapsto (y - c)/q$$

is a bijection, with inverse $z \mapsto c + qz$. It sends the progression difference to its exact integer quotient by q . Induction on t forces the original progression to be constant. This proof uses no primality assumption. QED.

Deleting excess generators yields $\Lambda_k \leq q^{1/|B|}$.

Lemma 5.2 (universal finite embedding). Every k -admissible B gives an element

$$(2S(B) + 1, B) \in M_k.$$

Proof. Put $q = 2S(B) + 1$. A modular k -AP among the displayed integer values in D has all second differences divisible by q . Each lies in $[-2S(B), 2S(B)]$, so each vanishes as an integer. The displayed values are therefore an integer k -AP and hence constant. Their modular step is zero. QED.

Let $\beta = \inf_{(q, B) \in M_k} q^{1/|B|}$. Lemma 5.1 gives $\Lambda_k \leq \beta$. Lemma 5.2 gives $\beta \leq F_k(n)^{1/n}$ for every n . Section 4 then gives $\beta \leq \Lambda_k$. This proves Theorem 1.

For each fixed q the literal search domain is finite: B is a subset of $[1, q-1]$ and $\sum B < q$. Define

$$m_k(q) = \max\{|B| : (q, B) \in M_k\}.$$

Ignore q with no nonempty certificate. Then

$$\Lambda_k = \inf_q q^{1/m_k(q)}.$$

The finite sequence obtained by restricting $q \leq Q$ decreases to Λ_k . This is a proved upper approximation; the identity alone supplies no computable error bound after a finite Q . Comparisons between candidate rates $q^{1/m}$ and $r^{1/s}$ are made exactly by comparing q^s and r^m .

6. Full carry graph, including nonstandard digits

Let D be any finite set of nonnegative integers containing 0; write $W = \max D$. Fix $q \geq 2$ and $k \geq 3$, and put

$$L = \left\lceil \frac{W}{q-1} \right\rceil.$$

A state is $(c_1, \dots, c_{k-1}, b)$, where b is a Boolean recording whether a nonzero digit of the progression difference has occurred, and

$$-L \leq c_i \leq L + i.$$

The initial state is $(0, \dots, 0, \text{false})$. The accepting state is $(0, \dots, 0, \text{true})$.

For a in D , e in $\{0, \dots, q-1\}$, and $b_1, \dots, b_{k-1}$ in D , draw a labeled edge when the integers c'_i satisfy

$$a + ie + c_i = b_i + qc'_i \quad (1 \leq i < k).$$

The target flag is b OR $(e \neq 0)$. Labels include the actual a, e, b_i , not merely their residues. When several actual digits have the same residue, all choices of b_i are retained.

6.1 Invariant bounds

The next carry is $(a+i e+c_i-b_i)/q$. For the lower bound its numerator is at least $-W-L$. Since $W \leq L(q-1)$, this quotient is at least $-L$. For the upper bound its numerator is at most $W+i(q-1)+L+i=W+L+iq \leq q(L+i)$. Thus the stated box is invariant. The graph has at most

$$V = 2 \prod_{i=1}^{k-1} (2L + i + 1)$$

states.

When D is contained in $[0, q-1]$, b_i is the unique remainder of $a+i e+c_i$. Starting from zero, $0 \leq c_i \leq i$ is invariant, so only

$$V = 2 \prod_{i=1}^{k-1} (i + 1) = 2k!$$

states are needed. The reduction of this bound uses uniqueness of ordinary digit remainders; the unrestricted graph retains signed carries.

6.2 Path-to-progression map

For a length- t accepting path with labels $a_j, e_j, b_{i,j}$, define

$$x_0 = \sum_{j=0}^{t-1} a_j q^j, \quad \delta = \sum_{j=0}^{t-1} e_j q^j, \quad x_i = \sum_{j=0}^{t-1} b_{i,j} q^j.$$

Multiply each edge identity by q^j and sum over j . Interior carry terms telescope; initial and terminal carries are zero. Hence

$$x_i = x_0 + i\delta.$$

The accepting flag gives $\delta > 0$, and all x_i belong to $L_t(D, q)$. This is an explicit witness map from labeled accepting paths to progression representations.

6.3 Progression-to-path map

Conversely, take any positive-step integer k -AP in L_t . Choose its actual D -word representations. Choose $\ell \geq t$ so q^ℓ exceeds its positive difference, and append zero digits to every representation. Expand the difference in its ordinary ℓ base- q digits e_j . Starting at zero, the identities in Section 6 define integer carries, because equality of the final numerical values implies divisibility of each low-prefix residual by the next power of q . Explicitly,

$$c_{i,j} = \frac{\sum_{h < j} (a_h + i e_h - b_{i,h}) q^h}{q^j}.$$

The value is integral: subtract the remaining high-position terms from $x_0 + i\delta - x_i = 0$. These carries satisfy the transition equation, stay in the invariant box, and end at zero. A nonzero difference produces the accepting flag. Negative-step progressions are handled by the involution

$$(x, d) \mapsto (x + (k-1)d, -d),$$

which reverses the tuple and has itself as inverse.

For ordinary digits, no padding beyond t is needed, and uniqueness of all digit expansions makes the two maps inverse bijections between accepting paths of length t and positive-step k -APs in L_t . For unrestricted D , the path labels preserve each chosen representation; the map to numerical progressions is surjective. Its fibers are

exactly the available representation choices, which are irrelevant to existence but have not been identified with distinct values.

Proof of Theorem 2. A forbidden progression exists at some finite length exactly when the accepting state is reachable. Reachability in the displayed finite graph is decidable. A shortest path never repeats a state, so a failure has a witness of length at most $V-1$. An accepting path returns the explicit digits and points above. A safe graph is certified by a reachable set containing the initial state, closed under every labeled transition, and excluding the accepting state. QED.

6.4 Relation to modular obstructions

For ordinary digits, a nonzero-step modular k -AP is exactly an initial edge with $e \neq 0$. Its endpoint consists of the carries $\text{floor}((a+i e)/q)$. Thus there is a typed bijection

$$\text{AP}_{k,\mathbb{Z}/q\mathbb{Z}}^*(D) \simeq \{\text{initial carry edges with } e \neq 0\}.$$

A modular certificate excludes all such initial edges and leaves only the zero/false state reachable. The full graph permits initial nonzero-step edges whose carry states never reach the accepting state.

For a concrete strict example, $B=\{2,7\}$, $D=\{0,2,7,9\}$, $q=11$, $k=3$. Its four modular progressions have $(\text{start}, \text{step})=(0,9), (2,9), (7,2), (9,2)$. The complete reachable set is

$$((0,0), \text{false}), \quad ((0,1), \text{true}), \quad ((1,1), \text{true}).$$

The initial state has four constant self-edges and four edges to the other two states. From $(0,1)$, the labels (a,e) are $(2,5), (7,6), (9,4)$, with targets $(0,1), (1,1), (1,1)$. From $(1,1)$, the labels are $(0,6), (2,4), (7,5)$, with targets $(0,1), (0,1), (1,1)$. These are all 14 valid edges. None reaches $((0,0), \text{true})$. Therefore every base-11 word with these digits is 3-AP-free despite those modular progressions.

Signed carries are also essential for the unrestricted theorem. With $D=\{0,2\}$, $q=2$, $k=3$, the actual progression $(0,2,4)$ is encoded by the two labels

$$(a, e, b_1, b_2) = (0, 0, 2, 0), \quad (0, 1, 0, 2).$$

The intermediate carries are $(-1,0)$, followed by $(0,0)$. Excluding negative carries would discard this witness.

7. From unrestricted carry certificates back to finite cyclic certificates

For distinct positive B and $q \geq 2$, write each actual b uniquely as $q^v u$ with $v \geq 0$ and q not dividing u . This is a bijection between positive integers and the pairs (v,u) , with exact inverse $(v,u) \rightarrow q^v u$. The pair, including v , is retained.

Call (q,B) separated when distinct members of B have distinct residual coordinates u . Then the sets $q^i B$ have no cross-level collisions: equality $q^i b = q^j b'$ is equivalent to $u=u'$ and $i+v=j+v'$. Conversely, equal residual coordinates for different b give a cross-level collision at a sufficiently long prefix.

Let C_k be the finite pairs (q,B) that are separated and whose full graph for $D=H(B)$ excludes acceptance. Section 6 proves that their prefixes A_t have exactly $t|B|$ positive distinct generators and are k -admissible. Their largest generator is $q^{t-1} \max(B)$, so

$$\Lambda_k \leq q^{1/|B|}.$$

Every M_k certificate is in C_k . There is also a concrete family of maps from C_k back to M_k :

$$(q, B) \mapsto \left(2S(B) \frac{q^t - 1}{q - 1} + 1, \bigcup_{j < t} q^j B \right).$$

The integer in the first coordinate is $2S(A_t)+1$. Lemma 5.2 proves modular admissibility. Its cost per generator satisfies

$$\left(2S(B)\frac{q^t-1}{q-1}+1\right)^{1/(t|B|)} \longrightarrow q^{1/|B|}.$$

Thus the unrestricted carry class and the simpler cyclic class have the same infimum of exponential costs:

$$\Lambda_k = \inf_{(q,B) \in C_k} q^{1/|B|} = \inf_{(q,B) \in M_k} q^{1/|B|}.$$

This equality keeps finite-level differences visible: a particular carry certificate may fail the modular test, while its explicitly constructed longer prefixes approach the same rate through valid modular certificates.

8. Applying the method: the six-generator block

Let

$$B_* = \{1, 4, 5, 17, 21, 22\}, \quad S(B_*) = 70.$$

Its actual subset-sum image is the following 38-element set:

$$D_* = \{0, 1, 4, 5, 6, 9, 10, 17, 18, 21, 22, 23, 25, 26, 27, 28, 30, 31, 32, \\ 38, 39, 40, 42, 43, 44, 45, 47, 48, 49, 52, 53, 60, 61, 64, 65, 66, 69, 70\}.$$

The certificate computes this image directly from all subset choices. It then checks every ordered pair (x, d) with $0 \leq x < q$ and $1 \leq d < q$. There are 9,312 pairs for $q=97$ and 8,556 pairs for $q=93$.

An independent chain calculation gives a compact description of the modular checks. For each nonzero step, decompose D_* into maximal chains under addition of that step. Full cosets are checked separately by verifying that the chain decomposition covers D_* ; hence repeating progressions in a proper subgroup are included. The histogram of the longest chain per step is:

q	longest chain 2	longest chain 3	longest chain 4	longest chain 5
97	6 steps	28 steps	62 steps	0 steps
93	4 steps	18 steps	68 steps	2 steps

The rows sum to $q-1$. Therefore D_* has no nonzero-step 5-AP modulo 97 and no nonzero-step 6-AP modulo 93. For 93 the argument includes all steps of nontrivial gcd with 93, rather than invoking a prime-modulus lemma.

Lemma 5.1 now proves the all-length constructions. Because $\max(B_*) = 22 < q \min(B_*)$, their elements are ordered by level and then by the displayed weights. Taking the first n generators gives exactly the bound in Theorem 3. This proves the theorem for every n , not just multiples of six.

The earlier successful blocks found on the same search route remain useful at small finite sizes:

k	q	B	number of subset sums	upper rate
4	19	$\{1, 7, 8\}$	7	$19^{1/3}$, existing contributor's control
5	23	$\{1, 3, 4, 7\}$	12	$23^{1/4}$
6	47	$\{1, 2, 6, 7, 14\}$	23	$47^{1/5}$
5	97	B_*	38	$97^{1/6}$
6	93	B_*	38	$93^{1/6}$

Taking the minimum of the applicable finite upper expressions is valid. Asymptotically, $97^{1/6} < 23^{1/4}$ because $97^2 < 23^3$; $93^{1/6} < 47^{1/5}$ follows from direct integer powers.

8.1 A correlated vector model explaining the block

Use the exact column map $V : \mathbb{Z}^6 \rightarrow \mathbb{Z}^3$ with columns

$$e_1, e_2, e_1 + e_2, e_3, e_2 + e_3, e_1 + e_2 + e_3.$$

The scalar map $L: \mathbb{Z}^3 \rightarrow \mathbb{Z}$, $L(x,y,z)=x+4y+17z$, satisfies $L(V \text{ epsilon})=\sum_i b_i \text{ epsilon}_i$ for every binary epsilon. Thus the square from binary subsets through the vector image or through the six numerical generators commutes exactly.

Every point of $V(\{0,1\}^6)$ has $0 \leq x, z \leq 3$ and $0 \leq y \leq 4$. A five-term integer-vector AP has zero x- and z-differences, since a nonzero such difference would span at least four. A nonconstant y progression must run from 0 to 4 or conversely. At $y=0$, only columns e_1, e_3 can occur, giving x, z in $\{0,1\}$. At $y=4$, all four columns with nonzero y must occur, giving x, z in $\{2,3\}$. These endpoint fibers are disjoint. Therefore the vector subset-sum image is 5-AP-free.

The scalar map can create additional additive relations; its action on each proposed modular progression is therefore checked by the complete finite certificate above, rather than assuming arbitrary projections preserve freeness. This correlated model motivates the six-generator choice and identifies the endpoint information lost by replacing its image with the full rectangular box.

8.2 Explicit failures of smaller bases

For this fixed B_* , every q from 71 through 96 fails at $k=5$, and every q from 71 through 92 fails at $k=6$. These are bounded statements about this exact block; the full graph returns integer witnesses. For example, $q=93$ at $k=5$ returns

$$(9, 990, 1971, 2952, 3933), \quad d = 981,$$

all in the two-block language. The $q=92$, $k=6$ witness is

$$(1, 967, 1933, 2899, 3865, 4831), \quad d = 966.$$

The $k=5$ base-22 version of $\{1,3,4,7\}$ fails with $(0,77,154,231,308)$. The published receipt retains the actual digit labels for all 57 nearby-base obstruction cases, including the familiar $k=4$ failures at bases 17 and 18.

9. A proved bound on the rectangular-digit route

Let $c_1, \dots, c_m$ be distinct nonzero vectors in $\mathbb{N}^r$ with coordinate totals $\sum_j c_{\{j,i\}} \leq \tau$. At most r columns have 11 mass one. Every other column has mass at least two. Consequently

$$2m - r \leq \sum_j \|c_j\|_1 \leq r\tau, \quad \boxed{m \leq \lfloor r(\tau + 1)/2 \rfloor}.$$

This is the count attained by the singleton-plus-edge construction in Korsky's appropriate nearly-regular graph range. For $p > \tau$ the evaluation

$$E_p : \{0, \dots, \tau\}^r \hookrightarrow \mathbb{N}, \quad x \mapsto \sum_i x_i p^i$$

is injective, with inverse its displayed digit vector. It sends the vector subset-sum image to the integer subset-sum image and retains the generator count. Its full-box envelope discards correlations between coordinates. Section 8.1 exhibits the precise lost data: the fibers at $y=0$ and $y=4$ have disjoint outer coordinates even though the full box would allow a five-term vertical progression.

Thus the column-mass bound is a proved obstruction to improving generator density while retaining that particular rectangular resource constraint. The carry graph and the correlated block operate on the actual image and keep the discarded endpoint constraints.

10. Exact remaining general question and research implications

Korsky's inspected bound gives

$$\frac{k-1}{k-2} \leq \Lambda_k \leq \min_{p \geq 3} p^{2/(\min\{p,k\}-1)}.$$

Theorem 1 replaces the separate lower/upper limits with an existing limit. Theorem 3 tightens the right side at $k=5$ and $k=6$. For large k , the inspected bounds still leave logarithmic rates between order $1/k$ and order $\log(k)/k$.

In the new exact finite formulation, the desired lower scale is the uniform statement

$$\frac{\log q}{|B|} \geq c \frac{\log k}{k} \quad \text{for every } (q, B) \in M_k$$

for some absolute positive c in the stated large- k regime. This is a research target, not a premise used in any proved claim here. A violation comes with a finite pair (q, B) and, by Section 5, an explicit infinite family of integer constructions. A proof must control every finite cyclic certificate; checking finitely many q cannot establish that quantifier. The exact maps in Sections 3, 5, 6, and 7 make both outcomes usable without discarding carries or changing the generator problem.

The next mathematical objective is therefore a uniform inverse or packing inequality for these actual subset-sum images. The rectangular count in Section 9 settles one restricted representation class; the six-generator examples demonstrate why its envelope alone cannot supply the general answer. The certificate search remains a productive way to test proposed lower inequalities and return concrete counterexamples when they fail.

11. Verification and reproducibility

The standard-library checker is `certificates/verify_general_k_capacity.py`. Run

```
python certificates/verify_general_k_capacity.py \
--output certificates/general_k_capacity_receipt.json
```

The default run does all of the following:

- audits the five displayed finite blocks by direct modular enumeration, independent maximal-chain calculations, full carry reachability, and direct actual generator subsets through two blocks;
- checks every D subset $[0, q-1]$ containing 0 for $q=2, \dots, 6$ and $k=3, \dots, 6$: 248 alphabets, 128 safe and 120 unsafe, with 744 direct language-length comparisons;
- checks every D subset $[0, 2q]$ containing 0 and $\max(D) \geq q$, for $q=2, 3, 4$ and $k=3, 4, 5$: 966 alphabets, 29 safe and 937 unsafe, with 1,932 direct comparisons;
- checks 738 literal compositions from 246 admissible left instances in the stated finite domain, preserving the exact sumset bijection, source cardinality, cost product, and progression exclusion;
- replays all 57 nearby-base failure witnesses, the prime-base carry-only example, signed carry failure, nonstandard-digit safe example, and cross-level generator-collision detection;
- performs a literal branch-and-bound search for modular blocks for every $q=3, \dots, 48$ and $k=5, 6$, with no rescaling of candidate weights.

Every safe graph is rechecked for closure under all valid labels. Failure witnesses are reconstructed by integer sums and checked against the actual language. No floating-point arithmetic is used in a proof check. Search upper-rate comparisons use integer powers. A transcript hash fixes each deterministic bounded replay.

Exploratory C++ scans additionally supplied the candidates at $q=93$ and 97 . Candidate discovery is separated from verification: the reproducible Python certificate checks those blocks directly without relying on the exploratory search or on a claim of optimality at those moduli. The command-line option `--search-max-base 127` permits a larger fully replayed search; the default receipt only claims the range it actually executes.

The receipt records its precise source hash. The companion summary records that hash and the full receipt hash, together with the principal results and chain profiles. The downloadable contribution bundle includes the full receipt. Existing workbench validation and Lean receipts have their earlier scopes; they are not silently extended to these files.

Formalization units

The direct finite modules are: composition and its inverse; second-difference decomposition; the cost product and infimum limit proof; modular lifting with composite q ; unrestricted carry-state invariance; accepting-path/represented-progression correspondence; separation of geometric generators; finite reachable-set closure; and the two six-generator certificates. All new units remain outside current Lean coverage. They can be formalized independently of PR #2's signed-block variance development.

References and source roles

Anonymous/deleted Reddit contributor, & The Clankers. (2026, September 13). *The exponential rate for four-term-progression-free subset-sum sets* [Workbench manuscript]. `paper/ep817_k4_rate.tex` at the base revision. Role: existing $k=4$ construction and algebraic comparison; original mathematical attribution retained.

Korsky, S. (2026, June 23). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/html/2606.24139v1>. Inspected: Theorems 1.2, 1.3 and 6.3; Sections 5–7. Role: benchmark general bounds, exact conventions, one-shift context, rectangular graph construction, and large- k target.

Costa, S. (2026, September 5). *A negative answer to the Erdős–Sárközy question* (arXiv:2609.06303v1). <https://arxiv.org/html/2609.06303v1>. Inspected: Theorem 1.1 and the reduction. Role: separates the historical $k=3$ positive-prefactor question from the broader all- k rate problem. Its construction is not an input to the proofs above.

sneed-and-feed. (2026, September 13). *Formalize finite upper bound theorem and core helper lemmas in Extended.lean* [Pull request #1]. <https://github.com/KokunoYumeto/erdos-problem-817-workbench/pull/1>. Role: existing formal upper-bound contribution, with credit separate from the anonymous mathematical construction.

The Clankers. (2026, September 9). *A handoff describes the research; it does not command the next researcher* [PolyClank design document]. `docs/polyclank/RESEARCH_STATE.md` in KokunoYumeto/yang-mills-interacting-workbench at 7ba7ea5a5dab84d88934477de93dda71f6af0662. Role: inspected research-record convention; no mathematical dependency or peer-validation claim.

Carpenter, R., Defant, C., & Kravitz, N. (2026, May 6). *Sets with few subset sums* (arXiv:2605.05498v1). <https://arxiv.org/html/2605.05498v1>. Abstract and source entry inspected as a possible inverse-theorem direction; not used in a proof here.

Croot, E., Mao, J., & Yip, C. H. (2026, March 15). *Hilbert cubes in sets with arithmetic properties* (arXiv:2603.14654v1). <https://arxiv.org/html/2603.14654v1>. Introduction/framework statements inspected; no theorem imported into the present proof.

Exact composition, finite carry certificates, and the general-k capacity

Source: `history/EP817_GENERAL_K_POLYCLANK_20260914/research/general_k/notes/general-k-carry-capacity.md` Identity: `1954d499b7c63eb3f0202366ea75ff7f63729e17be110e11e7c6e3e9b425f650` Erdős Problem 817 —

PolyClank research contribution, 14 September 2026.

Base revision: `23c0110c95b5a2036bdc04a1f352b6e5e27742c8` of KokunoYumeto/erdos-problem-817-workbench.

Status: complete ordinary proofs of the statements below, with reproducible integer checks and explicit finite certificates; independent review is requested. Lean coverage remains that of the existing Core/Extended modules. This work adds no Lean declarations or replacement receipts. The conclusions below use no conjectural block-classification lemma.

1. Scope and exact objects

For a finite set A of distinct positive integers, put

$$H(A) = \left\{ \sum_{a \in U} a : U \subseteq A \right\}, \quad S(A) = \sum_{a \in A} a, \quad R(A) = 2S(A) + 1.$$

Let $\mathcal{A}_k(n)$ be the set of such A with $|A|=n$ and with $H(A)$ containing no nonconstant k -term arithmetic progression. Here $k \geq 3$. Set $\mathcal{A}_k(0) = \{\emptyset\}$, and define

$$g_k(n) = \min_{A \in \mathcal{A}_k(n)} \max A \quad (n \geq 1), \quad h_k(n) = \min_{A \in \mathcal{A}_k(n)} R(A) \quad (n \geq 0).$$

For a subset X of an abelian group, $\text{AP}_k(X)$ denotes the set of ordered tuples $(x_0, \dots, x_{k-1}) \in X^k$ with $x_{i+2} - 2x_{i+1} + x_i = 0$ for every i . Constant tuples are included. Over a cyclic group, a nonconstant progression means that its common difference is a nonzero group element; its entries may repeat when that element has small order. All such orders are retained in the computations.

These definitions preserve every generator, its actual value, and every subset representation. R is an additional integer-valued observable. No gcd is removed, no generator is rescaled to 1, and no equality of subset sums is discarded. The original $k=4$ construction and signed-block proof remain credited to the anonymous/deleted Reddit contributor. The Lean finite upper bound remains separately credited to sneed-and-feed. Draft PR #2 supplies the earlier variance investigation; the present proofs have no dependency on its new lemma.

The symbol `lambda_k` below denotes the exponential growth rate. Korsky's Equation (1.5) uses the same letter for the polynomial-loss exponent $\log_2((k-1)/(k-2))$; that quantity is kept separately here. The map $\log_2:(1, \infty) \rightarrow (0, \infty)$, with inverse $x \mapsto 2^x$, relates his polynomial exponent to the lower exponential base $d_k = (k-1)/(k-2)$.

The broader problem requests estimates for all k . Costa (2026) establishes $\liminf g_3(n)/3^n = 0$, addressing the historical constant-factor question. The present target is the full family of exponential rates and constructions.

2. Exact product on generators and arithmetic-progression tuples

Theorem 2.1 (composition with explicit inverse maps). For arbitrary finite sets A, B of distinct positive integers, including the empty set, define

$$A \star B = A \cup R(A)B.$$

Then

$$|A \star B| = |A| + |B|, \quad S(A \star B) = S(A) + R(A)S(B), \quad R(A \star B) = R(A)R(B).$$

The maps

$$\begin{aligned} F_{A,B} : \mathcal{P}(A) \times \mathcal{P}(B) &\longrightarrow \mathcal{P}(A \star B), & (U, V) &\longmapsto U \cup R(A)V, \\ E_{A,B} : H(A) \times H(B) &\longrightarrow H(A \star B), & (x, y) &\longmapsto x + R(A)y \end{aligned}$$

are bijections. E induces a bijection

$$\text{AP}_k(H(A)) \times \text{AP}_k(H(B)) \longrightarrow \text{AP}_k(H(A \star B))$$

by componentwise evaluation. A target progression is constant exactly when both source progressions are constant. Consequently star sends $\mathcal{A}_k(n) \times \mathcal{A}_k(m)$ into $\mathcal{A}_k(n+m)$. It is associative and has identity $\emptyset$.

Proof. Put $R=R(A)$. Every member of A is less than R , while each member of RB is at least R . The two parts are disjoint, and multiplication by R is injective. The inverse of F sends W to $(W \cap A, \{b \in B : Rb \in W\})$. This proves the cardinality formula and the subset representation identity.

For x in $H(A)$, $0 \leq x \leq S(A) < R$. Thus the inverse of E is the restriction of the Euclidean division map

$$z \longmapsto (z - R\lfloor z/R \rfloor, \lfloor z/R \rfloor).$$

The first and second coordinates belong to $H(A), H(B)$ because z arose from a subset of $A \star B$. This proves the bijection on actual values. The subset-sum maps commute with F and E . In particular, writing $\mu_A(x)$ for the number of subsets of A summing to x , one has the exact fiber identity $\mu_{A \star B}(x + Ry) = \mu_A(x)\mu_B(y)$.

Take an arithmetic progression $z_i = x_i + Ry_i$ in the target and put $e_i = x_{i+2} - 2x_{i+1} + x_i$. The equation for z gives $e_i = -R(y_{i+2} - 2y_{i+1} + y_i)$. Meanwhile $-2S(A) \leq e_i \leq 2S(A)$, so $|e_i| < R$. Its divisibility by R forces $e_i = 0$, and the second difference of y is then zero as well. Conversely, evaluating two source progressions clearly gives a target progression. The pointwise inverse of E gives the inverse on tuples. If the target common difference is zero, its first two points have the same E -preimage, so both source differences are zero. The reverse implication follows by evaluation.

Finally $R(A \star B) = 2S(A) + 2R(A)S(B) + 1 = R(A)R(B)$. Both bracketings of a triple product are the same set $A \cup R(A)B \cup R(A)R(B)C$, and the empty set has $R=1$. QED.

Concrete contrapositive. A nonconstant k -AP in $H(A \star B)$, decoded by the displayed division map, gives a nonconstant k -AP in at least one of $H(A), H(B)$. Thus a failed product construction comes with a factor witness, rather than an unexplained failure of a heuristic.

The same product preserves the ternary image exactly. Put $T(A) = \{\sum_a x_a a : x_a \in \{0, 1, 2\}\}$. Since $0 \leq x \leq 2S(A) = R(A)-1$ for x in $T(A)$, division by R gives a bijection $T(A) \times T(B) \rightarrow T(A \star B)$. Consequently the unweighted image polynomials satisfy

$$\sum_{t \in T(A \star B)} z^t = \left(\sum_{x \in T(A)} z^x \right) \left(\sum_{y \in T(B)} z^{R(A)y} \right).$$

This supplies a direct connection to the earlier variance work while keeping its finite sets and their measures explicit.

3. Existence of every fixed-k exponential rate

Theorem 3.1. For every integer $k \geq 3$ the following limit exists:

$$\lambda_k = \lim_{n \rightarrow \infty} g_k(n)^{1/n} = \inf_{n \geq 1} h_k(n)^{1/n} = \inf_{\substack{A \in \mathcal{A}_k(|A|) \\ |A| > 0}} (2S(A) + 1)^{1/|A|}.$$

Also $1 < \lambda_k \leq 3$, and the sequence λ_k is nonincreasing in k .

Proof. Powers of 3 provide admissible sets for every n : their subset sums use only base-3 digits 0 and 1, and reduction of a second-difference relation modulo 3 followed by division forces each digit triple to be constant. A k -AP would have a nonconstant initial 3-AP. Thus the displayed minima exist, and $h_k(0)=1$.

Theorem 2.1 gives $h_k(n+m) \leq h_k(n)h_k(m)$. Put $b_n = \log h_k(n)$ and $L = \inf_{\{l \geq 1\}} b_l/l$. For fixed l write $n = ql + r$, $0 \leq r < l$. Repeated composition gives the exact inequality $b_n \leq q b_l + b_r$. Division by n and passage to the upper limit gives $\limsup b_n/n \leq b_l/l$, since the finitely many b_r are bounded. Taking the infimum over l gives $\limsup \leq L$. The definition of L gives $b_n/n \geq L$ for every $n \geq 1$. Therefore b_n/n tends to L .

The original maximum and the sum observable obey

$$2g_k(n) + 1 \leq h_k(n) \leq 2ng_k(n) + 1.$$

The left inequality follows from $S(A) \geq \max A \geq g_k(n)$. For the right one, use a set attaining $g_k(n)$. In particular

$$0 \leq \frac{\log h_k(n) - \log g_k(n)}{n} \leq \frac{\log(2n+1)}{n} \rightarrow 0.$$

Thus $\log g_k(n)/n$ tends to L as well. Exponentiating proves all three expressions. The elementary construction gives $\lambda_k \leq 3$. Inclusion $\mathcal{A}_k(n) \subseteq \mathcal{A}_{k+1}(n)$ follows by the explicit prefix map $\text{AP}_{k+1}(X) \rightarrow \text{AP}_k(X)$, which preserves a nonzero common difference. Hence λ_k is nonincreasing. Strict positivity above 1 follows from the next calculation. QED.

For completeness, the known elementary lower mechanism can be restated without an unproved input. Fix an ordering of an admissible A and let B be the partial subset-sum set just before a positive generator a is added. Split B into maximal chains with step a . A chain of length $k-1$ would extend in B union $(B+a)$ to a k -AP, so all chain lengths are at most $k-2$. Each chain adds one point. Consequently

$$|B \cup (B + a)| \geq \frac{k-1}{k-2} |B|.$$

Iteration from $\{0\}$ and $|H(A)| \leq S(A) + 1 \leq n \max A + 1$ gives

$$g_k(n) \geq \frac{((k-1)/(k-2))^n - 1}{n}, \quad \lambda_k \geq \frac{k-1}{k-2} > 1.$$

This chain mechanism and stronger adaptive finite estimates are due to the existing literature, in particular Korsky (2026, Section 5). It is restated here only to close the elementary existence theorem's bounds.

4. The exact finite carry graph

Let $b \geq 2$ and let D be a finite nonempty subset of $\{0, \dots, b-1\}$. Define

$$L_m(D; b) = \left\{ \sum_{j=0}^{m-1} d_j b^j : d_j \in D \right\}.$$

Words are written from least significant to most significant digit. The map $e_m : D^m \rightarrow L_m(D; b)$ is a bijection: its inverse is the m -digit base- b expansion. No digit identification occurs in this map.

For $k \geq 3$ define the directed, labeled graph with vertices

$$(c, f) \in \{-1, 0, 1\}^{k-2} \times \{0, 1\}.$$

For a digit column $d = (d_0, \dots, d_{k-1}) \in D^k$, let $\Delta_i(d) = d_{i+2} - 2d_{i+1} + d_i$. The column labels an edge

$$(c, f) \xrightarrow{d} (c', f')$$

precisely when, for every i ,

$$\boxed{bc'_i = c_i + \Delta_i(d), \quad f' = f \vee [d_0 \neq d_1].}$$

The initial vertex is $(0,0)$, and the accepting vertex is $(0,1)$.

Theorem 4.1 (all-length decision and witness bound). For every b, D, k as above, the following are equivalent:

- (a) Some $L_m(D; b)$, $m \geq 1$, contains a nonconstant k -AP.
- (b) The accepting vertex is reachable from the initial vertex.

Every such instance has a witness using at most 3^{k-2} digit columns. A finite vertex set containing the initial vertex, excluding the accepting vertex, and closed under every labeled transition certifies k -AP-freeness for every m . When 0 belongs to D , it also certifies the union over all m , since leading zero padding places any finite tuple in one common L_m .

Proof. A matrix of digit columns $d^{(0)}, \dots, d^{(m-1)}$ determines numbers $y_i = \sum_{j < m} b^j d_i^{(j)}$. Starting at $c^{(0)} = 0$, the edge equations say $\Delta(d^{(j)}) = bc^{(j+1)} - c^{(j)}$. Hence, exactly,

$$\sum_{j < m} b^j \Delta(d^{(j)}) = b^m c^{(m)}.$$

Ending with $c^{(m)} = 0$ is equivalent to all second differences of y vanishing. The flag is 1 exactly when the first two digit words differ; uniqueness of base- b expansion makes this equivalent to $y_0 \neq y_1$. Thus any accepting labeled path evaluates to a nonconstant k -AP.

Conversely, take a k -AP in L_m and use the unique base- b expansion of each term. Set

$$c^{(j)} = b^{-j} \sum_{t < j} b^t \Delta(d^{(t)}).$$

The full second difference is zero, so the prefix sum is divisible by b^j ; each $c^{(j)}$ is an integer vector. Since $|\Delta_i| \leq 2(b-1)$,

$$|c_i^{(j)}| \leq 2(1 - b^{-j}) < 2 \quad (j > 0).$$

It lies in $\{-1, 0, 1\}$. These carries satisfy the transition equations and end at zero. The nonconstant progression makes the flag 1. This is the inverse map from progressions, through their unique digits, to labeled paths.

The only reachable flag-0 vertex is $(0,0)$. Indeed, from zero with $d_0 = d_1$, the congruence for d_2 forces $d_2 = d_1$ because both are in $[0, b)$; successive congruences force all entries equal, and then every carry stays zero. The graph therefore has at most one reachable flag-0 vertex and 3^{k-2} reachable flag-1 vertices. A shortest accepting path repeats no vertex, so its length is at most 3^{k-2} . Closure of a proposed safe vertex set keeps every finite path inside that set, proving the last assertions. QED.

The graph may be explored using only $|D|^2$ starting digit pairs at each vertex. Having chosen d_0, d_1 , each subsequent digit is forced by

$$d_{i+2} \equiv 2d_{i+1} - d_i - c_i \pmod{b}, \quad 0 \leq d_{i+2} < b.$$

An edge exists exactly when all these forced digits belong to D . This is an exact enumeration of D^k columns satisfying the divisibility conditions. The checker also compares it with direct column enumeration on a smaller, fully specified domain. Parent edges retain actual digit columns so that negative results contain concrete integer witnesses.

4.2 Retaining digits above the base and every representation collision

The preceding canonical-digit map has a complete extension to arbitrary finite D contained in $[0, S]$, with S allowed to exceed b . Keep the same evaluation surjection $e_m: D^m \rightarrow L_m(D; b)$, and retain its entire fibers. For example, when $D = \{0, 7, 8, 15, 19, 26, 27, 34\}$ and $b = 19$, the words $(19, 7)$ and $(0, 8)$ both belong to D^2 and evaluate to 152. Both are retained. Tracking whether two digit words differ would flag these equal values incorrectly. The map to actual differences is therefore made explicit below.

Set

$$C_0 = \left\lceil \frac{2S}{b-1} \right\rceil, \quad H_0 = \left\lceil \frac{S}{b-1} \right\rceil.$$

The full graph has vertices

$$(c, h, f) \in \{-C_0, \dots, C_0\}^{k-2} \times \{-H_0, \dots, H_0\} \times \{0, 1\}.$$

For a column d in D^k require the exact transition equations

$$bc'_i = c_i + \Delta_i(d), \quad h + d_1 - d_0 = r + bh', \quad 0 \leq r < b, \quad f' = f \vee [r \neq 0].$$

Thus h' and r are respectively the Euclidean quotient and remainder of $h + d_1 - d_0$; negative values are handled by the floor quotient. Start at $(0, 0, 0)$. A vertex is accepting exactly when $c = 0$ and either $f = 1$ or $h = 0$.

Theorem 4.2 (full carry equivalence). For every $b \geq 2$, $k \geq 3$, and nonempty finite D contained in the nonnegative integers, an $L_m(D; b)$ contains a nonconstant k -AP exactly when this full graph has a reachable accepting vertex. A shortest witness has at most

$$2(2C_0 + 1)^{k-2}(2H_0 + 1) - 1$$

columns. A finite transition-closed set of vertices containing the initial vertex and excluding every accepting vertex certifies all word lengths.

Proof. For a labeled path of length m let $y_i = \sum_{j < m} b^j d_i^{(j)}$. The second-difference transitions telescope exactly as before to $\Delta(y) = b^m c^{(m)}$. Meanwhile the extra carry transitions give the separate identity

$$y_1 - y_0 = \sum_{j < m} b^j r_j + b^m h^{(m)}.$$

The sum of remainders lies in $[0, b^m)$. Consequently this difference vanishes exactly when all r_j vanish and $h^{(m)} = 0$. This is precisely the complement of the stated acceptance test at $c^{(m)} = 0$. Thus every accepting path gives an actual nonconstant progression, with no assumption about uniqueness of word representations.

Conversely, take any progression in L_m and any digit word representing each of its terms. For each j define the second-difference carry from the prefix formula in Theorem 4.1. It is integral because the full second difference vanishes. Its magnitude is at most $2S(1 - b^{-j})/(b - 1)$, hence at most C_0 . Starting with $h^{(0)} = 0$, perform the displayed quotient-remainder recursion. If $p_j = \sum_{t < j} b^t (d_1^{(t)} - d_0^{(t)})$, induction gives $h^{(j)} = \lfloor p_j / b^j \rfloor$ and a remainder in $[0, b^j)$. Since $|p_j| / b^j \leq S(1 - b^{-j})/(b - 1)$, one has $-H_0 \leq h^{(j)} \leq H_0$. These data define a full-graph path. The actual nonzero difference guarantees its accepting endpoint.

This construction is a bijection between labeled accepting paths and matrices of digit words whose evaluated rows form nonconstant progressions. Evaluation of the rows maps these matrices surjectively to the progression set, with exactly the original representation fibers. A shortest accepting path has no repeated vertex, proving the finite bound. Closure proves the all-length assertion. QED.

For canonical D , forgetting h gives a graph morphism on reachable states from this graph to the graph of Theorem 4.1: $(c, h, f) \rightarrow (c, f)$. Before the first differing canonical digit, $h = 0$; at the first such digit the difference has magnitude less than b and gives a nonzero remainder. Hence both flags become true on exactly the same column. This proves preservation of the transition labels and the acceptance condition on reachable vertices.

An explicitly checked positive full certificate uses $b = 19$, $A = \{7, 8, 19\}$, and $D = H(A) = \{0, 7, 8, 15, 19, 26, 27, 34\}$. Its full graph has 15 reachable states and 288 labeled transitions. Direct D^4 enumeration verifies closure. There is also an exact independent generator-set explanation:

$$\bigcup_{j < m} 19^j \{7, 8, 19\} \subseteq \bigcup_{j \leq m} 19^j \{1, 7, 8\}.$$

The terms 7 and 8 stay at their original levels, while each 19 at level j is the original 1-generator at level $j+1$. The subset-sum inclusion sends any progression witness to one in the original certified construction. This example uses digits above the base without replacing the input block.

5. Modular reduction, carry compatibility, and explicit contrapositives

Reduction $\rho_b : \mathbb{Z} \rightarrow \mathbb{Z}/b\mathbb{Z}$ is a group homomorphism. Applied coordinatewise to a digit column, it maps zero-start carry transitions to modular arithmetic progressions, since $\Delta(d) = bc'$. The target carry records the exact integer second difference divided by b . A sequence of such columns lifts to an integer progression precisely when the carries match and eventually return to zero, as proved by Theorem 4.1.

In particular, a modularly k -AP-free D has only constant zero-start columns; $\{(0,0)\}$ is then a closed safe certificate. This proves the usual modular digit lifting argument with no assumption that b is prime.

Two explicit examples retain the additional information carried by the graph. For $b=4$, $D=\{0,2\}$, $k=3$, the complete reachable set is

$$\{(0,0), (-1,1), (1,1)\}.$$

The modular tuple $(0,2,0)$ labels an edge from $(0,0)$ to $(-1,1)$. The vertex $(-1,1)$ has no outgoing edge: every integer second difference of D -digits is even, so adding -1 cannot be divisible by 4. The same holds at $(1,1)$. Consequently the whole digit language avoids 3-APs even though that modular tuple exists. This uses the original digits $\{0,2\}$ exactly as given.

For $b=8$, $D=\{0,1,3,4\}$, $k=5$, the complete reachable set is

$$\{((0,0,0),0), ((-1,1,-1),1), ((1,-1,1),1)\}.$$

Only constant columns occur as zero-to-zero transitions. The other two reachable vertices have no outgoing edge; their closure can be checked by the 16 possible first digit pairs at each vertex. The reduction map sends $(0,4,0,4,0)$ to a modular progression, and the exact carry $(-1,1,-1)$ records its terminal incompatibility. Thus arbitrary nonzero carries are retained rather than excluded by assumption.

The opposite outcome is equally explicit. For the four-generator block $\{1,3,4,7\}$ in base 16, the digit columns

$$(0,4,8,12,0), \quad (0,0,0,0,1)$$

form an accepting path and evaluate to $(0,4,8,12,16)$. For the original $\{1,7,8\}$ block in base 18, columns $(0,9,0,9), (0,0,1,1)$ evaluate to $(0,9,18,27)$. These are witnesses against those exact smaller-base lifts. They leave the valid base-23 and base-19 constructions intact.

6. A capacity formula covering every admissible generator set

A **modular certificate** is a pair (b,A) with $b \geq 2$, nonempty A a finite set of distinct positive integers, $S(A) < b$, and $H(A)$ modularly k -AP-free. A **carry certificate** has the same data, replacing the last property by absence of an accepting path in the graph of Section 4 for $D=H(A)$. Call their classes $\mathcal{M}_k$ and $\mathcal{C}_k$.

Theorem 6.1 (exact certificate capacity). For every $k \geq 3$,

$$\lambda_k = \inf_{(b,A) \in \mathcal{M}_k} b^{1/|A|} = \inf_{(b,A) \in \mathcal{C}_k} b^{1/|A|}.$$

Both classes admit finite exact verification of each member.

Proof. From either certificate form $A^{[m]} = \bigcup_{j=0}^{m-1} b^j A$. Its members are distinct: each a in A is in $[1,b)$, so two members at different levels have disjoint magnitude ranges. Thus $|A^{[m]}| = m|A|$ and $\max A^{[m]} = b^{m-1} \max A$.

The blockwise subset-sum map $\{0, 1\}^{m \times |A|} \rightarrow H(A)^m$ is surjective, with fiber cardinality $\prod_j \mu_A(d_j)$. Composing with e_m gives exactly $H(A^{[m]}) = L_m(H(A); b)$. The actual subset collisions therefore persist with known multiplicities, while the value-word map is bijective. Section 4 or 5 proves k -AP-freeness at every m . Deleting generators gives, for every $n \geq 1$,

$$g_k(n) \leq (\max A) b^{\lceil n/|A| \rceil - 1}.$$

Taking roots and using Theorem 3.1 proves $\lambda_k \leq b^{1/|A|}$, for either certificate class.

Conversely, for every nonempty admissible A , take $b = 2S(A) + 1$. Any modular k -AP in the integer digit set $H(A)$ has second differences divisible by b and of absolute value at most $2S(A) = b - 1$. They therefore vanish over the integers. Admissibility makes the progression constant. Thus $A \rightarrow (2S(A) + 1, A)$ is an explicit map from every admissible set to a modular certificate. Theorem 3.1 makes the infimum of the resulting costs exactly λ_k . Since every modular certificate is also a carry certificate, the opposite inequalities are proved. QED.

Finite countercertificate equivalence. For every positive real u ,

$$\lambda_k < u \iff \exists (b, A) \in \mathcal{M}_k [b < u^{|A|}].$$

The same equivalence holds with C_k . This follows from the two infimum identities, including the strict inequality on both sides. It is a proved witness statement, without an unproved extension premise. For an algebraic threshold $u = c^{1/t}$, positive integers c, t , the comparison is the exact integer inequality $b^t < c^{|A|}$.

There is a concrete monotone sequence approaching λ_k from above: minimize $b^{1/|A|}$ over certificates with $2 \leq b \leq B$. For each B this domain is finite, because A is a subset of $[1, b-1]$ with $S(A) < b$ and $|A|(|A|+1)/2 < b$. For $B \geq 3$ it is nonempty. Exhausting B exhausts all certificates, so its limit equals λ_k . The theorem supplies convergence; this contribution supplies no effective error bound for that convergence.

6.2 Full certificates and exact generator distinctness

A finite positive set A is **b -separated** when no two of its elements have ratio b^t for an integer $t \geq 1$. This is a finite decidable predicate: for each a , multiply successively by b until exceeding $\max A$, checking membership at each step. The map

$$q_m : \{0, \dots, m-1\} \times A \rightarrow \bigcup_{j < m} b^j A, \quad q_m(j, a) = b^j a$$

is surjective. It is injective for every m exactly when A is b -separated. Indeed, equality $b^i a = b^j a'$ either gives $i = j$ and $a = a'$, or, after taking the smaller index, gives precisely a power ratio. Conversely, $a' = b^t a$ gives the explicit collision $q_{t+1}(0, a') = q_{t+1}(t, a)$.

A **full certificate** consists of $b \geq 2$ and a nonempty b -separated set A of distinct positive integers, together with a safe full graph for $D = H(A)$. There is no restriction $S(A) < b$ in this definition. Write its class as F_k .

Theorem 6.2. For every $k \geq 3$,

$$\lambda_k = \inf_{(b, A) \in F_k} b^{1/|A|}.$$

Every full certificate gives, for all $n \geq 1$, $g_k(n) \leq (\max A) b^{\lceil n/|A| \rceil - 1}$.

Proof. Injectivity of q_m gives exactly $m|A|$ distinct generators. The map from subsets of these generators to their block digit words is surjective, and evaluation gives their actual subset-sum image. Because word evaluation may now have multiple fibers, the full multiplicity identity is

$$\mu_{A^{[m]}}(y) = \sum_{\substack{d \in H(A)^m \\ \sum_j b^j d_j = y}} \prod_{j < m} \mu_A(d_j).$$

This follows directly by partitioning the binary generator choices according to their block sums. Theorem 4.2 certifies the actual value image. The maximum is $(\max A) b^{(m-1)}$, and deletion gives the stated finite bound. Therefore $\lambda_k \leq b^{1/|A|}$ for each full certificate. Every canonical carry certificate is a full

certificate, because its generators lie in $[1, b)$ and its digit language has the same progression property under the full equivalence. Theorem 6.1 provides the reverse infimum inequality. QED.

The separation check is retained even when a digit-language check is available. For $b=19$ and $A=\{1,7,8,19\}$, q_m has $m-1$ duplicate pairs, and its image has $3m+1$ elements, compared with $4m$ indexed generator positions. For instance, $q_2(0,19)=q_2(1,1)=19$. The actual image embeds in the indexed system by choosing, for each image value, its least-level preimage; the resulting injection of finite subsets preserves their sums. It need not cover arbitrary indexed selections. The verifier detects this power alias before accepting a generator-cardinality certificate; it never silently deletes a generator and retains the original cardinality claim.

7. Explicit applications beyond $k=4$

The small certificates are

$$(k, b, A) = (5, 23, \{1, 3, 4, 7\}), \quad (6, 47, \{1, 2, 6, 7, 14\}).$$

Their digit sets are respectively

$$D_{23} = \{0, 1, 3, 4, 5, 7, 8, 10, 11, 12, 14, 15\}$$

and

$$D_{47} = \{0, 1, 2, 3, 6, 7, 8, 9, 10, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 27, 28, 29, 30\}.$$

To make the finite modular fact inspectable, for each positive step up to half the prime modulus the maximum number of consecutive terms in D is:

b	steps d in order	maximum run lengths in that order
23	1,...,11	3,4,4,4,4,2,3,4,3,4,4
47	1,...,23	5,3,4,4,3,5,5,4,3,4,3,3,3,3,2,4,2,4,4,4,3,4

Each row is verified by starting at every residue and iterating the stated step until leaving D . The reversal bijection $(x_0, \dots, x_{k-1}) \mapsto (x_{k-1}, \dots, x_0)$ exchanges d and $-d$, so the table covers every nonzero step. The executable additionally checks all $b(b-1)$ ordered start/step pairs directly, without using that reduction.

A larger exact search gives the six-generator block

$$A_* = \{1, 4, 5, 17, 21, 22\}, \quad S(A_*) = 70.$$

Its 38 distinct subset sums are

$$D_* = \{0, 1, 4, 5, 6, 9, 10, 17, 18, 21, 22, 23, 25, 26, 27, 28, 30, 31, 32, \\ 38, 39, 40, 42, 43, 44, 45, 47, 48, 49, 52, 53, 60, 61, 64, 65, 66, 69, 70\}.$$

This is modularly 5-AP-free at $b=97$ and modularly 6-AP-free at $b=93$. The second check includes all nonzero steps of the composite modulus, including steps of order 3 and 31. In each case the complete reachable carry certificate consists of the initial vertex alone. The exhaustive finite arithmetic is included in the checker and receipt.

Theorem 7.1 (new explicit upper constructions). For every $n \geq 1$,

$$g_5(n) \leq 22 \cdot 97^{\lceil n/6 \rceil - 1}, \quad g_6(n) \leq 22 \cdot 93^{\lceil n/6 \rceil - 1}.$$

Consequently $\lambda_5 \leq 97^{1/6}$ and $\lambda_6 \leq 93^{1/6}$. These statements follow by applying the fully proved lift to the displayed finite certificates. The smaller certificates also give $g_5(n) \leq 7 \cdot 23^{\lceil n/4 \rceil - 1}$ and $g_6(n) \leq 14 \cdot 47^{\lceil n/5 \rceil - 1}$, useful for some finite n . The minimum of the corresponding finite upper expressions may be used.

The target progression length is retained in the certificate. At base 93, the same block's two-level subset sums contain the five-term progression

9, 990, 1971, 2952, 3933

with difference 981. Its digit columns are (9,60,18,69,27) and (0,10,21,31,42), all entries in D_* . The prefix map from six-term progression tuples to five-term tuples is the restriction of $Z^6 \rightarrow Z^5$. It preserves the common difference. This particular five-term tuple has no preimage: its forced sixth term is 4914, whose residue 78 modulo 93 is absent from D_* . Thus the base-93 result is certified for $k=6$, while base 97 supplies the stated $k=5$ construction.

For a sharper remainder-sensitive version, list a certificate's generators as $a_1 < \dots < a_s$, write $n = sq + r$ with $0 \leq r < s$, and retain q complete levels and then the first r generators at level q . When $r > 0$ the maximum is $a_r b^q$; when $r=0$ it is $a_s b^{(q-1)}$. Positivity and disjoint level ranges prove the cardinality and the bound exactly.

The comparisons with Korsky's stated general graph bound are exact: for $k=5$ its rate is $\sqrt{5}$, and $97 < 5^3$; for $k=6$ its rate is $7^{2/5}$, and $93^5 < 7^{12}$. These comparisons identify improvement over that retrieved bound, without a claim of global literature priority.

8. An explicit correlated cube behind the six-generator block

Let the six columns in Z^3 be

$$v_1 = (1, 0, 0), \quad v_2 = (0, 1, 0), \quad v_3 = (1, 1, 0), \quad v_4 = (0, 0, 1), \quad v_5 = (0, 1, 1), \quad v_6 = (1, 1, 1).$$

Let C be the image of the binary cube under $\varepsilon \mapsto \sum_i \varepsilon_i v_i$. It lies in $[0,3] \times [0,4] \times [0,3]$.

Proposition 8.1. C has no nonconstant five-term arithmetic progression over Z^3 .

Proof. In a five-term progression, the first and third coordinates have integer common differences and span at least 4 times their absolute values. Their ranges have width 3, so both common differences are zero. A remaining nonzero middle difference must be ± 1 and use both middle levels 0 and 4. Middle level 0 forces the choices for columns 2,3,5,6 all to vanish; its first coordinate is at most 1. Middle level 4 forces all four of those choices to be present; its first coordinate is at least 2. They cannot share the required first coordinate. Thus every common difference is zero. QED.

The homomorphism $L : Z^3 \rightarrow Z$, $L(x,y,z) = x + 4y + 17z$, maps these columns to A_* . Direct enumeration gives $|C| = 38$ and $|L(C)| = 38$, so its restriction $C \rightarrow H(A_*)$ is a bijection. However the second-difference equations after projection require their own verification: $|C+C| = 201$ whereas $|L(C+C)| = 139$. The exact surjection $L : C+C \rightarrow T(A_*)$ retains this distinction; its fibers are included in the executable checks. The modular certificate in Section 7 verifies the required progression property after projection. The three-dimensional argument motivates the construction without silently asserting a Freiman equivalence that the sumset cardinalities contradict.

9. Overlapping relations of arbitrary extent already occur at $k=5$

Theorem 9.1. For every $r \geq 2$ the set

$$A_r = \{5^i : 0 \leq i < r\} \cup \{5^i + 5^{i+1} : 0 \leq i < r-1\}$$

has $2r-1$ distinct positive generators, has 5-AP-free subset sums, and has minimal signed relations whose intersection graph contains a path of $r-1$ vertices. Consequently the relation interaction has arbitrarily large connected extent for every $k \geq 5$.

Proof. Label generators $v_i = 5^i$ and $e_i = 5^i + 5^{i+1}$. Their standard base-5 digit vectors are respectively the unit vector u_i and $u_i + u_{i+1}$, which proves distinctness. The exact column homomorphism from $Z^{\{2r-1\}}$ to Z^r sends the generator labels to those columns; evaluation $\eta(x) = \sum_i x_i 5^i$ gives the actual integer weights. Each subset contributes at most three units at each position, one from its vertex and at most two from incident edges. Thus $H(A_r)$ lies in the base-5 language with digits $\{0,1,2,3\}$, which is modularly 5-AP-free and hence safe by Section 5.

The signed relation q_i has coefficients $+1$ on v_i, v_{i+1} , -1 on e_i , and zero elsewhere. It is minimal: a relation supported on a proper nonempty subset would have one or two coordinates, incompatible with positivity and distinctness of the weights. Consecutive q_i share v_{i+1} , producing the claimed path. The map $\Gamma : Z^{r-1} \rightarrow \ker(\Phi_{A_r})$, $\Gamma(t) = \sum_i t_i q_i$, is injective because its e_i coordinate is $-t_i$. This is an explicit

linear embedding of the entire path relation system into the actual generator kernel. Admissibility for larger k follows from the prefix map on progression tuples. QED.

The old four-term splitting algebra remains visible. For adjacent edges, $q_i + q_i(i+1)$ has magnitude-two layer consisting of the single nonzero weight $v_i(i+1)$. It gives the actual four-term progression

$$v_i + v_{i+2}, \quad v_i + v_{i+1} + v_{i+2}, \quad e_i + e_{i+1}, \quad e_i + e_{i+1} + v_{i+1},$$

with common difference $v_i(i+1)$. At $r=3$ these values are 26,31,36,41 in $H(\{1,5,25,6,30\})$. The restriction of the coordinate projection $Z^5 \rightarrow Z^4$ gives the prefix map $AP_5(H(A_r)) \rightarrow AP_4(H(A_r))$ preserves common differences; the displayed 4-AP lies outside its image. Its witness and the 5-AP-free proof identify exactly which step of a four-term-only decomposition cannot be imposed on the general- k domain.

10. A proved boundary for all positive interval-digit column models

Proposition 10.1 (full positive-column budget). Let M be an r by n nonnegative integer matrix with distinct nonzero columns. Let every row sum be at most τ . Then

$$2n \leq r(\tau + 1).$$

Proof. At most r columns have coordinate sum 1, since they must be unit vectors. Every other column has coordinate sum at least 2. The total column mass is therefore at least $2n-r$ and is exactly the total row mass, which is at most $r\tau$. Combining the exact counts proves the inequality. QED.

The column map $M:Z^n \rightarrow Z^r$ and the evaluation homomorphism $\eta_b(x) = \sum_{i=0}^{r-1} x_i b^i$ give the integer generators. When $\tau < b$, evaluation is injective on the full subset-coordinate box; its inverse is the actual base- b digit expansion. For prime b and $\tau = \min(b, k) - 2$ this is the positive interval-digit construction. The budget forces $r \geq \lceil 2n / (\min(b, k) - 1) \rceil$. With r active positions, some generator has value at least $b^{(r-1)}$. Thus, for each fixed prime b , that entire column class has exponential rate at least $b^{(2/(\min(b, k) - 1))}$; the graph construction attains that rate asymptotically. This is a statement about the precisely displayed column domain and maps.

An explicit failure of a proposed universal passage into that domain is supplied by $A = \{1, 3, 4, 7\}$. For any assignment of nonzero nonnegative columns w_1, w_3, w_4, w_7 preserving its subset equalities,

$$w_4 = w_1 + w_3, \quad w_7 = w_3 + w_4, \quad \sum_{a \in A} w_a = 3w_1 + 4w_3.$$

Some coordinate of w_3 is at least 1, and that row load is at least 4. Hence every such assignment violates a row-load bound of 3. The typed relation-preserving map $M:Z^4 \rightarrow Z^r$ has to send the two displayed relation vectors to zero; those equations prove the obstruction in every dimension. The actual scalar map (coefficients 1,3,4,7) and the modular projection to $Z/23$ are exhibited in Section 7 and produce a valid 5-AP-free construction. Thus the carried and correlated digit route has a concrete mathematical reason to be pursued beyond positive interval boxes.

11. What has been established and what remains to be evaluated

The proved all- k conclusions are the exact product, the existence of each λ_k , the complete certificate-capacity formulas, and the finite canonical and full carry decisions with an explicit witness-length bound. The explicit $k=5$ and $k=6$ certificates improve the retrieved general upper estimates. The path construction and positive-column count prove two precise structural boundaries for possible lower-bound arguments.

The exact numerical values of λ_k for general $k \geq 5$, and the sharp large- k behavior of $\log \lambda_k$, remain unevaluated here. The current lower and upper mechanisms leave a gap between order $1/k$ and order $\log(k)/k$. The new capacity identity identifies the exact optimization across all finite certificates, including every admissible integer set through its explicit map $A \rightarrow (2S(A)+1, A)$. A bounded search supplies exact finite data and decreasing upper bounds; it supplies no lower certificate for unsearched bases. Independent review should focus on Sections 2,4,6 before using the capacity theorem, and on the modular transcripts before citing new constants.

12. Reproducible verification and exact search domains

The standalone command, run from this contribution's directory, is

```
python certificates/verify_general_k.py --output certificates/general_k_receipt.json
```

The resulting receipt reports PASS for the following completely specified finite domains. The general implications are proved in Sections 2--6; these checks independently test their finite maps, graph transitions, and witnesses.

Check	Domain and result
Primary modular certificates	Every start and nonzero step: 342 pairs for base 19; 506 for base 23; 2,162 for base 47; 9,312 for base 97; 8,556 for base 93
Canonical carry graph	All D subset $[0, b)$ containing zero, $2 \leq b \leq 8$, $3 \leq k \leq 6$: 1,016 instances; 415 safe and 601 with explicit witnesses
Independent canonical closure	Direct enumeration of digit columns verifies 1,433 transitions across the safe certificates
Full carry graph	All D subset $[0, 6]$ containing zero, $b=2,3,4$, $k=3,4$: 384 instances; 30 safe and 354 with explicit witnesses
Independent full closure	122 transitions across the small safe cases, plus 288 transitions for the noncanonical $\{7,8,19\}$ example
Direct finite languages	All graph-domain cases at word lengths 1,2,3: 4,200 comparisons with separately generated integer sets
Exact composition	256 ordered pairs of empty, singleton, or two-element subsets of $[1,5]$; 1,024 full AP-tuple comparisons for $k=3,4,5,6$
Associativity	All 4,096 ordered triples from the same 16-set domain
Primary generator lifts	Two levels for each of five primary certificates; complete binary fiber multiplicities and remainder-sensitive deletions
Structural maps	The 38-point correlated cube, all 201 points in its sumset and all 139 projected values; path-relation family $r=2,3,4,5$
Regressions	Two modular/carry bridge examples, five decoded failed-base/progression-length constructions, power-alias cardinalities, and nine rejected invalid inputs or mutations

The canonical and full graph outputs are additionally compared on every canonical input with $b \leq 6$ and $k \leq 5$. Two independent transition generators are used: the search solver builds columns recursively by congruences, while the closure checker directly enumerates D^k . Negative outputs retain digit columns, which the checker evaluates independently to the displayed integer progression.

The constructive search is exhaustive over every positive generator set with sum strictly less than its base for $k=5,6$ and bases 3 through 160. A second search uses modular certificates for $k=7$ through 12 and bases 3 through 80. These are 784 base/length parameter records. The generic search has a depth cap of eight, but its largest returned generator counts are six in the first domain and seven in the second. The cap therefore leaves no larger admissible set hidden: any such set would have an admissible eight-element subset. The verifier separately rechecks all 784 recorded witnesses. It does not replay the entire maximality search by default.

The exhaustive-search pruning has explicit justifications. Generators are visited in strictly increasing order. For a partial tuple, the minimum sum needed to exceed the current best cardinality is an arithmetic-series sum; branches exceeding the base budget are rejected. A failed partial digit set cannot become safe after adding generators, because its progression word remains in every larger alphabet. The search retains the actual generators and only caches the progression property of an already computed digit set. No multiplication by units, gcd removal, or re-centering of an input set is used. All comparisons of exponential costs are exact cross-power integer comparisons.

The search can be replayed with

```
python search_carry.py --mode carry --min-base 3 --max-base 80 --k 5 6
python search_carry.py --mode carry --min-base 81 --max-base 160 --k 5 6
python search_carry.py --mode modular --min-base 3 --max-base 80 --k 7 8 9 10 11 12
```

A further retained full-carry probe fixes A_{**} and tests bases 23 through 70 for both $k=5$ and $k=6$. All 96 instances produce two-level progression witnesses. The complete witnesses are in `logs/noncanonical_fixed_block.json`; this is a failure of those precise base/block pairs, with the map to integer progressions supplied by Theorem 4.2. It provides no obstruction to different blocks at the same bases.

A repeated verifier run with Python's optimization flag `-O` gives the same receipt bytes. The program uses explicit exception-raising requirements, so optimization cannot remove its checks. Source and transcript SHA-256 hashes identify exact inputs and outputs. Lean elaboration was not run in this contribution; its certificate type is an ordinary mathematical proof plus finite exact verification, and existing Lean artifacts are preserved.

References, inspected sources, and contribution lineage

Costa, S. (2026). *A negative answer to the Erdős–Sárközy question* (Version 1) [Preprint]. arXiv:2609.06303. Source: <https://arxiv.org/html/2609.06303v1>. Inspected HTML, especially abstract and Section 1; used solely for the current historical $k=3$ question.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (Version 1) [Preprint]. arXiv:2606.24139. Source: <https://arxiv.org/html/2606.24139v1>. Inspected HTML, Sections 1,5,6; used for the problem definition, known chain mechanism, and the comparison upper bound.

Anonymous/deleted Reddit contributor, & The Clankers. (2026, September 13). *The exponential rate for four-term-progression-free subset-sum sets* [Workbench manuscript]. `paper/ep817_k4_rate.tex` at the base revision above. The original construction and proof route belong to the anonymous contributor.

sneed-and-feed. (2026, September 13). *Formalize finite upper bound theorem and core helper lemmas in Extended.lean* [Pull request #1]. Same workbench.

The Clankers. (2026, September 14). *An unweighted ternary-variance refinement of the $k=4$ lower bound* [Draft pull request #2], commit `612020848e4bbec5f5a723a64c5fdc85636b5e83`. Parallel predecessor contribution; its separate review and Lean status are retained.

PolyClank methodology: `erdos-strauss-foundation/POLYCLANK.md` (inspected) and `yang-mills-interacting-workbench/docs/polyclank/RESEARCH_STATE.md` at `7ba7ea5a5dab84d88934477de93dda71f6af0662` (inspected). These supply the public research-record convention, with claim-local evidence and source lineage; they supply no additional mathematical theorem used in this note.

Support-preserving outer control for Erdős Problem 817

Source: `history/EP817_SPLITZERO_OUTER_20260914/research/splitzero_outer/notes/outer-support-control.md`

Identity: `cf87ac7e695ad4b64b80e2e0049f8b2088539973426b49bc046e500fa7a1a543` Research contribution, 14

September 2026. Ordinary proofs and exact finite proof certificates; independent mathematical review remains appropriate. No new Lean elaboration is claimed.

1. Scope, sources, and mathematical outcome

For a finite set of distinct positive integers A , retain

$$H(A) = \left\{ \sum_{a \in U} a : U \subseteq A \right\}, \quad S(A) = \sum_{a \in A} a.$$

Zero is included as the value of the actual empty subset. A nonconstant ordered k -term arithmetic progression has the form $(x, x+d, \dots, x+(k-1)d)$, with an integer d different from zero. For k at least three,

$$g_k(n) = \min\{\max |A| : |A| = n, H(A) \text{ contains no such progression}\}.$$

The connected Zeta workbench was inspected at commit `42df8a2de002d5fc7090641fac46ea11be05fa71`. The interfaces used here are its support-index reconstruction and internal quotient, together with its original-source Gram and section-correction formulas. In particular, a zero at an admitted support retains that support, and every change of observation or metric has an explicit comparison map. The Zeta arithmetic determinant estimates themselves are not hypotheses in any result below.

A separate existing Erdős 817 branch was found at commit `98fd25802a387138053064e11ddd2f6d2007d55b`, path `research/splitzero_transfer/`. That branch already proves uniform mixed-radix relation inverses, arithmetic graph completion, and finite-controller optimization for a fixed generator block. Its results and attribution are retained as predecessor work. The contribution here controls the additional variation of the block and its integer parameters, and evaluates the resulting optimization through five generators per level.

For a fixed block A of size n and an infinite sequence of radices $b_j > S(A)$, define $P_0 = 1$ and $P_{j+1} = b_j P_j$. Its m -level generator set is the literal set

$$A_{\mathbf{b},m} = \bigcup_{j < m} P_j A.$$

It has exactly mn generators. A schedule is admissible when every finite prefix has k -AP-free subset sums. Define

$$\gamma_k(A) = \inf_{\text{admissible schedules}} \liminf_{m \rightarrow \infty} \left(\prod_{j < m} b_j \right)^{1/(mn)}, \quad \Gamma_{k,n} = \inf_{|A|=n} \gamma_k(A). \quad (1.1)$$

An empty infimum is infinity. The expression uses the original radix product. Its relation to the largest generator at the next level is exact:

$$\max A_{\mathbf{b},m+1} = P_m \max A. \quad (1.2)$$

Taking the indicated roots gives the same limiting inferior, including when it is infinite.

The main outcomes are the following.

Theorem O1 (complete bounded realization). Fix n positive and ℓ positive. Every canonical integer tuple $(a_1, \dots, a_n, b_1, \dots, b_\ell)$, with $a_1 < \dots < a_n$ and $\sum a_i < \min b_j$, has another tuple in precisely the same affine sign cell, coordinatewise no larger, with every coordinate at most

$$\boxed{3 \cdot 2^{n+\ell-1} (n + \ell)!}. \quad (1.3)$$

The sign cell records all labels (j, u, v, c) with u in $\{-2, -1, 0, 1, 2\}^n$ and v, c in $\{-1, 0, 1\}$, including every zero value of $u \cdot a - v b_j + c$. The two tuples have exactly corresponding binary representation fibers and complete labeled carry relations. Their arbitrary finite mixed-radix languages are order-preserving Freiman-2 isomorphic. The original tuple is recoverable from the bounded tuple and a retained finite list of nonnegative integer directions and coefficients.

Theorem O2 (finite global fixed-rank optimization). For every $k \geq 3$ and $n \geq 1$, $\Gamma_{\{k,n\}}$ is attained and is an exactly computable algebraic number. Set

$$V_k = 2^{3^{k-2}-1}.$$

A universal finite search suffices: all n -element positive blocks with $S(A) < 3^n$, all radices $S(A)+1$ through $2S(A)+2$, and periods of length at most V_k . The actual arithmetic upper bound 3^n may be replaced by any independently certified constant-radix benchmark Q of size n . This gives $S(A) < Q$ and radices at most $2Q$. The smaller optimization cutoff follows from the objective and is separate from the all-pattern realization bound (1.3).

Theorem O3 (exact viability by original return primitives). For every fixed canonical digit image, an explicitly constructed finite set of returning carries identifies precisely which reachable controller states have infinite admissible extensions. A returning carry has an actual next-column primitive producing a progression for every next radix. Every remaining controller state is returned to the initial state by the explicit tail radix $2S(A)+2$. The viable controller has at most 2 to the power of the number of nonreturning carries vertices.

Theorem O4 (evaluated global capacities). The values below optimize over every distinct positive block of the stated size and every infinite canonical mixed-radix schedule with that same block at each level:

generators per level n	$\Gamma_{\{5,n\}}$	$\Gamma_{\{6,n\}}$
1	$8^{(1/2)}$	$8^{(1/2)}$
2	$5^{(1/2)}$	$5^{(1/2)}$
3	$13^{(1/3)}$	$13^{(1/3)}$
4	$23^{(1/4)}$	$23^{(1/4)}$
5	$65^{(1/5)}$	$47^{(1/5)}$

The lower assertions use the proved finite cutoff and the complete exact certificates described in Section 8. Thus the finite enumeration domain is sufficient for the entire indicated infinite optimization. These are computational proof certificates rather than samples from the infinite domain.

The global rate still satisfies

$$\lambda_k = \lim_{n \rightarrow \infty} g_k(n)^{1/n} = \inf_{n \geq 1} \Gamma_{k,n}. \quad (1.4)$$

This contribution evaluates the displayed entries and gives a terminating algorithm for every additional fixed (k,n) . It does not evaluate the infimum over unbounded n for general k . Section 10 retains the exact remaining growth expression and the metric cost relevant to that last quantifier.

The anonymous/deleted Reddit contributor retains credit for the original $k=4$ mathematical construction and signed-block argument. **sneed-and-feed** retains separate credit for the existing finite-upper-bound Lean formalization. The present work does not assume the variance contribution in PR #2 and does not relabel any existing proof receipt.

2. The literal SplitZero support and quotient construction

The inspected Zeta source works with the split semiring $G(R) = R \hat{\cup} \text{disjoint union } \{\tau\}$. The element $0 \hat{\cup}$ is a supported coefficient zero; τ is the external additive zero. Its support-diagram presentation starts with a join-semilattice L and modules V_l with compatible transition maps. The total carrier is the disjoint union of the V_l , addition transports to the join, supported zero sends (l, x) to $(l, 0)$, and τ sends it to the bottom zero.

Here take $R = \mathbb{Q}$ and fix a finite horizon m . Let Ω_m be the literal set of binary choices at all n generators and all m levels. Define

$$\text{ev}_{A, \mathbf{b}}(\omega) = \sum_{j < m} P_j \sum_{i=1}^n \omega_{j,i} a_i.$$

Use $L_m = P(\Omega_m)$, with union as join and empty set as bottom. For S in L_m , put

$$V_S = \mathbb{Q}[S], \quad Y_S = \text{ev}(S), \quad W_S = \mathbb{Q}[Y_S], \quad \pi_S(e_\omega) = e_{\text{ev}(\omega)}, \quad R_S = \ker \pi_S. \quad (2.1)$$

These are free coefficient spaces on the displayed sets. Every word is still present, even when several words have the same numerical value. For S subset T , the original inclusions of words and values commute with π and map R_S into R_T . Appending a zero binary level gives the corresponding join-preserving transition from L_m to L_{m+1} . The horizon and its actual embedding are retained.

The two-term complex $R_S \rightarrow V_S$, with differential the inclusion, has

$$H^{-1} = 0, \quad H^0 \simeq W_S. \quad (2.2)$$

For each value y choose the first word ω_y in its fiber under a specified lexicographic ordering. The vectors $e_{\omega_y} - e_{\omega'_y}$, for all other words in that fiber, are a basis of R_S : subtracting their coefficients leaves the fiber sum at ω_y , which vanishes exactly for a relation. This proves both the relation description and the quotient isomorphism without deleting any original word.

Applying the source's reconstruction functor gives

$$\mathcal{T}(L_m, V) \rightarrow \mathcal{T}(L_m, W), \quad (S, z) \mapsto (S, \pi_S z). \quad (2.3)$$

A relation is sent to $(S, 0)$. The empty support has its separate bottom zero. The empty subset of generators is an actual word evaluating to zero; the support containing that word has a nonzero one-dimensional value fiber. This also records the empty-word case explicitly.

There are two observation maps in (2.1). The value-fiber map π is followed by the arithmetic amplitude map

$$w_S : W_S \rightarrow \mathbb{Q}, \quad w_S(e_y) = y.$$

Their composition is the original scalar numerical evaluation. Their relation spaces are connected by the exact sequence

$$0 \rightarrow \ker \pi_S \rightarrow \ker(w_S \pi_S) \xrightarrow{\pi_S} \ker w_S \rightarrow 0. \quad (2.4)$$

Surjectivity at the right follows by lifting each value-basis coefficient to any word in its fiber. Thus the additional scalar-observation kernel is retained, just as the Zeta source retains the difference between original boundaries and a larger observation kernel.

3. Source metrics, section corrections, and the tensor return cost

Give the literal word basis its original orthonormal metric. Define

$$\mu_S(y) = |\{\omega \in S : \text{ev}(\omega) = y\}|.$$

The minimum-norm section of π is

$$s_S(e_y) = \frac{1}{\mu_S(y)} \sum_{\text{ev}(\omega)=y} e_\omega. \quad (3.1)$$

It is orthogonal to R_S . Every lift z of v in W_S obeys the exact identity

$$\|z\|^2 = \|z - s_S v\|^2 + v^* G_S v, \quad G_S = \text{diag}_{y \in Y_S} (\mu_S(y)^{-1}). \quad (3.2)$$

The finite quotient Gram therefore records the actual representation multiplicities. This is the finite fiber version of the original-source Schur/orthogonal-section identity inspected in the Zeta workbench.

For S subset T and y in Y_S , the vector $s_T(e_y) - i s_S(e_y)$ belongs to the original relation space R_T . Its squared norm is exactly

$$\|s_T(e_y) - i s_S(e_y)\|^2 = \frac{1}{\mu_S(y)} - \frac{1}{\mu_T(y)}. \quad (3.3)$$

To verify this, expand in the original word basis. The two sections have norms squared $1/\mu_T$ and $1/\mu_S$, while their inner product is $1/\mu_T$. No independent change of metric is introduced.

For a full canonical m -level block, unique mixed-radix digits give

$$\mu_m(d_0, \dots, d_{m-1}) = \prod_{j < m} \mu_A(d_j), \quad G_m = G_A^{\otimes m},$$

$$\det G_m = (\det G_A)^{m|H(A)|^{m-1}}, \quad \lambda_{\min}(G_m) = \mu_{\max}^{-m}. \quad (3.4)$$

The identity on value coordinates, regarded as a map from metric G_m to the unit value metric, has norm exactly $\mu_{\max}^{\wedge(m/2)}$. For $A=\{1,7,8\}$, six digits have multiplicity one and digit 8 has multiplicity two. Thus

$$\det G_m = 2^{-m7^{m-1}}, \quad \|I : (W_m, G_m) \rightarrow (W_m, I)\| = 2^{m/2}. \quad (3.5)$$

For $A_*=\{1,4,5,17,21,22\}$, the multiplicity histogram is 24 digits of multiplicity one, eight of multiplicity two, and six of multiplicity four. All 64 subset choices are present. Its metric-return norm is $2^{\wedge m}$. Uniform boundedness of a carry inverse, in the predecessor's specified coefficient metric, does not remove this separately computed source-to-value cost. Equations (3.2)–(3.5) are the exact comparison maps and constants relevant here.

4. Complete affine signatures and an all-length arithmetic transport

Fix a positive block tuple $a=(a_1, \dots, a_n)$ in its displayed increasing order and an ordered list of ℓ canonical radices $b=(b_1, \dots, b_{\ell})$. For every label

$$(j, u, v, c), \quad 1 \leq j \leq \ell, \quad u \in \{-2, -1, 0, 1, 2\}^n, \quad v, c \in \{-1, 0, 1\},$$

record the sign of the literal affine form

$$F_{j,u,v,c}(a, b) = u \cdot a - v b_j + c. \quad (4.1)$$

There are $9 \ell 5^n$ labels, including forms that are identically zero and supported zero evaluations of nonzero forms. All labels remain in the signature. Identical inequality rows can be indexed together for a polyhedral calculation, while (4.1) retains every original label.

Proposition O5 (signature transport). Two tuples with the same complete signature have canonically corresponding binary value fibers and, for any finite word in the ℓ radix labels, order-preserving Freiman-2 isomorphic numerical digit languages. The correspondence retains every binary representation and every ordered progression tuple, at every length and every progression length k .

Proof. The signs with $v=c=0$ and u in $\{-1,0,1\}^n$ specify equality and order of all binary sums. The identity on binary masks therefore induces an order-preserving bijection $\theta: H(A) \rightarrow H(A')$. For a finite word of radix labels $j_0, \dots, j_{m-1}$, define

$$\Theta_m = \text{ev}_{\mathbf{b}'} \circ \theta^{\times m} \circ \text{ev}_{\mathbf{b}}^{-1}. \quad (4.2)$$

Both evaluations have their unique actual canonical-digit inverses. Inverse θ and the reverse evaluation give the explicit inverse of Θ_m . Ordering follows from the highest differing digit and the order of θ .

Consider any pair-sum equation between four evaluated words. Its column defect has the form $u.a$ with u in $\{-2,-1,0,1,2\}^n$. With initial and terminal carry zero, equality is equivalent to the integral equations

$$c_{r+1}b_{j_r} = c_r + u_r \cdot a. \quad (4.3)$$

For a zero-total relation, the prefix carries are integral by divisibility of the remaining suffix. Since each column defect has absolute value at most $2(b_{j_r}-1)$,

$$|c_r| \leq 2(1 - P_r^{-1}) < 2.$$

Thus c_r is in $\{-1,0,1\}$. Each equation (4.3) is a supported zero of a label in (4.1), so it holds for one tuple exactly when it holds for the other. This proves preservation and reflection of every pair-sum equation. In particular it proves preservation of all consecutive second-difference equations and of nonconstancy. The source mask tuples were unchanged, so every representation fiber and positive multiplicity is retained. $\square$

Linear extension of Θ_m gives an isomorphism of the value-basis spaces, and identity on the word basis gives a commuting square with the quotient maps (2.1). The fiber complexes, support diagrams, and original quotient metrics (3.2) are therefore explicitly transported.

The scalar numerical observation is retained alongside that square. Under the value-basis map J_m , the two scalar functionals are w_m and $w'_m J_m$, with difference

$$(w'_m J_m - w_m)e_y = \Theta_m(y) - y. \quad (4.4)$$

For the unrestricted generator relation lattices, the simultaneous map is

$$\Phi_{a,a'} : \mathbb{Z}^n \rightarrow \mathbb{Z}^2, \quad z \mapsto (a \cdot z, a' \cdot z).$$

It gives the exact comparison

$$0 \rightarrow \ker \Phi_a \cap \ker \Phi_{a'} \rightarrow \ker \Phi_a \xrightarrow{\Phi_{a'}} \Phi_{a'}(\ker \Phi_a) \rightarrow 0. \quad (4.5)$$

For example, $(a,b)=((1,100),301)$ and $(a',b')=((1,4),12)$ have the same complete signature. The unrestricted relation $(100,-1)$ has old amplitude zero and new amplitude 96. Equation (4.5) records that amplitude explicitly. The preserved pair-sum structure, the additional observation kernel, and the retained original integers consequently have distinct, explicitly connected roles.

5. A bounded realization with an explicit inverse code

Here is a self-contained integer-polyhedral bound used for Theorem O1. It is a deliberately elementary bound; no sharpness or priority claim is attached to it.

Lemma O6. Let $P = \{x \in \mathbb{R}^d : A x \geq r\}$ lie in the nonnegative orthant, with integer entries $|A_{ij}| \leq 2$ and $|r_i| \leq 2$. Every x in P intersect $\mathbb{Z}^d$ has a y in the same integer polyhedron satisfying

$$0 \leq y \leq x, \quad \|y\|_\infty \leq 3 \cdot 2^{d-1} d!, \quad (5.1)$$

and a representation

$$x = y + \sum_{i=1}^p m_i q_i, \quad p \leq d, \quad m_i \in \mathbb{N}_0, \quad q_i \in \mathbb{Z}_{\geq 0}^d \setminus \{0\}, \quad A q_i \geq 0. \quad (5.2)$$

The q_i can be chosen with every coordinate at most $2^{(d-1)}(d-1)!$.

Proof. Add the nonnegative-coordinate rows when needed. The homogenized cone consists of $(t, z) \geq 0$ with $Az \geq tr$. Its intersection with $t + \sum z_i = 1$ is a bounded polytope. Expressing a point as a convex combination of its finitely many vertices proves that P is a convex hull of its vertices plus a cone generated by its recession rays. Rays with positive t give vertices of P after retaining the t coordinate in the displayed division; rays with $t=0$ give its recession directions. This also proves that P has vertices when nonempty.

At a vertex of P , d independent active rows determine its coordinates. Cramer's rule and the determinant expansion bound each coordinate by $V_d = 2^d d!$: the denominator is a nonzero integer, hence has absolute value at least one, and the numerator has at most $d!$ terms bounded by 2^d . On an extreme recession ray, $d-1$ independent active rows have a one-dimensional nullspace. Their cofactor vector, oriented to have nonnegative entries, is an integral ray vector with coordinates at most $R_d = 2^{(d-1)}(d-1)!$. No division to a primitive ray and no rescaling of the input tuple are needed.

Write $x = z + \sum t_i q_i$ with z a convex combination of vertices and $t_i \geq 0$. The recession sum can use at most d rays: a linear dependence among more directions permits subtracting an appropriate multiple of that dependence from their nonnegative coefficients until one coefficient becomes zero. Repeating leaves at most d independent directions. Every direction is nonnegative because P is contained in the nonnegative orthant.

Set $m_i = \text{floor}(t_i)$ and $y = x - \sum m_i q_i$. Then y is integer and equals $z + \sum (t_i - m_i) q_i$, so it remains in P . Nonnegativity of the q_i gives $y \leq x$. Its coordinates are bounded by $V_d + dR_d = 3 \cdot 2^{d-1} \cdot d!$. This proves every displayed assertion. $\square$

For a complete signature (4.1), replace a positive sign by $F \geq 1$, a negative sign by $F \leq -1$, and zero by $F = 0$. Integer points have exactly the original signs. Include $a_1 \geq 1$, $a_{\{i+1\}} - a_i \geq 1$, and $b_j - \sum a_i \geq 1$. All coefficient magnitudes and all right-side magnitudes are at most two. The domain lies in the nonnegative orthant. Applying Lemma O6 with $d = n + \text{ell}$ proves Theorem O1.

The retained comparison is a genuinely reversible code:

$$x \mapsto (\text{complete labeled signature}, y, (m_i, q_i)_i), \quad (y, (m_i, q_i)_i) \mapsto y + \sum_i m_i q_i. \quad (5.3)$$

A deterministic encoding can enumerate all eligible y and directions within the proved bounds and coefficients $m_i \leq \|x\|_\infty$, retaining the first valid code in a specified finite ordering. The last bound follows because each nonzero nonnegative q_i has a coordinate at least one and $m_i q_i \leq x$. Thus this is an exact finite procedure for each original x , even when its arithmetic values are large.

For the example in Section 4, the complete code is

$$(1, 100, 301) = (1, 4, 12) + 96(0, 1, 2) + 97(0, 0, 1). \quad (5.4)$$

Both directions satisfy every recession inequality of the same signature cell. Another exact example is

$$10^6(1, 7, 8, 19) = 2(1, 7, 8, 19) + (10^6 - 2)(1, 7, 8, 19).$$

The factor two in the bounded point is retained because the constant terms $c = \pm 1$ distinguish that affine cell from the scale-one cell. The checker tests every one of the original affine labels in these examples.

For a single base, the all-pattern bound is $3 \cdot 2^n \cdot (n+1)!$. At $n=2$ it is 72. The exact replay enumerates all 29,190 positive increasing two-generator tuples with sum below a base at most 72, recording 198 signatures and all needed coordinatewise minimal witnesses. Every one of the 109,460 additional tuples with bases 73 through 120 has a coordinatewise smaller representative in the corresponding retained cell. These finite tests corroborate the proof; Lemma O6 provides the unbounded assertion.

6. Original return primitives and the exact viable controller

Let $D = H(A)$ be canonical at every permitted radix $b > S$. Put

$$C_k = \{-1, 0, 1\}^{k-2}, \quad \Delta(d)_i = d_{i+2} - 2d_{i+1} + d_i, \\ F(D, k) = \{c \in C_k : \exists d \in D^k, \Delta(d) = -c\}. \quad (6.1)$$

This is a constructed finite subset, not an assumed surjectivity statement. Choose the first actual column $r(c)$ for each c in F . The inclusion and section give the commuting identity

$$\Delta r = -\iota_F. \quad (6.2)$$

Here r and the inclusion are maps of the displayed finite sets. Their free linearization sends e_{-c} to $e_{-r(c)}$, and the linearized defect map sends that vector to $-c$ in $Q^{(k-2)}$. Every linear relation among the c remains in the kernel of this last amplitude map; no linear independence of the carry labels is assumed. Every missing carry remains in the explicitly retained complementary set $K = C_k$ minus F . The source of the equation is still the admitted digit tuples, with their binary representation fibers.

A true-flag carry c describes a represented prefix with a nonzero first difference and with $\Delta y = P_m c$. Appending $r(c)$ gives the literal numerical tuple

$$Y_i = y_i + P_m r(c)_i, \quad \Delta Y = P_m(c + \Delta r(c)) = 0. \quad (6.3)$$

The first difference stays nonzero. Its earlier nonzero value has absolute value less than P_m , while the appended correction is a multiple of P_m . This witness works for every next radix, since the actual digits $r(c)_i$ are at most S . Consequently every reachable true-flag support R meeting F has no infinite admissible extension.

Conversely, suppose $H(A)$ is integer k -AP-free and R is disjoint from F . At every $b \geq 2S+2$, the bound $|c_i + \Delta_i(d)| \leq 2S+1 < b$ makes a transition possible only when its target is zero and $\Delta(d) = -c$. No true carry in R has such a column. From the false initial state, only constant columns survive because D has no integer nonconstant k -AP. Hence

$$T_{2S+2}(R) = \emptyset. \quad (6.4)$$

Repeating that radix remains safe. Thus disjointness from F is sufficient as well as necessary for an infinite extension of a reachable controller state. This proves Theorem O3.

The viable controller therefore keeps subsets R of K , with their exact labeled transfers, and discards an edge precisely when its complete target support meets F . The discarded record retains that entire target support, one selected carry in the intersection, and its actual return column. Equations (6.2)–(6.3) reconstruct the obstruction rather than silently identifying that carry with absence.

All constant false-flag paths remain separately present. The empty true-flag support means there is no nonconstant prefix branch; it does not remove the constant zero-carry source. Positive binary representation multiplicities may be carried as edge weights. The map from nonnegative counts to their support is $n \mapsto [n > 0]$, a semiring homomorphism; no cancellation of signed coefficients is used to determine reachability.

For the five-generator block $\{2, 3, 5, 17, 34\}$ at $k=5$, exactly 25 of the 27 carries have return columns. The two others are $(0, -1, 0)$ and $(0, 1, 0)$. The original all-state controller can form many transient supports, whereas the exact viable controller has at most four possible supports. The calculations retain the return certificate for each removed support. This is the mechanism that made the complete rank-five calculation tractable.

7. Finite optimization over every block of a fixed size

For every fixed A , radices $b \geq 2S+2$ have exactly the same full labeled transition table: each numerator $c + \Delta(d)$ has magnitude below b , so divisibility is precisely equality to zero. Replace such a radix by $2S+2$, retaining its old value and the exact multiplicative cost ratio $b/(2S+2)$. The word-evaluation conjugacy gives the inverse finite-language map and preserves every progression tuple.

There are therefore only $S+2$ candidate radix labels, $S+1$ through $2S+2$. The viable controller is finite. Give each edge b the cost $\log b$ and the reward n . If a cycle C has ℓ edges, its rate is

$$\left(\prod_{e \in C} b_e \right)^{1/(n\ell)}. \quad (7.1)$$

Proposition O7. The optimum for the block is the minimum of (7.1) over its reachable directed simple cycles. A minimizing period is safe when started at the initial empty true-flag support.

Proof. Let ρ be the least log-cost per edge among those cycles. Removing cycles from any finite walk leaves a simple path of bounded length. Every removed cycle has nonnegative sum of $\log b - \rho$, and the finitely many remaining paths have a common finite lower bound. Division by walk length proves the lower limiting bound. Repetition of a minimum cycle attains it after a finite access path. The period also works from the initial state alone: transfer maps are monotone on supports, and the empty support is contained in the selected cycle state. Induction places every repeated-period prefix inside the corresponding safe cycle support. $\square$

There are at most V_k safe controller states even without return pruning, so $\ell \leq V_k$. Cycle rate comparisons are exact integer comparisons of products raised to integer powers. A period also supplies an original constant-radix block: take its actual union of $P_j A$ and base product Q . This block has $n \ell$ distinct generators and sum at most $Q-1$. Its repeated lifts are exactly the original periodic construction.

Now take a certified n -generator canonical benchmark (B, Q) with its constant schedule safe. It gives $\Gamma_{\{k, n\}} \leq Q^{1/n}$. Every block with $S(A) \geq Q$ has every radix at least $Q+1$, hence every schedule has rate at least $(Q+1)^{1/n}$, strictly exceeding the benchmark. Thus all minimizing blocks satisfy $S(A) < Q$. There are finitely many such literal positive sets. Combining this bound with Proposition O7 proves the benchmark form of Theorem O2.

For a universal benchmark, take $B = \{1, 3, \dots, 3^{n-1}\}$ and $Q = 3^n$. Its binary base-three digit language is three-term-progression-free by the least-column argument, and its modular language modulo 3^n has the same property: a nonzero modular step has a least nonzero base-three position, contradicting the binary digit condition there. It is consequently k -AP-free for every $k \geq 3$. Its sum is $(3^n - 1)/2 < Q$. This proves the advertised universal cutoff and attainment.

Theorem O1 supplies a different type of control: every full affine pattern has a bounded realization with an inverse code, including patterns far from optimal. The benchmark cutoff is sharper for optimization because it uses the actual rate objective. Both keep their own explicit comparison maps and scopes.

8. Exact finite proofs of the globally optimal small-rank capacities

The upper certificates used in the table are literal blocks:

n	k	block	base or period
1	5,6	$\{1\}$	(2,4)
2	5,6	$\{1,2\}$	5
3	5,6	$\{1,3,4\}$	13
4	5,6	$\{1,3,4,7\}$	23
5	5	$\{1,2,5,15,20\}$	65
5	6	$\{1,2,6,7,14\}$	47

The first periodic construction has rate $\sqrt{8}$ per generator. All other displayed upper bounds come from a single radix. Composite moduli are checked with every nonzero modular step, including steps whose order is smaller than k .

For the lower proofs, the benchmark theorem supplies an exhaustive integer domain. Every block in that domain is assigned one of three complete certificates.

An actual one-level integer k -AP excludes every radix schedule for that block. Otherwise the checker constructs F and r from (6.1)–(6.2). When $F = C_k$, every modularly bad radix produces the explicit two-level obstruction (6.3), independently of the following radix. Enumerating each radix from $S+1$ to the first modularly safe base therefore gives the exact block capacity. Every earlier base has its actual two-column witness, and the first safe base receives a separate full modular check.

For the remaining blocks, the verifier retains the entire viable controller, all its safe edges, every discarded support and return primitive, and an exact positive rational potential h . At a target base Q , every surviving edge satisfies

$$\boxed{bh(R') \geq Qh(R)}. \quad (8.1)$$

Multiplying along a path gives

$$\prod_{j < m} b_j \geq Q^m \frac{h(R_0)}{h(R_m)}. \quad (8.2)$$

The finitely many positive potentials retain a bounded endpoint factor, so every infinite schedule has rate at least $Q^{1/n}$. The larger-radix branch preserves (8.1) because its transfer is the same and its cost is larger. For a target $P^{1/\ell}$, the same computation uses $b^{\ell} h(R') \geq P h(R)$. This handles the $\sqrt[8]{8}$ singleton period without rounding its cost.

The exact rank-five domains and proof decomposition are:

k	all five-element blocks with $S < Q$	one-level admissible	complete return sections	remaining viable controllers
5, $Q=65$	41,185	2,403	2,172	231
6, $Q=47$	6,074	168	150	18

The $k=5$ controllers have at most two reachable viable states; the $k=6$ controllers have one. The retained safe-edge counts are 5,054 and 279, respectively. Their discarded edge records number 9,594 and 541. All returning primitive rows, including rows used in the controller branch, are retained: 64,163 rows at $k=5$ and 13,458 at $k=6$.

The independent auditor imports neither producer. It reconstructs the complete generator domain and each actual binary digit image, checks every integer obstruction and every return row, and scans the actual allowed next digits for every controller transfer. It checks all retained and discarded support records and all positive rational potential inequalities. It also independently enumerates modular start/step pairs at every claimed first safe base and at the final upper benchmarks. Consequently the two rank-five equalities in Theorem O4 have both complete production certificates and independent arithmetic audits.

For ranks at most four and additional small k , the other verifier checks 506 complete block instances, 2,995 safe controller edges, and 418 independent full-column transfer comparisons. Its record includes complete cycle data and rational lower potentials. In addition to Theorem O4, it establishes $\Gamma_{4,1} = \Gamma_{4,2} = 3$ and $\Gamma_{4,3} = 19^{1/3}$. It also calculates the singleton capacities through $k=9$; at $k=9$ the optimum is $\sqrt[8]{6}$, using period (2,3).

A consequence relevant to the existing best constructions is

$$\begin{aligned} \min_{1 \leq n \leq 5} \Gamma_{5,n} &= 23^{1/4} > 97^{1/6}, \\ \min_{1 \leq n \leq 5} \Gamma_{6,n} &= 47^{1/5} > 93^{1/6}. \end{aligned} \quad (8.3)$$

All comparisons can be checked by integer powers. Thus any improvement of those two six-generator record rates within the fixed-block canonical mixed-radix model must use at least six generators per level. The model keeps the same block at each level; changing the block itself from level to level is a separate, explicitly different dictionary optimization.

9. Return to the unrestricted exponential rate

For completeness, the passage to (1.4) retains the original generator quantity. For admissible A and B set

$$A \star B = A \cup (2S(A) + 1)B.$$

The map $(x,y) \rightarrow x + (2S(A)+1)y$ is a bijection from $H(A)$ times $H(B)$ to $H(A \star B)$, with remainder/quotient inverse. A second difference in the first factor lies between $-2S(A)$ and $2S(A)$, so divisibility by $2S(A)+1$ forces it to vanish. This pulls every progression back to progressions in both factors. A nonconstant composite progression therefore supplies a nonconstant factor witness.

Moreover

$$2S(A \star B) + 1 = (2S(A) + 1)(2S(B) + 1).$$

For the minimum $F_k(n)$ of $2S(A)+1$ at size n , submultiplicativity follows. Writing $n=t \cdot m+r$ proves convergence of $\log F_k(n)/n$ to its infimum, because the finitely many r -remainders contribute a bounded numerator. The exact inequalities

$$2g_k(n) + 1 \leq F_k(n) \leq 2ng_k(n) + 1$$

then prove existence of λ_k and

$$\lambda_k = \inf_{|A| \geq 1} (2S(A) + 1)^{1/|A|}.$$

Every admissible A has the canonical constant certificate $(2S(A)+1, A)$: all modular second differences lie strictly between minus and plus that base, so modular divisibility implies integer equality. This proves $\inf_n \Gamma_{k,n} \leq \lambda_k$. Conversely, each minimizing finite cycle in Section 7 is an actual integer generator construction at its aggregate radix, proving $\lambda_k \leq \Gamma_{k,n}$. Together these prove (1.4).

Noncanonical carry certificates from earlier work still enter the same global rate. A safe separated noncanonical block is sent to a finite lifted prefix C_t and then to $(2S(C_t)+1, C_t)$. This keeps the prefix's generators, has a valid canonical modular certificate, and its cost per generator tends to the original base cost. Its rank changes from n to tn and that change is retained. The new fixed-rank table consequently has its stated canonical scope, while the unrestricted global infimum receives the earlier noncanonical constructions through their actual maps.

10. The remaining uniform inequality and an exact cohomological growth defect

The last unbounded parameter in (1.4) is n . Neither the all-pattern bound nor the finite fixed-rank algorithm supplies a convergence modulus for that infimum. Their exact receiver is the universal assertion $\Gamma_{k,n} \geq u$ for every positive n , at a proposed lower threshold u . A strict failure $\lambda_k < u$ has a finite n and a finite periodic certificate by the definition of the infimum and the attained finite-rank optima. Such a witness is found by eventual enumeration. The converse lower assertion retains its full universal quantifier.

The source-growth expression below makes the remaining information concrete. Build an admissible block A in any fixed order. Before adding a new generator a , let B be the actual preceding subset-sum image, let $v=|B|$, and decompose B into maximal a -chains. Let their number be c and their lengths be $L_1, \dots, L_c$. Because $B \cup (B+a)$ is contained in the final k -AP-free image, every $L_i \leq h=k-2$.

The original relation sequence for this addition is

$$0 \rightarrow \mathbb{Q}[B \cap (B+a)] \rightarrow \mathbb{Q}[B] \oplus \mathbb{Q}[B] \rightarrow \mathbb{Q}[B \cup (B+a)] \rightarrow 0. \quad (10.1)$$

For x in the overlap, the first map is $e_x \rightarrow (e_x, e_{x-a})$; the last map sends $(e_x, 0)$ to e_x and $(0, e_y)$ to e_{y+a} . The first map is injective, and its image is precisely the kernel by grouping coefficients at each actual image value. This is another instance of the support-preserving quotient in Section 2.

Since $|B \cap (B+a)| = v-c$, the exact growth is $v' = v+c$. Retain the nonnegative defect

$$\delta = hc - v = \sum_{i=1}^c (h - L_i).$$

It gives

$$v' = \frac{h+1}{h}v + \frac{\delta}{h}. \quad (10.2)$$

For the successive stages $j=0, \dots, n-1$, with $v_0=1$, put $\gamma=(h+1)/h$. Iteration yields the exact identity

$$|H(A)| = \gamma^n + \frac{1}{h} \sum_{j=0}^{n-1} \gamma^{n-1-j} \delta_j. \quad (10.3)$$

Equivalently,

$$\log |H(A)| = n \log \gamma + \sum_{j=0}^{n-1} \log \left(1 + \frac{\delta_j}{(h+1)v_j} \right). \quad (10.4)$$

The first term recovers the known elementary exponential lower scale. The second term is an explicitly retained positive quantity in the original subset-sum image; it is not replaced by zero in the research target. The exact verifier checks (10.1)–(10.4) for 7,266 admissible sets in its stated finite domain, over 26,315 generator-addition stages. Those checks do not estimate the term uniformly in n . Obtaining a sufficiently strong uniform bound on this contribution, or another original-source invariant, remains necessary for a general matching lower law.

The current source-level obstruction to a free metric argument is equally explicit: the tensor metric cost (3.4) survives even when the carry relation inverse is uniformly bounded. Its growth and the actual feasible relation primitives must be controlled together. The present finite classifications exploit those primitives successfully through rank five, but do not assert a bound for every rank merely from cohomological exactness.

11. Reproducibility and evidence inventory

All code is standard-library Python. `verify_outer_control.py` produces the affine-signature, fiber-metric, transport, chain-defect, and small-rank certificates. `verify_rank_five.py` produces either a full rank-five record or an explicitly indexed shard. `audit_rank_five.py` independently verifies a full record or a shard, including its literal interval in the complete generator enumeration. No checker relies on Python assert statements.

The largest run was executed in disjoint exact index intervals, then joined only after checking their endpoints and their full generator lists against the complete independently enumerated domain. The full records are stored losslessly as gzip-compressed JSON. The accompanying replay/audit summary binds their uncompressed hashes, producer hashes, auditor hashes, and all interval coverage. Commands and output paths are given in the package README. Larger full JSON records can be regenerated or decompressed with the standard gzip module.

The affine-height replay includes 12,720 singleton tuples through base 160 and the complete two-generator domains in Section 5. Four reversible codes are checked against every literal affine label. Sixteen complete carry profiles are compared before and after the parameter transport, with all binary-mask fibers. Eight direct numerical-language transports check their full ordered pair-sum partitions. The support metric replay checks 13,122 support-inclusion pairs and 33,534 section-boundary identities, as well as exact tensor determinants through three levels.

These bounded replays have their specified roles. The all-rank polyhedral theorem has the proof in Section 5. The all-schedule and cutoff theorems have the proofs in Sections 6–7. The exact table uses those theorems to justify the finite domain, and the complete finite proof records then establish the arithmetic cases. Existing Lean receipts retain their original scope.

References and exact source roles

KokunoYumeto. (2026, September 14). *Reconstruction with changes of support index* [TeX source]. `workbenches/splitzero-tandem/tex/support_diagrams.tex`, Zeta Function Research Reader, commit 42df8a2de002d5fc7090641fac46ea11be05fa71. Used: full support-index reconstruction, internal quotient, and representative-kernel comparison. Source: https://github.com/KokunoYumeto/zeta-function-research-reader/blob/42df8a2de002d5fc7090641fac46ea11be05fa71/workbenches/splitzero-tandem/tex/support_diagrams.tex

KokunoYumeto. (2026, September 14). *Exact source-metric optimization and aggregate arithmetic control* [TeX source]. `workbenches/splitzero-tandem/tex/tau_chain.tex`, same pinned repository. Used: the original finite section/Schur identities TC7–TC8b; no arithmetic endpoint estimate imported. Source: https://github.com/KokunoYumeto/zeta-function-research-reader/blob/42df8a2de002d5fc7090641fac46ea11be05fa71/workbenches/splitzero-tandem/tex/tau_chain.tex

The Clankers. (2026, September 14). *SplitZero transfer to Erdős 817: uniform carry cohomology and exact mixed-base capacity* [Research note]. `research/splitzero_transfer/notes/splitzero-infinite-carry-control.md`, commit 98fd25802a387138053064e11ddd2f6d2007d55b. Used: original integral carries, uniform relation inverse, original metric return, finite fixed-block controller, and complete-return argument for A_* . This existing branch is a predecessor, not a new contribution claimed by the present note. Source:

https://github.com/KokunoYumeto/erdos-problem-817-workbench/blob/98fd25802a387138053064e11dd2f6d2007d55b/research/splitzero_transfer/notes/splitzero-infinite-carry-control.md

The Clankers. (2026, September 14). *Exact composition, finite carry certificates, and the general-k capacity* [User-delivered research package]. EP817_GENERAL_K_POLYCLANK_20260914.zip, `research/general_k/notes/general-k-carry-capacity.md`. Used: previous source maps and executable canonical carry implementation as a comparison; the new producer is standalone.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/html/2606.24139v1>. Role: original general-k conventions and benchmark lower/upper exponential bounds. The finite lower-growth term in Section 10 is derived here directly.

Borosh, I., & Treybig, L. B. (1992). A sharp bound on positive solutions of linear Diophantine equations. *SIAM Journal on Matrix Analysis and Applications*, 13(2), 454–458. <https://doi.org/10.1137/0613029>. Publisher abstract inspected as context for bounded integer-solution arguments. Its hypotheses or sharp bound are not imported into Lemma O6, whose weaker bound is proved in full here.

Anonymous/deleted Reddit contributor, & The Clankers. (2026, September 13). *The exponential rate for four-term-progression-free subset-sum sets* [Workbench manuscript]. Original mathematical construction credited to the anonymous contributor; reconstruction and verification recorded separately in the source repository.

sneed-and-feed. (2026, September 13). *Formalize finite upper bound theorem and core helper lemmas in Extended.lean* [Pull request #1]. Erdős Problem 817 Workbench. Role: existing upper-bound formalization and separate attribution; no extension of its Lean receipt is asserted.

A compact all-rank dual from the retained reset cost

Source: history/EP817_ZETA_INTEGRATION_20260914/payloads/ep817/research/variable_blocks/notes/compact-global-dual.md **Identity:** 1b82ffc587be0c025bac9d5195930b3f4bde1eb49e21387f338cc4af1b31482f Research continuation, 14 September 2026. This is an ordinary mathematical theorem with a complete proof. It does not evaluate the unrestricted Erdős 817 rate. The numerical calibrations below use the independently audited variable-block certificates; no new Lean elaboration is claimed.

1. The original infinite family and a finite set of coefficient variables

Fix $k \geq 3$. Retain every canonical level (b, A) , where A is a finite set of distinct positive integers, $S(A) = 2$; the empty block is also an admitted present zero-digit level. Let $C = \{-1, 0, 1\}^{(k-2)}$. As in the companion proof, true-flag supports exclude carry zero. The separately present false state $(0, \text{false})$ is retained at every support. The level transfer is the literal union of its initial nonconstant-column targets and all targets of its incoming true carries.

Let S_k be the finite set of ALL safe supports invariant under $c \rightarrow c$ and $c \rightarrow \text{reverse}(c)$. A support need not have been reached by a previously enumerated schedule to belong to S_k . The transfer preserves these symmetries. Write $T(b, A)(R)$ for the actual transfer, and keep an edge $R \rightarrow T$ only when T is safe. Every such edge retains the actual digit columns and their word fibers.

For $d = k - 2$ put

$$a_d = \frac{3^d + 3^{\lceil d/2 \rceil} + 1 + 3^{\lfloor d/2 \rfloor}}{4} - 1.$$

Then $|S_k| = 2^{(a_d)}$: the nonzero carry orbits can independently be present or absent. In particular $|S_5| = 512$ and $|S_6| = 2^{24}$. This is a finite coefficient space independent of generator count, generator magnitudes, and schedule length. The arithmetic family of edges is still infinite.

2. Exact compact dual theorem

Let λ_k be the original exponential rate, whose existence and finite-set formula are proved in the companion note. For every real u in $[1, 3]$, the following are equivalent:

$$u \leq \lambda_k$$

and the existence of a function $H: S_k \rightarrow [1, 2]$ such that, for EVERY canonical level and EVERY safe edge,

$$b H(T_{b,A}(R)) \geq u^{|A|} H(R). \tag{D1}$$

Consequently

$$\lambda_k = \max \{ u \in [1, 3] : \exists H \in [1, 2]^{S_k} \text{ satisfying (D1) for all actual levels} \}. \tag{D2}$$

The interval $[1,2]$ is a theorem furnished by the original reset cost. It is not a renormalization of a source metric, of the generator weights, or of the quotient. For a fixed u , (D1) is a family of linear inequalities in the finitely many coefficients $H(R)$.

Proof of necessity

Assume $u \leq \lambda_k$. Give an edge cost

$$w_u(b, A) = \log b - |A| \log u.$$

Consider ANY finite safe path, even one starting at a support not known to be reachable from the initial support. By monotonicity, its same sequence of labels can be read safely from empty true support, with each actual support contained in the displayed support. Append the supported zero reset $(2, \text{empty})$. The resulting word sends empty true support back to empty. Repeating it therefore gives an admissible periodic schedule.

Let P be the original path's product of radices and N its generator reward. If $N > 0$, the actual aggregated periodic generator block gives

$$\lambda_k \leq (2P)^{1/N}.$$

Since $u \leq \lambda_k$, the path cost obeys

$$\log P - N \log u \geq -\log 2. \quad (\text{D3})$$

If $N=0$, its cost is the sum of positive log radices and also satisfies (D3). No negative cycle or infinitely long approximation has been discarded; (D3) bounds every finite safe path uniformly.

For each R define $d(R)$ to be the infimum of costs of finite safe paths ending at R , allowing ANY starting vertex and the empty path of cost zero. Then

$$-\log 2 \leq d(R) \leq 0.$$

Appending an edge $R \rightarrow T$ to paths approaching $d(R)$ gives

$$d(T) \leq d(R) + \log b - |A| \log u.$$

Set $H(R) = \exp(-d(R))$. It belongs to $[1,2]$, and exponentiating the last inequality proves (D1). The infimum need not be achieved by a finite path; the inequality follows by taking the infimum, and continuity of the exponential is sufficient.

Proof of sufficiency

Assume (D1). Every admissible finite nonempty A has the modular-safe level $(2S(A)+1, A)$. Its transfer from empty true support is empty. Apply (D1) to that self-edge and cancel the positive $H(\text{empty})$:

$$2S(A) + 1 \geq u^{|A|}.$$

Taking the infimum over all such A and using the proved exact finite-set characterization gives $\lambda_k \geq u$. This proves both directions, including feasibility at $u = \lambda_k$, and hence the maximum in (D2). Alternatively, multiplying (D1) on every finite path retains the full endpoint ratio and proves its construction-rate lower bound directly. QED.

3. Exact finite-rank potentials and an explicit failed extrapolation

The same path argument applies to the finite bounded-rank dictionary, because its supported reset remains available and its positive-reward optimum is attained periodically. At an algebraic target $u = B^{1/p}$, one can retain rational potentials $h = H^p$ satisfying

$$1 \leq h(R) \leq 2^p, \quad b^p h(T) \geq B^{|A|} h(R). \quad (\text{D4})$$

The already independently audited eight variable-block certificates satisfy these inequalities on their full reachable state sets. The separate compact-dual checker verifies the bound 2^p on each actual rational coefficient and verifies (D4) on every retained source/target/reward cost representative. The preceding independent audits already verified every actual parallel label as well. Multiplication gives the uniform retained endpoint factor

$$(2P)^p \geq B^N$$

for every safe path in those controllers.

For the rank-four result $B=23, p=4$, the claim $u=23^{1/4}$ cannot be extended to all ranks. The six-generator modular certificate $A=\{1,4,5,17,21,22\}$, $b=97$ at $k=5$ gives an empty-support self-edge. It would require

$$97^4 \geq 23^6,$$

which is false because $97^2 < 23^3$. At $k=6$, $b=93$ fails the same proposed inequality because $93^2 < 23^3$. These are actual finite obstruction maps to an attempted all-rank potential, not a conclusion that other lower thresholds are impossible.

The reset factor cannot be omitted even in a finite certificate. For the rank-one target $u=\sqrt{8}$, the safe first level $(2, \{1\})$ has $P=2, N=1$, so $P^2=4 < 8$; appending the retained reset multiplies P by two and supplies the correct finite-path bound. No potential controls the cost of a path after dropping its endpoint or reset term.

4. Return to the general linear-pattern source

For the binary-rigid, translation-invariant integer matrix L in the companion Zeta note, the supported reset has radix $R_L = \max(2, \max_i m_i)$. Replace 2 by R_L throughout the proof. A finite safe path followed by that actual reset produces a periodic L -admissible generator set, giving path cost at least $-\log R_L$. The constructed coefficient function belongs to $[1, R_L]$. For the original λ_L , the exact dual is

$$u \leq \lambda_L \iff \exists H \in [1, R_L]^{S_L} : bH(T_{b,A}(Q)) \geq u^{|A|} H(Q)$$

on all safe supports and all actual canonical levels. Here S_L can be the full finite carry-support set; no AP-specific symmetry is assumed. The actual finite-set formula from the original product with radix $MS(A)+1$ proves the converse. The binary-obstruction case has no nonempty admissible generators and remains separately identified by its explicit witness.

This result supplies a compact, dimension-fixed coefficient formulation of the all-rank control problem. The infinitely many block inequalities still have to be controlled arithmetically. Nothing in the compactness argument evaluates their feasible maximum, bounds the original tensor-metric return for free, or proves a statement about the zeros of the Riemann zeta function.

Sources and scope

The complete carry correspondence, original support conventions, reset, and finite-set rate formula are proved in the two companion notes. The source-metric and original-observation-kernel interfaces are those of the inspected Zeta reconstruction at commit [42df8a2de002d5fc7090641fac46ea11be05fa71](#). The shortest-path potential construction is classical in form; the uniform interval here is obtained from the retained arithmetic reset and the original extremal-rate characterization. No global literature-priority claim is made.

Supported zero resets for translation-invariant integer patterns

Source: history/EP817_ZETA_INTEGRATION_20260914/payloads/ep817/research/variable_blocks/notes/supported-reset-generalization.md **Identity:** a36db5a19e605ccb8559a99c2a1c5722f75377ef9fde0f07a21db173833b2d06
Research contribution returned from the Erdős 817 workbench to the SplitZero workbench. The original support reconstruction, internal relation quotient, and source-metric distinctions are retained. All general statements below have written proofs; the accompanying computations are exact finite calibrations and EP817 certificates, not new Lean elaboration or a theta-source weight estimate.

1. A precise all-pattern dichotomy

Let L be any finite integer matrix with $t \geq 2$ columns and row sums zero. Keep every row, including redundant and zero rows. For a subset X of the integers, a forbidden pattern is a tuple y in X^t with $Ly=0$ whose entries are not all equal. For a finite positive generator set A , $X=H(A)$ includes zero. Put $m_i = \sum_j \max(L_{ij}, 0)$; the zero row has $m_i=0$. Set $M = \max_i m_i$, with $M=0$ when there are no rows.

There are two exhaustive, finitely decidable cases.

Case I. There is a nonconstant epsilon in $\{0,1\}^t$ with $L \text{ epsilon} = 0$. Then every nonempty generator set A has a forbidden pattern. For any actual a in A , the typed map $\text{epsilon} \rightarrow a$ belongs to $H(A)^t$, has nonconstant entries, and is killed by L . The domain of nonempty admissible generator sets is empty. This conclusion is obtained from the displayed witness, not from a claim that a general additive-pattern problem is impossible.

Case II. The only binary kernel vectors are constant. Call this finite property binary rigidity. Put

$$q = \max(2, M + 1), \quad R = \max(2, M). \quad (1)$$

Then all the finite and infinite constructions in Sections 2–5 below exist, and the infinite canonical scheduling optimization at every fixed maximum block size is exactly computable.

This is an unconditional dichotomy for the stated matrix class. Binary rigidity is tested by the finite enumeration of 2^t columns, rather than assumed as an unknown global extremal hypothesis. The AP second-difference matrices belong to Case II, with $M=2$, $q=3$, $R=2$. The row $(2,-3,1)$, describing an affine three-point pattern with unequal spacing, also belongs to Case II and gives $q=4$, $R=3$. The row $(1,1,-1,-1)$ belongs to Case I: $(1,0,1,0)$ is a retained nonconstant binary witness. A Sidon formulation that excludes such trivial patterns would have a different forbidden set and needs its own flag definition; that different problem is not silently substituted here.

2. The original integral relation and bounded carries

At level j let $b_j \geq 2$, let D_j be an actual subset-sum image contained in $[0, b_j - 1]$, and let $P_0 = 1$, $P_{(j+1)} = b_j P_j$. The map from digit rows to numerical rows is

$$\text{ev}_{\mathbf{b}}(d)_i = \sum_{j < m} P_j d_i^{(j)}.$$

It is bijective on the product of the distinct canonical digit images. All original binary representations of each digit remain in its fiber. Multiplication by L commutes with evaluation:

$$Lev_{\mathbf{b}}(d) = \sum_{j < m} P_j Ld^{(j)}. \quad (2)$$

For an actual forbidden-pattern row define prefix carries by

$$c_{u,i} = P_{\nu}^{-1} \sum_{j < \nu} P_j (Ld^{(j)})_i. \quad (3)$$

The numerator is integral-divisible by P_{ν} when the terminal numerical equation holds, since the remaining tail is divisible by that power. Consecutive prefixes give the exact relation

$$b_j c_{j+1} = c_j + Ld^{(j)}, \quad c_0 = c_m = 0. \quad (4)$$

Conversely, (4) telescopes to $Ly=0$. Because the row sums vanish and the positive and negative masses both equal m_i ,

$$|(Ld^{(j)})_i| \leq m_i(b_j - 1), \quad |c_{\nu,i}| \leq m_i(1 - P_{\nu}^{-1}) < m_i. \quad (5)$$

For $m_i > 0$, the complete integer carry coordinate lies in $\{1-m_i, \dots, m_i-1\}$; a zero row keeps coordinate zero. The finite source retains all these coordinates, not merely a basis of the rational row space.

A flag records whether any digit column so far is nonconstant across the t rows. Canonical digit uniqueness makes this exactly nonconstancy of the numerical tuple. The only reachable false-flag state is zero carry: a constant column is killed by every original row and preserves zero carry. Acceptance is zero carry with true flag. Equations (2)–(5) give explicit inverse maps between accepting labeled paths and represented numerical patterns.

3. The reset is a present zero, with positive cost

A level with no generators has digit image $\{0\}$. Its word source $Q[\{\text{empty word}\}]$ is one-dimensional. Its scalar amplitude observation kills the basis vector, while the original value-basis source remains present. The extra kernel of that scalar observation is the whole one-dimensional space. This is a literal instance of the original-boundary/further-observation-kernel distinction in the inspected SplitZero reconstruction.

Apply such a level with radix R from (1). Its relation is $Rc'=c$. Every allowed nonzero carry has a coordinate of magnitude between one and $R-1$; it has no integral outgoing transition. At a safe controller state there is no true zero carry, so all true paths disappear. The false zero path has its actual all-zero label and survives. Hence the reset sends every safe true-flag support to empty.

The full active support still contains the false zero state. It is a nonbottom support in the reconstruction semilattice of active states. The reconstructed coefficient zero at that support and external absence remain distinct. The controller's shorthand 'empty true support' never discards the constant paths.

The level has positive logarithmic cost $\log R$ and no generator reward. It multiplies the position of every later generator by R . Omitting this factor would alter the integer construction. For an internal empty level, the exact absorption map multiplies the preceding nonempty radix by R . A leading empty product C sends the first level (b,A) to (Cb,CA) . These maps preserve actual generator values, original subset masks, and total radix product. Thus zero levels can be retained for optimization and then removed with their full cost stored in neighboring radices.

4. Finite anchors and the complete bounded-rank reduction

In Case II, the singleton digit image $\{0,1\}$ is pattern-free modulo q . Indeed $|(L \text{ epsilon})_i| \leq m_i < q$, so modular zero forces actual zero, and binary rigidity then forces a constant column. The same argument at the least base- q digit and induction proves that

$$C_n = \{1, q, \dots, q^{n-1}\}, \quad Q_n = q^n \quad (6)$$

is an n -generator modular anchor for every n . Its sum is smaller than Q_n , and all repeated-root or coefficient relations in L remain part of the same row equations.

Fix a maximum nonempty block size r . A schedule may change every radix, every positive weight, and every block size from one through r . At any safe prefix, replace a level (b, A) of size n with the reset (R, empty) followed by (Q_n, C_n) . The result is safe, has reward n , cost $R Q_n$, and outgoing true support empty. That support is contained in any original safe outgoing support. The exact deterministic carry transfer is monotone under support inclusion, so the original remaining suffix stays safe. If $b > R Q_n$, the replacement strictly lowers cost.

The word-level substitution is a new construction, not an equality of old and new amplitudes. The anchor has distinct binary subset sums, so the identity on masks gives a surjection from its value-basis space to the original $Q[H(A)]$. The added kernel consists exactly of old representation collisions. This quotient and the outgoing support inclusion are the two stated comparison maps.

After all replacements the dictionary is finite:

$$(R, \emptyset), \quad 1 \leq |A| = n \leq r, \quad S(A) < b \leq Rq^n. \quad (7)$$

A better explicitly verified size- n modular anchor replaces q^n by its actual modulus. Equations (4) and (5) give a finite deterministic controller on sets of true carries. Every safe support has a reset edge back to the initial support, so all reachable safe supports communicate through that support. Assign to an edge cost $\log b$ and reward $|A|$. Zero-reward cycles have positive cost; cycles with positive reward have exact cost $\exp(\sum \log b / \sum |A|)$. Removing cycles from any finite path bounds its excess over the smallest cycle ratio by the bounded residual simple path. Repeating a least-ratio cycle attains the bound. Support monotonicity allows its periodic word to start at the initial support.

Theorem. For every binary-rigid row-sum-zero integer matrix L and every $r \geq 1$, the optimum over all positive-reward canonical schedules with block sizes at most r equals the minimum positive-reward cycle ratio in (7). It is attained periodically and is an exactly computable algebraic number. This conclusion quantifies over arbitrary original radices and integer weights, not only a preset finite collection.

An attaining period is mapped back to one ordinary finite generator block by $C = \bigcup_j P_j A_j$, $Q = \prod_j b_j$. The source has exactly $\sum_j |A_j|$ distinct generators and $S(C) < Q$. The explicit labeled path correspondence certifies its infinite repetitions.

5. Exact relation to the unrestricted growth rate

For binary-rigid L , $M \geq 1$. For finite L -admissible A, B define

$$A \star_L B = A \cup (MS(A) + 1)B.$$

Euclidean division identifies their actual subset-sum image with $H(A)$ times $H(B)$. If a target tuple satisfies $Lz = 0$, the first coordinate row amplitude has magnitude at most $m_i S(A) \leq M S(A)$, strictly below the base. Divisibility therefore makes each factor tuple satisfy its original L equations. Admissibility in both factors forces a constant target tuple. The new cost is exactly multiplicative:

$$MS(A \star_L B) + 1 = (MS(A) + 1)(MS(B) + 1). \quad (8)$$

Let $g_L(N)$ be the minimum largest generator. Minimize the sum observable in (8), and use

$$Mg_L(N) + 1 \leq h_L(N) \leq MNg_L(N) + 1.$$

The submultiplicative log argument proves existence of $\lambda_L = \lim_N g_L(N)^{1/N}$. Every admissible A has the modular certificate $(MS(A)+1, A)$, by the same row-amplitude divisibility. Thus

$$\lambda_L = \inf_{A \neq \emptyset, L\text{-admissible}} (MS(A) + 1)^{1/|A|} = \inf_{r \geq 1} \delta_{L,r}. \quad (9)$$

Formula (9) does not evaluate that infimum or give its convergence modulus. It supplies an exact increasing-rank family of finite problems and finite construction witnesses for every strict rate improvement. The numerical evaluations in the attached certificates concern L equal to the AP matrices for $k=5$ and $k=6$, with maximum block size at most four.

6. Original cohomology and metric interfaces

For a supported finite set S of original words, let $V_S = Q[S]$, $W_S = Q[\text{ev}(S)]$, and $\pi_S(e_w) = e_{\text{ev}(w)}$. Its original relation module is $R_S = \ker \pi_S$. These are transport-compatible under support inclusion and reconstruct over the original split scalar using `LinearDiagram.Total`. The quotient map sends a boundary to its receiving supported zero. A later amplitude observation $W_S \rightarrow Q$ may have its own additional kernel; it is not substituted for R_S .

The word fibers also give the exact minimum-norm quotient metric. In the original orthonormal word basis, the section of e_y is the average of the $\mu_S(y)$ actual words with value y , and its Gram is $\text{diag}(1/\mu_S(y))$. For S subset T , the two sections differ by an actual T -relation; its squared norm is $1/\mu_S(y) - 1/\mu_T(y)$. At several canonical levels the quotient Gram is the tensor product of the actual level Grams. The norm of the identity into unit value coordinates is $\prod_j \sqrt{\max_y \mu_j(y)}$. The reset has $\mu(0) = 1$ but retains its radix cost. These formulas separate value-space metrics, original relation directions, and numerical construction costs.

This contribution returns a finite-source control technique to the Zeta workbench. It does not assert that the theta quotient has become a finite carry system. The source-comparison map used is the original support/quotient reconstruction applied to these explicitly defined word spaces. No arithmetic Mellin observation kernel, spectral packet, or source metric is replaced. The Zeta programme's separate growing-source metric estimate retains its own scope.

The companion compact-global-dual.md proves a further exact all-rank dual: the original rate is the largest u admitting a coefficient function H in $[1, R]$ on the full finite safe-support set, with $bH(T) \geq u^{|A|} H(S)$ for every actual canonical edge. The uniform interval follows from appending the retained reset to any finite safe path. Its remaining constraints quantify over all generator blocks; no finite rank truncation is substituted for them.

7. Applications and evidence

For the AP matrix with $k=5$ or $k=6$, the companion EP817 note proves

$$\delta_{k,1} = \sqrt{8}, \quad \delta_{k,2} = \delta_{k,3} = \sqrt{5}, \quad \delta_{k,4} = 23^{1/4}.$$

These permit changes among every block of every admitted size. Both rank-four lower certificates cover the full 41,571-label dictionaries. A separate auditor verifies their original binary fibers, every carry profile, all safe and rejected support transfers, positive rational potential inequalities, and an attaining periodic word. The reusable calibration checker also tests ten integer matrices, 1,102 reset carry states, 512 rigid binary columns, 1,364 literal empty-level absorptions, seven orbit counts, and a nontrivial collision quotient with Gram entry $1/2$. Finite calibrations corroborate the written all-matrix proof; they are not a substitute for it.

No original SplitZero or Lean file is changed by this add-only workbench. The producer, independent auditor, complete compressed proof records, and exact source hashes are included so that the arithmetic application can be replayed without checking out the EP817 repository. No independent human review or new Lean success is claimed.

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Classical ratio-objective context, not a source of the supported-reset arithmetic construction.

Supported zero resets and exact variable-block capacity

Source: `history/EP817_ZETA_INTEGRATION_20260914/payloads/ep817/research/variable_blocks/notes/variable-block-control.md` **Identity:** `e7045ee59c2b86eb72caab28f7b1e46104bde3924dda7d1dfa8f31cb3d937213` Research continuation, 14 September 2026. Ordinary mathematical proofs and exact finite certificates; independent mathematical review pending. No new Lean elaboration is claimed. This note extends the preceding fixed-block work to schedules in which both the generator block and its size may change at every level.

1. Original problem and scope

For a finite set A of distinct positive integers, retain its actual subset-sum set $H(A)$, including zero, and its sum $S(A)$. A is k -admissible when $H(A)$ has no ordered nonconstant k -term arithmetic progression. The extremal quantity $g_k(N)$ minimizes $\max(A)$ among N -element admissible sets. The original mathematical construction for $k=4$ remains attributed to the anonymous/deleted Reddit contributor; sneed-and-feed retains separate credit for its finite upper-bound Lean formalization.

A canonical level is a pair (b, A) , where $b \geq 2$ is an integer and $S(A) < b$. In an infinite schedule retain

$$P_0 = 1, \quad P_{j+1} = b_j P_j, \quad G_m = \bigcup_{j < m} P_j A_j, \quad N_m = \sum_{j < m} |A_j|.$$

Generators are distinct: each one at level j is strictly below P_{j+1} , and all generators at later levels are at least P_{j+1} . A subset chooses one actual digit in $D_j = H(A_j)$ at every level. Consequently its numerical image is exactly

$$H(G_m) = \left\{ \sum_{j < m} P_j d_j : d_j \in D_j \right\}. \quad (1)$$

Mixed-radix digit evaluation is bijective on the product of distinct digit images. Subset representations within each image retain their multiplicities. No distinctness of all binary subset sums is assumed.

Fix $r \geq 1$. Permit every level to have at most r generators, and require N_m to tend to infinity. For the moment the empty block is also permitted. Define

$$\delta_{k,r} = \inf_{\substack{|A_j| \leq r, \, N_m \rightarrow \infty \\ H(G_m) \text{ } k\text{-AP-free for every } m}} \liminf_{m \rightarrow \infty} P_m^{1/N_m}. \quad (2)$$

Indices before the first positive N_m have no effect on this limit. The objective is the retained radix cost per actual generator. Empty levels have positive radix cost and zero generator reward. Section 3 proves that they can be absorbed exactly into nonempty levels; their introduction does not enlarge the infimal value beyond that attainable by nonempty canonical schedules.

The previous quantity $\Gamma(k, n)$ uses the SAME n -element block at every level. The inclusion of those schedules in (2) gives $\delta_{k,r} \leq \Gamma(k, n)$ for every $n \leq r$. The current theorem treats changes in the actual weights and in block size, as well as arbitrary changing radices.

Main theorem. For every $k \geq 3$ and $r \geq 1$, $\delta_{k,r}$ is an attained, exactly computable algebraic number. It is the minimum cost/reward ratio on an explicit finite controller using only

$$(2, \emptyset), \quad 1 \leq |A| = n \leq r, \quad S(A) < b \leq 2 \cdot 3^n. \quad (3)$$

A better verified modular anchor of size n and radix q_n replaces the corresponding cutoff by $2q_n$. The minimum is attained by a periodic schedule. Moreover

$$\delta_{k,r+1} \leq \delta_{k,r}, \quad \lambda_k := \lim_{N \rightarrow \infty} g_k(N)^{1/N} = \inf_{r \geq 1} \delta_{k,r}. \quad (4)$$

The equality in (4) is a characterization, not a numerical evaluation of the unbounded-rank infimum. The sharp general lower law remains unproved in this contribution.

2. Carry paths with all support and column data retained

For a k -tuple of actual digits d , put

$$\Delta(d)_i = d_i - 2d_{i+1} + d_{i+2}, \quad 0 \leq i < k-2.$$

The original relation at consecutive levels is

$$b_j c_{j+1} = c_j + \Delta(d^{(j)}), \quad c_0 = 0. \quad (5)$$

Multiplying by P_j and summing gives

$$\sum_{j < m} P_j \Delta(d^{(j)}) = P_m c_m. \quad (6)$$

Thus terminal carry zero is exactly the integer second-difference condition. Every integral prefix carry satisfies

$$|c_{m,i}| \leq 2(1 - P_m^{-1}) < 2,$$

because each column defect has magnitude at most $2(b_j - 1)$ and $\sum_{j < m} P_j(b_j - 1) = P_m - 1$. The exact carry domain is therefore contained in $C_k = \{-1, 0, 1\}^{(k-2)}$, independently of the sizes of the blocks or radices.

A flag records whether the first two digit rows have ever differed. Canonical evaluation makes this precisely numerical inequality of their first two values. When the flag remains false, the first two current digits agree; starting from carry zero, divisibility in (5) then forces each later digit of that column to agree too. The only reachable false-flag state is therefore $(0, \text{false})$. Its actual constant-column paths are always retained.

For a level (b, A) let $I(b, A)$ be the targets from $(0, \text{false})$ with a nonconstant column. For c in C_k , let $T(b, A)(c)$ be the complete set of targets from c , using all actual columns in $H(A)^k$. A controller state R records the reachable true-flag carries, with deterministic update

$$\mathcal{T}_{b,A}(R) = I_{b,A} \cup \bigcup_{c \in R} T_{b,A}(c). \quad (7)$$

The separately retained false state is understood in (7); the full active support is $\{(0, \text{false})\} \cup \{(c, \text{true}) : c \text{ in } R\}$. In particular $R = \emptyset$ does not mean that the original source is absent. The controller is safe exactly when 0 is not in R . Every accepted path gives a numerical progression by (1) and (6); every numerical progression recovers its unique rows and its integral prefix carries by division. The path/tuple correspondence is exact.

The transfer (7) is monotone: $R \subset R'$ implies $T(R) \subset T(R')$. This follows from the displayed union and is the precise simulation relation used in the replacement argument. The finite alphabet can be evaluated without choosing all k digits independently. For fixed c and the first two actual digits, the remaining digits are forced modulo b by (5); each residue has at most one representative in the canonical digit image. The verifier retains every actual binary-mask fiber of every digit.

3. A supported zero level is a universal reset

Lemma 1. For every safe true-flag support R , the level $(2, \text{empty})$ sends R to the empty true-flag support and retains the false state.

Proof. Its digit set is exactly $\{0\}$. Equation (5) becomes $2c' = c$. For integral c, c' in $\{-1, 0, 1\}^{(k-2)}$, this is possible only when $c = c' = 0$. Since safe R excludes true carry zero, no true path survives this level. The false zero path has the actual all-zero column and survives. QED.

An empty block is a PRESENT level with one subset choice and one numerical digit. Its word space is $Q[\{\text{empty word}\}]$, of dimension one. The scalar amplitude observation sends its basis vector to zero; its kernel is the whole one-dimensional word space. Sending that space to the zero object is the explicit additional observation, not the original subset-sum image. The radix multiplier two remains in $P_{(j+1)} = 2P_j$. Erasing it would change every later generator.

Empty levels can also be eliminated without discarding their effect. Each empty level after a nonempty level multiplies that preceding level's radix. A leading run of empty levels with product L is absorbed by sending the first nonempty (b, A) to (Lb, LA) . The literal generator sets and final radix product are unchanged, $S(LA) < Lb$, and block sizes are unchanged. Internal and trailing empty runs are handled by the same rule. For completeness, write a_i for $\log P$ just after the i -th nonempty level, e_i for the sum of $\log$ radices in the following empty run, and N_i for its cumulative generator reward. Then $a_i/N_i \leq (a_i + e_i)/N_i \leq (a_{(i+1)})/N_{(i+1)}$ ($N_{(i+1)}/N_i$). Since $1 \leq N_{(i+1)} - N_i \leq r$ and N_i tends to infinity, the outer sequences have the same limit inferior, also when it is infinite. These are respectively the pre-empty and absorbed-endpoint costs. Thus elimination preserves the infimal objective in (2), even for empty runs of unbounded length. For a periodic schedule with positive reward, cyclically start at a nonempty level and absorb the finitely many empty runs in its period.

This is also a concrete instance of the original SplitZero distinction: the level has a nonbottom active support even though its numerical coefficient is zero. The inherited reconstruction and original-versus-further-kernel square are the interfaces used here; no theta-source estimate is imported.

4. A rank-dependent finite cutoff even when all blocks vary

Let C_n be a verified n -element block whose subset-sum digits are k -AP-free modulo q_n , with $S(C_n) < q_n$. Its transfer from empty true support returns empty support. Such anchors exist for every n : take $C_n = \{1, 3, \dots, 3^{(n-1)}\}$ and $q_n = 3^n$. At the least base-3 digit, a modular 3-AP in the binary digits would have to be constant; stripping common digits proves modular 3-AP-freeness modulo 3^n . Its first three entries exclude a nonconstant k -AP for any $k \geq 3$.

Lemma 2 (cost-bounded replacement). An arbitrary canonical level (b, A) with $|A| = n$ can be replaced, from any safe incoming support, by

$$(2, \emptyset), (q_n, C_n). \quad (8)$$

The replacement has the same generator reward n , total radix cost $2q_n$, and final true support empty. When $b > 2q_n$ its cost is strictly smaller.

Proof. Lemma 1 resets the incoming support to empty; modular admissibility of the anchor returns empty. Its intermediate support is safe. If the original outgoing support was R' , empty is a subset of R' ; by monotonicity, every remaining original suffix that was safe from R' is safe from the replacement support. QED.

This compares two explicitly constructed schedules. It does not identify their scalar generator values. At the n binary choices, the ternary anchor has distinct subset sums, so the identity on masks induces a surjection from its value-basis space to $Q[H(A)]$. Its kernel is spanned by differences of masks with the same original A -value. Choosing the mean of each fiber gives the original quotient $\text{Gram diag}(1/\mu_A(y))$; the corresponding source metric correction is retained. At the controller boundary the map is the explicit inclusion $\text{empty} \rightarrow R'$. Those are the two distinct comparison maps, not an asserted arithmetic isomorphism of arbitrary integers.

Replace every overlarge level by (8), and every empty level's radix by two. At aligned original block boundaries generator reward is unchanged, accumulated cost is no larger, and the new support is contained in the old one. All inserted intermediate levels are safe. The resulting schedule belongs to the finite dictionary

$$\mathcal{D}_{k,r} = \{(2, \emptyset)\} \cup \{(b, A) : 1 \leq n = |A| \leq r, S(A) < b \leq 2q_n\}. \quad (9)$$

Taking the limit inferior along the aligned boundaries proves that it cannot have a larger optimal rate. Conversely the finite dictionary is a subclass of (2); empty absorption supplies an original nonempty schedule when needed. Thus (9) has exactly the same infimum as the unrestricted bounded-rank class.

Every positive-weight tuple in (9) is literally enumerated; no rescaling of candidate weights is performed. Invalid one-level blocks receive an actual integer progression witness. In the applied calculations the independently checked anchors are

$$(q_1, C_1) = (3, \{1\}), \quad (q_2, C_2) = (5, \{1, 2\}), \quad (q_3, C_3) = (13, \{1, 3, 4\}), \quad (q_4, C_4) = (23, \{1, 3, 4, 7\}). \quad (10)$$

They work for $k=5$ and $k=6$ and give the exact cutoffs 6,10,26,46. These cutoffs control every larger original radix through (8), not through finite empirical evidence.

5. Exact symmetry transport and the finite controller

Subset-sum images have the involution $d \rightarrow S(A)-d$. On column paths it maps c to $-c$ and preserves all binary representations by complementing the chosen subsets. Reversal of the k rows maps c to its reversed coordinate tuple. Equation (5) commutes with both operations. At the initial state, a nonconstant modular progression has unequal consecutive digits at both ends; on true states the flag is already retained. Hence all reachable true supports are invariant under

$$c \mapsto -c, \quad c \mapsto \text{rev}(c). \quad (11)$$

No coefficient vector is being identified with zero. A symmetric support is in bijection with the set of full orbits that it contains; the inverse takes their union. Transfer images are evaluated on all original vectors in each source orbit before this reversible support encoding.

Put $d=k-2$. Burnside counting gives the number of nonzero orbits

$$a_d = \frac{3^d + 3^{\lceil d/2 \rceil} + 1 + 3^{\lfloor d/2 \rfloor}}{4} - 1. \quad (12)$$

The four fixed-point counts are respectively $3^{\lceil d \rceil}$, $3^{\lceil d/2 \rceil}$ for reversal, one for negation, and $3^{\lfloor d/2 \rfloor}$ for negated reversal. Removing the zero orbit gives (12). Thus the safe support controller has at most 2^{a_d} states, including the empty true support. For $k=5$ this is $2^9=512$, rather than the unrestricted powerset count 2^{26} . For $k=6$ it is 2^{24} . Actual reachable state sets can be much smaller.

Every safe state has a reset edge to the initial support. The entire reachable safe graph is therefore strongly connected. Edges retain their radix b , generator reward n , and original block A . The empty reset edge has $n=0$ and strictly positive cost $\log 2$.

For a directed cycle C with total reward $N(C)>0$ and radix product $B(C)$, define rate $B(C)^{1/N(C)}$. There are finitely many simple cycles. Zero-reward cycles have strictly positive cost and do not improve any cost/reward ratio. If ρ is the least $\log B(C)/N(C)$, all cycles have nonnegative sum of $\log b - \rho n$. Remove cycles from a finite path; the residual simple path has bounded length and bounded remaining cost. Dividing by cumulative generator reward and taking the limit inferior proves the lower bound ρ for every infinite positive-reward path. Repeating a minimizing cycle attains it. Monotonicity permits its period to start from empty true support even when the cycle was first encountered at a larger support.

All cycle comparisons are exact integer-power comparisons. This proves the main finite-optimization theorem and periodic attainment. The underlying minimum-cycle-ratio principle is classical; the supported reset and the resulting complete arithmetic cutoff are the additional construction-specific ingredients.

6. Exact evaluations for variable blocks of sizes at most four

The results are

$$\delta_{5,1} = \delta_{6,1} = 8^{1/2}, \quad \delta_{5,2} = \delta_{6,2} = \delta_{5,3} = \delta_{6,3} = 5^{1/2}, \quad \delta_{5,4} = \delta_{6,4} = 23^{1/4}. \quad (13)$$

These allow every block value, every size at most the stated bound, and every infinite canonical schedule. Attaining periods are $(2, \{1\}), (4, \{2\})$ at rank bound one; the constant level $(5, \{1, 2\})$ at bounds two and three; and $(23, \{1, 3, 4, 7\})$ at bound four. The first period is verified from the actual initial support; it need not return to that support after one period, so the full eventual periodic support trace is retained.

Each lower certificate gives a positive rational h on every reachable safe support and integers B, p such that

$$b^p h(R') \geq B^n h(R) \quad (14)$$

for every actual safe edge, including $n=0$. Multiply along the path to obtain

$$P_m^p \geq B^{N_m} h(R_0) / h(R_m).$$

There are finitely many positive rational potentials. Their ratio disappears under the N_m -th root, proving rate at least $B^{1/p}$ for every schedule. The attaining period has exactly that cost. This is a finite proof certificate for (13), not a numerical approximation to a spectral radius.

The rank-four dictionaries each contain 41,571 literal levels including the reset. For $k=5$, 17,846 levels have an immediate integer obstruction, leaving 23,725 usable levels. Their complete controller has 16 reachable safe states, 49,506 safe labeled transfers and 330,094 rejected transfers. For $k=6$, there are 13,080 immediate obstructions, 28,491 usable levels, 128 reachable safe states, 259,832 safe transfers and 3,387,016 rejected transfers. Every valid source carry is tested; the two column transcripts respectively contain 9,722,718 and 18,969,820 labeled nonzero-source columns.

Parallel edges with the same source, target and reward may use the cheapest radix for finding a potential, but all original labels and transfers remain in the certificate and are checked against (14). The producer uses rational arithmetic only. A separate auditor imports no producer code: it reconstructs the literal domain with combinations rather than recursive partitions, generates columns by an independent closed-prefix formula, and checks all source fibers, all carry transitions, the complete support graph, every potential inequality, and the attaining period.

The known six-generator blocks still give upper rates $97^{1/6}$ for $k=5$ and $93^{1/6}$ for $k=6$. Since $23^3 > 97^2$ and $23^3 > 93^2$, (13) rules out beating those rates with ANY canonical schedule whose individual blocks have at most four generators. The previous fixed-block rank-five classification does not cover arbitrary changes among rank-five blocks; that larger class remains unevaluated here.

7. Return to the unrestricted rate and source metrics

For any periodic safe schedule, aggregate one period into $C = \text{union}_j P_j A_j$ and $Q = \text{product}_j b_j$. Its generators are distinct, $|C|$ is the total period reward, and $S(C) < Q$. Repeating the period is the literal constant- Q lift of C . Consequently every cycle upper construction bounds the original λ_k , so $\lambda_k \leq \delta_{k,r}$.

Conversely, for every k -admissible finite A , the modulus $2S(A)+1$ is safe: each modular second difference of actual values in $[0, S(A)]$ is divisible by that modulus and strictly smaller in magnitude, hence zero over the integers. Thus $\delta_{k,|A|} \leq (2S(A)+1)^{1/|A|}$. The exact product $A \star B = A \text{ union } (2S(A)+1)B$ satisfies

$$2S(A \star B) + 1 = (2S(A) + 1)(2S(B) + 1).$$

Euclidean division identifies its subset-sum image with $H(A)$ times $H(B)$ and splits every progression into factor progressions. Minimizing this sum observable gives submultiplicativity; its comparison $2g_k(N)+1 \leq h_k(N) \leq 2N g_k(N)+1$ proves existence of λ_k and its equality with $\inf_A (2S(A)+1)^{1/|A|}$. These maps prove (4). No assumption of an unknown lower bound occurs. The separate note compact-global-dual.md additionally constructs an exact all-rank dual whose finitely many potential coefficients lie in $[1, 2]$; its block inequalities remain an infinite arithmetic family.

Across arbitrary levels, representation multiplicity remains the product of the level multiplicities. If $\mu_j(y)$ counts subsets of A_j with value y , the original orthonormal word-space quotient has Gram $\text{diag}(1/\mu_j(y))$; at m levels its Gram is the tensor product of those original Grams. The identity into the unit value-coordinate metric has norm

$$\prod_{j < m} \sqrt{\max_y \mu_j(y)}. \quad (15)$$

Neither the finite controller nor the reset removes (15). They control feasible supports and explicit constructions; a proof of the matching all-rank lower rate must also control the original arithmetic information that grows with rank. The present result does not evaluate $\inf_r \delta_{\mathbf{L}}(k, r)$, give its convergence modulus, or prove the Riemann hypothesis.

8. Reusable extension to linear-pattern cohomology

The companion note in the Zeta payload proves an unconditional dichotomy for every integer matrix L with equal row length $t \geq 2$ and row sums zero. If L has a nonconstant binary kernel vector, every nonempty positive generator set has a nonconstant forbidden pattern; the witness map is $\epsilon \rightarrow a$ for an actual generator a . Otherwise the pattern is binary-rigid and admits a uniform singleton anchor. Writing m_i for the sum of the positive entries in row i , canonical carries satisfy $|c_i| < m_i$ (a zero row keeps carry zero). The supported zero level of radix $\max(2, \max_i m_i)$ resets all safe nonzero carries. Its positive cost is retained. A modular anchor and reset then give the finite bounded-rank dictionary exactly as above.

This is a transfer of the original support, relation, and additional-observation-kernel interfaces. It is not an arithmetic weight estimate for the theta quotient. The source-metric return remains explicit, including the one-dimensional word source at a zero-digit level.

9. Reproduction and publication boundary

Run the producer, then the independent auditor, for example:

```
python certificates/verify_variable_blocks.py --k 5 --rank 4 --output certificates/variable_k5_r4.json.gz
python certificates/audit_variable_blocks.py certificates/variable_k5_r4.json.gz --producer
→ certificates/verify_variable_blocks.py --output certificates/audit_k5_r4.json
```

The two largest records are losslessly compressed. Their source, compressed, and plaintext hashes are bound by the release verifier. The same commands apply to $k=6$ and rank bounds one through three. The separate reset-pattern checker tests the general matrix dichotomy, all state coordinates in the stated calibration domains, original generator-preserving absorption of empty levels, the Burnside orbit count, and a representation-collision quotient metric. Negative tests must reject deliberately modified domain, support, fiber, potential, and reset data.

The publication payload also carries the preceding research/splitzero_outer directory byte-for-byte with its original evidence labels and its 47,259-case audit bindings. Existing Lean sources, pins, receipts, and accepted claim statuses are untouched. GitHub access in this execution was read-only; actual publication status belongs in the release record, not in the mathematical statements. The supplied publisher opens draft PRs only and never merges them or updates main.

References and precise roles

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/html/2606.24139v1> . Role: original conventions and benchmark general bounds; not a source of (3), (9), or (13).

Karp, R. M. (1978). A characterization of the minimum cycle mean in a digraph. *Discrete Mathematics*, 23(3), 309–311. [https://doi.org/10.1016/0012-365X\(78\)90011-0](https://doi.org/10.1016/0012-365X(78)90011-0) . Role: classical graph-optimization background; the cost/reward proof used here is included explicitly.

Megiddo, N. (1978). Combinatorial optimization with rational objective functions. In *Proceedings of the tenth annual ACM symposium on Theory of computing* (pp. 1–12). <https://doi.org/10.1145/800133.804326> . Role: classical minimum-ratio-cycle context, not a claim of novelty for cycle optimization.

The Clankers. (2026, September 14). *Support-preserving outer control for Erdős Problem 817*. Retained research/splitzero_outer source from EP817_SPLITZERO_OUTER_20260914.zip. Role: original word-to-value quotient, source metrics, and fixed-block classification through rank five. That fixed-block domain remains separately specified.

The Clankers. (2026, September 14). *SplitZero transfer to Erdős 817: Uniform carry cohomology and exact mixed-base capacity*. EP817 revision 98fd25802a387138053064e11ddd2f6d2007d55b, research/splitzero_transfer. Role: prior uniform inverse and fixed-block/mixed-radix interfaces.

KokunoYumeto. (2026). *Reconstruction with changes of support index*. Zeta revision 42df8a2de002d5fc7090641fac46ea11be05fa71, workbenches/splitzero-tandem/tex/support_diagrams.tex. Role: original reconstructed support diagrams, internal quotients, and the distinction between original boundaries and a further observation kernel. Later Zeta main was observed separately; no newer analytic conclusion is imported.

Graphical obstruction propagation and a stronger six-term construction

Source: `history/EP817_OBSTRUCTION_GRAPH_20260915/payloads/ep817/research/graphical_obstructions/notes/graphical-obstructions.md` **Identity:** `1e7dda55215faf90696066ab8690a6136518f343635fefaf739e857e8ab933e7f` Research contribution, 15 September 2026. Ordinary mathematical proofs and exact finite certificates; independent mathematical review pending. No new Lean elaboration or global priority claim is made.

1. The original problem and the result obtained

For a finite set A of distinct positive integers let

$$H(A) = \left\{ \sum_{a \in U} a : U \subseteq A \right\}, \quad S(A) = \sum_{a \in A} a.$$

The empty subset is an actual source choice with numerical value zero. Let $g_k(n)$ be the least possible maximum of an n -element A for which $H(A)$ has no nonconstant ordered k -term arithmetic progression. The preceding work proves existence of

$$\lambda_k = \lim_{n \rightarrow \infty} g_k(n)^{1/n} = \inf_{A \neq \emptyset, k\text{-admissible}} (2S(A) + 1)^{1/|A|}. \quad (1.1)$$

This continuation obtains

$$\lambda_6 \leq 1651^{1/10} = 2.0978513292766505 \dots \quad (1.2)$$

The actual ten-generator block is

$$A_{\sharp} = \{3, 4, 7, 34, 37, 41, 216, 250, 253, 257\}. \quad (1.3)$$

It is the set of pairwise distances of the five actual marks (0,4,7,41,257). Its 291 distinct subset sums are certified six-progression-free modulo 1651, including every nonzero step for this composite modulus, $1651=13 \cdot 127$. Writing $n=10j+s$ with $1 \leq s \leq 10$ and a_s the s -th increasing member of (1.3),

$$g_6(n) \leq a_s 1651^j \leq 257 1651^{\lceil n/10 \rceil - 1}. \quad (1.4)$$

The previous workbench bound was $93^{1/6}$. The strict improvement is the integer inequality

$$1651^3 = 4,500,297,451 < 6,956,883,693 = 93^5.$$

The modulus 1651 is also optimal over all infinite canonical mixed-radix schedules repeating this exact block. This fixed-block optimality is distinct from an assertion that (1.2) is the exact unrestricted λ_6 .

The structural results are stronger than the finite record. They give an all-rank collision-to-progression theorem for complete-graph sources, prove a sharp leading asymptotic within that entire source family, identify an infinite

two-parameter family of failed extensions, and give a precise infinite family of successful replacements. The companion note constructs a canonical monotone Bellman form of the earlier compact dual.

The original k=4 construction and signed-block proof remain credited to the anonymous/deleted Reddit contributor. sneed-and-feed retains separate credit for PR #1's Lean finite-upper-bound extension. The new bounds and proofs do not inherit that Lean verification status.

2. Recognizing the complete-graph source without changing its arithmetic

The old blocks have a common exact description:

$$\begin{aligned} \{1, 7, 8\} &= \{t_j - t_i : i < j\}, & T &= (0, 1, 8), \\ \{1, 4, 5, 17, 21, 22\} &= \{t_j - t_i : i < j\}, & T &= (0, 1, 5, 22). \end{aligned}$$

Let $G=(V,E)$ be a finite simple graph and t_v distinct integer marks. Order vertices by their marks and orient each edge $i < j$. Its positive generator is $a_{ij}=t_j-t_i$. When these values are pairwise distinct, edge subsets are exactly subsets of the original generator set. In all applications asserting a cardinality for A , distinctness is checked rather than assumed from the number of edges.

Retain the integral boundary and scalar observation

$$\begin{aligned} \partial : \mathbb{Z}^E &\longrightarrow \mathbb{Z}^V, & e_{ij} &\longmapsto e_j - e_i, \\ L_T : \mathbb{Z}^V &\longrightarrow \mathbb{Z}, & x &\longmapsto \sum_v t_v x_v. \end{aligned}$$

The subset-sum map factors exactly as

$$\boxed{\Phi_A = L_T \partial.} \tag{2.1}$$

Translation of all marks by a specified integer t changes L_T by t times the coordinate-sum functional. That functional vanishes on $\text{im}(\partial)$, so it leaves every actual edge generator unchanged. No rescaling of generators is used.

The two relation spaces are compared by the exact sequence

$$0 \longrightarrow \ker \partial \longrightarrow \ker \Phi_A \xrightarrow{\partial} \ker(L_T|_{\text{im } \partial}) \longrightarrow 0. \tag{2.2}$$

Surjectivity means exactly that an element of $\text{im}(\partial)$ killed by L_T has an original edge-chain lift. The kernel of that restriction is $\ker(\partial)$. For a connected graph on m vertices with e edges, the corresponding rational dimensions are $e-m+1$, $e-1$, and $m-2$. For the new K_5 example they are 6, 9, and 3. These full linear kernels are not identified with their intersections with the binary difference box.

For each admitted edge support, use the free module on its actual binary choices, the free module on its boundary-image points, and then the free module on its numerical-image points. Their maps send each basis element to its actual image. Inclusions of admitted supports commute with both maps. These are diagrams to which the original SplitZero reconstruction and internal-quotient construction apply: a relation goes to the zero of its original support fiber, while external absence remains the bottom support. The second map retains its additional observation kernel separately, as in (2.2).

3. A general obstruction from disjoint lifts of one observed boundary

Theorem 3.1. Let a finite free integral source have basis E , positive observed weights a_e , a boundary map ∂ , and observation ℓ with $\Phi = \ell \partial$. Suppose $c^1, \dots, c^r$ are vectors in $\{-1, 0, 1\}^E$ with pairwise disjoint supports and

$$\partial c^h = z \quad (1 \leq h \leq r), \quad d = \ell(z) \neq 0.$$

Then $H(A)$ contains an $(r+1)$ -term progression of difference d , with explicit actual subset representatives.

Proof. Write $c^h = c_+^h - c_-^h$ with disjoint positive and negative supports. For Q subset $\{1, \dots, r\}$, choose $c^h_+ +$ when h belongs to Q and c^h_- otherwise. Their union is an actual subset because the supports of the different c^h are disjoint. Its value is

$$X + |Q|d, \quad X = \sum_h \Phi(c_-^h).$$

The map Q to this union to its numerical value is the required typed source map. Choosing $Q = \{1, \dots, j\}$ for $0 \leq j \leq r$ gives the progression. Negative d is permitted; reversal gives a positive-step tuple. QED.

For a graph, every oriented u - v path gives a signed edge chain of boundary $e_v - e_u$. Thus r edge-disjoint u - v paths force an $(r+1)$ -term progression of difference $t_v - t_u$. On K_m there are $m-1$ explicit paths: the direct edge and one path through each of the other vertices. Hence every distinct-distance K_m source has an m -term progression.

For example, the new K_5 block has the five-term progression

$$0, 257, 514, 771, 1028.$$

The four disjoint increments are the direct edge of length 257 and the three two-edge paths through marks 4, 7, 41. This proves that the same block cannot be used as a five-progression-free construction.

A further graph consequence is $|E| \leq (k-2)(|V| - c(G))$ for a k -admissible distinct-distance graph, where $c(G)$ is its component count. Indeed each vertex pair has at most $k-2$ edge-disjoint paths. An integral unit-capacity flow either finds another path or exhibits a cut of size at most $k-2$: when no augmenting path remains, the vertices reachable from its source give such a cut. Split a connected component along that cut, apply induction to its two induced graphs, and add at most $k-2$ cut edges. Disconnected components are treated separately. Thus $|E| \leq (k-2) \text{rank}(\text{partial})$. This is a bound on this specified graphical source, not a claim that arbitrary generator relations are graphical.

The converse to Theorem 3.1 does not hold for an arbitrary scalar observation. For $A = \{1, 2\}$, the map $\Phi_{(1,2)}: \mathbb{Z}^2 \rightarrow \mathbb{Z}$ sends the actual masks 00, 10, 01, 11 to 0, 1, 2, 3. Three disjoint nonzero lifts cannot fit on two basis supports. The numerical progression is instead retained by this explicit projection and its relation $2e_1 - e_2$. The theorem supplies a source obstruction, not an equivalence for arbitrary projections.

4. Tournament-score fibers of the complete graph

Write $m = |V|$ and $n_m = m(m-1)/2$. Begin with the tournament in which vertex i has score i . Choosing an edge reverses its orientation. The score of a binary edge vector ϵ is

$$s(\epsilon) = \rho - \partial\epsilon, \quad \rho = (0, 1, \dots, m-1). \quad (4.1)$$

Let Σ_m be the image of this map. It consists of the actual tournament-score vectors, with all orientation multiplicities retained in its fibers. Each coordinate lies in $[0, m-1]$, and the coordinate sum is n_m . Consequently Σ_m contains no nonconstant $(m+1)$ -term vector progression: any nonzero integral coordinate difference would span at least m , exceeding its width.

The scalar square is

$$\Phi_A(\epsilon) = C_T - L_T(s(\epsilon)), \quad C_T = \sum_i it_i. \quad (4.2)$$

Every permutation of $(0, 1, \dots, m-1)$ is the score of exactly one transitive tournament, giving $m!$ distinct source score points. Enumerating all original orientation masks gives $|\Sigma_m| = 2, 7, 38, 291, 2932$ for $m = 2, 3, 4, 5, 6$. These finite counts are independently computed here; no general forest-count formula is imported.

Injection at the first value level alone does not certify progression preservation. For $T = (0, 1, 5, 22, 257)$, (4.2) is bijective on all 291 score points, but the numerical image has a six-term progression. Section 7 gives the actual rows and the receiving-kernel defect. The next theorem proves the converse implication needed for a lower bound: a collision at the first level forces an additional progression.

5. All-rank collision rigidity, with its original integer correction

5.1 Integral orientation allocation

The following criterion is the classical generalized Landau orientation theorem; see Kolesnik and Sanchez (2024, Corollary 4.2). The complete integral-flow proof is included to fix the source, capacities, and the actual allocation map used in the new prolongation. The criterion itself is not claimed as a new result.

Lemma 5.1. For a finite loopless multigraph M , an integer vector d is the score vector of an orientation if and only if

$$\sum_v d_v = |E(M)|, \quad \sum_{v \in U} d_v \geq |E(M[U])| \quad (U \subseteq V). \quad (5.1)$$

Proof. Every edge internal to U contributes one win in U , proving necessity. For sufficiency, make a network with source-to-edge-node capacity one, an edge-node-to-each-endpoint capacity $|E(M)|+1$, and vertex-to-sink capacity d_v . Singleton inequalities make d_v nonnegative. A cut of capacity below $|E(M)|$ cannot separate an edge node from either endpoint. Optimizing its chosen edge nodes for a given vertex set U makes its capacity $|E(M)| - |E(M[U])| + \sum_{v \in U} d_v$, at least $|E(M)|$. Thus there is no smaller cut.

Integral augmenting paths produce an integral flow. If they stop below $|E(M)|$, the reachable set in the residual network is a cut of that smaller value, a contradiction. Every original edge therefore sends its one unit to precisely one endpoint. These assigned endpoints are its winners, with scores d . This also gives the explicit construction. QED.

5.2 Root-budget prolongation

Theorem 5.2. Let $m \geq 2$ and let r in $\mathbb{Z}^m$ be nonzero, have coordinate sum zero, and satisfy

$$r(U) := \sum_{i \in U} r_i \leq |U|(m - |U|) \quad (U \subseteq \{0, \dots, m-1\}). \quad (5.2)$$

Choose i where r is maximal and j where it is minimal, and put $\alpha = e_j - e_i$. There are actual tournament-score vectors $x_0, \dots, x_m$ with first difference α and

$$\Delta(x_0, \dots, x_m) = (0, \dots, 0, r). \quad (5.3)$$

The construction uses integer endpoint allocations and actual edge reversals. In particular the defect r is retained rather than silently declared zero.

Proof. Nonzero integrality and sum zero give $r_i \geq 1$ and $r_j \leq -1$. Put $w = r + \alpha$. Let F_{ij} be the score face with $s_i = 0$ and $s_j = m-1$; its remaining coordinates are one plus the scores of a tournament on the other $m-2$ vertices.

Form the multigraph M by taking one K_m and one additional K_{m-2} on those other vertices. Put

$$b_i = 0, \quad b_j = m-1, \quad b_h = m-2 \quad (h \neq i, j), \quad d = w + b.$$

The sum of d is $|E(M)|$. For a subset U of size a , the upper score cut bound on w is

$$M(U) = |E(M)| - |E(M[V \setminus U])| - b(U).$$

Counting the two complete graphs gives the following four exact values:

membership of i, j in U	$M(U)$
neither	$a(m-a) - a$
i only	$a(m-a)$
j only	$a(m-a) - (m-1)$
both	$a(m-a) - (m-a).$

(5.4)

We prove $w(U) \leq M(U)$. Empty and full subsets are immediate. When neither endpoint belongs to U , maximality gives $r_i \geq r(U)/a$. Apply (5.2) to U union $\{i\}$:

$$r(U) + r_i \leq (a+1)(m-a-1),$$

so $r(U) \leq a(m-a-1)$. When both endpoints belong to U , apply the same argument to $-r$ on the complement, whose maximum is at j . This gives $r(U) \leq (m-a)(a-1)$. When only i belongs to U , (5.2) and $w(U) = r(U) - 1$ suffice.

It remains to treat U containing j but not i . At $a=1$ or $a=m-1$, $r_j \leq -1$ or $r_i \geq 1$ gives $r(U) \leq a(m-a)-m$. For $2 \leq a \leq m-2$, let P_a be the sum of the a largest coordinates, ordered with i first and j last. Then

$$r(U) \leq P_a - r_i + r_j, \quad r_i \geq P_a/a, \quad r_j \leq -P_a/(m-a), \quad P_a \leq a(m-a).$$

Since $1-m/[a(m-a)] \geq 0$ in this range,

$$r(U) \leq \left(1 - \frac{m}{a(m-a)}\right) P_a \leq a(m-a) - m.$$

Adding the $+1$ from alpha proves (5.4) also in this last case. These four cases cover every original cut; no averaging of the relation coordinates changes r .

Because w has total zero, the upper cuts for w are equivalent by complements to the lower cuts (5.1) for d . Lemma 5.1 constructs an orientation of M . Split its score as $t + \tau$, where t is the score of the first K_m and τ is the score of the internal $K_{(m-2)}$. Define

$$s_i = 0, \quad s_j = m-1, \quad s_h = m-2 - \tau_h \quad (h \neq i, j).$$

Reversing the internal tournament shows s belongs to F_{ij} . The actual equality is

$$\boxed{t - s = r + \alpha.} \tag{5.5}$$

In the tournament for s , j wins against every other vertex and i loses against every other vertex. The direct j - i path and the paths j - h - i are pairwise edge-disjoint. Reverse the first h such paths, for $h=0, \dots, m-1$. They yield actual scores $s-h\alpha$. Order these scores as

$$s - (m-1)\alpha, \dots, s - \alpha, s,$$

and append t . The first m rows have difference α , and their final second difference is $t-s-\alpha=r$ by (5.5). This proves (5.3), with actual orientation masks for every row. QED.

5.3 Arithmetic consequence and bounded-kernel equality

For s, t in Σ_m , their difference $r=t-s$ satisfies (5.2): for any a -vertex set U , a tournament has between $\text{binom}(a,2)$ and $\text{binom}(a,2)+a(m-a)$ wins in U . Hence the difference of two such scores is at most $a(m-a)$.

Theorem 5.3 (collision rigidity). Let T have distinct real marks, and let the binary edge-subset image of its complete graph contain no nonconstant $(m+1)$ -AP. Then L_T is injective on Σ_m .

Proof. A collision gives distinct score vectors with $r=t-s$ nonzero and $L_T r=0$. Apply Theorem 5.2. The scalar image of its packet has zero second differences, while its common difference is $L_T \alpha = t_j - t_i$, nonzero because the marks are distinct. Equation (4.2) carries it to an actual progression of edge-subset sums, with the opposite nonzero difference. Reversal gives a positive difference when desired. This contradicts admissibility. QED.

For an EP817 ruler, all edge weights are additionally distinct positive integers. Its first-level representation fibers are therefore exactly the tournament-score fibers. In particular $H(A)$ has $|\Sigma_m|$ values and the same full multiplicity distribution, independent of the admitted ruler.

The precise bounded-relation assertion is

$$\boxed{\ker \Phi_A \cap \{-1, 0, 1\}^E = \ker \partial \cap \{-1, 0, 1\}^E} \tag{5.6}$$

for every such $(m+1)$ -admissible ruler. To prove it, decompose c in the displayed box into its positive and negative binary vectors. Their score difference is partial c , and numerical equality forces it to vanish by Theorem 5.3. The reverse inclusion follows from (2.1). Full linear kernels keep their separate dimensions in (2.2); coefficients outside the stated box have not been removed.

A complete finite calibration constructs (5.3) for every nonzero difference of score points at $m=2,3,4,5$: respectively 2,18,200,3080 literal directions. It retains all 3300 packets and their binary masks. Another 200 explicitly seeded cases cover $m=6,\dots,10$. The all-rank assertion follows from the cut proof above, not extrapolation of those tables.

6. A sharp all-rank asymptotic within the critical graphical family

For fixed $m \geq 2$, permit arbitrary canonical mixed-radix schedules whose levels are complete-graph distance blocks of m -mark integer rulers, with distinct positive edge weights, and require every prefix image to be $(m+1)$ -AP-free. The ruler and radix may change at every level. Each level has $n_m = \binom{m}{2}$ generators. Let Ξ_m be the infimal exponential product cost per generator over this whole family.

Every individual level is $(m+1)$ -admissible, since its progression would survive with all other level choices zero. By Theorem 5.3 it has at least $m!$ distinct subset sums. Its canonical radix therefore satisfies

$$b > S(A), \quad b \geq S(A) + 1 \geq |H(A)| \geq m!.$$

Multiplying over any schedule gives

$$\boxed{\Xi_m \geq (m!)^{2/[m(m-1)]}}. \quad (6.1)$$

There is a corresponding explicit general upper construction. For a prime $p > m$, take

$$T = (0, 1, p, \dots, p^{m-2}), \quad q = p^{m-1}. \quad (6.2)$$

All pairwise distances are distinct. Edges from zero give powers of p ; the other edges are $p^a(p^d - 1)$. Their p -adic valuations and residual factors distinguish them, since $p \geq 3$ and $p^d - 1$ differs from one. Moreover

$$S(A) \leq C_T = \sum_{i=1}^{m-1} ip^{i-1} \leq (m-1) \frac{q-1}{p-1} < q.$$

Under (4.2), the reflected image $C_{T-H}(A)$ lies in the actual base- p digit box with every digit in $\{0, \dots, m-1\}$. A nonzero-step modular $(m+1)$ -AP modulo q , examined at the least p -adic position where its difference is nonzero, would have $m+1$ distinct residues modulo p in a set of only m allowed residues. Common lower digits are removed by their exact residue-and-quotient maps. This contradiction proves modular safety, including all word lengths after base- q lifting. The map to the digit coordinates is injective; the omitted zero-mark score is recovered from the fixed score sum.

The upper rate is

$$\boxed{\Xi_m \leq p^{2/m}}. \quad (6.3)$$

Bertrand's theorem supplies $m < p \leq 2m$ for every $m \geq 2$. Its classical prime-existence statement is the only prime-distribution input here. The elementary integral estimate $\log(m!) \geq m \log m - m + 1$ now gives

$$\frac{2(m \log m - m + 1)}{m(m-1)} \leq \log \Xi_m \leq \frac{2 \log(2m)}{m}.$$

Consequently

$$\boxed{\lim_{m \rightarrow \infty} \frac{m \log \Xi_m}{\log m} = 2}. \quad (6.4)$$

This leading law covers all arithmetic choices and infinite canonical schedules in the critical complete-graph family, without an assumed score-injectivity hypothesis: admissibility itself forces that injection. It is not a lower bound for the unrestricted λ_{m+1} , whose generators need not arise from this graph.

For general k , choose $m = \min(p, k) - 1$ in (6.2); a forbidden $(m+1)$ -AP is already excluded, so every longer k -AP is excluded. This independently rederives the prime upper bound $p^{2/(\min(p, k) - 1)}$ in Korsky's inspected v1. That numerical bound is prior work, not a novelty claim. The new role of (6.4) is to identify a complete source family where its leading coefficient cannot be improved by changing rulers or radices.

The implication for further work is specific. Finite constants can improve substantially inside this family, as Section 9 shows. A reduction of the leading coefficient below two for the full problem must use a different source family or a different relation structure; collapsing score classes in the critical complete graph creates a progression by Theorem 5.3. The next obstruction analysis therefore has to retain which parts of the complete-graph cut argument fail for a new source, rather than assuming all scalar collisions can be exploited safely.

7. An obstruction hyperplane that survives every appended mark

The old successful K_4 ruler is $T = (0, 1, 5, 22)$. A direct extension to five marks looks natural but fails for every appended mark for which the ten distances are distinct.

More generally, take integers

$$0 < a < b, \quad b \neq 2a, \quad c = 5b - 3a, \quad z > 2c. \quad (7.1)$$

The ten distances of $(0, a, b, c, z)$ are distinct positive integers. Among the six old distances, the only possible collision under these inequalities is $a = b - a$, excluded by $b \neq 2a$; the other three large old distances exceed b . Every new distance exceeds c because $z > 2c$, so it is distinct from every old one, and the new distances are mutually distinct.

Their actual subset sums contain

$$\begin{aligned} y_0 &= z - c, \\ y_1 &= z - b, \\ y_2 &= z + c - 2b, \\ y_3 &= z + 2b + c - 3a, \\ y_4 &= z + b + 2c - 3a, \\ y_5 &= z + 3c - 3a. \end{aligned} \quad (7.2)$$

Each has an explicit edge-subset representation. The first two use a single new edge. The next uses $(c-b) + (z-b)$. The last three use respectively

$$\begin{aligned} [b + (b - a) + (c - a)] + (z - a), \\ [c + (b - a) + (c - a)] + (z - a), \\ [c + (b - a) + (c - a) + (c - b)] + (z - a). \end{aligned}$$

All bracketed terms use distinct old edges. The consecutive differences are $c-b$ except for the middle one, which is $4b-3a$. Equation (7.1) makes that difference equal to $c-b$ as well, and $c-b > 0$. This proves a six-term progression for the entire integer family, not just sampled z .

At $a=1, b=5, c=22$ the row is

$$z - 22, z - 5, z + 12, z + 29, z + 46, z + 63, \quad d = 17.$$

The source kernel direction is visible independently of the appended z . The six tournament-score rows are

$$\begin{aligned} (0, 1, 2, 4, 3), \quad (0, 1, 3, 3, 3), \quad (0, 1, 4, 2, 3), \\ (1, 4, 0, 2, 3), \quad (1, 4, 1, 1, 3), \quad (1, 4, 2, 0, 3). \end{aligned}$$

Their four vector second defects are $0, r, -r, 0$, where

$$r = (1, 3, -5, 1, 0), \quad L_T(r) = 3a - 5b + c. \quad (7.3)$$

The last coordinate is zero, so changing z never repairs this relation. At $z=257$, scalar evaluation is still a bijection on all 291 first-level score values, but (7.3) creates the numerical progression 235,252,269,286,303,320. The second-difference comparison, rather than first-level injectivity alone, is the exact place where the obstruction survives.

The comparison of coefficient levels is explicit. The map $(x,y) \rightarrow (x,y,y)$ from Σ_5^2 to Σ_5^3 intertwines the pair difference $x-y$ with the second-defect map $x-2y+z$. Every pair difference has each coordinate in $[-4,4]$, whereas the displayed r has a coordinate -5 . The obstruction thus occurs at the larger, specified three-row source, while its scalar image still vanishes. This explains how first-level injectivity and the failed progression can coexist without equating their kernels.

Thus the failed extension identifies an arithmetic hyperplane to exclude, and it explains why searching larger appended marks on the same prefix cannot help. The successful replacement changes the prefix itself.

8. Affine source fibers give a complete infinite tail test

For fixed prefix $T=(t_0, \dots, t_{(m-1)})$, append $z > \max T$. Write A_T for the old distances. Selecting r of the m new edges contributes rz minus the sum of their r old endpoints. Hence the exact numerical image is

$$H(A_{T \cup \{z\}}) = \bigcup_{r=0}^m (rz + F_r),$$

$$F_r = H(A_T) - \left\{ \sum_{i \in I} t_i : |I| = r \right\}. \quad (8.1)$$

The original old subset and new-edge subset masks remain attached to every (r,v) . The free value-source projection sends its basis vector to $e_{(rz+v)}$; its further scalar-amplitude map is a separate observation. Equality in a projection does not erase the underlying support.

For a general finite packet C subset $\mathbb{Z}^2$, put $W = \max_C v - \min_C v$ and $E_z(r,v) = rz + v$. For every integer $z > 2W$, E_z is injective and preserves and reflects the k -AP relation, for every $k \geq 3$. To prove reflection, lift the numerical rows uniquely. Their second differences obey

$$z\Delta r_i + \Delta v_i = 0, \quad |\Delta v_i| \leq 2W < z,$$

so both integer differences vanish. Nonconstant vector progressions stay nonconstant because injectivity holds. Injection follows from $|v-v'| \leq W < z$. At smaller z , any scalar AP whose lift is not a vector AP satisfies the explicit equation $z = -\Delta v_i / \Delta r_i$ for at least one nonzero Δr_i . The actual finite packet and its full kernels are retained on both sides.

When $k=m+2$ and C is the packet in (8.1), its slopes lie in $[0,m]$, so a k -term vector progression has constant slope. Therefore the following gives the complete tail dichotomy: a k -AP in any F_r persists under translation rz for every valid z ; when all F_r are k -AP-free, every $z > 2W$ is safe. Distinct generator weights are checked separately at finite z and hold automatically sufficiently far out.

For the new prefix $T=(0,4,7,41)$, $S(A_T)=126$, $\sum T=52$, and $W=178$ is a valid whole-packet width. All five F_r are six-term-free by exact finite enumeration. Therefore every $z > 356$ gives a distinct ten-generator, integer-six-term-free distance block. The remaining 315 integers $z=42, \dots, 356$ were completely classified: 270 have an explicit six-AP, six have repeated generator weights, and 39 are integer-six-term-free. The full rows, original masks, and all successful parameter values are in the receipt. The record parameter $z=257$ belongs to the latter class.

This proves an infinite construction family and a complete finite exceptional range. It does not assert that every such z has the same efficient modular base. The modulus is separately tested, as follows.

9. The base-1651 certificate and its exact canonical fixed-block optimum

For (1.3), the verifier reconstructs all 1024 binary masks and their 291 value fibers. It checks all $1651 \cdot 1650 = 2,724,150$ ordered start/nonzero-step modular parameter pairs. No six-term progression occurs. A separate implementation scans the 480,150 pairs whose first term is actually a digit; the other 2,244,000 pairs are excluded by that first-term condition. Both computations include all steps for the composite modulus.

Canonical lifting is proved directly. A modular-safe digit set D subset $[0, q-1]$ gives a safe language at every length. Reduce a proposed integer AP modulo q . Its step must vanish modulo q and its lowest digits must agree. Remove the common actual digit c by $y \rightarrow (y-c)/q$, with inverse $x \rightarrow qx+c$. Induction on the length forces the original progression to be constant. Generator distinctness follows because all base weights lie strictly between zero and q . Taking an initial n -generator subcollection proves (1.4).

The certificate also constructs a returning section on all 81 carries:

$$r : \{-1, 0, 1\}^4 \longrightarrow D^6, \quad \Delta r(c) = -c. \quad (9.1)$$

Every smaller canonical base $q=1103, \dots, 1650$ has a stored nonconstant modular row d . Its carry $c = \Delta d/q$ belongs to $\{-1, 0, 1\}^4$. The original integer row

$$y_i = d_i + q r(c)_i \quad (9.2)$$

has zero second differences. It is nonconstant: unequal low digits in $[0, q-1]$ cannot be canceled by an integer multiple of q . Each value is checked against a mask of the actual twenty-generator set A_{sharp} union $q A_{\text{sharp}}$.

At the next radix, whatever integer $b > S(A_{\text{sharp}})$ is chosen, the carry equation is $c + \Delta r(c) = b \cdot 0$. Thus no subsequent radix repairs the obstruction. If the bad q appears later in a schedule, prepend actual constant-zero columns; the same witness is multiplied by its original preceding radix product. All earlier representations and the empty-word choice are retained.

An all-prefix-safe canonical schedule repeating A_{sharp} consequently has no radix below 1651. Constant radix 1651 is certified. Hence

$$\gamma_6(A_{\sharp}) = 1651^{1/10}. \quad (9.3)$$

The statement is quantified over all infinite canonical schedules for this fixed block. Radices $\leq S(A_{\text{sharp}})$, other blocks, and the unrestricted global λ_6 are not covered by its optimality assertion.

10. Original quotient metrics and the further additive kernel

Let $V = \mathbb{Q}[\{0, 1\}^E]$ with its actual orthonormal mask basis, and let $W = \mathbb{Q}[H(A)]$. The map π sends each basis mask to its numerical-value basis vector. For each value y let $\mu(y)$ be its number of masks. Its least-norm section is the average of those $\mu(y)$ masks, and the original quotient Gram is

$$G = \text{diag}_y(1/\mu(y)). \quad (10.1)$$

For any source coefficients in one fiber, subtracting their mean proves the exact Pythagoras identity; no metric is declared Euclidean after quotienting. On the new ruler the multiplicity histogram is

μ	1	2	4	8	14	24
number of values	120	60	60	30	20	1.

Its total source mass is 1024. The largest fiber has size 24, so the identity from quotient coordinates with Gram G to unit value coordinates has norm $\sqrt{24}$. For t canonical levels, the actual tensor quotient Gram is $G^{\otimes t}$, and the corresponding return norm is $24^{t/2}$. These costs remain separate from the unit-cost Boolean observation used only for existence of a path.

At the next additive level, the map

$$\Sigma_5 + \Sigma_5 \longrightarrow H(A_{\sharp}) + H(A_{\sharp})$$

has 3081 source values and 2103 numerical values. Its free linearization is onto and has kernel dimension 978. The addition square from $\mathbb{Q}[\text{Sigma_5}]$ tensor $\mathbb{Q}[\text{Sigma_5}]$ commutes with the numerical addition square. This identifies the extra observation kernel at that level explicitly. The first-level rigidity theorem does not delete this second-level kernel, and the modular certificate is what controls its feasible six-row intersections.

11. Obstruction handling and the global objective

The companion Bellman theorem distinguishes two consequences of a failed trial inequality. A particular coefficient potential may need correction even when its trial threshold is achievable in the corresponding restricted family. A closed cycle of actual levels with cost below its generator threshold gives an arithmetic construction and rules that threshold out. The new base-1651 empty-support self-edge is of the second type: it violates the old $93^{1/6}$ target independently of every potential coefficient.

The graphical progression mechanism makes other failures constructive. A nonzero scalar observation on disjoint lifts gives a progression by Theorem 3.1. A collapsed complete-graph score direction gives the original source packet (5.3). A bad append parameter is detected in the exact affine fibers (8.1); the hyperplane (7.3) proves that an entire attempted extension family fails. A bad smaller radix has the stored return primitive (9.2). These failures change the next construction instead of being treated as evidence that the general problem is impossible.

The unbounded result (6.4) is a matching leading law for the critical complete-graph source class. The unrestricted block family remains broader. The present proof does not supply a comparison embedding every admissible block into that class with controlled generator reward and original metric; none is asserted. The full general problem therefore still requires control of the infinite arithmetic inequalities in the compact dual, beyond this source family.

12. Verification and source roles

The standalone producer is `certificates/verify_graphical_obstructions.py`. The independent arithmetic auditor is `certificates/audit_graphical_receipt.py`, which imports no producer module. The commands and delivered input hashes are in the contribution README. Their scopes include the complete new modular certificate, 81 original return rows, 548 bad-radix witnesses, the full 315-parameter exceptional range, all 3300 complete collision packets through five vertices, eight prior finite-rank Bellman systems, and mutation tests. The larger-rank collision cases are explicitly identified as samples. The general proofs are in this note.

The graph calibration additionally checks 1098 simple graphs on two through five vertices, 10,650 terminal pairs, and 9044 nonempty disjoint path packings. The maximum flows are compared with all cuts in that finite domain. The affine-packet calibration checks 2730 parameter tests. Eight deliberately false formulas and eight corrupted proof-data records must be rejected. Exact integers or rational numbers are used in every proof check.

The original workbench main was read at `23c0110c95b5a2036bdc04a1f352b6e5e27742c8`. The observed outer-control branch is `74ba7dcc4bd623b3555046feb6d45ccac38417a3`. The preceding variable-block/compact-dual package remains a distinct local source with its own hashes and prior audit scope. No new remote write or inherited Lean claim is implied by these references.

References

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- Kolesnik, B., & Sanchez, M. (2024). The geometry of random tournaments. *Discrete & Computational Geometry*, 71, 1343–1351. <https://doi.org/10.1007/s00454-023-00571-4>. Published online September 21, 2023. Sections 1–4, especially Corollary 4.2, inspected for the classical multigraph score criterion used in Lemma 5.1. The specific root-budget correction and scalar-obstruction construction are proved in this note.
- Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/html/2606.24139v1>. Inspected general bounds and graph construction; the prime upper benchmark is prior work. No theorem from an uninspected paper is substituted for the proofs here.

Postnikov, A. (2009). Permutohedra, associahedra, and beyond. *International Mathematics Research Notices*, 2009(6), 1026–1106. <https://doi.org/10.1093/imrn/rnn153>. Original preprint math/0507163. Abstract inspected for graphical/permutohedral context. The root-budget and orientation arguments above are proved explicitly; no forest-count identity is borrowed from an unread body.

Erdős, P. (1932). Beweis eines Satzes von Tschebyschef. *Acta Litterarum ac Scientiarum, Sectio Scientiarum Mathematicarum*, 5, 194–198. <https://acta.bibl.u-szeged.hu/13396/>. Archival metadata checked; historical source of the elementary Bertrand theorem. The precise prime-existence statement is also exposed in the official Mathlib documentation below.

The mathlib community. (n.d.). *Mathlib.NumberTheory.Bertrand* [Formal source documentation]. https://leanprover-community.github.io/mathlib4_docs/Mathlib/NumberTheory/Bertrand.html. Inspected declaration `Nat.exists_prime_lt_and_le_two_mul`; used only for the standard theorem $m < p \leq 2m$. This citation does not certify any new result in this contribution in Lean.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. `workbenches/splitzer o-tandem/tex/support_diagrams.tex`, Zeta Function Research Reader, revision 42df8a2de002d5fc7090641fac46ea11be05fa71. Original diagram reconstruction, supported zeros, and internal quotient interfaces.

The Clankers. (2026, September 14). *A compact all-rank dual from the retained reset cost* [Prior research note]. EP817_ZETA_INTEGRATION_20260914, `research/variable_blocks/notes/compact-global-dual.md`. Exact predecessor; its accepted scope is not expanded by reference alone.

Canonical monotone potentials and compilation of arithmetic obstructions

Source: history/EP817_OBSTRUCTION_GRAPH_20260915/payloads/ep817/research/graphical_obstructions/notes/monotone-obstruction-dual.md **Identity:** f65163e7b4e1c1f1fb3216fc629d82f97024a4042a5f77f44ce6c001bfe88a9d
Research continuation, 15 September 2026. This note continues the exact all-rank dual in the retained 14 September integration package. All statements below have ordinary proofs. The finite calibrations use the existing independently audited bounded-rank certificates; they are not new all-rank Lean checks.

1. The original support controller and the retained reset

Fix $k \geq 3$. A canonical arithmetic level is (b, A) , with $b \geq 2$, A a finite set of distinct positive integers, and $\sum A < b$. The empty block is allowed as a present zero-digit level. Its digit source has one actual word, not an absent object.

Let $C_k = \{-1, 0, 1\}^k$. A state R is a safe sign-and-reversal-invariant support of true-flag carries, so 0 does not belong to R . The separately present false state is $(0, \text{false})$. The actual column equation is

$$bc' = c + \Delta d, \quad \Delta d_i = d_i - 2d_{i+1} + d_{i+2}, \quad d \in H(A)^k.$$

The transfer $T_-(b, A)(R)$ is the union of all actual true-flag successors and all nonconstant successors of the false state. It is monotone under inclusion of R . Keep an edge only when its target is safe. Every edge retains its radix b , generator reward $n = |A|$, full source columns, and representation fibers.

The original level $(2, \text{empty})$ sends every safe true support to empty true support. Its only actual digit column is zero, so $2c' = c$ forces $c = c' = 0$; no nonzero true carry survives. The false zero state survives. Its radix cost is two. Erasing that cost changes every subsequent generator and is not allowed.

The earlier exact finite-set characterization is

$$\lambda_k = \inf_{A \neq \emptyset, k\text{-admissible}} (2S(A) + 1)^{1/|A|}. \quad (1.1)$$

A canonical modular-safe block is an empty-true-support self-edge. Every admissible A supplies such an edge at radix $2S(A) + 1$. These facts, and the represented path/AP correspondence, are proved in the predecessor notes and retained in this contribution's source inventory.

2. The greatest bounded subsolution is an explicit future value

For $u > 0$ and a finite safe path p , retain its actual product $P(p)$ of radices and total generator reward $N(p)$. Define

$$F_u(R) = \inf_{p \text{ finite safe starting at } R} \frac{P(p)}{u^{N(p)}}. \quad (2.1)$$

The empty path is included and has value one. There is no assumption that an infimum is attained. Every finite path has a strictly positive value.

Theorem 2.1. For every u in $[1, 3]$, the following statements hold.

If $u \leq \lambda_k$, then $1/2 \leq F_u(R) \leq 1$ at every safe support. The coefficient function $H_u = 2F_u$ belongs to $[1, 2]^{\wedge}(S_k)$, is monotone under inclusion of supports, and satisfies the exact equation

$$H_u(R) = \min \left\{ 2, \inf_{R \xrightarrow{(b,A)} T} \frac{b}{u^{|A|}} H_u(T) \right\}. \quad (2.2)$$

It is the greatest nonnegative function bounded above by two that satisfies all original edge inequalities

$$bH(T) \geq u^{|A|} H(R). \quad (2.3)$$

If $u > \lambda_k$, then $F_u(R) = 0$ at every safe support. In particular (2.2) has greatest bounded subsolution zero.

Proof. A finite path from an arbitrary safe R can be read from empty true support with the same labels, since every successor then remains contained in the original support. Append the actual reset. The resulting period is admissible, with product $2P(p)$ and reward $N(p)$. If the reward is positive,

$$\lambda_k \leq (2P(p))^{1/N(p)}.$$

For $u \leq \lambda_k$ this gives $P(p)/u^{\wedge}(N(p)) \geq 1/2$. Zero-reward paths have product at least one. Taking infima proves the bounds.

If R is a subset of T , every label path safe from T is safe from R , with unchanged cost. Therefore $F_u(R) \leq F_u(T)$. Separating empty paths from nonempty paths by their first edge proves (2.2), including the possibility that the inner infimum is not achieved. Positive multiplication commutes with these infima.

Let G take values in $[0, 2]$ and satisfy (2.3). Multiplication along any finite path gives

$$G(R) \leq \frac{P(p)}{u^{N(p)}} G(\text{end } p) \leq 2 \frac{P(p)}{u^{N(p)}}.$$

Infimizing proves $G \leq 2F_u$. The function $2F_u$ itself satisfies (2.2), so it is the greatest subsolution.

Finally, if $u > \lambda_k$, (1.1) provides a literal modular-safe self-edge (q, A) with $\theta = q/u^{\wedge}(|A|) < 1$. From any R apply the reset and then repeat this edge h times. Its cost is $2\theta^h$, tending to zero. Thus $F_u(R) = 0$. Every path in this sequence retains its positive numerical cost and actual source words. QED.

This is an exact dichotomy over all generator ranks, not an assumption about an unknown general lower estimate. It does not evaluate the threshold λ_k .

3. A stronger form of the all-rank dual

The theorem gives the equivalent compact formulation

$$\lambda_k = \max \{ u \in [1, 3] : \exists H : S_k \rightarrow [1, 2], H \text{ is inclusion-monotone and satisfies (2.3) on all actual safe edges} \}. \quad (3.1)$$

Necessity uses H_u . For sufficiency, an admissible A has the modular-safe self-edge $(2S(A)+1, A)$ at empty true support. Cancel its positive $H(\text{empty})$ in (2.3); then use (1.1). The same argument proves feasibility at $u = \lambda_k$, so the maximum is attained.

The monotonicity is proved from future feasibility, rather than imposed on the earlier past-path construction. The coefficient interval still comes from the exact reset cost; no source Gram or arithmetic weight has been rescaled. The number of coefficients depends on k alone. The infinitely many actual block inequalities in (3.1) remain the arithmetic task.

When $u > \lambda_k$, the limit $H_u(R) = 0$ is a zero coefficient at the same admitted support. The coordinate observation $R^{\wedge} S_k \rightarrow R$ at that support sends the limiting vector to zero; it does not remove the false state or turn its label into external absence. This distinguishes a vanishing infimal cost from a missing source object. A finite violating arithmetic period, rather than the notation for that limit alone, is the useful obstruction certificate.

4. Compiling a low-cost path into an actual modular block

Suppose a finite safe path has

$$2P(p) < u^{N(p)}. \quad (4.1)$$

Read its labels from empty true support and append the present zero reset. If its nonempty levels are (b_j, A_j) , set

$$C = \bigcup_j P_j A_j, \quad Q = 2P(p),$$

where P_j is the retained product of all preceding radices, including empty levels. Canonical inequalities imply that the generators are distinct and $|C|=N(p)$. Also

$$S(C) \leq \sum_j P_j (b_j - 1) = P(p) - 1 < Q.$$

The reset makes the whole period's true transfer empty. A nonconstant modular k -AP in $H(C)$ modulo Q would give its nonconstant initial carry path through that period, contradicting the empty transfer. Hence (Q, C) is an actual modular-safe certificate. Inequality (4.1) gives

$$\lambda_k \leq Q^{1/|C|} < u.$$

This is a typed compiler from a finite supported path to a finite arithmetic certificate and then to an infinite family of distinct integer generators. It preserves every intermediate radix and zero-word choice.

If a closed safe cycle has product P and positive reward N with $P < u^N$, it already produces an admissible periodic integer construction from empty true support by monotonicity. To obtain a modular certificate with the reset retained, repeat it any integer h with $2(P/u^N)^h < 1$ and then apply the compiler. At $u = B^{1/p}$, the strict test is entirely integral:

$$(2P^h)^p < B^{hN}.$$

Conversely, $u > \lambda_k$ supplies such a low-cost path by repeating the self-edge used in the proof of Theorem 2.1. Thus the failure of a threshold has an actual finite certificate. The theorem gives neither a uniform bound on its rank nor a stopping bound for a search at a correct threshold.

5. Which obstruction changes coefficients, and which changes the rate

A failed inequality for a chosen H need not rule out its threshold. In the restricted rank-one controller for $k=5$ or $k=6$, $u = \sqrt{8}$ is the proved optimum. The choice H identically one fails the safe first level $(2, \{1\})$ because $2 < \sqrt{8}$. Its incoming/outgoing coefficient ratio can be repaired: (2.3) requires $H(T)/H(R) \geq \sqrt{2}$. The canonical future potential supplies that change. This example concerns the specified restricted controller, not a claim that $\sqrt{8}$ is globally feasible.

A self-edge has no coefficient ratio available to repair. The new ten-generator block has an empty-support self-edge at radix 1651. At the old global upper threshold $u = 93^{1/6}$, (2.3) would require $1651^3 \geq 93^5$, which fails. This obstruction changes the arithmetic upper bound independently of every proposed H . Its actual block and modular proof are delivered in the companion note.

For the eight previously audited finite controllers, future values can be computed in exact rational powers. At $u = B^{1/p}$, write $f = F_u^p$. The recurrence is

$$f(R) = \min\{1, \min_{R \rightarrow T} (b^p / B^{|A|}) f(T)\}.$$

The new checker computes these values, proves $2^{-p} \leq f \leq 1$, checks every retained cost representative, verifies equality in the recurrence, and checks support-inclusion monotonicity. The original full profile audits retain their separate scope. All eight $k=5,6$ and $\text{rank}=1, \dots, 4$ calibrations pass, including states that need a nonconstant potential.

6. General pattern matrices and the preserved boundary

For the preceding binary-rigid translation-invariant integer pattern matrix L , replace the AP second-difference matrix by L and the reset radix by its established value $R_L = \max(2, \max_i m_i)$, where m_i is the positive row mass. The same proof gives

$$F_u(Q) \in [1/R_L, 1], \quad H_u = R_L F_u \in [1, R_L]$$

at every feasible threshold. The monotone Bellman equation has terminal value R_L , and the path compiler retains the factor R_L . This transfer uses the literal support diagram, actual pattern columns, integral carries, and old quotient maps. It does not claim a new analytic estimate for the Zeta source.

The new graph packet gives another source-level obstruction mechanism in that framework. A scalar relation among complete-graph score points is prolonged to an actual packet whose sole last second defect is that same relation. Its scalar observation is zero while its original source defect remains visible. The resulting arithmetic progression is the obstruction returned by the compiler at the source level.

References and proof scope

The Clankers. (2026, September 14). *Supported zero resets and exact variable-block capacity; A compact all-rank dual from the retained reset cost* [Research notes]. Exact prior package `EP817_ZETA_INTEGRATION_20260914`. These supply the original controller and finite-set interface. The complete notes and their hash-bound certificates are retained as predecessors.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. Zeta Function Research Reader, revision `42df8a2de002d5fc7090641fac46ea11be05fa71`. Original support and internal quotient conventions.

The Clankers. (2026, September 15). *Graphical obstruction propagation and a stronger six-term construction* [Companion research proof]. This contribution. No new Lean verification or completed evaluation of the all-rank feasible maximum is claimed.

Returning graphical obstruction control to the original SplitZero interfaces

Source: `history/EP817_OBSTRUCTION_GRAPH_20260915/payloads/zeta/workbenches/ep817-graphical-obstructions/notes/splitzero-return.md` **Identity:** `bd31cc477b88ac3312bef19b7f813fd10308609e1aa01acd1c78fc93a7522061`
 Research contribution, 15 September 2026. This is an additive transfer of proved algebraic and combinatorial results from Erdős Problem 817. It uses the original support-reconstruction and internal-quotient conventions. No theta-source relation, zero packet, analytic metric, or arithmetic weight estimate is redefined. Written proofs and exact finite certificates are supplied; no new Lean elaboration is claimed.

1. Two original relation complexes and the actual comparison

Fix an oriented finite graph $G=(V,E)$, integer marks t_v , and the original integral map

$$\partial : \mathbb{Z}^E \rightarrow \mathbb{Z}^V, \quad e_{ij} \mapsto e_j - e_i, \quad \ell_t(x) = \sum_v t_v x_v.$$

Let $\Omega = \{0,1\}^E$ be the actual binary source. For each admitted subfamily S subset Ω , define

$$V_S = \mathbb{Z}[S], \quad X_S = \partial(S), \quad Y_S = \ell_t(X_S),$$

$$q_S : V_S \rightarrow \mathbb{Z}[X_S], \quad e_\epsilon \mapsto e_{\partial\epsilon}, \quad J_S : \mathbb{Z}[X_S] \rightarrow \mathbb{Z}[Y_S], \quad e_x \mapsto e_{\ell_t x}.$$

Write $B_S = \ker q_S$ and $K_S = \ker(J_S q_S)$. Their inclusion is literal, and

$$\boxed{0 \rightarrow B_S \rightarrow K_S \xrightarrow{q_S} \ker J_S \rightarrow 0.} \quad (1)$$

Surjectivity follows by lifting each source basis value x to an actual binary word in its fiber. The kernel is exactly B_S . Thus the induced map on the degree-zero cohomologies of $[B_S \rightarrow V_S]$ and $[K_S \rightarrow V_S]$ has the additional kernel $K_S/B_S = \ker J_S$. Source fibers and the extra observation kernel remain separate.

For S subset T the basis inclusions commute with all maps. These are diagrams over the actual join-semilattice of admitted word supports. Applying the original SplitZero reconstruction sends a boundary to its own supported zero. The nonbottom support $\{\text{empty binary word}\}$ has a one-dimensional value basis e_0 . The further amplitude map $e_0 \rightarrow 0$ is explicit; it is not external absence.

The free-word relation complex and the graph-chain complex also have an explicit comparison. Let $a_S : \mathbb{Z}[S] \rightarrow \mathbb{Z}^E$ send e_ϵ to ϵ , and let $b_S : \mathbb{Z}[X_S] \rightarrow \mathbb{Z}^V$ send e_x to x . Then

$$\partial a_S = b_S q_S. \quad (2)$$

For the full source Ω , a_Ω maps B_Ω onto $\ker(\partial)$. To see surjectivity, select a spanning forest. Each nonforest edge gives an integral fundamental cycle with coefficients in $\{-1,0,1\}$. Its positive and negative binary parts have the same boundary, so their word difference belongs to B_Ω and augments to that cycle. Subtracting fundamental cycles expresses every integral graph cycle. Consequently

$$0 \rightarrow B_\Omega \cap \ker a_\Omega \rightarrow B_\Omega \xrightarrow{a_\Omega} \ker \partial \rightarrow 0. \quad (3)$$

No equality between these two kernels is asserted. Formula (3) retains the relations among word representatives that the graph amplitude map forgets.

At the graph amplitude level the second observation has its own exact sequence

$$0 \rightarrow \ker \partial \rightarrow \ker(\ell_t \partial) \xrightarrow{\partial} \ker(\ell_t|_{\text{im } \partial}) \rightarrow 0. \quad (4)$$

Equations (1)–(4) are the comparison maps between the different source types used in the arithmetic application.

2. A generic disjoint-boundary obstruction

If $c^1, \dots, c^r$ in $\{-1, 0, 1\}^E$ have disjoint supports, the same boundary z , and $\text{ell}_t(z) \neq 0$, choose the positive part of c^h for h in Q and its negative part otherwise. This defines a map

$$\{0, 1\}^r \longrightarrow \Omega \longrightarrow \mathbb{Z}$$

with value $X + |Q| \text{ell}_t(z)$. Thus $r+1$ distinct progression values have original binary witnesses. This applies to any specified free source, boundary map, and scalar observation, not only graphs. The full proof and exact graph specialization are in Section 3 of `graphical-proof-provider.md`.

For graphs, edge-disjoint paths are disjoint lifts of $e_v - e_u$. Their original scalar increment is $t_v - t_u$. For a complete graph on m marked vertices there are $m-1$ such paths. They generate an m -term progression, explaining the critical $(m+1)$ -term threshold in the complete-graph construction.

3. Root-budget prolongation retains the whole original defect

For the complete graph K_m , the actual tournament-score image is

$$\Sigma_m = \{\rho - \partial\epsilon : \epsilon \in \Omega\}, \quad \rho = (0, 1, \dots, m-1).$$

If r is a nonzero integral vector of coordinate sum zero with

$$r(U) \leq |U|(m - |U|) \quad (U \subseteq V), \quad (5)$$

the new theorem constructs actual binary words whose $m+1$ score rows $x_0, \dots, x_m$ satisfy

$$\boxed{x_1 - x_0 = e_j - e_i, \quad \Delta(x_0, \dots, x_m) = (0, \dots, 0, r)}, \quad (6)$$

where i maximizes r and j minimizes r . The construction retains r itself and does not divide it by its gcd or replace it by a support label.

The proof constructs an integral endpoint allocation on K_m plus an additional $K_{(m-2)}$. Its cut inequalities are deduced from (5) by four exact membership cases. The resulting correction is $t_s = r + (e_j - e_i)$, with $s_i = 0$ and $s_j = m-1$. Reversing the $m-1$ disjoint $j-i$ paths gives the original root string through s , and appending t gives (6). Section 5 of the proof provider contains all cut bounds, the integral flow construction, and the binary path maps.

Every difference of two actual score points satisfies (5). Therefore a scalar collision $\text{ell}_t(r) = 0$ between distinct score classes produces an $(m+1)$ -term progression. The original source defect r is nonzero, while its receiving scalar defect is zero; the nonzero scalar increment $t_j - t_i$ makes the resulting progression nonconstant.

This is a constructive obstruction map, not a declaration that the original cohomology class is absent. It uses the exact additional kernel in (1) and (4) to produce the forbidden arithmetic witness.

4. A supportwise cohomology equivalence forced by admissibility

Assume the complete-graph marks are distinct, its positive edge weights are distinct, and its binary numerical image has no nonconstant $(m+1)$ -AP. The previous theorem proves that ell_t is injective on the full boundary image X_Ω . It follows on every admitted word support S that J_S is a bijection on its actual value bases.

Consequently

$$\boxed{B_S = K_S \text{ for every } S \subseteq \Omega,} \quad (7)$$

and the maps induced by J_S give a natural isomorphism of the two quotient diagrams in (1). The inverse sends e_y to the unique original e_x with $\ell_t(x)=y$. On reconstructed totals the map preserves the original support, its coefficient zero, and global absence; the scalar semiring is the inherited $G(Z)$. This is a genuine diagram/cohomology statement for the specified finite binary source, proved uniformly for every graph order m .

At the integral edge-coordinate level the corresponding bounded statement is

$$\ker(\ell_t \partial) \cap \{-1, 0, 1\}^E = \ker \partial \cap \{-1, 0, 1\}^E. \quad (8)$$

The full linear kernel in (4) retains its additional $m-2$ rational dimensions. At the doubled value source, further collisions can also occur. The new ten-generator example has 3081 doubled score classes and 2103 numerical ternary classes; the additional free-value kernel has dimension 978. Its addition square is retained in the proof provider. Thus (7) is not extended without proof to a higher coefficient box, tensor observation, or the original infinite theta source.

5. An all-rank obstruction with its quantitative return

At the critical progression length $m+1$, every admissible complete-graph projection is score-injective by (6). There are at least $m!$ transitive score classes. Therefore every canonical radix for such a block is at least $m!$, even when its ruler changes at every level. With $n_m = \text{binom}(m, 2)$, this gives the exact lower rate $(m!)^{1/n_m}$.

A prime $p > m$ and marks $(0, 1, p, \dots, p^{m-2})$ give a valid modular construction in base p^{m-1} , hence upper rate $p^{2/m}$. Bertrand's bound $p \leq 2m$ and the elementary lower estimate $\log(m!) \geq m \log m - m + 1$ show that the optimum Ξ_m throughout this complete critical source family satisfies

$$\boxed{\lim_{m \rightarrow \infty} \frac{m \log \Xi_m}{\log m} = 2.} \quad (9)$$

This is a matching leading asymptotic for a specified unbounded-rank family. It does not bound the unrestricted Erdős 817 rate from below. Its implication is that further large-rank improvement must alter the critical source geometry, rather than attempt to collapse these particular score classes: their collapse has the explicit forbidden packet (6).

For the metric calculation extend the displayed free integral presentations to real coefficients; their free basis quotients and specified maps are retained. The source metric stays equally explicit. For the orthonormal binary-word basis, if $\eta(x)$ is the number of words with boundary x , the first quotient metric is $\text{diag}(1/\eta(x))$. If $\mu(y) = \sum_t (\ell_t x = y) \eta(x)$, the original minimum-norm quotient metric at the numerical image is $\text{diag}(1/\mu(y))$. The sections are the actual averages in each word fiber. These formulas follow by completing the square in each finite fiber and commute with the original quotient maps.

For the new base-1651 block the largest multiplicity is 24, so its t -level tensor return to unit value coordinates has norm $24^{t/2}$. No uniform arithmetic gain is inferred after omitting this factor. On smaller supports the same formulas use their own retained fiber counts.

6. Exact affine failure loci and the successful replacement

The old successful ruler $(0, 1, 5, 22)$ lies on the obstruction hyperplane $c = 5b - 3a$ for prefixes $(0, a, b, c)$. Every valid appended mark z gives a six-term progression; for $0 < a < b$, $b \neq 2a$ and $z > 2c$ the ten edge weights are automatically distinct. Its source defects are $0, r, -r, 0$ with $r = (1, 3, -5, 1, 0)$, so the appended coordinate never changes the obstructing observation $\ell_t(r) = 3a - 5b + c$.

For the replacement prefix $(0, 4, 7, 41)$, retain the complete affine packet (r, v) of old subset sums minus the r selected endpoints of new edges. Its projection is $(r, v) \rightarrow rz + v$. The whole packet width in v is 178. All constant- r fibers avoid six-term progressions, so the exact second-difference bound proves safety at every $z > 356$. The remaining 315 values are completely classified, with all masks retained. The admitted value $z = 257$ gives the stronger modular construction at base 1651.

The generic affine-packet comparison is also reusable: for finite C subset Z^2 of second-coordinate width W , the evaluation $(r,v) \rightarrow rz+v$ is injective and preserves and reflects all k -APs at every integer $z > 2W$. A failure below that bound retains a nonzero coefficient equation $z \Delta r + \Delta v = 0$. No parameter is replaced by its asymptotic class without the displayed inverse and finite remainder domain.

7. Canonical monotone dual and finite obstruction compiler

The second proof provider, `monotone-obstruction-dual.md`, constructs the exact future value

$$F_u(R) = \inf_{p \text{ safe from } R} P(p)/u^{N(p)}.$$

For $u \leq \lambda_k$ it lies in $[1/2, 1]$, is inclusion-monotone, and $H_u = 2F_u$ is the greatest bounded Bellman subsolution. For $u > \lambda_k$ it vanishes at every support, through explicitly represented positive-cost paths. A finite path with $2P < u^N$, followed by the retained zero reset, produces an actual modular-safe block of rate below u . The general pattern-matrix version retains reset cost R_L and coefficient interval $[1, R_L]$.

The false state remains active throughout this limiting calculation. A zero coefficient in the future-value limit has its supported-zero meaning, not a missing source label. Finite graph closure and finite source packets are tools to control that problem, not claims that the arithmetic inequalities over all blocks have already been evaluated.

8. Verification, provenance, and scope

The complete proof providers and standalone verifier/auditor are included in this same workbench directory. The 3300 complete root-budget packets are checked against actual binary masks in two separate implementations. Another 200 higher-rank instances calibrate the general cut construction. The applied modular and affine records have complete stated domains; every omitted larger rank or parameter is covered only where a written theorem supplies it.

The original source interface is `workbenches/splitzero-tandem/tex/support_diagrams.tex` at Zeta revision `42df8a2de002d5fc7090641fac46ea11be05fa71`. The current research source and all inherited analytic scopes remain unchanged. The original EP817 $k=4$ contributor and sneed-and-feed retain their distinct credits. No private identity material is included.

The present toolset supplies no GitHub write action. This directory is a tested additive publication payload; it has not been represented as a merged or independently Lean-certified contribution. The generic algebraic/cohomological results are separate from the specific finite record at base 1651.

Exact graphical direction modules and supported arithmetic visibility

Source: history/EP817_SUPPORTED_DEFECT_20260915/payloads/ep817/research/supported_defects/notes/graphic-visibility-filtration.md **Identity:** 2d443b8fed3a39951bedbe233458734993909828a480b78297cac94032df88ca
15 September 2026. Research continuation with ordinary proofs and bounded exact certificates; no new Lean elaboration or priority claim. The cut-tree existence theorem used here is classical and is attributed below. The original graph, its orientations, the integral cut capacities, and the arithmetic observation remain in every comparison.

1. The original obstruction and the stronger question

The preceding complete-graph argument uses edge-disjoint paths: r paths joining u to v give $r+1$ subset sums in progression, provided the observed endpoint difference $t_v - t_u$ is nonzero. This contribution asks what happens when that difference is a supported zero and how to account for *every* progression direction of an arbitrary finite graph, rather than only a chosen path family.

Let $G=(V,E)$ be a finite loopless undirected multigraph. Parallel edges are separate source labels. Choose and retain an orientation of each edge, with incidence

$$\partial : \mathbb{Z}^E \rightarrow \mathbb{Z}^V.$$

Every binary choice reverses the corresponding edge of a reference orientation. Let $\text{Sigma}(G)$ be the set of its actual orientation-score vectors; their coordinates count wins, and their sum on each connected component is its number of edges. Let $\mu_G(s)$ be the number of reference reversal masks producing score s . Those multiplicities are retained.

For integer vertex marks $T=(t_v)$, define

$$L_T(s) = \sum_v t_v s_v.$$

If every chosen edge orientation goes from smaller to larger mark, and all edge distances are distinct positive integers, the original subset-sum image is a reflection/translation of $L_T(\text{Sigma}(G))$. Specifically, with reference score ρ ,

$$\Phi_A(\epsilon) = L_T(\rho) - L_T(\rho - \partial\epsilon). \quad (1.1)$$

Equal marks at nonadjacent vertices are allowed by this identity. Cardinality claims about the EP817 generators still require all actual edge distances to be distinct; no multigraph edge count is substituted for that condition.

Write $\lambda_G(u,v)$ for the maximum number of edge-disjoint u - v paths, equal to the minimum cardinality of an original separating edge cut. Put $\lambda_{\max}(G)=\max \lambda_G(u,v)$, with value zero for an edgeless graph.

For $q \geq 1$, define the integral module generated by actual source progression directions:

$$\mathcal{D}_q(G) = \langle z \in \mathbb{Z}^V : \exists x \in \Sigma(G), \quad x, x+z, \dots, x+qz \in \Sigma(G) \rangle_{\mathbb{Z}}. \quad (1.2)$$

The bracket is explicitly the integral span; the actual feasible direction set is retained before taking that span.

2. The classical cut tree and its exact source-coordinate map

For each nontrivial connected component, the Gomory-Hu theorem supplies a weighted tree on the *same original vertices*. Removing a tree edge e gives an actual vertex partition U_e and its complement. The weight is

$$c_e = |\delta_G(U_e)|,$$

and for all vertices u, v ,

$$\lambda_G(u, v) = \min_{e \in \text{tree path } u \rightsquigarrow v} c_e. \quad (2.1)$$

Each fundamental tree cut is an actual minimum cut for the endpoints of that edge. This existence/representation theorem is the imported classical result (Gomory & Hu, 1961; Gusfield, 1990). No new algorithmic complexity claim is made. The finite checker supplies and independently verifies the entire cut tree on each enumerated graph.

Fix a root and choose U_e as the subtree away from that root. For a score vector s , retain the cut coordinates $s(U_e)$, together with the fixed component score total. These coordinates have an explicit integral inverse: at a nonroot vertex subtract the cut sums of its child subtrees from the sum of its own subtree; at the root subtract all child subtree sums from the retained total. Thus the cut-coordinate map is an integral affine isomorphism on the component-total hyperplane, not an untracked reduction of dimension.

Every orientation satisfies

$$|E(G[U_e])| \leq s(U_e) \leq |E(G[U_e])| + c_e. \quad (2.2)$$

Internal edges contribute exactly one win to U_e ; crossing edges contribute zero or one. Hence the displayed interval is the exact original range, of width c_e .

3. The full direction filtration

Delete from the cut forest every tree edge with $c_e < q$. Let P_q be the resulting partition of V . By (2.1), two vertices belong to the same class exactly when $\lambda_G(u, v) \geq q$. The partition therefore does not depend on which valid cut tree was chosen.

Define

$$K_q(G) = \left\{ z \in \mathbb{Z}^V : \sum_{v \in C} z_v = 0 \text{ for each } C \in P_q \right\}. \quad (3.1)$$

It is the direct sum of the zero-sum lattices of the actual classes, of rank $|V| - |P_q|$.

Theorem 3.1 (exact direction module). For every finite G and every $q \geq 1$,

$$\boxed{\mathcal{D}_q(G) = K_q(G)}. \quad (3.2)$$

Every root generator $e_v - e_u$ in one class of P_q has an explicitly constructible $(q+1)$ -term progression in the original orientation source.

Proof of the upper inclusion. Take an actual source progression $x, x+z, \dots, x+qz$. For a cut-tree edge of weight $c_e < q$, (2.2) gives

$$q|z(U_e)| \leq c_e < q.$$

The integer $z(U_e)$ is therefore zero. Component totals also give zero total charge on every original component. Contract the high-weight tree edges. The remaining graph is a tree on the classes P_q , and every one of its fundamental cut charges is zero. The explicit leaf/subtree inverse of Section 2 now gives zero charge on each class. Thus z belongs to K_q , and the same holds for its integral span.

Proof of the lower inclusion. If u, v belong to one class, unit-capacity max-flow supplies q edge-disjoint u - v paths. It can be decomposed into q integral path chains c_h with boundary $e_v - e_u$; a standard augmenting-path construction supplies actual paths, rather than fractional capacities. Write each $c_h = c_h^+ - c_h^-$ using its positive and negative edge supports relative to the retained reference orientation.

Choose c_h^+ on all paths, then replace them one at a time by c_h^- . Disjointness of the paths makes every union an actual binary edge choice. Its boundary decreases by $e_v - e_u$ at every step, so its score increases by $e_v - e_u$. This constructs the promised $(q+1)$ -term progression. The vectors $e_v - e_u$ in each class generate its zero-sum lattice, proving the lower inclusion and the theorem. QED.

The distinction between directions and their integral span remains explicit. The proof supplies actual progressions for the class-root generators; it does not claim that every integral combination is itself a feasible direction.

Corollary 3.2. The longest progression in $\text{Sigma}(G)$, allowing a singleton when no nonconstant progression exists, has length

$$\boxed{1 + \lambda_{\max}(G)}. \quad (3.3)$$

The lower witness is the path construction. Above that length, P_q is discrete, $K_q=0$ and (3.2) excludes a nonzero direction. The entire unbounded q -filtration changes at only the finitely many cut-tree weights; the original q label is retained on any zero or constant plateau.

4. Exact supported visibility under the arithmetic observation

For each q retain

$$N_{q,T} = K_q \cap \ker L_T, \quad I_{q,T} = L_T(K_q) \subseteq \mathbb{Z}.$$

The actual image ideal is

$$\boxed{I_{q,T} = a_q \mathbb{Z}, \quad a_q = \gcd\{|t_v - t_u| : u, v \text{ in the same class of } P_q\}}, \quad (4.1)$$

with $\gcd$ of the empty/all-zero family equal to zero. This follows from the root generators in Theorem 3.1. The full original sequence is

$$\boxed{0 \rightarrow N_{q,T} \rightarrow K_q \xrightarrow{L_T} a_q \mathbb{Z} \rightarrow 0.} \quad (4.2)$$

The terminal ideal has its original inclusion in $\mathbb{Z}$. It is not divided by a_q and then silently treated as a fixed metric copy of $\mathbb{Z}$. The $\gcd$ describes the image of the integral span; individual witnessed progressions retain their actual endpoint differences.

Theorem 4.1 (visibility). A $(q+1)$ -term progression in the original score source has nonconstant numerical image under L_T if and only if $a_q > 0$. Equivalently, the largest visible source progression has length

$$\boxed{1 + \max_{u,v:t_u \neq t_v} \lambda_G(u,v)}, \quad (4.3)$$

with maximum zero when there is no distinct observed pair.

Proof. A nonconstant observed source progression has a direction z with $L_T z \neq 0$. Theorem 3.1 puts z in K_q , so its image ideal is nonzero. Conversely, $a_q > 0$ gives u, v in a common class with unequal marks. The root progression constructed in Theorem 3.1 has observed difference $t_v - t_u \neq 0$. The equivalence and the maximum formula follow. QED.

When $a_q = 0$, equation (4.2) says $N_q(T) = K_q$; it does not say that K_q has disappeared. On an active support its observed coefficient is the supported zero. This is a direct instance of the Zeta source's receiving-boundary and observation-kernel formula (D7)–(D8), with all source directions still present.

For q increasing, $K_{(q+1)}$ is a sublattice of K_q . Every inclusion commutes with L_T , giving morphisms of the original kernel-image sequences. The entire infinite index family has a literal SplitZero realization: take

$$\mathcal{L} = (\{1, 2, \dots\} \cup \{\infty\}, \text{ reverse usual order}), \quad \perp = \infty, \quad q \vee p = \min(q, p), \quad K_\infty = 0. \quad (4.4)$$

For $q \geq p$, the transport $K_q \rightarrow K_p$ is the original lattice inclusion. Use the same supports for $N(q, T)$ and $I(q, T)$, and their original inclusions. Reconstructing these diagrams with the source's $G(Z)$ functor retains every finite q label. In particular, $K_q = 0$ at a finite q is its supported zero fibre; it is not the bottom at infinity.

There is also a finite support-changing map. Let P be the finite chain of actually occurring cut partitions, with a *separate* external bottom adjoined. Send infinity to that bottom and a finite q to the active partition P_q . This preserves bottom and joins, since $P_{\min(q, p)}$ is the coarser of P_q and P_p . Its coefficient map is the literal identity $K_q = K(P_q)$; on reconstructed totals it is

$$(q, z) \mapsto (P_q, z). \quad (4.5)$$

Two images agree exactly when their cut partitions agree and their coefficient vectors agree. Thus the map's support coalescences are specified, while the original unbounded q values remain on the source. The active discrete partition at $q > \lambda_{\max}$ is distinct from the separately adjoined bottom. No finite supported zero is sent to external absence. The observation and all kernel-image maps commute with (4.5). This is finite control of the original infinite filtration with its comparison map and support fibres exposed; no analytic theta estimate is used.

5. The positive arithmetic consequences and the retained extra kernel

If all vertex marks are globally distinct and the observed subset-sum image is k -AP-free, (4.3) forces

$$\lambda_{\max}(G) \leq k - 2. \quad (5.1)$$

For a simple graph let $r = |V| - c(G)$, the rank of its incidence lattice. Map each original edge to its signed path in the cut forest. A column has at least one nonzero entry. At most r columns have exactly one entry, since a simple original edge can connect the endpoints of a given tree edge at most once. Counting every entry in both orders gives

$$2|E| - r \leq \sum_{a \in E} |\text{tree path of } a| = \sum_{e \in \text{cut forest}} c_e \leq \lambda_{\max}(G)r.$$

Consequently

$$\boxed{2|E| \leq (\lambda_{\max}(G) + 1)r, \quad 2|E| \leq (k - 1)r \text{ under (5.1).}} \quad (5.2)$$

This workbench deduction uses the classical cut-tree representation. It strengthens the earlier elementary graphical edge-count consequence, but no priority claim is made for the graph inequality itself. Parallel original edges would repeat unit columns; the simple-graph condition is therefore retained in this count.

A scalar projection can create progressions whose chosen source lifts are not progressions. The comparison is the following exact pullback, for $q \geq 2$:

$$\mathcal{F}_T = \{(s_0, \dots, s_q) \in \Sigma(G)^{q+1} : \Delta(s_0, \dots, s_q) \in (\ker L_T)^{q-1}\}. \quad (5.3)$$

The actual vector-progression set is the subfibre $\Delta = 0$. Evaluation maps F_T onto the complete scalar progression set; each numerical tuple has its actual score-lift fibre, and each score in turn has its original orientation masks. The visibility theorem calculates the $\Delta = 0$ part with nonzero observed difference. The remaining feasible fibre in (5.3) is retained rather than being identified with that part.

For example a two-edge star with marks 0,1,2 has source maximum progression length two, whereas its scalar subset sums are $\{0,1,2,3\}$. The source tuples with scalar three-APs are accounted for by the nonzero receiving defects in (5.3). This explicit comparison prevents using (5.1) as an unproved converse.

5.1 The observation-faithful quotient, with all original edges retained

There is a further exact comparison when the graph actually represents an EP817 set. Assume every edge has nonzero mark difference and all absolute edge differences are pairwise distinct. Identify original vertices exactly when their arithmetic marks agree. Let $p:V \rightarrow \bar{V}$ be this quotient, and retain each original edge as a separately labelled edge joining its two image vertices.

No edge becomes a loop, because an edge has nonzero mark difference. No two retained edges become parallel: that would give the same unordered pair of endpoint marks and therefore the same absolute difference, contrary to the stated distinctness. Thus $\bar{G}$ is a simple graph with the original edge set, and its vertex marks are globally distinct. Its generator set is literally the same set of positive integers, not just an image with the same cardinality.

The chain square is

$$\begin{array}{ccc} \mathbb{Z}^E & \xrightarrow{\partial_G} & \mathbb{Z}^V \\ \text{id} \downarrow & & \downarrow p_* \\ \mathbb{Z}^E & \xrightarrow{\partial_{\bar{G}}} & \mathbb{Z}^{\bar{V}}, \end{array} \quad L_T = L_{\bar{T}} p_*. \quad (5.4)$$

For every orientation mask, its quotient score is exactly p_*^* s: the wins at vertices in one observed class are added, with every incident edge still present. All quotient orientations lift by the identity on the edge-mask set. Consequently

$$p_* : \Sigma(G) \rightarrow \Sigma(\bar{G}), \quad \mu_{\bar{G}}(y) = \sum_{s: p_* s = y} \mu_G(s). \quad (5.5)$$

The cycles created by this vertex identification are retained by the exact sequence

$$0 \rightarrow H_1(G; \mathbb{Z}) \rightarrow H_1(\bar{G}; \mathbb{Z}) \xrightarrow{\partial_G} \text{im } \partial_G \cap \ker p_* \rightarrow 0. \quad (5.6)$$

Here graphs are their original one-dimensional chain complexes, so H_{-1} is the literal incidence kernel. A quotient cycle has original boundary in the displayed intersection. The kernel of that boundary is $H_{-1}(G)$. Every z in the intersection has an original edge lift c with $\partial_G c = z$, and $p_*^* z = 0$ makes c a quotient cycle; this proves surjectivity over the integers, not only equality of rational dimensions.

If $r_G = |V| - c(G)$ and $r_{\bar{G}} = |\bar{V}| - c(\bar{G})$, the last free lattice has rank $r_G - r_{\bar{G}}$. Its arithmetic observation vanishes, since $L_T = L_{\bar{T}} p_*^*$. These are present original zero actions, not deleted edge choices. For the Theta_r family, (5.6) adds exactly one cycle direction when the two equally marked terminals are identified; the quotient is the original bouquet of r triangles.

The word metric also commutes with this comparison. On the score values use the original quotient Gram $\text{diag}(1/\mu_G(s))$; then the least-norm section of (5.5) is

$$e_y \mapsto \sum_{p_* s = y} \frac{\mu_G(s)}{\mu_{\bar{G}}(y)} e_s. \quad (5.7)$$

Its squared norm is $1/\mu_{\bar{G}}(y)$. Thus the minimum-norm quotient is precisely the one obtained from the same original edge masks, with no unweighted-score metric silently substituted.

The visibility theorem now applies to $\bar{G}$ with its globally distinct marks. In particular every k -admissible edge-distance set with this presentation obeys

$$2|E| \leq (k-1)(|\bar{V}| - c(\bar{G})). \quad (5.8)$$

It counts the actual distinct observed vertex marks after retaining the full relative sequence (5.6). The earlier bound used globally distinct marks at the source; (5.8) covers coincident nonadjacent marks through their explicit quotient. A general positive generator set still has a star presentation with $|\bar{V}| - c = n$, so this necessary inequality alone does not yield the sought sharp global exponential rate.

6. Application to an unbounded supported-zero obstruction

Use the graph Θ_r of the companion note: r internally disjoint paths of length three joining u to v , with the two internal vertices on path j marked L_{19}^j and $8L_{19}^j$. Initially $t_u = t_v = 0$. The actual edge weights are L times the original base-19 triples, all positive and distinct.

For $r \geq 2$, every vertex pair has at least two edge-disjoint paths. Pairs containing an internal vertex have at most two because that vertex has degree two. The terminal pair has exactly r disjoint paths. Thus for $r \geq 3$:

$$P_1 = P_2 = \{V\},$$

and for $3 \leq q \leq r$ the only nonsingleton class of P_q is $\{u, v\}$. Above r every class is a singleton. The source module in that middle interval is exactly

$$K_q = \mathbb{Z}(e_v - e_u). \quad (6.1)$$

It is nonzero at arbitrarily large q as r increases. Its observation is nevertheless zero because $t_v = t_u$. Theorem 4.1 proves that the longest numerically visible *source* progression has length three, uniformly over all r . The original mod-19 lift independently excludes any additional scalar four-AP in (5.3), so the actual arithmetic image is four-AP-free at every r .

Now change t_v to $-t$. On the entire middle filtration,

$$L_T(e_v - e_u) = -t, \quad I_{q,T} = t\mathbb{Z}. \quad (6.2)$$

For $t \neq 0$ all these retained directions become visible; their path witnesses produce a progression with $r+1$ terms. With $L > 2r|t|$ and $t > 0$, the companion deformation theorem proves the exact maximal length, retaining every original scalar value and representation fibre.

This controls the infinite family rather than inferring from a single rank. The graphical direction module, its supported zero observation, the relative homology under terminal identification, and the first-order specialization map are all explicit. The terminal identification leaves the literal generator set unchanged. Conversely, changing its arithmetic mark changes the scalar operator, and (6.2) records exactly which long directions become visible.

7. Full finite replay

The proof does not extrapolate from computations. The finite suite independently calibrates its integer constructions on every labelled simple graph on 1.5 vertices: 1,099 graphs. It rebuilds all 40,398 orientation-score values across these instances, examines 1,391,037 endpoint pairs, and supplies 1,152 valid cut trees for nontrivial components. Every fundamental cut is checked in the original graph and against all required pair minima.

The producer also constructs 15,147 root-direction witnesses, verifies 8,493 original mark observations, and checks 23,118 exact image ideals. For up to four vertices the observation domain is every mark list in $\{0,1,2\}^V$. At five vertices the three stated lists are all zero, $(0,1,2,3,4)$, and $(0,0,1,2,3)$. The complete records retain this domain, including the globally unequal and supported-zero cases. The independent auditor reconstructs source orientations by direct binary enumeration, uses a separately written cut calculation, and checks all these observation records. It does not import the producer.

A separate observation-quotient replay checks 1,211 admitted marked graphs among 2,197 literal candidates (all labelled graphs through five vertices, with the two stated power-mark families), 60,585 original orientation masks, and the relative incidence sequence for Θ_r through $r=12$. Full theta word fibres are enumerated through $r=4$. The quotient producer and its separate auditor retain original edge labels, positive distinct weights, all score-fibre masses and their quotient metrics.

The complete compressed records for the first suite are `certificates/supported_defect_proof.json.gz`. False image counts, missing supports, altered cut capacities and zeroed actual witnesses are rejected. The exact source identities and replay outputs are separately hashed. The full generality of Theorems 3.1 and 4.1 comes from the cut-tree and path proofs above.

References and source roles

- Gomory, R. E., & Hu, T. C. (1961). Multi-terminal network flows. *Journal of the Society for Industrial and Applied Mathematics*, 9(4), 551–570. <https://doi.org/10.1137/0109047> . Role: existence and exact fundamental-cut representation of the cut-equivalent tree, not a source of an EP817 numerical bound.
- Gusfield, D. (1990). Very simple methods for all pairs network flow analysis. *SIAM Journal on Computing*, 19(1), 143–155. <https://doi.org/10.1137/0219009> . Role: classical cut-tree construction and interpretation. No implementation from this reference is imported into the finite checker.
- Hartmann, T., & Wagner, D. (2013). *Dynamic Gomory-Hu tree construction—fast and simple* (arXiv:1310.0178) [Preprint]. <https://arxiv.org/abs/1310.0178> . Role: independently inspected primary-source description of the cut tree as a minimum-cut basis.
- KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source, Zeta main 58626a62cd5648e8ae7450fb54d4a4aab2330981]. `workbenches/splitzero-tandem/tex/support_diagrams.tex`, (D6)–(D8). Role: original supported reconstruction and quotient/observation-kernel comparison.
- The Clankers. (2026, September 15). *Graphical obstruction propagation and a stronger six-term construction* [Predecessor workbench contribution]. Role: disjoint path-to-subset witness map and complete-graph context; retained with original attribution and evidence scope.

Supported zero actions, exact defect transport, and all-rank deformations

Source: history/EP817_SUPPORTED_DEFECT_20260915/payloads/ep817/research/supported_defects/notes/supported-defect-universality.md **Identity:** 3fcd9ccf780ae3fa05ba6691a7246086c52572e0c78729283b014bde6b57ea
Research continuation, 15 September 2026. Ordinary mathematical proofs and exact bounded certificates; independent mathematical review pending. No new Lean elaboration or literature-priority claim is made.

1. Objective, original source, and conclusions

For a finite set A of distinct positive integers, retain its actual binary source, numerical image and representation multiplicities:

$$\Omega_A = \{0, 1\}^A, \quad H(A) = \{\Phi_A(\epsilon) : \epsilon \in \Omega_A\}, \quad \Phi_A(\epsilon) = \sum_{a \in A} \epsilon_a a, \quad \mu_A(y) = |\Phi_A^{-1}(y)|.$$

The empty binary choice is an actual source element. It has numerical value zero and is never removed. A positive-step k -progression is an ordered tuple $(y, y+d, \dots, y+(k-1)d)$, $d > 0$; negative steps are recovered by reversing the tuple. Let $g_k(n)$ be the original EP817 maximum-generator minimum, and λ_k its previously established n th-root limit. The new work does not improve the recorded numerical bound $\lambda_6 \leq 1651^{1/10}$. It gives exact all-rank deformation and visibility results, and identifies where the unrestricted arithmetic instance survives those maps.

The inherited block is (1,7,8). Its construction and mathematical route remain attributed to the anonymous/deleted Reddit contributor; sneed-and-feed retains separate credit for the finite-upper-bound Lean extension. At EP817 main 23c0110c95b5a2036bdc04a1f352b6e5e27742c8, Core.lean contains the bounded mod-19 kernel and the all-length four-progression-free digit lift. That exact source was inspected, not rebuilt here.

The Zeta source was inspected at 58626a62cd5648e8ae7450fb54d4a4aab2330981. The used file `workbenches/splitzerotandem/tex/support_diagrams.tex`, especially (D6)–(D8), retains the internal quotient, the receiving-boundary kernel and the distinction between fibre zero and external absence. Its original source blob is d3493f891291ee6e94dbf2c77649f7d85d240df2. No analytic zeta estimate is imported.

The principal conclusions are these.

Theorem 1 (all signed perturbations). Fix $r \geq 1$ and integers p_j, q_j, s_j , $0 \leq j < r$. Put

$$M = \sum_{j < r} (|p_j| + |q_j| + |s_j|), \quad L > 2M,$$

where L is an integer, and define the literal $3r$ -generator set

$$A_{L,\eta} = \bigcup_{j < r} \{L19^j + p_j, 7L19^j + q_j, 8L19^j + s_j\}, \quad d_j = p_j + q_j - s_j. \quad (1.1)$$

Its generators are distinct and positive. For every $k \geq 4$,

$$\boxed{H(A_{L,\eta}) \text{ is } k\text{-AP-free} \iff H(d_0, \dots, d_{r-1}) \text{ is } k\text{-AP-free.}} \quad (1.2)$$

The right side is the image of the indexed binary cube, so repeated or zero d_j are retained as source coordinates. An explicit translation/reflection map relates it to the absolute-value list. Every k -AP on the left decodes to an actual k -AP in the right-hand image. Conversely, a fixed original fibre embeds every such right-hand witness back into the left side.

For $k=3$ the exact criterion is injectivity of the indexed ternary map $\sum t_j d_j$, t_j in $\{0,1,2\}$. A zero defect is therefore treated differently from an absent coordinate: it prevents that injectivity.

Theorem 2 (exact finite data transport). For $k \geq 4$, the number ap_k of positive-step progressions, the distinct image size, and all representation multiplicities satisfy explicit subset transforms in Section 4. Two perturbations with the same entire defect vector have an explicit value-space bijection preserving those k -AP relations and the original quotient metrics. Every original parameter is retained in that map and its inverse.

Theorem 3 (active-support rate). For $k \geq 4$, minimize the largest generator in the special family $p_j = q_j = 0$, $s_j = h_j \geq 0$, $L > 2 \sum h_j$, subject to exactly a positive, pairwise distinct defect amplitudes and k -admissibility. Call the result $C_k(r, a)$. For any integers $0 \leq a_r \leq r$ with $a_r/r \rightarrow \alpha$,

$$\lim_{r \rightarrow \infty} C_k(r, a_r)^{1/(3r)} = 19^{1/3} \lambda_k^{\alpha/3}. \quad (1.3)$$

This is an exact optimum within this defined deformation family, not a formula asserting that the unrestricted λ_k has been evaluated.

Theorem 4 (supported degeneration). The same zero-defect construction admits a graph representation with r edge-disjoint paths between two distinct vertices having equal arithmetic marks. Its nonzero terminal boundary remains present, and its scalar image is zero. A common perturbation t changes that action to $-t$. At $t=0$ the original relation complex gains one explicitly computed specialization class. At every $t > 0$ in the stated separated range, the actual image has a longest progression of length $r+1$. The common-defect classification is proved for every r , not inferred from samples.

The companion `graphic-visibility-filtration.md` gives an all-graph theorem determining exactly which integral source direction modules are visible under an arbitrary integer mark observation. It explains the graph-side infinite obstruction in Theorem 4 without requiring globally distinct vertex marks.

2. General separation through the original word fibres

The following argument precedes the specialized construction. Let $b_1, \dots, b_n$ be distinct positive integers, let ϵ_i be any integers, and set

$$M_- = \sum_i \max\{-\eta_i, 0\}, \quad M_+ = \sum_i \max\{\eta_i, 0\}, \quad M = M_- + M_+.$$

Choose $L > 2M$ and set $a_i = Lb_i + \epsilon_i$. Positivity follows from $a_i \geq L - M > 0$. If $b_j > b_i$ then $a_j - a_i \geq L - |\epsilon_i| - |\epsilon_j| \geq L - M > 0$. Thus original labels correspond to distinct actual generators without any repeated-weight convention.

On the unchanged binary source $\Omega = \{0,1\}^n$ retain

$$B(\epsilon) = \sum_i b_i \epsilon_i, \quad E(\epsilon) = \sum_i \eta_i \epsilon_i, \quad \Phi_A(\epsilon) = LB(\epsilon) + E(\epsilon). \quad (2.1)$$

For each original value x in $H(b)$, let

$$\Omega_x = B^{-1}(x), \quad F_x = E(\Omega_x).$$

Then

$$H(A) = \coprod_{x \in H(b)} (Lx + F_x). \quad (2.2)$$

The notation denotes a genuinely disjoint union of numerical sets. All low values lie in $[-M_-, M_+]$, an interval of width $M < L$, and the exact inverse is

$$x = \left\lfloor \frac{y + M_-}{L} \right\rfloor, \quad u = y - Lx. \quad (2.3)$$

It recovers both coordinates (x, u) , not just x . The word-to-joint-value map and the joint-value-to-integer map factor the original Φ_A exactly.

Suppose $H(b)$ is k -AP-free. Write a target progression as $y_i = Lx_i + u_i$ using (2.3). Every low second difference has absolute value at most $2M$. Consequently

$$L\Delta x_i + \Delta u_i = 0, \quad |\Delta u_i| \leq 2M < L \implies \Delta x_i = \Delta u_i = 0.$$

Admissibility makes the high progression constant. The low progression is nonconstant because the full one is. Therefore

$$\text{AP}_k^+(H(A)) \simeq \coprod_{x \in H(b)} \text{AP}_k^+(F_x), \quad (x, (u_i)) \mapsto (Lx + u_i)_i. \quad (2.4)$$

Formula (2.3) is its inverse. This proves both existence and full tuple correspondence.

The same proof works for an integer pattern matrix P with $P1=0$. For row i let m_i be its positive coefficient sum, also the absolute negative coefficient sum. Then $|(Pu)_i| \leq m_i M$. Requiring $L > (\max_i m_i)M$ proves $Px = Pu = 0$. When $H(b)$ has only constant solutions of that specified pattern, every target solution is in one high fibre. This is a general finite-source comparison theorem with all actual offsets retained.

3. The nineteen-adic source and its complete short kernel

Let

$$B_r = \bigcup_{j < r} 19^j \{1, 7, 8\}, \quad D = \{0, 1, 7, 8, 9, 15, 16\}.$$

The elementary bounded-kernel identity is

$$u + 7v + 8w \equiv 0 \pmod{19}, \quad u, v, w \in [-2, 2] \cap \mathbb{Z} \iff (u, v, w) = (t, t, -t), \quad -2 \leq t \leq 2. \quad (3.1)$$

For completeness, put $p = u + w$ and $q = v + w$. Then $|p|, |q| \leq 4$ and $|p + 7q| \leq 32$, so a multiple of 19 is $-19, 0$ or 19 . The value 19 forces $(p, q) = (-2, 3)$. The latter q forces $w \geq 1$, whence $p = u + w \geq -1$, a contradiction. Negation excludes -19 . The value zero forces $q = p = 0$, proving (3.1).

The seven digits have no nonconstant four-AP modulo 19. This finite claim is included in the inherited Lean Core and is replayed directly over all 342 start/nonzero-step pairs. A symbolic proof can also project each binary block through $(a, b, c) \rightarrow (a + c, b + c)$: (3.1) makes both projected second differences zero over the integers; four values in $\{0, 1, 2\}^2$ have zero common difference. This forces their original digit values to agree.

Reduce an integer four-AP in $H(B_r)$ modulo 19. Its low digits agree; remove that exact common digit and divide by 19. Induction proves four-AP-freeness at every r . Therefore $H(B_r)$ is also k -AP-free for every $k \geq 4$.

Apply (3.1) at the least block of any integer short relation in B_r . Its three coefficients equal $(t, t, -t)$ and its integer block contribution is zero, not an unrecorded multiple of 19. Divide by 19 and repeat. Thus

$$\ker \Phi_{B_r} \cap [-2, 2]^{3r} = \{(t_j, t_j, -t_j)_{j < r} : t_j \in [-2, 2] \cap \mathbb{Z}\}. \quad (3.2)$$

For $A(L, \eta)$ in (1.1), a short relation c obeys

$$L\Phi_{B_r}(c) + \sum_i c_i \eta_i = 0, \quad \left| \sum_i c_i \eta_i \right| \leq 2M < L.$$

The integer high relation vanishes and (3.2) applies. The exact resulting map is

$$\boxed{\{t \in [-2, 2]^r \cap \mathbb{Z}^r : \sum_j t_j d_j = 0\} \xrightarrow{\sim} \ker \Phi_{A_{L,\eta}} \cap [-2, 2]^{3r}, \quad t \mapsto (t_j, t_j, -t_j)_{j < r}.} \quad (3.3)$$

Its inverse reads the first coefficient of each triple. This is an identity of the original integral coefficient boxes; it does not identify unrestricted scalar kernels with incidence kernels.

A positive indexed generator family is three-AP-free exactly when its ternary evaluation is injective. One direction follows by expressing a three-AP as a binary second difference. Conversely a nonzero relation in $[-2, 2]^n$ can be written coordinatewise as $u-2v+w$ with u, v, w binary. Its three numerical values either give a nonconstant AP, or are equal. In the latter case two distinct binary representatives, after removing their common chosen indices, give disjoint nonempty subsets of equal positive value a . The original image then contains $0, a, 2a$. This proves the criterion without assuming distinct subset sums. Applying it to (3.3) proves the three-term statement in Theorem 1.

4. Every perturbation is controlled by its retained circuit defects

A high digit eight has exactly two representatives: the first two generators or the third one. The other six digits have unique representatives. For a word $z = (z_0, \dots, z_{(r-1)})$ in D^r , set

$$J(z) = \{j : z_j = 8\}, \quad x(z) = \sum_j 19^j z_j.$$

Choose the third generator as the specified representative at digit eight. The low anchor contribution of one digit is

High digit	Actual low anchor
0	0
1	$p_{_j}$
7	$q_{_j}$
8	$s_{_j}$
9	$p_{_j} + s_{_j}$
15	$q_{_j} + s_{_j}$
16	$p_{_j} + q_{_j} + s_{_j}$

Let $a_{_eta}(z)$ be the sum of these anchors. Choosing the other representative at a digit eight adds exactly $d_{_j} = p_{_j} + q_{_j} - s_{_j}$. Therefore the complete fibre formula is

$$\boxed{H(A_{L,\eta}) = \prod_{z \in D^r} (Lx(z) + a_{\eta}(z) + H(d|_{J(z)}))}. \quad (4.1)$$

An original word in that fibre is specified by the unique non-eight choices and one binary choice for each eight position. Consequently, including all coincident low values,

$$\mu_{A_{L,\eta}}(Lx(z) + a_{\eta}(z) + y) = \mu_{d|_{J(z)}}(y). \quad (4.2)$$

The map from each original subset mask to its high word and its actual low mask is a bijection; (4.2) follows from its fibres. This retains each zero $d_{_j}$: both binary choices still exist when they have the same low value.

Since $H(B_{_r})$ is k -free for $k \geq 4$, (2.4) applies. Every $H(d|_{_J})$ is a subset of $H(d)$. Conversely, the high word with every digit eight gives exactly the translated copy

$$L 8 \sum_{j < r} 19^j + \sum_j s_j + H(d). \quad (4.3)$$

Embedding a low witness in (4.3) and decoding arbitrary target witnesses by (2.3) prove (1.2). A witness in another high fibre decodes, after subtracting its anchor, to $H(d|_{_J})$ and then to $H(d)$. The resulting

section/retraction is on actual numerical APs, with original mask representatives available before passing to values.

Each fixed J has $6^{\wedge(r-|J|)}$ high words. Thus

$$|H(A_{L,\eta})| = \sum_{J \subseteq [r]} 6^{r-|J|} |H(d|_J)|, \quad (4.4)$$

and for every $k \geq 4$,

$$\text{ap}_k(H(A_{L,\eta})) = \sum_{J \subseteq [r]} 6^{r-|J|} \text{ap}_k(H(d|_J)). \quad (4.5)$$

These count distinct values and positive-step numerical tuples. The full representation profile is the disjoint union of the profiles in (4.2). In particular

$$\max_y \mu_{A_{L,\eta}}(y) = \max_y \mu_d(y). \quad (4.6)$$

For the upper inequality, a subset-family representation injects into the full defect family by fixing all other choices to zero. Equality occurs in (4.3).

Negative defects have the explicit cube involution: flip the chosen bit at every negative coordinate. It satisfies

$$\sum_j d_j \epsilon_j = - \sum_{d_j < 0} |d_j| + \sum_j |d_j| (\kappa \epsilon)_j. \quad (4.7)$$

It preserves every representation multiplicity and is its own inverse on the indexed cube. Zero coordinates are fixed, not deleted.

4.1 Same-defect perturbations and the exact isomorphism they induce

Let (L, η) and (L', η') satisfy their respective separation inequalities and have the same literal defect vector d . For y in the first image, recover its high word z by (2.3) followed by its base-19 digits, and put

$$\Psi(y) = L'x(z) + a_{\eta'}(z) + (y - Lx(z) - a_{\eta}(z)). \quad (4.8)$$

The inverse uses the primed division data and exchanges the parameters. Formula (4.1) proves both inverse laws. On each fibre this is a translation, so (2.4) proves preservation and reflection of all k -AP tuples for every $k \geq 4$. Formula (4.2) shows that the representation multiplicities and quotient metrics are retained exactly.

No global linearity on all integers is asserted. Its domain is the actual finite value image; its complete high-label decomposition is part of the map. In particular the kernels and offsets of the original arithmetic parameters are not discarded.

For the parameter homomorphism

$$D : \mathbb{Z}^{3r} \rightarrow \mathbb{Z}^r, \quad (p, q, s) \mapsto (p_j + q_j - s_j)_j,$$

one literal section is $d \rightarrow (0, 0, -d)$. Subtracting this section from η produces the retained element of $\ker D$. Equations (4.8) and its inverse explain exactly what is preserved along that parameter fibre. The supported circuit action descends through the old high-value quotient precisely when every $d_j = 0$. Indeed the old fibre differences are generated, as word relations, by the choice of the two representatives of each high digit eight, and their low observation is d_j .

4.2 The defect vector is the full original relative cohomology class

There is a geometric identification of D , not only a parameter analogy. Let Θ_r have terminals u, v and, for each j , a path with two new vertices a_j, b_j . Orient its three edges as

$$u \rightarrow a_j, \quad a_j \rightarrow b_j, \quad v \rightarrow b_j.$$

Take the original integral relative cochain complex for the pair $(\Theta_r, \{u, v\})$. A relative vertex potential vanishes at both terminals. In its actual internal-vertex coordinates (α_j, β_j) , the coboundary is

$$\delta^0(\alpha_j, \beta_j)_{j < r} = (\alpha_j, \beta_j - \alpha_j, \beta_j)_{j < r}.$$

There are no two-cells. Therefore the entire relative cohomology sequence is

$$0 \longrightarrow \mathbb{Z}^{2r} \xrightarrow{\delta^0} \mathbb{Z}^{3r} \xrightarrow{D} H^1(\Theta_r, \{u, v\}; \mathbb{Z}) \cong \mathbb{Z}^r \longrightarrow 0. \quad (4.9)$$

Indeed $D \delta^0 = 0$. If $p_j + q_j - s_j = 0$, the unique relative potential primitive is $\alpha_j = p_j$, $\beta_j = s_j$. Surjectivity of D is witnessed by the section $(0, 0, -d_j)$. For an arbitrary η , its retained decomposition is

$$\eta = \delta^0(p_j, p_j + q_j)_{j < r} + (0, 0, -d_j)_{j < r}. \quad (4.10)$$

Both summands and the original coordinates are recoverable; the first summand is an actual relative coboundary with its specified primitive.

The nineteen-adic high cochain itself is the relative coboundary of the potential $(L19^j, 8L19^j)$. Thus the actual generator cochain $A_-(L, \eta)$ has relative class exactly d , with all original integer edge values retained. Theorems 1–2 say that, within this explicitly separated family, k -AP-freeness for $k \geq 4$ is determined exactly by this complete relative H^1 class. Formula (4.8) gives the numerical and metric comparison for changes by relative coboundaries. The proof uses the full constrained binary source; an abstract isomorphism of free groups alone is not substituted for that feasible-set comparison.

The absolute H^1 is the further quotient

$$0 \longrightarrow \mathbb{Z} \xrightarrow{t \mapsto (t, \dots, t)} H^1(\Theta_r, \{u, v\}; \mathbb{Z}) \xrightarrow{d \mapsto (d_j - d_0)_{1 \leq j < r}} H^1(\Theta_r; \mathbb{Z}) \cong \mathbb{Z}^{r-1} \longrightarrow 0. \quad (4.11)$$

To verify the map, the original absolute cycles are $c_j - c_0$, where c_j has edge coefficients $(1, 1, -1)$ along its path; evaluation gives exactly $d_j - d_0$. Its kernel is the diagonal terminal-action line and its surjectivity is explicit by choosing $d_0 = 0$. A common nonzero defect therefore has zero absolute H^1 class but a nonzero relative class. It is exactly the terminal-action component that creates the long arithmetic progression in Section 6. Forgetting it would lose an actual obstruction, despite leaving the absolute cohomology unchanged.

For inclusions of path supports, extension by zero on internal vertices, edges and defects commutes with (4.9)–(4.10). These are original support-diagram maps, with no assumed injectivity of an unrelated observation. The componentwise formulas work over any coefficient ring; the arithmetic statements retain $\mathbb{Z}$, while the parameter specialization below uses the displayed $\mathbb{Q}[t]$.

5. Original supported quotient, relative kernel, and metric

Fix the original word set Ω and any admitted word support $S \subset \Omega$. Use the original base ring $\mathbb{Q}$ and its split semiring $G(\mathbb{Q})$. Define

$$V_S = \mathbb{Q}[S], \quad W_S = \mathbb{Q}[B(S)], \quad Z_S = \mathbb{Q}[(B, E)(S)].$$

The maps send basis words to their actual high values or joint values:

$$q_S : V_S \rightarrow W_S, \quad p_S : V_S \rightarrow Z_S, \quad f_S : Z_S \rightarrow W_S, \quad q_S = f_S p_S.$$

All maps commute with inclusions of admitted word supports. Set $R'_S = \ker q_S$ and $R'_S = \ker p_S$. The actual exact sequences are

$$0 \rightarrow R'_S \rightarrow R_S \xrightarrow{p_S} \ker f_S \rightarrow 0, \quad (5.1)$$

$$0 \rightarrow \ker f_S \rightarrow V_S/R'_S \longrightarrow V_S/R_S \rightarrow 0. \quad (5.2)$$

Surjectivity in (5.1) follows by lifting a joint-value vector whose high observation vanishes. Its kernel is exactly R'_S . The two-term relation complexes therefore have a chain map from the refined to the original coarser complex, and (5.2) calculates the induced quotient kernel. A nonzero low defect has not been asserted to kill an old boundary; (5.1) retains its failure to do so.

Under the inspected SplitZero reconstruction, a relation maps to $(S,0)$. External absence is the bottom support with its zero coefficient. The empty subset word belongs to every full binary source and remains an active basis vector e_0 . Its final numerical-amplitude observation may vanish; that is a further specified map, with its own kernel.

Give the word basis its original orthonormal metric. For a high value x , let M_x be the number of its words, and $m_{(x,u)}$ the size of its joint-value cells. Then

$$G_{x,x}^{\text{high}} = 1/M_x, \quad G_{(x,u),(x,u)}^{\text{joint}} = 1/m_{x,u}. \quad (5.3)$$

The minimum-norm section of f_S is

$$e_x \mapsto \sum_u \frac{m_{x,u}}{M_x} e_{x,u},$$

and its squared norm is

$$\sum_u \left(\frac{m_{x,u}}{M_x} \right)^2 \frac{1}{m_{x,u}} = \frac{1}{M_x}. \quad (5.4)$$

Thus the quotient of the refined metric is exactly the original coarse metric. On word space the two averaging projectors satisfy

$$P_{\text{high}} P_{\text{joint}} = P_{\text{joint}} P_{\text{high}} = P_{\text{high}}.$$

Their difference is the orthogonal projector onto the restored relative classes and has rank $|(B,E)(S)| - |B(S)|$. Orthogonality is proved by these projectors, not inferred from different support labels.

If exactly a of the r defects are nonzero, (4.6) and the retained zero bits give

$$\mu_{\max}(A) = 2^{r-a} \mu_{\max}(d_{\text{active}}), \quad \|I : (\text{quotient metric}) \rightarrow (\text{unit value metric})\| = 2^{(r-a)/2} \sqrt{\mu_{\max}(d_{\text{active}})}. \quad (5.5)$$

This is the original metric return cost. In particular sublinear active support does not make the many zero-action representation choices disappear.

6. Zero terminal actions, actual graph homology, and specialization

Take a graph with terminals u, v and r internally disjoint paths of length three. For path j use vertices a_j, b_j and edges $(u, a_j), (a_j, b_j), (v, b_j)$, with these displayed incidence orientations. Let c_j be the edge chain $(1, 1, -1)$ on that path. Its original boundary is

$$\partial c_j = e_v - e_u =: z \neq 0. \quad (6.1)$$

Assign marks $t_u = t_v = 0$, $t_{(a_j)} = L19^j$ and $t_{(b_j)} = 8L19^j$. The actual positive edge weights are $L19^j, 7L19^j, 8L19^j$, all distinct. The nonzero source vector z has numerical observation zero. For all r , this is exactly the inherited four-AP-free arithmetic set L_{B_r} .

Identifying u and v gives a bouquet of r triangles. The map on edge chains is the identity and the vertex map sends e_u and e_v to the same basis vector. The relative homology calculation is

$$0 \longrightarrow H_1(\Theta_r; \mathbb{Z}) \longrightarrow H_1(\text{triangle bouquet}; \mathbb{Z}) \xrightarrow{\partial} \mathbb{Z}z \longrightarrow 0. \quad (6.2)$$

The bouquet cycles are the c_j . Their linear combination has original theta boundary $(\sum \alpha_j)z$. Hence the first kernel is the free span of $c_j - c_0$, and the last map is their augmentation. This proves exactness and all generators explicitly. The vertex identification is licensed by equality of their actual marks, so every original edge value and every binary source mask is unchanged.

Choosing the positive or negative part of each c_j gives 2^r distinct subset masks of the common value $8L \sum_j 19^j$. A supported zero action thus retains a fibre of size 2^r . Choosing neither part, one part, or both parts also gives an explicitly represented ternary image in the disjoint supports. No such choices are external absence.

Now change only the terminal mark to $t_v = -t$. The edge weights become

$$L19^j, \quad 7L19^j, \quad 8L19^j + t, \quad \Phi_t(c_j) = -t. \quad (6.3)$$

For fixed integer $t > 0$ choose $L > 2rt$. Formula (4.3) gives the actual progression

$$8L \sum_j 19^j, \quad 8L \sum_j 19^j + t, \quad \dots, \quad 8L \sum_j 19^j + rt. \quad (6.4)$$

For $r \geq 3$, (1.2) applied at $k = r + 2$ proves there is no longer progression, since the defect image is the $r + 1$ -point interval $\{0, t, \dots, rt\}$. At $r = 2$ the same argument excludes four-APs and (6.4) has length three. At $r = 1$, (3.3) has no nonzero short relation, so the image is three-AP-free and its longest progression has length two. Thus the longest progression has exactly $r + 1$ terms for every $r \geq 1$ and $t > 0$ in the stated separated range. At $t = 0$ it has exactly three terms: the original block gives $0, 8L, 16L$, and the inherited theorem excludes four terms.

For any $\epsilon > 0$ and $k \geq 4$, choose $r = k - 1$ and $L > \max(2r, 1/\epsilon)$. Perturbing only each third weight by one changes every weight relatively by less than ϵ , but changes a k -admissible original family into a family containing (6.4). All original masks and all common-progression second-difference equations are retained; the terminal action changes from supported zero to a nonzero difference. This is an exact arithmetic statement about the effect of the perturbation, not an obstruction to studying the source by its full quotient data.

6.1 The specialization class and its first-order detector

Let $R = \mathbb{Q}[t]$. Restrict to the original path-chain span $U_J = R^{\wedge} J$ for a finite nonempty active path support J . Its differential to the scalar action is

$$d_t = -t\varepsilon_J, \quad \varepsilon_J(x) = \sum_{j \in J} x_j. \quad (6.5)$$

Over R , $H^{\wedge}(-1) = \ker \varepsilon_J$ and $H^{\wedge}0 = R/(t)$. On specialization $t = 0$, the differential is zero, so $H^{(-1)} = Q^J$ and $H^{\wedge}0 = Q$. The exact additional class is

$$0 \rightarrow \ker \varepsilon_J \rightarrow Q^J \xrightarrow{-\varepsilon_J} Q \rightarrow 0. \quad (6.6)$$

The rightmost Q is the Tor_1 contribution calculated by the literal free resolution $R \dashrightarrow R$ of $R/(t)$. Its sign in (6.6) is the actual $-t$ in (6.5). A source lift x satisfies $d_t x/t = -\varepsilon_J(x)$; this is the connecting map. Thus the first parameter derivative detects exactly the new class. The statements use a computed resolution, not an unspecified derived-functor comparison.

For J subset K , source inclusion and identity on the receiving scalar commute with augmentation. All maps and the added class therefore form the original type of support diagram. One may index by the finite active supports with a separately adjoined external bottom. At the active empty path set, U is zero but the receiving scalar remains R ; it is not assigned the nonempty-support cokernel $R/(t)$. Its map to a nonempty receiving quotient is the actual reduction $R \rightarrow R/(t)$. This retains the empty action chart and global absence separately.

The actual edge-chain norm of c_j is squared norm three, and different paths have disjoint supports. The quotient of $Q^{\wedge} J$ by $\ker \varepsilon_J$ consequently has Gram $3/|J|$, attained by the section with every coefficient $1/|J|$. Its change with $|J|$ remains in the comparison; no uniform Euclidean norm is substituted for it. At the word-fibre level the corresponding common-value Gram at $t = 0$ is $2^{\wedge}(-|J|)$, consistent with the independently counted $2^{\wedge}|J|$ representatives.

7. Unbounded active-support control and its exact extremal cost

Consider the special deformation

$$A_{L,h} = \bigcup_{j < r} \{L19^j, 7L19^j, 8L19^j + h_j\}, \quad h_j \geq 0, \quad L > 2 \sum_j h_j. \quad (7.1)$$

There are r active generator triples even when some actions h_j are zero. Let exactly a entries of h be positive and pairwise distinct; retain their positions and the zero entries. For $k \geq 4$, (1.2) says exactly that the positive active set is k -admissible.

Let $C_k(r, a)$ minimize the largest generator among (7.1) with these conditions. At $a=0$,

$$C_k(r, 0) = 8 \cdot 19^{r-1}.$$

For $a \geq 1$, the original definition of $g_k(a)$ gives the exact comparison

$$8(2g_k(a) + 1)19^{r-1} \leq C_k(r, a) \leq 8(2ag_k(a) + 1)19^{r-1} + g_k(a). \quad (7.2)$$

For the lower bound, $\sum h \geq \max h \geq g_k(a)$, and the last high triple has generator at least $8L19^{r-1}$. For the upper bound choose an actual minimizing active set for $g_k(a)$, put it in any a distinct positions, and choose $L = 2 \sum h + 1$. The separation and distinctness statements have already been proved, and the last perturbation is at most $g_k(a)$. Thus (7.2) is a bound on the actual original integers, uniform in the positions of the active support.

For completeness, λ_k exists by the exact composition $A \star B = A \cup (2S(A)+1)B$. Euclidean division identifies the distinct subset-sum product. The low second difference is smaller than the radix, so every progression decodes to one in each factor. The cost $2S+1$ is multiplicative. Its minimum $F_k(n)$ is submultiplicative; writing $n = qm + s$ and dividing its logarithm by n proves convergence to its infimum. Finally $2g_k(n)+1 \leq F_k(n) \leq 2n g_k(n)+1$ transfers that limit to g_k . Powers of three give existence and the bound $g_k(a) \leq 3^{a-1}$.

Take $a_r/r \rightarrow \alpha$. When a_r tends to infinity, $\log g_k(a_r) = a_r \log \lambda_k + o(a_r)$. When it stays bounded its contribution divided by r is zero; if $\alpha=0$, the uniform powers-of-three bound handles any sublinear unbounded subsequence. Taking logarithms in (7.2), retaining the $O(\log(a_r+1))$ error, proves

$$\lim_{r \rightarrow \infty} \frac{\log C_k(r, a_r)}{3r} = \frac{\log 19}{3} + \frac{\alpha \log \lambda_k}{3}.$$

This proves (1.3), including $\alpha=0$ and $\alpha=1$. Infinitely many nonzero defect actions are therefore explicitly controlled even when their support grows; their density determines the precise additional exponential cost in this family.

The formula is a reduction to the original λ_k , not a calculation of its value. At every $\alpha > 0$ the unknown arithmetic rate is still present with its stated exponent. The exact source-to-quotient return norm (5.5) also remains. The new theorem neither asserts an improved unrestricted upper record nor a matching general lower bound.

8. Mathematical implication for the general programme

Every indexed nonnegative defect list appears in the fixed high fibre (4.3). For positive distinct defect lists, this embeds every original EP817 instance. For $k \geq 4$ the correspondence has a witness decoder in the other direction; for $k=3$ the exact short-kernel map gives the same freeness equivalence for positive lists. Thus the full arithmetic problem is present inside separated deformations of one fixed, verified nineteen-adic source.

The successful infinite-family statement is not just an estimate for finitely many horizons. Sections 2–7 quantify over every r , every admitted integer perturbation, all original word fibres, and every k in the stated range. The common action $t=0$ versus $t \neq 0$ is completely classified, and its new class, derivative, and metrics are computed. The companion cut-tree theorem controls all integral graphical direction modules and their observation ideals.

The remaining general EP817 obligation is to control the actual defect images uniformly as the arithmetic input varies. A purely unconstrained linear identification is insufficient to perform that step, and its exact reason is visible rather than asserted: the row action (1,3,9) equals the augmentation row composed with $\text{diag}(1,3,9)$, but

that invertible rational map takes the binary box to $\{0,1\}$ times $\{0,3\}$ times $\{0,9\}$. The different original boxes are retained. The defect lists $(1,1,1)$ and $(1,3,9)$ have rational row complexes of the same ranks, while the first gives a four-AP and the second does not. Their literal feasible source fibres distinguish them.

This identifies the part on which a stronger all-rank inequality must act: the constrained arithmetic image and its original metric, together with the explicitly retained supported classes. It does not rule out such an inequality or claim a complete resolution of the general extremal problem.

9. Checks and exact evidence boundaries

The standalone producer and separate auditor use Python integers and Fraction. The auditor does not import or execute the producer. Both rebuild original source words. All new infinite statements rest on the written proofs; finite computations are corroboration and certificates for displayed finite constants.

The finite suite checks 43 full deformations through three triples, including all defect lists in $\{0,1,2\}^r$ for $r=1,2,3$ and four stated additional lists; 4,002 admitted general separation instances; all 756 signed perturbations in $\{-1,0,1\}^{3r}$, $r=1,2$; all 256 word supports and 6,561 inclusions for the one-triple refinement; the parameter complex and actual path incidences; and the original 125-coefficient, 342-progression mod-19 checks. The separate auditor reconstructs 14,562 distinct deformation values and 162 positive-step APs in its 43 displayed cases. Its signed-perturbation scope is reported separately in the final receipt.

The companion graph suite covers all 1,099 labelled simple graphs on 1 through 5 vertices, 40,398 source-score values, 1,391,037 endpoint pairs, 1,152 explicitly checked component cut trees, 8,493 observations, 23,118 image-ideal calculations, and 15,147 original root-direction witnesses. The original graph cut theorem is classical; the full source visibility argument and exact observation filtration are provided in the companion note.

Corrupted source counts, missing supports, zeroed witnesses, false cut capacities, and false metric/zero-action assertions are required to fail. Source and receipt hashes and the normal/optimized replay results are in the delivery manifest. No earlier large arithmetic search was silently rerun or assigned a new verification label. No inherited Lean source or receipt is changed.

References and exact roles

Anonymous/deleted Reddit contributor, & The Clankers. (2026, September 13). *The exponential rate for four-term-progression-free subset-sum sets* [Workbench manuscript]. EP817 main 23c0110c95b5a2036bdc04a1f352b6e5e27742c8. Role: original (1,7,8) construction and mathematical attribution.

sneed-and-feed. (2026, September 13). *Formalize finite upper bound theorem and core helper lemmas in Extended.lean* [Pull request #1]. Role: inherited formal upper-bound coverage; not verification of this continuation.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. Zeta Function Research Reader, `workbenches/splitzero-tandem/tex/support_diagrams.tex` at 58626a62cd5648e8ae7450fb54d4a4aab2330981. Role: original reconstruction, internal quotient, and additional receiving-kernel interfaces, especially (D6)–(D8).

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/html/2606.24139v1>. Role: problem conventions and the general upper/lower benchmark, not a source of the new perturbation or specialization formulas.

The Clankers. (2026, September 15). *Graphical obstruction propagation and a stronger six-term construction* [Predecessor contribution supplied in this conversation]. Role: disjoint-path witness map, complete-graph rigidity and the retained base-1651 upper record. Its exact package is retained unchanged in the new delivery; its full audit was not rerun here.

Relative defect classes and the retained infinite visibility filtration

Source: history/EP817_SUPPORTED_DEFECT_20260915/payloads/zeta/workbenches/ep817-supported-defects/notes/splitzero-receiving-map.md **Identity:** b7e90f40ecc6fa2d93245c2e12ada2a4ca756e640af0504ba3f65e749b5690e8 15 September 2026. Additive receiving contribution for the Zeta SplitZero workbench. The full new proof bodies and standalone integer/rational verification programs accompany this note. No arithmetic theta estimate, Riemann-hypothesis conclusion, or new Lean elaboration is claimed.

1. Original source and the exact transferred statement

The receiving source is `workbenches/splitzero-tandem/tex/support_diagrams.tex` at Zeta revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, blob d3493f891291ee6e94dbf2c77649f7d85d240df2. Equations (D6)–(D8) give the original internal quotient, the additional receiving-observation kernel and the supported-zero versus global-absence comparison. They are used here as stated, with ordinary integer or rational module fibres and the original reconstruction over $G(R)$. No substitute scalar or homology definition is introduced. The existing source file and all its verification records stay unchanged.

The new calculation identifies an actual relative cohomology class controlling an unbounded family of finite arithmetic sources. It also computes a complete infinite graphical-direction filtration with its support-changing map and original observation ideals. All statements have proofs in the two copied companion notes; their bodies are byte-identical in the EP817 and Zeta payloads.

2. Relative cochains, original primitives and arithmetic feasible sets

Let Θ_r have terminals u, v and r paths of length three, with internal vertices a_j, b_j and oriented edges $u \rightarrow a_j, a_j \rightarrow b_j, b_j \rightarrow v$. Relative vertex potentials vanish at u, v . The actual relative cochain sequence is

$$0 \rightarrow R^{2r} \xrightarrow{\delta^0} R^{3r} \xrightarrow{D} H^1(\Theta_r, \{u, v\}; R) \cong R^r \rightarrow 0,$$

where

$$\delta^0(\alpha_j, \beta_j) = (\alpha_j, \beta_j - \alpha_j, \beta_j), \quad D(p_j, q_j, s_j) = p_j + q_j - s_j.$$

The inverse on the boundary image reads $\alpha_j = p_j$ and $\beta_j = s_j$. The specified class section is $d \rightarrow (0, 0, -d)$, and the exact decomposition is

$$\eta = \delta^0(p_j, p_j + q_j)_j + (0, 0, -d_j)_j.$$

The original high cochain consisting of $L19^j, 7L19^j, 8L19^j$ is itself a relative coboundary. When $L > 2 \sum_j (|p_j| + |q_j| + |s_j|)$, every perturbed edge value stays positive and distinct. The full original binary source then satisfies, for $k \geq 4$,

$$H\left(\bigcup_{j < r} \{L19^j + p_j, 7L19^j + q_j, 8L19^j + s_j\}\right) \text{ is } k\text{-free} \iff H(d_0, \dots, d_{r-1}) \text{ is } k\text{-free}.$$

This uses more than an abstract cohomology isomorphism. The proof partitions the original words by their exact high digits, retains all low anchors and representation counts, and constructs inverse numerical maps for representative changes. Every defect progression embeds in the high fibre whose digits are all eight; every full progression decodes to an actual defect progression. These are explicit constrained-source maps. For three terms, the separate criterion is injectivity of the indexed ternary defect evaluation.

On the original absolute cycles c_{-j-c_0} , restriction sends d to d_{-j-d_0} . Consequently there is the exact sequence

$$0 \rightarrow R \xrightarrow{t \mapsto (t, \dots, t)} H^1(\Theta_r, \{u, v\}; R) \rightarrow H^1(\Theta_r; R) \rightarrow 0.$$

The common-action line is therefore visible in relative cohomology even when its absolute class vanishes. This is the precise class producing the long arithmetic progression under a common terminal perturbation.

3. Supported specialization and first-order detection

The actual path chains c_j have boundary $e_v - e_u$. Pairing an edge cochain with these chains is precisely the defect row d . For the common third-edge change t , the pairing is $-t$ times augmentation. Over $R = \mathbb{Q}[t]$ and a nonempty admitted path support J , retain the complex

$$R^J \xrightarrow{-t\varepsilon_J} R, \quad \varepsilon_J(x) = \sum_{j \in J} x_j.$$

Its cohomology is $\ker \varepsilon_J$ in the source degree and $R/(t)$ in the receiving degree. At $t=0$ the source kernel becomes $\mathbb{Q}^J$. The exact additional class is computed from the literal principal-ideal resolution:

$$0 \rightarrow \ker \varepsilon_J \rightarrow \mathbb{Q}^J \xrightarrow{-\varepsilon_J} \mathbb{Q} \rightarrow 0.$$

The last $\mathbb{Q}$ is $\text{Tor}_1^R(R/(t), \mathbb{Q})$, and the displayed map is obtained by dividing the actual differential by t before specializing. It is the first-order detector, not an assumed spectral or analytic comparison.

Zero extension on the admitted path supports and identity on the receiving scalar give the original support diagram. At the active empty path support, the receiving fibre is still R and the source is zero. Its receiving quotient is R , not the nonempty-support $R/(t)$. Global absence is an additional bottom label. The nonzero terminal chain with zero arithmetic pairing remains in its own admitted fibre throughout.

The edge-chain norm of each c_j has square three. The augmentation quotient Gram is therefore $3/|J|$. The common binary word fibre instead has $2^{|J|}$ representatives and quotient Gram $2^{|J|}$. These are separate original presentations, with their computed metrics retained.

4. Infinite graphical filtration and finite receiving support

For any finite loopless multigraph G , write $\text{Sigma}(G)$ for its actual orientation scores. The integral span of directions of $(q+1)$ -term source progressions is

$$K_q(G) = \{z \in \mathbb{Z}^V : \sum_{v \in C} z_v = 0 \text{ for every } q\text{-edge-connected class } C\}.$$

The proof uses the classical Gomory–Hu cut-tree theorem. Low cut capacities force the indicated zero charges, and edge-disjoint paths supply actual progressions for every class-root generator. Integral-span combinations are not declared feasible directions without their original source witnesses.

For an actual mark observation L_T , its image is the ideal

$$I_{q,T} = a_q \mathbb{Z}, \quad a_q = \gcd\{|t_u - t_v| : u, v \text{ belong to the same } q\text{-class}\}.$$

The original exact sequence is

$$0 \rightarrow K_q \cap \ker L_T \rightarrow K_q \xrightarrow{L_T} a_q \mathbb{Z} \rightarrow 0.$$

A nonconstant visible source progression exists exactly when $a_q > 0$. When all its source directions have zero observed action, the source K_q remains; it has not become an absent support.

The entire q -family is reconstructed on the semilattice of positive integers with reverse order, together with bottom infinity and join min. The fibre at infinity is zero; a zero fibre at any finite q remains supported. There is a finite support map $q \rightarrow P_q$, with infinity sent to a separately adjoined bottom. It preserves joins and uses the identity on each coefficient lattice. Its support coalescences are exactly the equal cut partitions. Thus every q value and original coefficient is retained on the source while a finite target controls all changes of the filtration.

When actual edge weights are positive and distinct, identifying only vertices with equal arithmetic marks preserves every edge and creates a simple quotient graph. Its relative cycle sequence is

$$0 \rightarrow H_1(G; \mathbb{Z}) \rightarrow H_1(\bar{G}; \mathbb{Z}) \xrightarrow{\partial_G} \text{im } \partial_G \cap \ker p_* \rightarrow 0.$$

The observation vanishes on the displayed relative module. The source masks are unchanged and the quotient score masses are summed over their actual p_* fibres. The induced minimum-section metric is the original weighted quotient metric, not unit mass on the coarser score values.

5. Original word quotients and what still requires arithmetic control

For any admitted word support S , define $V_S = Q[S]$, its high-value image W_S , and its joint high/defect-value image Z_S . The maps $q_S = f_S p_S$ commute with all support inclusions. Their original relation families satisfy

$$0 \rightarrow \ker p_S \rightarrow \ker q_S \xrightarrow{p_S} \ker f_S \rightarrow 0.$$

The high quotient Gram is $1/M_x$ and the joint quotient Gram is $1/m_{(x,u)}$. The minimum section has coefficients $m_{(x,u)}/M_x$. In particular representative changes and the additional quotient kernel have exact source metrics. For r path defects, with a nonzero active actions, the return to unit mass on numerical values has norm

$$2^{(r-a)/2} \sqrt{\mu_{\max}(d_{\text{active}})}.$$

This factor is not discarded. The companion active-support theorem gives an exact asymptotic within its defined separated family, $19^{1/3} \lambda_k^{(\alpha/3)}$, for active density α . The unrestricted arithmetic λ_k remains in that formula; no general EP817 solution follows merely from an unconstrained module equivalence or finite dimension count.

The present construction is a worked transfer of original supported quotient, relative cohomology, specialization and observation-kernel techniques. It does not replace the Zeta workbench's infinite theta quotient by a finite graph, and it imports no unevaluated arithmetic weight estimate.

6. Sources and checking scope

The complete proofs, their primary references and exact finite domains are in `supported-defect-universality.md` and `graphic-visibility-filtration.md`. The cut-tree theorem is attributed to Gomory and Hu (1961) and Gusfield (1990). The original nineteen construction remains credited to the anonymous/deleted Reddit contributor, and the separate Lean upper-bound extension to sneed-and-feed.

The standalone producer and separate auditor are copied unchanged into this receiving workbench, along with their complete compressed proof records and receipts. A second independently implemented pair checks the faithful graph quotient with original masks and relative cycle classes. Their exact source hashes and evidence boundaries are in `polyclank/verification.json`. No existing Zeta file or formal verification record is modified by this addition.

Recurrence sources with saturated short relations: exact progression control and a seven-term obstruction

Source: history/EP817_RECURRENCE_CONTROL_20260915/payloads/ep817/research/recurrence_control/notes/recurrence-control.md Identity: 8f389209ad2fec65f7a5498c26fc3c78cbb68a6b3bf5b8629fa22c74e43b8346 15
September 2026. PolyClank research continuation for Erdős Problem 817.

1. Scope, definitions, and outcome

For a finite positive generator set A retain its binary source, numerical image, and all representations:

$$\Omega_A = \{0, 1\}^A, \quad \Phi_A(\epsilon) = \sum_{a \in A} \epsilon_a a, \quad H(A) = \Phi_A(\Omega_A).$$

A nonconstant k -term arithmetic progression is a tuple $(x, x+d, \dots, x+(k-1)d)$ with d nonzero. Counts below use positive d ; reversing a tuple recovers the negative-step version. The empty subset is an actual source point, with numerical value zero.

This contribution develops an arithmetic source outside the previously used fixed integer-base blocks. It proves progression exclusion at every length for a quantitative gap family, computes its exact optimal cost, and specializes to the Pell weights $1, 2, 5, 12, 29, \dots$. For those weights it gives a complete five-state algebraic carry alphabet, an exact rational formula for four-term progression counts, and the complete integral short-relation presentation. The infinite relation operator has a sharp inverse norm $1/2$. A delayed generalization has inverse norm of order the square of its delay, with explicit bounds uniform in source length and coefficient-space dimension.

An attempted acceleration of these recurrences has an unconditional obstruction: a six-generator relation pattern constructs seven actual subset sums in progression. The obstruction excludes every longer-lag recurrence of the specified form from supplying a five-, six-, or seven-term-free infinite construction, irrespective of its starting positive seeds.

No improved global numerical bound is claimed. In particular $1+\sqrt{2}$ is larger than the preceding workbench upper rates for $k=5$ and $k=6$. The point is an all-rank test of which information an extremal lower argument must retain: an arbitrarily large admissible source can have a rank-one scalar quotient, an integrally saturated relation lattice generated by coefficient-two relations, a uniformly bounded inverse, and no five-term progression. Its actual feasible binary source and arithmetic observation cannot be recovered from those linear invariants alone.

All general statements have the proofs below. The finite graphs and calibration domains have exact executable checks and a separately implemented audit. No new Lean elaboration or global literature-priority claim is made. The prior mathematical construction attributed to the anonymous/deleted contributor and sneed-and-feed's Lean extension keep their separate credits.

2. A gap theorem for the complete infinite family

Let $a_0, a_1, \dots$ be positive integers. Put $S_j = \sum_{i=0}^j a_i$ and $S_j = 0$ for $j < 0$. For a fixed integer $r \geq 1$ require

$$a_j > S_{j-1} \quad (1 \leq j < r), \quad a_j > S_{j-1} + S_{j-r} \quad (j \geq r). \quad (2.1)$$

These inequalities retain the actual weights and all finite sums. They imply $a_j > S_{j-1}$ for every $j \geq 1$, so the binary evaluation is injective: the largest differing coordinate outweighs the total contribution below it.

Theorem R1 (uniform gap exclusion). Every finite prefix of a sequence satisfying (2.1) has subset sums free of nonconstant progressions of length 2^{r+1} . The conclusion covers all choices of the positive gaps and all prefix lengths simultaneously.

Proof. Reverse a negative-step tuple, so its step is positive. If a purported tuple has some equal highest binary coordinates, remove those common generator choices by the actual translation map. Let h be its largest differing coordinate. If $h < r$, the remaining image has at most $2^{h+1} \leq 2^r$ values and cannot contain the tuple.

Suppose $h \geq r$. Set $T = S_{h-r}$. Decompose every actual binary word into its top r coordinates, indexed $h-r+1, \dots, h$, and its remaining lower word. This gives a cover of the numerical image by 2^r translates of $H(\{a_0, \dots, a_{h-r}\})$, each contained in an interval of width T . The word evaluation, with both parts retained, is the specified bijection onto $H(\{a_0, \dots, a_h\})$; its inverse reads the unique binary representation.

The tuple uses both values of the h -th binary coordinate. Its consecutive entries that cross from coordinate zero to coordinate one cross the gap between S_{h-1} and a_h . By (2.1), that gap is strictly larger than T . Hence its common difference exceeds T . On the other hand, 2^{r+1} entries in the 2^r top-word classes place two entries in a single interval of width T ; their positive difference is an integer multiple of the common difference, which is then at most T . This contradiction proves the theorem. Every class label and its lower source remains in the partition. QED.

For $r=2$ the exact condition is

$$a_1 > a_0, \quad a_j > a_{j-1} + 2S_{j-2} \quad (j \geq 2). \quad (2.2)$$

It is useful to retain the gap itself:

$$g_j = a_j - a_{j-1} - 2S_{j-2}, \quad j \geq 1.$$

Then the identity

$$g_j - g_{j-1} = a_j - 2a_{j-1} - a_{j-2} \quad (2.3)$$

holds for $j \geq 2$. Thus every sequence with $0 < a_0 = 2a_{j-1} + a_{j-2}$ is in the five-term-free family. Its gaps are nondecreasing and positive. More generally the positive gaps may decrease or oscillate; only their actual positivity is used in (2.2).

3. Exact extremizers inside the gap family

Define

$$p_j^{(r)} = 2^j \quad (0 \leq j < r), \quad p_j^{(r)} = 2p_{j-1}^{(r)} + p_{j-r}^{(r)} \quad (j \geq r). \quad (3.1)$$

For $r=1$ this is $p_j = 3^j$. For $r \geq 2$ it is a recurrence with local coefficients of absolute value at most two.

Theorem R2 (sharp cost for the specified gap family). At every coordinate j , (3.1) is the smallest possible positive integral value among all sequences satisfying (2.1). For a prefix of length n , its longest progression has exactly $\min(2^r, 2^n)$ terms. Its exponential maximum-generator rate is the unique real root $\rho_r > 2$ of

$$\rho_r^r - 2\rho_r^{r-1} - 1 = 0. \quad (3.2)$$

Proof. During the first r positions, the smallest positive strict superincreasing choice is $1, 2, \dots, 2^{r-1}$. At $j=r$ the gap $p_j - S_{j-1} - S_{j-r}$ equals one. Subtracting successive gaps gives exactly $p_j - 2p_{j-1} - p_{j-r}$, so all later gaps remain one. Coordinatewise induction in (2.1), with positive integer gaps at least one, proves minimality.

The first $\min(r, n)$ generators have all subset sums from 0 through $2^{\min(r, n)} - 1$. Theorem R1 excludes one more term when $n \geq r$, and cardinality excludes one more when $n < r$. This proves the exact longest length.

The function $x^{(r-1)}(x-2)$ is strictly increasing on $x > 2$, starts at zero, and tends to infinity, giving the unique root. Choose positive constants c, C bounding the finitely many initial ratios p_j/ρ_r^j . The positive recurrence and its characteristic identity preserve $c \cdot \rho_r^j \leq p_j \leq C \cdot \rho_r^j$. Taking j -th roots proves the rate. QED.

The generating function is literally

$$\sum_{j \geq 0} p_j^{(r)} z^j = \frac{1}{1 - 2z - z^r}. \quad (3.3)$$

As r grows, ρ_r decreases to two; the exact equation $\rho_r - 2 = \rho_r^{(1-r)}$ gives $0 < \rho_r - 2 < 2^{(1-r)}$. This decrease comes with an increasing permitted progression length 2^r . It does not improve the known large- k global upper scale, which is much smaller than a rate tending to two.

For $r=2$ write $p_j = p_j^{(2)}$, so

$$p_0 = 1, \quad p_1 = 2, \quad p_{j+2} = 2p_{j+1} + p_j, \quad \alpha = 1 + \sqrt{2}, \quad \beta = 1 - \sqrt{2},$$

$$p_j = \frac{\alpha^{j+1} - \beta^{j+1}}{2\sqrt{2}}. \quad (3.4)$$

Thus $g_5(n) \leq p_{(n-1)}$, while the exact minimum within (2.2) is $p_{(n-1)}$. The inclusion of the gap family into all five-admissible sources proves this upper comparison, not its reverse.

4. Full integral saturation by short original relations

For $r \geq 2$ and a length n , keep $U_n = Z^{(n-1)}$, $V_n = Z^n$, and

$$\Phi_n(z) = \sum_{j < n} p_j^{(r)} z_j.$$

Let the j -th original boundary column, $1 \leq j < n$, be

$$b_j = e_j - 2e_{j-1} - \mathbf{1}_{j \geq r} e_{j-r}, \quad d_n(c) = \sum_{j=1}^{n-1} c_j b_j. \quad (4.1)$$

Each column has at most three actual coordinates and coefficients in $[-2, 2]$.

Theorem R3 (original scalar presentation). For every n and $r \geq 2$,

$$0 \longrightarrow U_n \xrightarrow{d_n} V_n \xrightarrow{\Phi_n} \mathbb{Z} \longrightarrow 0 \quad (4.2)$$

is split exact over the integers. In particular the entire scalar kernel is generated by coefficient-two relations, the quotient is torsion-free of rank one, and these facts coexist with the all-length exclusion in Theorem R1.

Proof and primitive. The recurrence gives $\Phi_n b_j = 0$. Eliminate the largest coordinate of z using its actual column b_j , proceeding downwards. The residual is $\Phi_n(z) e_0$ because $p_0 = 1$. The same operation records unique integral coefficients c_j . Therefore

$$z = d_n c + \Phi_n(z) e_0. \quad (4.3)$$

The uppermost nonzero c_j gives a nonzero j -th coordinate in $d_n c$, proving injectivity and uniqueness. Equation (4.3) supplies both the scalar section and the original-boundary primitive. QED.

At $r=2$ this gives, for every n , a five-admissible set with

$$\text{span}_{\mathbb{Z}}(\ker \Phi_n \cap [-2, 2]^n) = \ker \Phi_n, \quad \text{rank}(V_n / \ker \Phi_n) = 1. \quad (4.4)$$

There can therefore be no inequality $n \leq C(k) \cdot \text{rank}(V_n / \ker \Phi_n)$ for all k -admissible sets, even when the full kernel is generated by the stated short relations. This is an explicit counterexample family to that proposed dimension route, not a counterexample to Erdős Problem 817. The feasible binary source has 2^n distinct image values and remains part of the presentation.

5. A uniform infinite inverse, with its delay dependence proved

Let E be any real or complex Hilbert coefficient space with its specified inner product, and let S be the unilateral right shift on $\ell^2(N_0; E)$. Identify x_j with the original $c_{(j+1)}$. For $r \geq 2$, put $m=r-1$. The zero-padded boundary operator is exactly

$$\mathcal{D}_r = -2I + S - (S^*)^m. \quad (5.1)$$

Its finite-support restriction is (4.1), including the initial columns where the shifted coordinate is absent. It is a bounded map on the entire specified Hilbert space.

Theorem R4 (two-sided uniform control). The operator is bijective and

$$\|\mathcal{D}_r^{-1}\| \leq \frac{(r-1)^2}{2}. \quad (5.2)$$

For a nonzero coefficient space E and $r=2$, equality holds: the inverse norm is exactly $1/2$. For nonzero E and $r \geq 3$,

$$\|\mathcal{D}_r^{-1}\| \geq \frac{1}{2(1 - \cos(\pi/(r-2)))} \geq \frac{(r-2)^2}{\pi^2}. \quad (5.3)$$

The constants are independent of the length, the coefficient dimension, and the chosen orthonormal coordinates. The original coefficient inner product is retained. On the zero coefficient fibre the unique map is still a bijection and has norm zero; the lower bounds and norm-equality assertions explicitly require E nonzero.

Proof. Since S is an isometry, direct expansion gives

$$-\operatorname{Re}\langle \mathcal{D}_r x, x \rangle = \frac{1}{2} (\|(I - S)x\|^2 + \|(I + S^m)x\|^2). \quad (5.4)$$

Also

$$I - S^m = (I + S + \cdots + S^{m-1})(I - S),$$

so $\|(I - S^m)x\| \leq m\|(I - S)x\|$. The parallelogram identity gives

$$\|(I - S^m)x\|^2 + \|(I + S^m)x\|^2 = 4\|x\|^2.$$

As $m \geq 1$, these identities imply that (5.4) is at least $2\|x\|^2/m^2$. Cauchy-Schwarz gives $\|D_r x\| \geq 2\|x\|/m^2$, proving injectivity and closed range. The real quadratic form of its adjoint is identical; hence $\ker D_r^* = 0$ and the range is dense. Thus the closed range is the whole space, proving (5.2).

For $r=2$, $S-S^*$ is skew-adjoint and the bound is $1/2$. Constant vectors truncated to longer intervals have boundary energy divided by their squared norm tending to zero, so $\|D_r x\|/\|x\|$ tends to two; the inverse bound is sharp.

For $r \geq 3$ choose $\theta = \pi/(m-1)$ and truncate the sequence $\exp(i \cdot j \cdot \theta)v$ for a fixed unit coefficient v . Away from the finitely many cutoff rows, D_r acts by

$$-2 + e^{-i\theta} - e^{im\theta} = -2 + 2\cos\theta,$$

because $m\theta = \pi + \theta$. The number and magnitudes of boundary errors are fixed as the interval length grows. Their squared norm divided by the interval length tends to zero. Consequently the ratio $\|D_r x\|/\|x\|$ tends to $2(1 - \cos\theta)$, proving the first lower bound. The inequality $1 - \cos t \leq t^2/2$ proves the second. In a real coefficient space the same bound follows from the real and imaginary parts, or from the equality of an operator's real and complexified norms. QED.

The inverse also has a convergent operator integral with an explicit uniform remainder:

$$D_r^{-1} = -\int_0^\infty e^{tD_r} dt, \quad \left\| D_r^{-1} + \int_0^T e^{tD_r} dt \right\| \leq \frac{(r-1)^2}{2} e^{-2T/(r-1)^2}. \quad (5.4a)$$

Indeed, differentiating $\|\exp(tD_r)x\|^2$ and applying (5.4) gives $\|\exp(tD_r)\| \leq \exp(-2t/(r-1)^2)$. The integral converges in operator norm. Integrating its derivative proves both inverse laws, and integrating the exponential bound gives the displayed tail. This is a uniform calculation on the entire specified infinite source, not an interchange of a fixed-length calibration with an unproved limiting claim.

For $r=2$ and a finite primitive $c_1, \dots, c_{n-1}$, put $c_0=c_n=c_{n+1}=0$. The exact finite identity is particularly simple:

$$\|d_n c\|^2 = 4 \sum_{j=1}^{n-1} \|c_j\|^2 + \sum_{j=0}^{n-1} \|c_{j+2} - c_j\|^2. \quad (5.5)$$

Every boundary term is present. In matrix form the relation Gram has diagonal $5, 6, \dots, 6$, zero first off-diagonals and minus one second off-diagonals. The identity includes $n=1$ and $n=2$ with their literal zero-dimensional or one-dimensional source.

5.1 The quotient-completion comparison

On finite coefficient supports the arithmetic quotient in (4.2) is E . The Hilbert completion of (5.1) is onto and has zero quotient. The actual comparison map of complexes therefore induces $E \rightarrow 0$. It does not say that the original supported arithmetic class was absent.

The lost observation is explicit. For fixed v in E ,

$$z^{(N)} = (p_N^{(r)})^{-1} e_N v \rightarrow 0 \quad \text{in the raw Hilbert norm,} \quad \Phi z^{(N)} = v.$$

Retain it through the graph source $z \rightarrow (z, \Phi z)$. Its closure is $\ell^2(E)$ direct-sum E : truncate any first component, then correct its amplitude at a sufficiently high coordinate using the displayed vector. The completed relation map and contraction are

$$\begin{aligned} \widehat{dx} &= (D_r x, 0), \quad \pi(z, v) = v, \quad s(v) = (0, v), \quad K(z, v) = D_r^{-1} z, \\ K\widehat{d} &= I, \quad \widehat{d}K + s\pi = I. \end{aligned} \quad (5.6)$$

The forgetful map to the raw Hilbert completion has kernel 0 direct-sum E . These are maps on the original completed relation and observation spaces.

At finite length retain $W_n = \sum_{j < n} (p_j^{(r)})^2$. The least-norm scalar section is $a \rightarrow (p_j a / W_n)_j$. The raw and graph quotient Grams are

$$G_n^{\text{raw}} = G / W_n, \quad G_n^{\text{graph}} = (1 + 1/W_n)G. \quad (5.7)$$

For a coefficient space of finite dimension h , the original return cost is

$$\log \det G_n^{\text{raw}} = \log \det G_n^{\text{graph}} - h \log(1 + W_n). \quad (5.8)$$

For Pell weights, $2W_n = p_{n-1}p_n$, proved by telescoping $p_{j+1}p_{j+1} - p_{j-1}p_{j-1} = 2p_j^2$ (and the initial endpoint). Thus the raw quotient Gram decays at the explicit exponential rate α^{-2n} , while the infinite inverse norm remains $1/2$. The source's uniform inverse and the arithmetic observation's return cost are simultaneously present.

6. The algebraic observation and its supported zero directions

For the Pell case let $E_Z = \mathbb{Z}[\alpha]$, $\alpha^2 = 2\alpha + 1$. Use the literal basis $(1, \alpha)$ and the map

$$J_n : Z^n \rightarrow E_Z, \quad J_n e_j = \alpha^j, \quad \nu(u + v\alpha) = u + 2v. \quad (6.1)$$

By (3.4), $\text{nu}(\alpha^j) = p_j$, so $\Phi_n = \text{nu } J_n$. For $n \geq 2$, J_n is onto. Its kernel is generated by the shifted polynomial relations

$$e_{j+2} - 2e_{j+1} - e_j, \quad 0 \leq j < n - 2.$$

Monic polynomial division proves this over $\mathbb{Z}$, with the actual quotient coefficients retained. The additional scalar relation $b_1 = e_1 - 2e_0$ maps to $\alpha - 2$. Hence

$$\boxed{0 \rightarrow \ker J_n \rightarrow \ker \Phi_n \xrightarrow{J_n} \mathbb{Z}(\alpha - 2) \rightarrow 0.} \quad (6.2)$$

The scalar observation is $\mathbb{Z}$ -linear, not a ring homomorphism. The element $\alpha - 2$ is a nonzero element of $E_{\mathbb{Z}}$ with supported scalar image zero. Replacing the terminal condition $\text{nu}(z) = 0$ by $z = 0$ discards actual arithmetic tuples.

For example the binary representations of 0,1,2,3 in the weights 1,2 have algebraic second differences $\alpha - 2$ and $2 - \alpha$. Both are nonzero and both have scalar image zero. This example is retained as a negative control in the finite certification.

7. A complete five-state carry computation for every word length

For a coefficient word c_j in $[-2, 2]$, evaluate its polynomial in α from the highest coefficient to the lowest by Horner's rule. In integer coordinates the transition is

$$(u, v) \mapsto (v + c, u + 2v). \quad (7.1)$$

The actual terminal arithmetic condition is $u + 2v = 0$.

Lemma R5 (exact bounded carry alphabet). Every Horner prefix of a scalar-zero word belongs to the following five-element set:

$$\boxed{\mathcal{C} = \{0, 1, -1, \alpha - 2, 2 - \alpha\}.} \quad (7.2)$$

Proof. Conjugation sends α to $\beta = 1 - \sqrt{2}$, with $|\beta| = \sqrt{2} - 1$. Every finite Horner prefix z obeys

$$|z'| < \frac{2}{1 - |\beta|} = 2 + \sqrt{2}.$$

If the full word z has $\text{nu}(z) = 0$, then $z = v(\alpha - 2)$, and its conjugate is $-\alpha^*v$. Therefore $|v| < \sqrt{2}$, so v is $-1, 0$, or 1 and $|z| \leq \alpha - 2$.

If a prefix has t remaining digits, the full-word equation is $z = \alpha^t q + R$, with $|R| \leq 2(\alpha^{t-1})/(\alpha - 1)$. Hence

$$|q| \leq \frac{2}{\alpha - 1} + \frac{|z| - 2/(\alpha - 1)}{\alpha^t} < \sqrt{2}.$$

Write $q = (u + v) + v \sqrt{2}$, $q' = (u + v) - v \sqrt{2}$. The two strict bounds give $|v| < 1 + \sqrt{2}$, and more sharply $|v| < (2 + 2\sqrt{2})/(2\sqrt{2}) < 2$, so v is $-1, 0, 1$. For $v = 0$ the possible u are $-1, 0, 1$. For $v = 1$, intersecting the two bounds gives $u = -2$; the other case gives $u = 2$. This is exactly (7.2). All strict endpoints matter. QED.

For k rows a state is $(z_0, \dots, z_{k-3}, \text{flag})$, with z_i in $\mathcal{C}$. A binary column ϵ in $\{0, 1\}^k$ updates the carries by (7.1) using $c_i = \epsilon_i - 2\epsilon_{i+1} + \epsilon_{i+2}$. The flag records whether the first two binary words have ever differed. A target is admitted only when every updated carry belongs to $\mathcal{C}$. Initial state is all zero with false flag. Acceptance requires true flag and $\text{nu}(z_i) = 0$ for every row.

The binary numerical evaluation is injective by Section 2. Consequently the flag is exactly numerical nonconstancy for a terminal arithmetic tuple. The path-to-tuple map evaluates the actual columns in the reversed Pell weights. The inverse reads each unique binary representation and its Horner prefixes. Lemma R5 proves that every accepted arithmetic tuple stays in the finite carrier; no real progression is removed by that bound.

The complete reachable data are:

k	states	labeled transitions	accepting states
3	8	26	3
4	22	46	3
5	27	28	0
6	49	50	0

Every listed transition is independently re-enumerated from actual binary columns. The 27-state five-term exclusion is a second proof of the Pell specialization of Theorem R1. The finite graph, together with the prefix lemma and the exact terminal map, proves every length. It is not an extrapolation from tested prefixes.

7.1 Exact four-term progression counts

Let F_n be the number of positive-step four-term progressions among the subset sums of the first n Pell weights, with $F_0=0$. The weighted path count gives

$$\sum_{n \geq 0} F_n z^n = \frac{z^2}{(1-2z)(1-z-4z^2)}. \quad (7.3)$$

Equivalently, $F_0=F_1=0$, $F_2=1$, and

$$F_n = 3F_{n-1} + 2F_{n-2} - 8F_{n-3} \quad (n \geq 3). \quad (7.4)$$

The first values are 0,0,1,3,11,31,91,247,675,1791,4747,12423,32435. The dominant counting rate is $(1+\sqrt{17})/2$, strictly larger than the image-cardinality rate two. Thus the infinite family is richly four-progressive while remaining five-progressive-free.

For a small transparent certificate, retain the eight states that can reach acceptance. In order they are

$$(r, -r; 1), (-1, 1; 0), (-1, 1; 1), (0, 0; 0), (0, 0; 1), (1, -1; 0), (1, -1; 1), (-r, r; 1), \quad r = \alpha - 2.$$

Their labeled-edge multiplicity matrix is

$$T = \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

The initial row is index 3, counting from zero; the accepting indices are 0,4,7. Thus $F_n = (e_3^T T^n (e_0 + e_4 + e_7))/2$. Reversal pairs the two step signs. Solving $(I-zT)f = e_0 + e_4 + e_7$ gives (7.3). The executable also certifies the denominator on the full 22-state graph: its recurrence-residual row has zero outputs for the first 22 powers; Cayley–Hamilton then proves all subsequent outputs vanish. The full discarded states remain in the receipt and are excluded from this smaller counting matrix only by their proved inability to reach an accepting state.

Pisot-base finite-state arithmetic is established prior methodology (Frougny & Pelantová, 2018). Here the complete small alphabet, actual terminal observation, all labeled transitions and integer arithmetic checks are supplied for this specific recurrence. No generic literature theorem is substituted for these calculations.

7.2 An explicit scalar-observation carrier for every delay

The finite arithmetic control extends to every $r \geq 2$, not only the quadratic case. The bounds below deliberately retain their degree dependence rather than declaring a uniform compact coefficient set as r increases.

Put

$$Q_r(X) = X^r - 2X^{r-1} - 1, \quad E_r = \mathbb{Z}[X]/(Q_r), \quad \nu_r \left(\sum_{i < r} z_i X^i \right) = \sum_{i < r} 2^i z_i. \quad (7.5)$$

This is the actual scalar observation satisfying $\nu_r(X^j) = p_j^{(r)}$ for every j . It is $\mathbb{Z}$ -linear, and its kernel is retained. Let $M \geq 1$ be an integer coefficient bound, and define the explicit positive rational numbers and integer cutoff

$$\delta_r = \frac{2}{(r-1)^2(3r-2)}, \quad B_r(M) = \frac{M(r-1)(r+2)}{2r\delta_r^2},$$

$$C_r(M) = \left\lceil \frac{M(r-1)(r+2)2^{r-1}3^{r-2}}{r\delta_r^3} \right\rceil. \quad (7.6)$$

Theorem R5u (all-delay finite arithmetic carrier). Every Horner prefix of every finite coefficient word in $[-M, M]$ whose full scalar observation nu_r is zero has all of its original coefficients in $[-C_r(M), C_r(M)]$. Consequently, for fixed r and k , all progression equations on every recurrence prefix are decidable in a finite graph with at most

$$2(2C_r(2) + 1)^{r(k-2)}$$

states. Actual terminal membership in $\ker \text{nu}_r$ is used, rather than zero in the larger algebra. This coarse bound is a proved finite domain, not a claim that its whole graph was enumerated in the delivered computation.

Proof of the root and observation estimates. Q_r has exactly one root $\alpha > 1$ outside the unit circle, with $2 < \alpha < 3$, and its other $r-1$ roots lie strictly inside. To verify the count, for $0 < t < 1$ apply Rouché's theorem on the unit circle to $X^r - 2X^{r-1} - t$: the term $-2X^{r-1}$ has magnitude 2, greater than $1+t$. Thus $r-1$ roots lie inside. At $t=1$ there is no unit-circle root: equality in $2|z|^{r-1} = |z^r - 1|$ at $|z|=1$ would force both $\hat{z}^r = -1$ and $\hat{z}^{r-1} = -1$, hence $z=1$, a contradiction. The root count is unchanged on reaching $t=1$. The positive root is unique by (3.2). The roots are simple because a common nonzero root of Q_r and Q_r' would be $2(r-1)/r < 2$, where Q_r is negative. In particular α is a Pisot number. Irreducibility also follows: a nonconstant monic integer factor excluding α would have nonzero integral constant coefficient of modulus less than one, since all its roots are inside the unit circle.

On $|z|=1$, with $m=r-1$,

$$|Q_r(z)| \geq 2 - \operatorname{Re} z + \operatorname{Re} z^m = \frac{1}{2}(|1-z|^2 + |1+z^m|^2) \geq 2/m^2.$$

The final inequality is the scalar version of (5.4): use $|1-z^m| \leq m|1-z|$ and $|1-z^m|^2 + |1+z^m|^2 = 4$. Inside the unit disk $|Q_r'| \leq 3r-2$. Applying the derivative bound along a radius from a stable root λ to the unit circle proves

$$1 - |\lambda| \geq \delta_r. \quad (7.7)$$

The constant coefficient of Q_r has modulus one, $\alpha < 3$, and all other roots have modulus less than one. Hence each stable root has modulus greater than $1/3$.

Let $\lambda_0 = \alpha$ and $\lambda_1, \dots, \lambda_{r-1}$ be the other roots. Partial fractions of the literal generating function (3.3) give the scalar observation formula

$$\nu_r(q) = \sum_{a=0}^{r-1} w_a q(\lambda_a), \quad w_a = \frac{\lambda_a^{r-1}}{Q_r'(\lambda_a)} = \frac{\lambda_a}{r\lambda_a - 2(r-1)}. \quad (7.8)$$

Indeed expansion at zero gives this equality first on X^j , hence on every original remainder. The expanding coefficient satisfies $w_0 \geq 2/(r+2) > 0$. For the stable coefficients,

$$|w_a| \leq 1/(r\delta_r),$$

because $|r\lambda_a - 2(r-1)| \geq r-2+r\delta_r \geq r\delta_r$.

Every finite Horner prefix has $|q(\lambda_a)| \leq M/\delta_r$ at a stable root. If its full word z has $\text{nu}_r(z)=0$, equation (7.8) therefore gives $|z(\alpha)| \leq B_r(M)$. A prefix q with t remaining digits satisfies $z(\alpha)=\alpha^t q(\alpha)+R$ with $|R| \leq M(\alpha^t-1)/(\alpha-1)$. Since $B_r(M) > M$ and $\alpha > 2$,

$$|q(\alpha)| \leq B_r(M), \quad |q(\lambda_a)| \leq M/\delta_r \leq B_r(M). \quad (7.9)$$

Finally retain the exact interpolation map back to the original monomial coordinates:

$$q(X) = \sum_{a=0}^{r-1} q(\lambda_a) \frac{Q_r(X)}{(X - \lambda_a)Q'_r(\lambda_a)}. \quad (7.10)$$

Every coefficient of $Q_r(X)/(X-\lambda_a)$ has magnitude at most 2^r , since the product of $1+|\lambda_b|$ over the other roots is at most $4 \cdot 2^{r-2}$. At stable roots, $|Q'_r(\lambda_a)| \geq 3^{-(r-2)} \cdot r\delta_r$. The same common reciprocal bound $3^{-(r-2)}/(r\delta_r)$ bounds the expanding derivative reciprocal as well. Equations (7.9)-(7.10) yield the coefficient bound

$$|[X^i]q| \leq B_r(M) 2^r 3^{r-2}/\delta_r \leq C_r(M).$$

This proves the finite carrier in the literal Z -basis; the complex embeddings serve only as specified estimates and are not replacements for that basis. QED.

Proof of the progression decision statement. Use Horner multiplication by X in E_r , reducing by the original relation $X^r = 2X^{r-1} + 1$, and add each second-difference coefficient from the actual binary k -column. Retain only target coefficient vectors in the proved box, plus the first-two-word difference flag. Every genuine progression stays in the box by the theorem; every accepted path evaluates to its actual scalar progression because $\text{nu}_r(X^j)=p_j$. The binary evaluation is injective by Theorem R1, so the flag is the exact nonconstancy test. The graph is finite with the stated size bound, and reachability is equivalent to existence at some finite length. All support labels, algebraic carries, and original scalar-zero terminal classes remain explicit.

For $r=2$, $M=2$ the coarse formula gives $C_2(2)=64$. Lemma R5 and its two-embedding rectangle refine that entire coarse lattice box to the exact five values used in the fully enumerated graphs. For $r>2$ the delivered finite checks test original polynomial reduction and scalar-zero prefix containment in bounded word domains; the complete higher-degree graph is not claimed to have been constructed. The theorem gives its explicit stopping domain.

8. The faster-delay obstruction: seven actual subset sums

Consider the six distinct positive generators

$$x, \ b, \ u = x + 2b, \ c, \ v = b + 2c, \ w = u + 2v = x + 4b + 4c. \quad (8.1)$$

Theorem R6 (seven-term relation motif). Their subset sums contain a seven-term progression of positive difference $b+c$. In the displayed generator order, its binary masks are

$$1, \ 11, \ 19, \ 28, \ 32, \ 42, \ 50. \quad (8.2)$$

The actual values are

$$x, \ x+b+c, \ x+b+v, \ u+c+v, \ w, \ w+b+c, \ w+b+v,$$

which equal $x+j(b+c)$, $0 \leq j \leq 6$. This proves the assertion by direct substitution. The second differences of the masks are integer combinations of the original three relation rows

$$(-1, -2, 1, 0, 0, 0), \ (0, -1, 0, -2, 1, 0), \ (0, 0, -1, 0, -2, 1).$$

Those coefficients are included in the finite proof data.

Now take any strictly increasing positive initial seeds $a_0, \dots, a_{(r-1)}$, $r \geq 3$, and extend by

$$a_j = 2a_{j-1} + a_{j-r} \quad (j \geq r).$$

At the six distinct indices

$$0, r-1, r, 2r-2, 2r-1, 2r$$

the weights have exactly the form (8.1). Thus the first $2r+1$ generators always contain the displayed seven-term progression. This is a theorem for every choice of positive increasing seeds, not a bound on a finite seed search. In particular these longer delays cannot yield infinite k -admissible constructions for $k=5,6,7$, even though their characteristic rates approach two.

The index distinctness is indispensable. At $r=2$ the positions r and $2r-2$ coincide. The original six-label map to the five actual generator coordinates is

$$(0, 1, 2, 3, 4, 5) \mapsto (0, 1, 2, 2, 3, 4). \quad (8.3)$$

It is a linear map on free coefficient modules, but its pushforward sends the fourth mask in (8.2) to a vector with coefficient two at the third generator. Thus it does not send the whole binary feasible source into the five-generator binary source. For the actual Pell weights 1,2,5,12,29, the candidate seven values are 1,8,15,22,29,36,43, and 22 is absent. It lies in the retained gap (20,29). All other displayed values are present. The same scalar relation equations therefore yield an admissible witness for $r \geq 3$ but an unsupported binary request at $r=2$; the exact support-changing map records the distinction.

8.1 Equal-difference packing for arbitrary progression length

The motif also gives an all- k constraint with its support hypothesis retained. Suppose s copies of (8.1) have pairwise disjoint actual generator sets and the same positive difference $D=b+c$. Let their initial values be $x_1, \dots, x_s$. For every integer h with $0 \leq h \leq 6s$, choose actual integers h_j in $[0,6]$ with sum h (fill the first coordinate up to six, then the next, and so on). In motif j take the subset with mask (8.2) at index h_j . Their disjoint union is an actual generator subset of value

$$\sum_{j=1}^s x_j + hD.$$

Thus s such motifs force a $(6s+1)$ -term progression. A k -admissible set can contain at most $\text{floor}((k-2)/6)$ pairwise disjoint copies with a fixed common D . This is a support-sensitive packing condition, not a bound on the number of overlapping symbolic relations. The complete incidence-labelled witnesses make the restriction usable in a candidate search or in a later matching argument.

9. Consequences for the general research programme

Theorem R3 excludes a tempting all-rank strategy: the number of generators cannot be bounded using only the rank of the saturated short-relation quotient, even with primitive coefficient-two relations. Theorem R4 strengthens that test: a uniform source inverse does not add the missing information, since its norm is exactly $1/2$ in a five-admissible family of arbitrary length. The true binary source and its observation remain 2^n -point arithmetic data, and (5.8) gives the quantitative cost lost by forgetting the original observation in Hilbert completion.

The gap theorem supplies a working replacement in this family: retain the original high-coordinate partition and its actual separation budget. It handles arbitrary positive gap sequences and proves the exact best recurrence within that class. The finite algebraic state calculation supplies another replacement, capable of retaining supported scalar-zero directions such as α_2 rather than discarding them at a stronger observation.

The seven-term motif then tests an attempted escape from the gap optimum. The faster delayed recurrence has an explicit all-seed obstruction after at most $2r+1$ generators. Any new recurrence intended to improve the $k=5$ or $k=6$ bounds must avoid this motif with its six distinct original supports, or supply a different admissibility mechanism. This is a necessary arithmetic constraint on candidate constructions; it is not asserted sufficient for general admissibility.

The unrestricted rates and their sharp large- k lower law remain unevaluated by this contribution. The inspected general benchmark gives a lower logarithmic scale of order $1/k$ and an upper scale of order $\log(k)/k$ (Korsky, 2026). The exact maps above identify concrete information that a proposed uniform lower argument must use, and a concrete relation motif it must recognize.

10. Verification and source record

Run the standard-library producer and separately implemented auditor from this directory's certificates folder, supplying the paths shown in its README. No software download, floating-point comparison, or external service is needed to check the finite data.

The producer constructs all reachable graphs from the exact five-element carry alphabet, retains every binary edge label, and checks its counts against independent integer-bitset progression counts through thirteen weights. The algebraic carry alphabet is verified by exact signs in $\mathbb{Q}(\sqrt{2})$; its finiteness proof is in Lemma R5. The finite checks also cover all 2,186 specified positive-gap sequences through length seven, the stated dyadic-gap prefixes, 3,240 triangular basis identities through length eighty, exact coercivity and section metrics, 90 delayed-recurrence applications through delay 32, 690 finite general-delay operator identities, and complete scalar-zero coefficient-word checks for delays two through six and word lengths zero through six. These are bounded calibrations; they are not labelled proofs by exhaustion of infinite parameter families.

The auditor imports no producer module. It reconstructs the complete carry transitions and their accepted scalar observation, compares arithmetic AP counts by direct pairs, checks original basis primitives and finite Grams independently, and verifies the seven-term masks and their alias failure. The completed audit is recorded in certificates/independent_audit.json. Normal/optimized comparisons, mutation controls, and precise evidence boundaries are bound in CHECKS.json.

The existing EP817 main was read at 23c0110c95b5a2036bdc04a1f352b6e5e27742c8. No existing main file, prior receipt, or accepted status is modified. The inspected Zeta interface is support_diagrams.tex, equations (D6)–(D8), at 58626a62cd5648e8ae7450fb54d4a4aab2330981, blob d3493f891291ee6e94dbf2c77649f7d85d240df2. The companion receiving note spells out the original diagrams and distinguishes word-value quotients from coefficient-amplitude quotients.

References

- Frougny, C., & Pelantová, E. (2018). *Two applications of the spectrum of numbers* (Version 3; original preprint 2015) [Preprint]. arXiv:1512.04234v3. <https://arxiv.org/abs/1512.04234v3>. Retrieved abstract used for attribution of the prior finite-state Pisot methodology; no uninspected theorem body is an input to these proofs.
- Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1) [Preprint]. <https://arxiv.org/html/2606.24139v1>. Inspected Theorems 1.2–1.3 and Sections 4–6 for the general benchmark, the scalar ternary observation, and source-layout constraints.
- KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. Zeta Function Research Reader, revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, workbenches/splitzero-tandem/tex/support_diagrams.tex, (D6)–(D8). Used for the original supported quotient and additional observation-kernel interface.
- The Clankers. (2026, September 15). *Supported zero actions, exact defect transport, and all-rank deformations* [Predecessor research contribution]. Supplied EP817_SUPPORTED_DEFECT_20260915 archive. Used as the retained source and methodological predecessor, not as a new independent validation claim.

Recurrence cohomology, supported arithmetic-zero carries, and the unchanged source metric

Source: `history/EP817_RECURRENCE_CONTROL_20260915/payloads/zeta/workbenches/ep817-recurrence-control/notes/splitzer-receiving-map.md` **Identity:** `a99ec2576fc0a2b69f96761920d8781ff48add647973b11723c498dab2082ef6` 15 September 2026. Additive receiving contribution to the Zeta SplitZero workbench. Ordinary proofs; no new Lean or arithmetic-zeta growth conclusion.

1. The exact interfaces used

The source remains the support-diagram reconstruction and internal quotient in `workbenches/splitzer-tandem/tex/support_diagrams.tex`, (D6)–(D8), at revision `58626a62cd5648e8ae7450fb54d4a4aab2330981`. Its source blob is `d3493f891291ee6e94dbf2c77649f7d85d240df2`.

For a coefficient field K , retain $G(K)=\{\tau\}$ disjoint-union $\{r^\bullet:r \text{ in } K\}$. On the diagram total, supported zero sends (n,x) to $(n,0)$; τ sends it to $(\text{bottom},0)$. The new coefficient maps below are ordinary K -linear maps on their original fibres and commute with the stated transports. Their reconstructed maps therefore use that existing $G(K)$, rather than a new scalar convention.

The EP817 application has two different sources: the free coefficient module K^n , whose arithmetic map is a linear form, and the free module on the 2^n actual binary words. Both are retained. A proof about one kernel does not identify the two source modules.

2. Finite recurrence complexes and the supported quotient

Fix delay $r \geq 2$ and let p_j be the exact integers

$$p_j = 2^j \ (j < r), \quad p_j = 2p_{j-1} + p_{j-r} \ (j \geq r).$$

At positive support n put $U_n=K^{(n-1)}$, $V_n=K^n$ and

$$d_n e_j = e_j - 2e_{j-1} - \mathbf{1}_{j \geq r} e_{j-r}, \quad \Phi_n(z) = \sum_{j < n} p_j z_j.$$

The formula for d_n uses the actual source basis indexed $1 \leq j < n$. At support zero use zero coefficient spaces. The support semilattice is N_0 with join max and bottom 0. The transports in both U and V are zero padding; the positive-support quotient transport is the identity on K .

Triangular elimination supplies a unique original primitive h_n satisfying

$$d_n h_n(z) + \Phi_n(z) e_0 = z, \quad h_n d_n = I.$$

Thus each positive-support complex $U_n \rightarrow V_n$ has receiving cohomology K . Its coefficient-zero class remains in that positive support. Every original arithmetic relation is generated by columns of support at most three

and coefficient magnitudes at most two. No rational saturation or unrecorded replacement of the integer lattice occurs: the same exact sequence already holds over $\mathbb{Z}$.

The finite p_j source is nevertheless (2^r+1) -AP-free for every length. For $r=2$ it is five-term-free, for all n , even though the scalar quotient has rank one and the short relations generate the whole kernel. This demonstrates that these cohomological rank and generator-size data do not bound the original binary source dimension.

The exact replacement that proves exclusion is a high-word partition with its literal interval budget. More generally, every positive sequence with

$$a_j > \sum_{i < j} a_i \quad (j < r), \quad a_j > \sum_{i < j} a_i + \sum_{i \leq j-r} a_i \quad (j \geq r)$$

has the same exclusion. The complete EP817 note proves it by comparing the gap crossed by a progression with the widths of its 2^r admitted top-word fibres. The recurrence source is the coordinatewise least positive integral member of this specified gap family.

3. The extra arithmetic observation kernel in the Pell case

For $r=2$, use the literal algebraic coefficient ring $E_{\mathbb{Z}} = \mathbb{Z}[\alpha]$ with $\alpha^2 = 2\alpha + 1$. The original two observation maps are

$$J_n(e_j) = \alpha^j, \quad \nu(u + v\alpha) = u + 2v, \quad \Phi_n = \nu J_n.$$

The receiver of J_n is its actual image: it is zero at support 0, $\mathbb{Z}^*1$ at support 1, and $E_{\mathbb{Z}}$ at every support $n \geq 2$. Transports are the original inclusions. The second map is $\mathbb{Z}$ -linear; it is not asserted to preserve multiplication.

Monic polynomial division and the original first boundary column give the exact sequence

$$0 \rightarrow \ker J_n \rightarrow \ker \Phi_n \xrightarrow{J_n} \mathbb{Z}(\alpha - 2) \rightarrow 0 \quad (n \geq 2).$$

The nonzero element $\alpha - 2$ is thus a supported class that the arithmetic observation sends to its own receiving zero. It is not the external bottom. Its presence is essential: the actual four-term tuple $(0, 1, 2, 3)$, represented with weights $(1, 2)$, has algebraic second differences $\alpha - 2$ and $2 - \alpha$.

The all-length finite automaton retains both embeddings of $E_{\mathbb{Z}}$. In original coordinates (u, v) , the Horner step for a coefficient c is

$$(u, v) \mapsto (v + c, u + 2v).$$

A scalar-zero word with coefficients in $[-2, 2]$ can visit only

$$0, 1, -1, \alpha - 2, 2 - \alpha.$$

The proof bounds every prefix in both real embeddings; it is independent of word length. All terminal carries with $nu=0$ are retained, including the two nonzero ones. Coupling $k-2$ such carries through actual binary k -columns gives a complete graph with 27 reachable states and 28 labeled edges for $k=5$, and no accepting state. It is an exact admissibility calculation, not a claim that vanishing of an unrestricted linear amplitude makes a binary witness present.

The four-term graph supplies the exact numerical count

$$\sum_{n \geq 0} \text{ap}_4(H(p_0, \dots, p_{n-1})) z^n = \frac{z^2}{(1 - 2z)(1 - z - 4z^2)}.$$

Every original binary word has a unique value in this source. Thus weighted path counting equals numerical tuple counting; no representation weights were silently discarded.

3.1 An explicit degree-dependent carrier on every original recurrence algebra

For every delay $r \geq 2$, use $E_r = \mathbb{Z}[X]/(X^r - 2X^{r-1} - 1)$ and the literal scalar observation

$$\nu_r \left(\sum_{i < r} z_i X^i \right) = \sum_{i < r} 2^i z_i.$$

The recurrence gives $\text{nu}_r(X^j) = p_j$ for every j , so the original evaluation square commutes at every support. The polynomial has one expanding real root α in $(2,3)$, and all other roots are strictly inside the unit circle. The finite-prefix bound is derived in these embeddings and returned by the actual interpolation map to the original monomial coefficients.

With a coefficient alphabet $[-M, M]$, put

$$\delta_r = \frac{2}{(r-1)^2(3r-2)}, \quad C_r(M) = \left\lceil \frac{M(r-1)(r+2)2^{r-1}3^{r-2}}{r\delta_r^3} \right\rceil.$$

Every prefix of a word whose complete scalar observation is zero has each original coefficient in $[-C_r(M), C_r(M)]$. The proof retains the full nonzero terminal kernel of nu_r . It uses the exact spectral expression

$$\nu_r(q) = \sum_{Q_r(\lambda)=0} \frac{\lambda}{r\lambda - 2(r-1)} q(\lambda),$$

then the inverse interpolation formula back to Z^r . The scalar unit-circle energy inequality from the original recurrence gives $1 - |\lambda| \geq \delta_r$ at each stable root; hence no unspecified spectral separation constant is used.

For arithmetic-progression second differences take $M=2$. The resulting finite graph bound is $2(2C_r(2)+1)^{r(k-2)}$ for each specified r, k . Its coefficients grow with r ; they are not represented as one uniformly bounded coefficient cube at unbounded degree. The theorem is proved for every degree, while the delivered higher-degree computations only replay the explicitly recorded finite coefficient-word domains. The sharp quadratic five-state graph remains the complete applied case.

4. Infinite cochain recovery with an evaluated uniform remainder

For a Hilbert coefficient space E with its given form G , let S be the unilateral right shift on $\ell^2(\mathbb{N}_0; E)$. The full boundary operator is

$$D_r = -2I + S - (S^*)^{r-1}.$$

Direct expansion in the original inner product gives, with $m=r-1$,

$$-\text{Re}\langle D_r x, x \rangle = \frac{1}{2} (\|(I - S)x\|^2 + \|(I + S^m)x\|^2) \geq 2\|x\|^2/m^2.$$

The inequality follows from $I - S^m = (I + \dots + S^{m-1})(I - S)$ and the parallelogram identity for x and $S^m x$. It also holds for D_r^* . Therefore D_r has closed dense range and is invertible, with

$$\|D_r^{-1}\| \leq m^2/2.$$

For a nonzero coefficient space, the $r=2$ norm is exactly $1/2$. For $r \geq 3$, truncated plane waves give the lower bound $(r-2)^2/\pi^2$. These two explicit bounds identify quadratic delay growth. The zero coefficient fibre retains its unique bijection with norm zero; a lower-norm assertion is made only for a nonzero fibre.

The original full inverse is also

$$D_r^{-1} = - \int_0^\infty e^{tD_r} dt, \quad \left\| D_r^{-1} + \int_0^T e^{tD_r} dt \right\| \leq \frac{m^2}{2} e^{-2T/m^2}.$$

Differentiate $\|\exp(tD_r)x\|^2$ to get its stated exponential bound, integrate the derivative to verify both inverse laws, and integrate the bound for the remainder. All constants retain their delay dependence and are uniform in coefficient dimension and the infinite horizon.

The onto Hilbert map has zero degree-one quotient. The finite-support arithmetic complex has quotient E . Its graph source retains the exact observation $z \rightarrow (z, \Phi z)$; its closure is $\ell^2(E)$ direct-sum E . The completed relation is $x \rightarrow (D_r x, 0)$, with contraction

$$K(z, a) = D_r^{-1}z, \quad \pi(z, a) = a, \quad s(a) = (0, a), \quad K\hat{d} = I, \quad \hat{d}K + s\pi = I.$$

The projection that forgets arithmetic amplitude has kernel 0 direct-sum E . This is the precise receiving kernel; no class is declared absent merely because raw Hilbert completion kills it.

5. The two original source metrics remain different objects

For the coefficient-amplitude map Φ_n on K^n with the ordinary coordinate metric, put $W_n = \sum p_j^2$. Its quotient Gram and graph quotient Gram are

$$G_n^{\text{raw}} = G/W_n, \quad G_n^{\text{gr}} = (1 + 1/W_n)G.$$

For a finite-dimensional coefficient space of dimension h , the exact return is

$$\log \det G_n^{\text{raw}} = \log \det G_n^{\text{gr}} - h \log(1 + W_n).$$

In the Pell case $W_n = p_{(n-1)}p_n/2$, so the return is explicitly exponential in horizon before logarithms. Uniform inverse control has not made that cost disappear.

Separately, on $V_{\text{word}} = K[\Omega_n]$ with its original orthonormal word basis, binary evaluation is bijective onto its 2^n actual value set. The value-basis quotient Gram is the identity. The further scalar amplitude observation has all differences and cancellations in its own kernel. The original coefficient map, free word map, and value-amplitude map are connected by the explicit map $e_{\epsilon} \rightarrow \epsilon$ and the commuting evaluation square. They are not identified by equality of the final scalar values.

More generally, whenever two words share a value, the original supported quotient metric is $1/\mu(y)$, as in the earlier workbench. Here the gap theorem proves $\mu(y)=1$; the coefficient-amplitude Gram $1/W_n$ still has a different source and a different rank. These facts are compatible through the stated maps.

6. A supported-versus-unsupported obstruction to changing the recurrence

Six distinct actual generators

$$x, b, x + 2b, c, b + 2c, x + 4b + 4c$$

carry a seven-term progression with masks 1,11,19,28,32,42,50 and values $x+j(b+c)$, $0 \leq j \leq 6$. The three original coefficient relations are retained, and every second difference of the masks has its explicit primitive in their span.

For every delay $r \geq 3$ in $a_j = 2a_{(j-1)} + a_{(j-r)}$, the indices $0, r-1, r, 2r-2, 2r-1, 2r$ are distinct and instantiate that motif. Thus every increasing positive seed gives a forbidden seven-term progression in the first $2r+1$ generators.

At $r=2$, the free six-label map identifies the third and fourth coordinates. Its fourth motif mask has coefficient two after pushforward, so it leaves the actual binary source. For the Pell prefix 1,2,5,12,29 the requested central value 22 lies in the original numerical gap (20,29). This is an unsupported action request, not a supported scalar-zero value. The linear coefficient map still exists; its restricted-domain failure is computed rather than hidden.

This supplies a general methodological test: retain the exact feasible carrier when transporting relation primitives. A supported zero under a weaker observation may be essential to a real witness, while an unadmitted binary source is not created merely by an algebraic relation.

7. Evidence, attribution, and exclusions

The full EP817 theorem-proof note and both executable programs are included in this additive receiving folder, together with their generated receipts. The auditor imports no producer module and reconstructs all finite edge labels and arithmetic observations. Its scope is the recorded finite tests and source identities; it is not an analytic-zeta certificate.

No published reader, scalar declaration, original relation quotient, dependency pin, Lean source, or accepted claim status is changed. The generic Pisot finite-state methodology is attributed to prior literature; the specific constants, finite graphs, progression identities and recurrence proofs are given in full here without a priority assertion.

The original EP817 $k=4$ construction remains credited to the anonymous/deleted contributor; sneed-and-feed retains separate credit for the Lean upper theorem. The current work depends on the predecessor's support discipline and problem definition, not on its unformalized all-rank arithmetic conclusions.

The unrestricted EP817 rates and the Zeta programme's original arithmetic weight bound remain separate targets. The result supplied here is an exactly controlled infinite recurrence source, a complete supported observation kernel, its metric return, and a witness-producing obstruction to a natural faster recurrence.

Defect closure, synchronized witnesses, and isolated graphical cores

Source: `history/EP817_DEFECT_CLOSURE_20260915/payloads/ep817/research/defect_closure/notes/closure-and-core-isolation.md` **Identity:** 08a1e01d182e1062f119d0ca3c78cc69d2067d04e2bfbf71372e038f49053443 15

September 2026. The Clankers. Ordinary mathematical arguments and exact finite proof certificates; independent mathematical review pending. No new Lean elaboration or literature-priority claim.

1. Original problem and substantive continuation

For a finite set A of distinct positive integers, retain the original binary source, its numerical image, and its multiplicities:

$$\Omega_A = \{0, 1\}^A, \quad \Phi_A(\epsilon) = \sum_{a \in A} a\epsilon_a, \quad H(A) = \Phi_A(\Omega_A), \quad \mu_A(y) = |\Phi_A^{-1}(y)|.$$

An ordered k -term progression has the form $(x, x+d, \dots, x+(k-1)d)$, and is nonconstant when d is nonzero. The original extremal function is $g_k(n) = \min \max A$ over n -element positive distinct A with $H(A)$ k -AP-free. The empty choice and the numerical value zero are retained.

This continuation makes three advances. First, it constructs the least relation closure forced by progression-freeness and proves its exact universal property, including a positive-integer realization criterion. Second, it certifies a new local source, the complete bipartite graph $K_{\{3,3\}}$, using *binary-switch* defect witnesses rather than only the former single terminal defect. Third, those witnesses synchronize through an arbitrary additional finite source. Consequently a five-admissible A has no binary-value coupling between a disjoint collection of $K_4/K_{\{3,3\}}$ distance cores and any remaining generators. The number of cores and the size of the remainder are unrestricted.

The resulting value-product bijection supplies exact cardinality, representation and metric identities. It also gives a finite second-moment lower bound for arbitrarily many $K_{\{3,3\}}$ cores in their original integer coordinates. This contribution does not improve the previously constructed global upper bound $\lambda_6 \leq 1651^{1/10}$, evaluate the unrestricted λ_5 or λ_6 , or close the large- k lower-bound gap.

The original $k=4$ construction remains attributed to the anonymous/deleted Reddit contributor; sneed-and-feed retains the separate finite-upper-bound Lean credit. The predecessor `EP817_SUPPORTED_DEFECT_20260915.zip` is retained unchanged. The directly inspected Zeta interface is `workbenches/splitzero-tandem/tex/support_diagrams.tex`, (D6)–(D8), at commit `58626a62cd5648e8ae7450fb54d4a4aab2330981` and blob `d3493f891291ee6e94dbf2c77649f7d85d240df2`. Its source/receiving quotient distinction is used below; no analytic arithmetic-weight estimate is imported.

2. Finite-source relation closure with an exact universal property

Let E be a finite-dimensional rational vector space, S a finite subset, W_0 a specified rational subspace, and $k \geq 3$. For $x = (x_0, \dots, x_{k-1})$ in S^k , set

$$\Delta_i x = x_i - 2x_{i+1} + x_{i+2}, \quad v(x) = x_1 - x_0.$$

Define the following increasing sequence on the *same* ambient E :

$$W_{j+1} = W_j + \text{span}_{\mathbb{Q}}\{v(x) : x \in S^k, \Delta_i x \in W_j \text{ for every } i\}. \quad (2.1)$$

Each strict increase raises dimension. Thus the sequence stabilizes after at most $\dim(E) - \dim(W_0)$ strict increases. Write $\text{Cl}_k(S, W_0)$ for its final value W_* .

Theorem 2.1 (exact universal closure).

$$W_* = \bigcap_{\substack{\ell: E \rightarrow \mathbb{Q} \text{ linear} \\ \ell(W_0)=0; \ell(S) \text{ k-AP-free}}} \ker \ell. \quad (2.2)$$

The zero observation is allowed in this intersection. The quotient image of S in E/W_* is k-AP-free as a vector-valued set. The construction is extensive, monotone and idempotent. For a specified linear $f: E \rightarrow E'$ with $f(S)$ subset S' and $f(W_0)$ subset W'_0 , it satisfies

$$f(\text{Cl}_k(S, W_0)) \subseteq \text{Cl}_k(S', W'_0). \quad (2.3)$$

Proof. If ℓ kills W_j , a tuple admitted in (2.1) becomes a scalar arithmetic progression. Admissibility makes its first difference zero. Thus ℓ kills W_{j+1} ; induction proves that the left side of (2.2) is contained in the right side.

Let $q: E \rightarrow E/W_*$ be the displayed quotient. If a tuple in $q(S)$ is a progression, choose actual lifts x_i in S . Its defects lie in W_* . By the fixed-point identity its first difference belongs to W_* , so the quotient tuple is constant.

For the reverse inclusion fix z outside W_* . On E/W_* avoid the following finite collection of nonzero linear-observation hyperplanes: those defined by $q(z)$, all nonzero $q(x-y)$, and all nonzero $q(x-2y+w)$, with x, y, w in S . A rational linear functional outside their union exists. To verify this constructively, use a rational basis $b_1, \dots, b_d$ of the dual and the curve $\ell_t = \sum_{j=1}^d t^j b_j$. Every forbidden nonzero vector gives a nonzero polynomial of degree at most d . If there are M vectors, one of the integers $1, \dots, dM+1$ avoids all roots. The resulting ℓ detects z , is injective on $q(S)$, and preserves and reflects every second-difference equation on that finite set. Its image is k-AP-free. Pullback through q gives an observer in (2.2) detecting z . This proves equality.

Monotonicity follows by induction in (2.1); extensiveness is immediate; a fixed point stays fixed. Equation (2.3) follows at each stage from $f(\Delta_i x) = \Delta_i(fx)$ and $f(v(x)) = v(fx)$. These equations also prove the stated functoriality. QED.

The iteration is implemented by finding vector progressions in the finite quotient image, without enumerating arbitrary coefficient magnitudes. A new step records the original tuple and an exact rational expression for each of its defects in the old retained basis. Gaussian elimination only calculates that expression; the original tuples, basis rows and all rational coefficients remain in the certificate.

2.1 What an unsuccessful proposed observation returns

Every derived vector comes with an implication: if an observation kills all earlier relation vectors but does not kill this step, its recorded tuple is an actual forbidden progression. Therefore a certificate ending in a vector that a proposed observation must detect produces a concrete witness by taking the first nonvanishing derived step. This is a finite witness algorithm with at most $\dim(E) - \dim(W_0)$ strict derivation steps. It does not infer a numerical progression merely from a dimension count.

2.2 Exact positive distinct integer realization criterion

Specialize to $E = \mathbb{Q}^n$ and $S = \{0, 1\}^n$. Let W_* be (2.1). There exists a positive integer vector a with pairwise distinct coordinates such that

$$a \cdot W_0 = 0, \quad H(a) \text{ is k-AP-free}$$

exactly when

$$W_* \cap \mathbb{R}_{\geq 0}^n = \{0\}, \quad e_i - e_j \notin W_* \quad (i \neq j). \quad (2.4)$$

Here W_* in the real intersection means its scalar extension, with its original rational lattice retained.

Proof. Necessity follows from (2.2): a positive vector cannot annihilate a nonzero nonnegative vector, and distinct coordinates cannot annihilate $e_i - e_j$.

For sufficiency, the simplex $\Delta = \{x \geq 0 : \sum x_i = 1\}$ is disjoint from the real space W_* . Project it orthogonally to $W_*^\perp$ in the specified ordinary coordinate space. The compact convex image excludes zero. If v is its least-norm point, then $v \cdot q(x) \geq \|v\|^2$ for all x in Δ , by taking the one-sided derivative along the segment to $q(x)$. Since v belongs to $W_*^\perp$, this implies $v_i \geq \|v\|^2 > 0$ for every i . A nearby rational vector in that rational subspace remains positive; multiply by an explicit denominator to obtain a positive integral a_0 .

Choose an integral basis $b_1, \dots, b_d$ of the rational annihilator. Form all nonzero quotient vectors from $S \cdot S$ and $S \cdot 2S + S$, and also all coordinate differences $e_i - e_j$. Let their number be M , put $T = \max(1, dM + 1)$, and retain

$$C = 1 + \sum_{j=1}^d T^j \|b_j\|_\infty, \quad a(t) = Ca_0 + \sum_{j=1}^d t^j b_j. \quad (2.5)$$

Every coordinate is positive for $1 \leq t \leq T$. Each forbidden vector gives a nonzero polynomial in t of degree at most d . One admitted t avoids all zero sets. Its $a(t)$ has distinct coordinates and preserves the relevant finite-value and second-difference kernels of E/W_* . The fixed-point conclusion of Theorem 2.1 gives k -AP-freeness. All parameters in (2.5) are part of the realization. QED.

Formula (2.5) is an effective existence construction, not an optimized height bound. Its large integer coordinates are not asserted to improve $g_k(n)$. The supplied 592 small-source cases include 140 actual positive-distinct realizations and 452 explicit obstructions. Every rejected case has either a displayed coordinate-equality vector or a displayed nonzero nonnegative relation, with rational coefficients in the retained closure basis.

3. The original quotient and the additional forced-observation kernel

For an integral original relation lattice $L_0 \subset \mathbb{Z}^d$, set $W_0 = \mathbb{Q} L_0$. Its saturation is $L_0^{\text{sat}} = W_0 \cap \mathbb{Z}^d$. The original map

$$\mathbb{Z}^d / L_0 \longrightarrow \mathbb{Z}^d / L_0^{\text{sat}}$$

has the explicitly retained torsion kernel L_0^{sat} / L_0 . An integer-valued observation kills this kernel because $D \text{ ell}(x) = 0$ implies $\text{ell}(x) = 0$ for every nonzero integer D . No source relation is divided without recording this receiving-kernel map.

At the rational closure level the relevant exact sequence is

$$0 \rightarrow W_*/W_0 \rightarrow E/W_0 \rightarrow E/W_* \rightarrow 0. \quad (3.1)$$

The first term is the additional kernel forced on *admissible observations*. It is not redeclared to be an original boundary. Each of its generators comes with the progression implication in Section 2.1. On integral data, multiplying a recorded rational span identity by its common denominator gives the actual integral identity and the same torsion-free receiving implication.

For nested finite admitted point supports $S \subset T$, (2.3) gives $W_*(S) \subset W_*(T)$. Identity on E and the corresponding quotient maps form a natural diagram. The original SplitZero reconstruction therefore applies: a relation at support S maps to that support's coefficient zero. The map to external absence is a separate bottom-support map. Exact sequence (3.1) is retained at each support.

This is a transfer of the inspected algebraic quotient interface. It makes no identification between this finite arithmetic closure and the complete analytic theta quotient. Nor does finite-dimensional stabilization in E claim a bound uniform in an unbounded number of original generator coordinates. The uniform-rank result below instead uses synchronized *finite local witnesses*, not a suppressed dimension parameter.

4. Literal graphical sources and binary-switch packets

For a finite graph $G = (V, E)$, let $\text{Sigma}(G)$ consist of the actual orientation score vectors: each edge contributes one to exactly one endpoint. An edge-head mask supplies a source representative. For integer marks $T = (t_v)$,

let

$$A_G(T) = \{|t_v - t_u| : uv \in E\}, \quad C_T = \sum_{uv \in E} \max(t_u, t_v), \quad L_T(s) = \sum_v t_v s_v.$$

When the edge distances are nonzero and distinct, orientations correspond bijectively to the binary subsets of this generator set: select an edge exactly when its chosen head has the smaller mark. The original numerical equality is

$$\boxed{\Phi_A(\text{subset mask}) = C_T - L_T(s)}. \quad (4.1)$$

An orientation-score fibre may have several masks; all are retained. In particular K_4 has 64 masks and 38 scores, and $K_ \{3,3\}$ has 512 masks and 328 scores.

Fix a score difference $r=t-s$. A binary-switch packet consists of a in $\text{Sigma}(G)$, a vector v , and a five-entry word σ in $\{0,1\}^5$ with $\sigma_0=0$, such that

$$\boxed{x_i = a + iv + \sigma_i r \in \Sigma(G) \quad (0 \leq i \leq 4)}. \quad (4.2)$$

It obeys

$$\Delta_i x = (\sigma_i - 2\sigma_{i+1} + \sigma_{i+2})r. \quad (4.3)$$

Thus any observation killing r maps the packet to a five-term progression of difference $\text{ell}(v)$. Multiple actual defect positions, including their signs, are kept. A zero value of one $\text{ell}(v)$ can be compared with another packet rather than being interpreted as the absence of the original r .

The binary restriction on σ is the essential additional information: it lets the packet synchronize with *any* other source pair, as shown in Section 6.

5. Complete local certificates for K_4 and $K_ \{3,3\}$

5.1 K_4

For each of the 200 nonzero differences of its 38 score vectors, the certificate supplies a packet with

$$\sigma = (0, 0, 0, 0, 1), \quad v = e_j - e_i.$$

The original tuple is found by taking a root string of four scores and an actual final score. This is the previously developed single-final-defect mechanism, independently replayed here on its complete finite domain. No older all-rank theorem is needed for this local input.

5.2 $K_ \{3,3\}$: complete domain and symmetries

Label the two parts $\{0,1,2\}$ and $\{3,4,5\}$. Direct enumeration of the original 512 head masks gives 328 scores and 5,299 distinct differences including zero. Permutations within parts, exchange of the parts, and global sign give 99 nonzero difference orbits.

The source transformations are explicit. A graph automorphism permutes score coordinates and transports each edge head. Global sign on differences is induced by the score involution $s \rightarrow 3*1-s$, reversing every head. These maps preserve the actual source and all packet identities. Each orbit representative, full orbit cardinality and masks for every packet row are retained in `proof_data.json.gz`. The auditor reconstructs all 5,298 nonzero differences and verifies that the 99 disjoint orbits cover them exactly.

For 91 orbits the single-final-defect root packet works. There are seven two-switch cases and one retained local-division case. Write 00001 and 00011 for the literal binary switch words. Each pair in the following table specifies (anchor a , step v , switch σ); its five source rows are exactly $a+i v+\sigma_i r$.

Difference r	First packet (a; v; sigma)	Second packet (a; v; sigma)
(-3,-3,-3,3,3,3)	(0,0,0,3,3,3); (1,1,1,-1,-1,-1); 00001	(1,1,3,0,2,2); (1,1,0,0,-1,-1); 00011
(-3,-3,-2,2,3,3)	(0,0,0,3,3,3); (1,1,1,-1,-1,-1); 00001	(0,1,2,1,2,3); (1,1,0,0,-1,-1); 00011
(-3,-3,-1,1,3,3)	(0,0,0,3,3,3); (1,1,1,-1,-1,-1); 00001	(0,0,1,2,3,3); (1,1,0,0,-1,-1); 00011
(-3,-3,-1,2,2,3)	(0,0,0,3,3,3); (1,1,1,-1,-1,-1); 00001	(0,1,1,1,3,3); (1,1,0,0,-1,-1); 00011
(-3,-3,0,1,2,3)	(0,0,1,2,3,3); (1,1,0,0,-1,-1); 00011	(0,1,1,3,1,3); (1,1,0,-1,0,-1); 00011
(-3,-3,0,2,2,2)	(0,1,1,1,3,3); (1,1,0,0,-1,-1); 00011	(0,1,1,3,1,3); (1,1,0,-1,0,-1); 00011
(-3,-3,1,1,1,3)	(0,0,3,2,2,2); (1,1,-1,0,0,-1); 00011	(0,1,0,2,3,3); (1,1,0,0,-1,-1); 00011

In each row the difference of the two displayed steps is a root, up to sign. With distinct vertex marks, both step observations cannot vanish. The packet membership checks are finite integer equalities and actual orientation-mask checks; this table does not appeal to a relaxation of the score source.

The remaining representative is

$$r = (-2, -2, -2, 2, 2, 2), \quad v = (1, 1, 1, -1, -1, -1), \quad a = (0, 0, 0, 3, 3, 3), \quad \sigma = 00001. \quad (5.1)$$

It is a valid binary-switch packet and $r=2v$. If $\text{ell}(v)$ is nonzero it already provides the progression. If $\text{ell}(v)=0$, retain the original identity $r=2h$ with $h=(-1,-1,-1,1,1,1)$. This h is an actual score difference and has a certified single-final-defect root packet. Since $\text{ell}(h)=0$, that local packet has a nonzero step under distinct marks. Both r and h , the factor two, and the auxiliary original masks are in the record. No global tensor collision is divided componentwise.

Proposition 5.1. For either graph, every integer mark assignment with nonzero pairwise distinct edge distances and five-AP-free $H(A_G(T))$ is injective on the original orientation-score image.

Proof. Distinct edge distances imply distinct vertex marks for these graphs. For K_4 every vertex pair is an edge. For $K_{\{3,3\}}$, equal marks across parts give a zero edge, while equal marks within one part give two equal edges to any vertex of the other part. An observed collision gives a nonzero source difference r with $L_T(r)=0$. A root packet, a two-switch pair or the two alternatives in (5.1) supplies a nonconstant five-term numerical progression. Equation (4.1) converts it into an actual subset-sum progression. QED.

As a concrete distributed-defect example, take

$$T = (0, 60, 4, 7, 13, 52), \quad r = (-3, -3, 1, 1, 1, 3), \quad L_T(r) = 0.$$

The nine distinct edge distances are $\{3,7,8,9,13,47,48,52,53\}$. The last row of the table gives numerical progressions 152,156,160,164,168 and 55,60,65,70,75, after reversing the first tuple. These are witnesses of the same original collision, not claims that this marked graph is admissible.

6. Synchronization with arbitrary other sources

Suppose a finite source product X_1 times ... times X_h times Y has observation

$$\ell(x_1, \dots, x_h, y) = \sum_j \ell_j(x_j) + \ell_Y(y).$$

Every X_j is one of the two graphical score sources with its actual mark observation, and Y is any finite subset of any rational vector space with its stated linear observation. In the generator application $Y=H(B)$ subset Z and ell_Y is identity.

Take two product source points whose observed values agree. Write their difference as

$$r = (r_1, \dots, r_h, r_Y), \quad \ell(r) = 0.$$

Select any j with r_j nonzero. A binary-switch packet $a+i v+\sigma_i r_j$ in that factor produces the following actual product points:

$$X_i^{(j)} = a + iv + \sigma_i r_j, \quad X_i^{(l)} = x_l + \sigma_i r_l \ (l \neq j), \quad Y_i = y + \sigma_i r_Y. \quad (6.1)$$

Because σ_i is zero or one, every other coordinate is literally one of the original two source points. Numerical evaluation gives

$$\ell(X_i, Y_i) = C + i\ell_j(v) + \sigma_i\ell(r) = C + i\ell_j(v). \quad (6.2)$$

A root packet or two-switch row guarantees a nonzero $\ell_j(v)$, so it supplies a forbidden progression.

For (5.1), if $\ell_j(v)$ is nonzero, use (6.1). Otherwise $\ell_j(h)=0$ and the recorded auxiliary root packet gives a progression entirely inside X_j ; fix all the other factors at their original points. This branch is local and uses no claim that each individual component r_1 of a global collision has observed value zero.

Theorem 6.1 (core isolation at every rank). Let A be any five-admissible set of distinct positive integers. Choose any disjoint collection of generator subcollections which are the actual edge-distance sets of K_4 or $K_{\{3,3\}}$. Each K_4 core has exactly six edge labels and six distinct positive edge weights; each $K_{\{3,3\}}$ core has exactly nine edge labels and nine distinct positive edge weights. Let B consist of the remaining generators, and let there be a K_4 cores and b $K_{\{3,3\}}$ cores. Then the literal addition map is a bijection

$$\prod_{j=1}^{a+b} H(A_j) \times H(B) \xrightarrow{\sim} H(A), \quad (y_1, \dots, y_{a+b}, z) \mapsto z + \sum_j y_j. \quad (6.3)$$

No spacing, radix, scale ordering, or separation hypothesis is imposed on these original integer generators.

Proof. Surjectivity comes from the disjoint generator labels. For injectivity, lift each core value to its unique score, using Proposition 5.1; alternatively apply (6.1) directly to any unequal score components. If a collision differs at a core, (6.2) supplies a five-AP in $H(A)$, a contradiction. If every core coordinate agrees, the two actual numbers in $H(B)$ agree as well. Thus the tuples are equal. QED.

The proof handles an arbitrary finite number of cores and an arbitrary finite reservoir. The number of product points need not be enumerated. The local finite certificates are used once, and (6.1) is the uniform proof for all product sizes.

For every value under (6.3),

$$\mu_A \left(z + \sum_j y_j \right) = \mu_B(z) \prod_j \mu_{A_j}(y_j), \quad |H(A)| = 38^a 328^b |H(B)|. \quad (6.4)$$

The inverse of (6.3) is the unique finite-source lookup determined by the literal generators. It is not asserted to be Euclidean digit division when the generators were not separated.

The unweighted numerical generating polynomial factors with coefficients exactly zero or one:

$$\sum_{x \in H(A)} z^x = \left(\sum_{x \in H(B)} z^x \right) \prod_j \left(\sum_{x \in H(A_j)} z^x \right). \quad (6.5)$$

This isolation result is stronger than injectivity within each individual core. It prohibits any new binary-value coupling to the unrestricted reservoir. It does not prohibit all higher-coefficient scalar relations.

7. Exact moments, finite bounds, and metric return

For uniform measure on the *distinct* $K_{\{3,3\}}$ score image, symmetry and direct full enumeration give mean $(3/2)^*1$ and covariance

$$\text{Cov}(s) = \frac{25}{82} \mathcal{L}_{K_{3,3}} - \frac{1}{164} uu^T, \quad u = (1, 1, 1, -1, -1, -1). \quad (7.1)$$

Here the graph Laplacian is in the original six vertex coordinates. Its diagonal is three, same-part off-diagonal entries are zero, and cross-part entries are minus one. The covariance diagonal is $149/164$, same-part off-diagonal is $-1/164$, and cross-part off-diagonal is $-49/164$. All entries are checked from the 328 actual scores.

Writing $Q_j = \sum_{e \in A_j} e^2$ and $D_j = \sum_{\{l\}t_i - \sum_{\{r\}t_i}$, (7.1) gives

$$\text{Var}(\text{uniform } H(A_j)) = \frac{25}{82}Q_j - \frac{D_j^2}{164} \quad (7.2)$$

for a $K_{\{3,3\}}$ core. For K_4 the corresponding exact formula is

$$\text{Cov}(s) = \frac{25}{76}\mathcal{L}_{K_4}, \quad \text{Var}(\text{uniform } H(A_j)) = \frac{25}{76}Q_j. \quad (7.3)$$

Theorem 6.1 transports product uniform measure to uniform measure on $H(A)$. Therefore variances add with the *actual* unweighted reservoir variance:

$$\text{Var}(H(A)) = \text{Var}(H(B)) + \frac{25}{76} \sum_{K_4} Q_j + \sum_{K_{3,3}} \left(\frac{25}{82}Q_j - \frac{D_j^2}{164} \right). \quad (7.4)$$

For M distinct integers $x_1 < \dots < x_M$, the pair-difference identity gives

$$M^2 \text{Var}(x) = \sum_{i < j} (x_j - x_i)^2 \geq \sum_{i < j} (j - i)^2 = \frac{M^2(M^2 - 1)}{12}. \quad (7.5)$$

In particular, if A is entirely partitioned into h $K_{\{3,3\}}$ cores, let $Q = \sum_{a \in A} a^2$ and D_j retain the actual marked imbalances. Then

$$41(328^{2h} - 1) \leq 150Q - 3 \sum_j D_j^2. \quad (7.6)$$

With $N = \max A$, $Q \leq 9hN^2$, hence

$$N \geq \left\lceil \sqrt{\frac{41(328^{2h} - 1)}{1350h}} \right\rceil. \quad (7.7)$$

For $h=1,2,3$ this yields $N \geq 58, 13,258$ and $3,550,471$. Distinctness permits the further exact bound $Q \leq nN^2 - n(n-1)N + n(n-1)(2n-1)/6$ with $n=9h$. The source imbalances in (7.6) are retained when this refinement is evaluated.

The exponential lower base within this arbitrarily coupled pure-core family is $328^{1/9}$, approximately 1.9034502848. Inclusion of the family into the unrestricted admissible sets gives the upper comparison direction for their extremal minima; this family lower bound does not give that lower bound for unrestricted λ_5 .

For a reservoir of u positive generators, the elementary one-shift chain argument gives $|H(B)| \geq (4/3)^u$: before adding a remaining generator, every chain in its direction has length at most three, and adjoining the generator adds one new point per chain. Thus (6.4) gives the exact structural lower estimate

$$|H(A)| \geq 38^a 328^b (4/3)^u, \quad n = 6a + 9b + u,$$

$$N \geq \left\lceil \frac{38^a 328^b (4/3)^u + n(n-1)/2 - 1}{n} \right\rceil. \quad (7.8)$$

The one-shift ingredient is also present in Korsky's Theorem 1.2 context; it is not a new claimed general exponent. The new factorization makes its combination with arbitrarily many actual cores exact.

7.1 Original representation metric

The $K_{\{3,3\}}$ multiplicity histogram is

representations	1	2	3	5	6
distinct scores	230	54	24	18	2

It sums to 328 values and 512 original masks. For K_4 the histogram is 24 values of multiplicity one, eight of multiplicity two, and six of multiplicity four.

The original quotient metric on value basis e_y remains $\text{diag}(1/\mu_A(y))$. Equation (6.4) proves its tensor-product formula, including the reservoir. The identity to the unit value metric has norm

$$\boxed{4^{a/2} 6^{b/2} \sqrt{\max_z \mu_B(z)}} \quad (7.9)$$

The same 328-point uniform score covariance is not the covariance of 512 uniformly weighted orientations. The former is required in (7.1); the latter has covariance one quarter of the graph Laplacian. This distinction and all cross terms are independently checked.

8. Original supported quotient maps for all core counts

At a fixed set of admitted core labels, take the original free word module, the free module on the product of core score images and reservoir values, and the free module on $H(A)$. The word-to-score map records graph orientation fibres and reservoir subset fibres. The second map is induced by (6.3), with (4.1)'s exact affine constants. It is an isomorphism on the displayed finite value bases.

Consequently the kernel of the composite numerical-*value-basis* map equals the original word-to-product relation kernel. That kernel is the sum of the lifted local relation modules. An explicit primitive for a difference of two words in one fibre changes one core coordinate at a time; the differences telescope. This proves the tensor relation claim over Z as well as Q , without a flatness assumption on an unspecified module.

Support extension appends the actual empty subset in a new core; in score coordinates it appends the orientation that chooses the larger marked endpoint of every edge. The affine constant in (4.1) then makes its numerical contribution exactly zero. These embeddings commute with the word, score and value maps. Applying the original SplitZero reconstruction preserves each active support label and its fibre zero.

A further amplitude observation $e_y \rightarrow y$ has its own kernel. Neither (6.3) nor the equality of value-basis relation kernels sets that additional amplitude kernel to zero. Likewise, the exact closure extension (3.1) remains an added receiving kernel, rather than being silently relabelled an original graph boundary.

9. What the result changes, and what it does not determine

The first failed extension of the complete-graph method was informative. Its single final defect did not cover 196 of the $K_{\{3,3\}}$ score differences. The binary-switch packets cover those differences while retaining multiple defect positions. Their switch coefficients, not just their spans, are what permit synchronization with an arbitrary reservoir. The all-core theorem therefore follows from the actual packet forms rather than from a slogan about tensoring finite obstructions.

The closure theorem provides a complete implication and realization calculus at every specified finite source. The core-isolation theorem gives a genuine uniform conclusion as the number of such cores and the reservoir size increase without bound. Together they restrict how an efficient general construction can use these relations: it cannot gain binary-image compression by coupling K_4 or $K_{\{3,3\}}$ cores to the rest of a five-admissible set.

This does not show that every five-admissible set has such a core packing. It does not classify all closure fixed points as n grows, bound their arithmetic height at the optimal exponential scale, or replace the universal block inequalities in the preceding compact dual with finitely many verified inequalities. Those remain the precise missing global controls. The new positive-integer realization criterion proves that a surviving finite relation pattern is genuinely realizable; it does not bound that realization by the current best $g_k(n)$.

The original global problem is pursued here through arithmetic consequences of the source relations, not by treating a finite quotient dimension as a universal lower bound.

10. Executed checks and reproduction

From this directory's `certificates/`, run:

```
python build_certificate.py
python audit_certificate.py
python verify_reservoir.py
```

All programs use the standard library only. The producer replays all 512 $K_{\{3,3\}}$ orientations, 99 disjoint difference orbits covering all 5,298 nonzero differences, all 200 K_4 differences, and 35 actual integer tensor-collision cases with two, three or five $K_{\{3,3\}}$ cores. The arbitrary number of factors in Theorem 6.1 is proved by (6.1), not by those 35 examples.

The general closure domain consists of every nonzero initial relation in $[-2,2]^n$ for $n=2,3$ and every $k=3,4,5,6$: 592 cases. There are 140 positive-distinct integer realizations, 232 forced-equality certificates and 220 positive-relation obstructions. Every derivation coefficient and each original AP tuple is retained. The separate audit reconstructs finite sources and tests quotient fixed points using coset reduction, a different map from the producer's annihilator projection. It imports neither `exact_closure.py` nor `build_certificate.py`.

The separate direct reservoir replay checks 294 distinct singleton extensions of the K_4 block $\{1,4,5,17,21,22\}$ through added weight 300, and 4,991 extensions of the $K_{\{3,3\}}$ block from marks $(0,1,5,25,125,625)$ through weight 5,000. Respectively 201 and 1,884 extensions are admissible; every admissible one has the proved image-product cardinality. Two specified two-generator reservoirs check exact multiplicities and unweighted variance addition. The singleton scan records a deterministic transcript hash; the full local proof certificates, not all exploratory searches, are included in the compressed proof file.

Eight mutations reject a wrong orientation mask, a missing orbit, a discarded step relation, an incorrect local saturation factor, a removed covariance term, a false numerical progression, an unsupported closure premise, and a nonpositive claimed realization.

The generation and independent arithmetic audits have their own scopes. No result here is described as Lean-checked or as externally reviewed. Source hashes and actual run outcomes are bound in the delivery's integration record.

References and source roles

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/abs/2606.24139v1>. Inspected Sections 1--4 for exact problem conventions, general exponential bounds, and the three-term source-injectivity comparison. No new upper-rate record is claimed against that source in this continuation.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [Original TeX source]. `workbenches/splitzerotandem/tex/support_diagrams.tex` at commit 58626a62cd5648e8ae7450fb54d4a4aab2330981, especially (D6)--(D8). Used for the original internal quotient and its additional receiving kernel. Fetched in this session; no whole-repository analytic re-audit is claimed.

The Clankers. (2026, September 15). *Graphical obstruction propagation and a stronger six-term construction* [Predecessor note]. Supplied EP817_OBSTRUCTION_GRAPH_20260915.zip. Used as motivation for the single-final-defect test. The finite K_4 instance needed here is replayed directly.

The Clankers. (2026, September 15). *Supported zero actions, exact defect transport, and all-rank deformations* [Predecessor note]. Supplied EP817_SUPPORTED_DEFECT_20260915.zip. The source-preserving treatment of relation observations and the original representation metrics is retained.

Receiving progression closure and synchronized source witnesses

Source: history/EP817_DEFECT_CLOSURE_20260915/payloads/zeta/workbenches/ep817-defect-closure/notes/splitzero-receiving-map.md **Identity:** b530efe6075f73b8a9bf086cacbc78a02d24554a1d452fc75963a77752d99654 15

September 2026. Ordinary mathematical contribution; independent mathematical review pending. No new Lean elaboration, theta-source estimate, or remote merge is claimed.

Original interface

The directly inspected source is the original support reconstruction and internal quotient in `workbenches/splitzero-tandem/tex/support_diagrams.tex`, equations (D6)--(D8), at Zeta commit 58626a62cd5648e8ae7450fb54d4a4aab2330981. Its blob is d3493f891291ee6e94dbf2c77649f7d85d240df2. In particular a relation at an admitted support maps to that support's zero, and the additional kernel of a receiving observation is retained separately from the original boundary family.

The EP817 contribution applies this interface to two different, explicitly connected sources. The arithmetic-coordinate source is a rational vector space E containing a finite point set S . The word source is the free module on the actual binary masks, with the intermediate free module on their distinct score values. These modules are not identified merely because their points share numerical observations.

1. A least receiving relation closure

For $k \geq 3$ and an original rational relation subspace $W_0 \subset E$, define

$$W_{j+1} = W_j + \text{span}\{x_1 - x_0 : x_i \in S, x_i - 2x_{i+1} + x_{i+2} \in W_j \text{ for every } i\}.$$

There are at most $\dim(E) - \dim(W_0)$ strict rank increases. The final W_* has the exact universal description

$$W_* = \bigcap_{\ell|_{W_0}=0, \ell(S) \text{ k-AP-free}} \ker \ell.$$

The intersection is over rational linear observations. Its proof is given in full in the companion EP817 note: necessity is induction through the recorded source tuples; sufficiency separates a specified surviving vector by a rational functional avoiding finitely many original pair-difference and second-difference hyperplanes. A bounded integer-polynomial search constructs that functional.

Every derived step therefore carries a source proof: either its scalar observation is zero, or its actual tuple is a nonconstant forbidden progression. A failed final constraint returns the first observed nonzero step, together with its numerical witness. Stabilization is an algebraic finite-rank statement, not an analytic compactness or weight estimate.

The relation extension remains the exact sequence

$$0 \rightarrow W_*/W_0 \rightarrow E/W_0 \rightarrow E/W_* \rightarrow 0.$$

W_*/W_0 is an additional receiving kernel forced on progression-free observations. It is not asserted to equal an original theta boundary. Integral lattices have the separate saturation arrow $Z^d/L \rightarrow Z^d/(Q \cap L)$ intersect

$\mathbb{Z}^d$), with its exact torsion kernel. Clearing the denominators in every displayed span identity retains its integral implication for torsion-free integer observations.

For S subset T , closure is monotone. More generally a specified linear f with $f(S)$ subset T and $f(W_0)$ subset U_0 satisfies $f(W_*) \subset U_*$. This proves naturality of the quotient maps, so the original reconstructed SplitZero totals apply without introducing a replacement scalar.

2. The word source, coordinate observation, and their different kernels

For the graph application write

$$M = \mathbb{Q}[\Omega], \quad V = \mathbb{Q}[\Sigma], \quad \pi : M \rightarrow V, \quad p : V \rightarrow E, \quad p(e_s) = s.$$

The original orientation boundary is $B = \ker \pi$. For a closure stage W_j define $I_j = p^{-1}(W_j)$ and $K_j = (p \pi)^{-1}(W_j)$. Then

$$0 \rightarrow B \rightarrow K_j \xrightarrow{\pi} I_j \rightarrow 0.$$

These arrows explicitly locate the coordinate-observation kernel in the original word presentation. The original fibre relation is not silently replaced by all scalar or coordinate cancellations. All maps commute with the admitted support inclusions where the corresponding source and relation families are included.

For an actual numeric value map $J: V \rightarrow \mathbb{Q}[H(A)]$, $J(e_s) = e_{\{C-T-L_T(s)\}}$, the new $K_4/K_{\{3,3\}}$ theorem proves J injective under five-admissibility and distinct positive edge generators. Consequently $\ker(J \pi) = B$ at that *value-basis* level. The later amplitude observation $\mathbb{Q}[H(A)] \rightarrow \mathbb{Q}$, $e_y \rightarrow y$, still has its separately retained kernel. The equality of word/value relation kernels does not assert injectivity of amplitude on a free module.

3. Why binary-switch data survive an unbounded tensor source

A local certificate keeps an actual source tuple

$$x_i = a + iv + \sigma_i r \in S, \quad \sigma_i \in \{0, 1\}.$$

Its consecutive defects are $(\Delta \sigma_i) r$. If a collision in a product source has full defect $r = (r_1, \dots, r_h, r_Y)$, place this packet in one factor j and choose $x_{l+\sigma_i r_1}$ in every other factor. Those are exactly one of the two original source points, so no convexification or linear-span substitution occurs. Total evaluation is $C + i \ell_j(v) + \sigma_i \ell(r)$, hence a progression whenever $\ell(r) = 0$.

The original last-defect form uses $\sigma = 00001$. The new $K_{\{3,3\}}$ proof requires seven additional orbit types using 00011, usually with two step vectors whose difference is a vertex root. The one exceptional doubled defect records its factor two and a separate local packet. The scalar alternatives are proved before any tensor step; the full collision is never divided componentwise.

This produces an all-rank core-isolation theorem: an arbitrary number of disjoint K_4 or $K_{\{3,3\}}$ distance blocks have no new binary-value collision with each other or with an arbitrary remaining generator set in a five-admissible A . The resulting addition bijection and original multiplicity formula are

$$\prod_j H(A_j) \times H(B) \xrightarrow{\sim} H(A), \quad \mu_A(z + \sum y_j) = \mu_B(z) \prod_j \mu_{A_j}(y_j).$$

This theorem makes no scale-separation or radix-schedule assumption. The unrestricted number of factors is handled by the displayed synchronized formula, not by extrapolation from a finite product enumeration.

4. Original metrics and an arithmetic consequence

The word quotient metric is still $\text{diag}(1/\mu)$. For a $K_{\{3,3\}}$ core the score multiplicities are 230 ones, 54 twos, 24 threes, 18 fives and two sixes. The uniform measure on 328 distinct scores is therefore different from the measure on 512 uniformly weighted masks. The exact unweighted covariance is

$$\frac{25}{82}\mathcal{L}_{K_{3,3}} - \frac{1}{164}uu^T, \quad u = (1, 1, 1, -1, -1, -1).$$

For a K_4 core it is $(25/76)$ times its graph Laplacian. All cross terms are computed in the original vertex coordinates. The core-isolation bijection makes unweighted variances additive, including the actual reservoir variance.

For h $K_{\{3,3\}}$ cores and no remaining generators, $Q = \sum a^2$ and the original part-sum imbalances D_j obey

$$41(328^{2h} - 1) \leq 150Q - 3 \sum_j D_j^2.$$

The quotient-to-unit-value metric return has norm $6^{h/2}$. For a mixed core collection it is $4^{a/2}6^{b/2}\sqrt{\max \mu_B}$. These costs remain present despite the exact value-space bijection.

5. Scope and evidence

The full companion note, four standard-library verification files, compressed certificate and receipts accompany this receiving note. The certificate covers all 512 $K_{\{3,3\}}$ head masks, all 5,298 nonzero score differences in 99 explicitly transported orbits, and all 200 K_4 differences. A separate auditor imports neither producer nor closure library. It also checks all 592 stated small closure instances, 140 positive-distinct models, 452 exact nonexistence alternatives and 35 multi-core integer witness cases. The direct reservoir scan covers 5,285 singleton extensions and two specified larger reservoirs. Eight corrupted proof records are rejected.

No new file is asserted to be Lean-checked. The source identities and normal/optimized execution outcomes belong to the integration record. These finite verifications certify the local data and stated replay domains; the arbitrary-rank conclusions use the explicit proofs.

The global EP817 rate remains unevaluated. Not every admissible generator family has a $K_4/K_{\{3,3\}}$ core packing, and the general closure fixed points have not been classified at unbounded generator rank. Those are separate arithmetic obligations, not consequences of a finite-dimensional auxiliary representation.

Source attribution

The original SplitZero source retains its existing attribution and formalization scope. The original EP817 $k=4$ construction belongs to the anonymous/deleted Reddit contributor, with sneed-and-feed separately credited for the Lean upper extension. The local K_4 data here are independently replayed; the $K_{\{3,3\}}$ two-switch certificates and their reservoir synchronization are the present workbench contribution. No identity inference, private correspondence or copied private transcript is included.

Canonical arithmetic factors and integral isolation at every clique rank

Source: history/EP817_CANONICAL_ISOLATION_20260916/payloads/ep817/research/canonical_isolation/notes/canonical-isolation.md **Identity:** b7b1b12d3944ae3b3e868112ee263fdbcf9e463fba127e08a27db0455d10dfd0
 Research contribution of The Clankers, 16 September 2026. Ordinary mathematical proofs and exact finite certificates; independent mathematical review pending. No new Lean elaboration is asserted.

The original problem is Erdős 817: for fixed $k \geq 3$, minimize the largest member of an n -element set A of distinct positive integers whose set of binary subset sums contains no nonconstant k -term arithmetic progression. The original $k=4$ construction remains attributed to the anonymous/deleted Reddit contributor; the checked finite upper-bound Lean extension remains separately attributed to sneed-and-feed.

This contribution continues the delivered EP817_DEFECT_CLOSURE_20260915 package. It strengthens its clique isolation to every rank and every rational observation, supplies a triangular-prism core, calculates the integral obstruction layers for $K_{3,3}$, and makes the generator decomposition canonical. The all-rank results are proved below. The two finite graph statements are additionally accompanied by complete original-mask certificates, not sampled graph searches. The conformal-minimal relations used for the canonical partition are the classical Graver basis elements of the original one-row integer matrix; that terminology and method are credited to the existing literature, not introduced as a new algebraic object.

1. Exact statements and their scope

For a finite indexed list $A=(a_i)$ of distinct positive integers and $q \geq 2$, retain

$$\Phi_A : \mathbb{Z}^n \longrightarrow \mathbb{Z}, \quad \Phi_A(x) = \sum_i a_i x_i, \quad H_q(A) = \Phi_A(\{0, \dots, q-1\}^n).$$

Write $H(A)=H_2(A)$. Numerical values are counted once in H_q , while the multiplicity of their original digit representations is denoted $\mu_-(A,q)$. No distinct-representation assumption is built into these definitions.

The main all-rank result is the following.

Theorem CI1. Let Σ_m be the set of actual orientation scores of the complete graph K_m , for $m \geq 2$. For every rational linear map $\ell: \mathbb{Q}^m \rightarrow \mathbb{Q}$ and every finite Y subset Q , if $\ell(\Sigma_m)+Y$ is $(m+1)$ -AP-free, the addition map

$$\ell(\Sigma_m) \times Y \longrightarrow \ell(\Sigma_m) + Y$$

is a bijection. For every nonzero r in $\Sigma_m - \Sigma_m$, the forced relation r is obtained in the original source Σ_m union $(\Sigma_m + r)$ by at most $\lfloor m^2/4 \rfloor$ integral root steps. All tuple representatives and all preceding-boundary coefficients remain explicit. There is no distinctness assumption on the coordinate values of ℓ .

Theorem CI2. The triangular-prism score source and the $K_{3,3}$ score source have the same isolation property at progression length five, for every rational linear observation. For the prism, every source difference is the negative sum of at most four actual first-round progression steps in the two-translate source. For $K_{3,3}$, the complete proof has 94 integral first-round types, three genuine order-two first-round classes, and two types requiring a second progression-closure round.

Theorem CI3. Every positive distinct generator list has a unique finest partition $\text{Pi}_q(A)$ of its literal coordinates for which

$$\prod_{C \in \text{Pi}_q(A)} H_q(A_C) \xrightarrow{+} H_q(A)$$

is bijective. Its classes are the connected components of the supports of those Graver elements g of Phi_A with $\text{norm_infinity}(g) \leq q-1$. Increasing q can merge classes but cannot split them. The partition is a one-class partition by $q = \max(A) + 1$ whenever $n \geq 2$.

Theorem CI4. In a k -AP-free subset-sum set, every full $K_{(k-1)}$ edge-distance core with positive distinct edge values is an entire component of Pi_2 . For $k=5$, the same holds for full $K_{3,3}$ and triangular-prism edge-distance cores. In particular any two such certified cores are disjoint or are the same set. At $k=5$, the three different graph types cannot have the same generator set. Removing all the certified cores therefore leaves a uniquely specified remainder, independent of a chosen packing.

These statements do not classify all components of Pi_2 . Nor do they identify Pi_2 with Pi_3 : Section 8 supplies an admissible seven-generator example in which a six-generator binary component and a singleton merge in the ternary image, with a calculated 84-dimensional additional receiving kernel. There is no claimed improvement to the unrestricted numerical record for λ_5 or λ_6 in this contribution.

2. A complete criterion for isolation from every finite remainder

Let S be a nonempty finite subset of a finite-dimensional rational vector space E . Say that S is universally k -isolating if, for every linear $\text{ell}: E \rightarrow Q$ and every finite Y subset Q , k -AP-freeness of $\text{ell}(S) + Y$ forces addition $\text{ell}(S) \times Y \rightarrow \text{ell}(S) + Y$ to be injective. The factors are the actual distinct numerical images. This definition does not demand that ell be injective on S .

For any finite U subset E , let $\text{Cl}_k(U)$ be the following rational progression closure. Start with $W_0 = 0$ and put

$$W_{j+1} = W_j + \text{span}_{\mathbb{Q}}\{x_1 - x_0 : x \in U^k, x_i - 2x_{i+1} + x_{i+2} \in W_j \text{ for all } i\}. \quad (2.1)$$

A strict step increases dimension. Let W_{∞} be the resulting fixed space. An observation whose image of U is k -AP-free annihilates every W_j : apply it to each actual tuple in (2.1), and use the fact that an admitted progression must be constant.

Conversely, in E/W_{∞} every vector-valued k -AP of the image of U is constant by the fixed-point equation. Given z outside W_{∞} , choose a rational functional on that quotient avoiding the finitely many hyperplanes attached to nonzero pair differences, nonzero second differences, and z . Such a functional exists because a finite union of proper rational hyperplanes does not exhaust the rational dual. It detects z and its numerical image of U is k -AP-free. Thus

$$\text{Cl}_k(U) = \bigcap_{\ell(U) \text{ } k\text{-AP-free}} \ker \ell. \quad (2.2)$$

The quotient map $E \rightarrow E/W_{\infty}$ and every new subspace $W_{(j+1)}/W_j$ remain part of this construction. An integral tuple certificate can be stronger than the rational conclusion (2.2); later sections state explicitly when integer multiples rather than the original vector are derived.

Proposition 2.1 (two-translate criterion).

$$\boxed{S \text{ is universally } k\text{-isolating} \iff r \in \text{Cl}_k(S \cup (S + r)) \text{ for every } r \in S - S.} \quad (2.3)$$

Proof. Suppose the right side holds and $\text{ell}(s_1) + y_1 = \text{ell}(s_2) + y_2$ is a collision with distinct factor pairs. Then $d = \text{ell}(s_1 - s_2) = y_2 - y_1$ is nonzero. Put $r = s_1 - s_2$. A translate of $\text{ell}(S \cup (S + r))$ is contained in $\text{ell}(S) + Y$. Formula (2.2) forces $\text{ell}(r) = 0$, a contradiction.

Conversely, suppose r lies outside the indicated closure. The separating observation constructed after (2.1) has $\text{ell}(r)$ nonzero and $\text{ell}(S \cup (S + r))$ k -AP-free. Set $Y = \{0, \text{ell}(r)\}$. The two source representatives realizing r give a collision of factor pairs, so universal isolation fails. Both directions construct the actual maps and, in the failure direction, the entire finite numerical counterexample. QED.

Universal isolation is preserved by specified linear images: compose any receiving observation with the given source map. It is also preserved by finite Cartesian products. For the product, apply isolation of the first factor with all other numerical factors and Y as its finite remainder; then repeat for the next factor. Every resulting addition bijection is the restriction of the original sum map.

3. Integral clique packets and the all-rank proof

3.1 Original orientation sources and their cut criterion

An orientation assigns each edge of a loopless multigraph to one endpoint. The score is the vector of the numbers assigned to each vertex. For a target integer score d , an orientation exists precisely when

$$\sum_v d_v = |E|, \quad d(U) \geq |E(U)| \quad \text{for every vertex set } U. \quad (3.1)$$

The equivalent upper inequality is $d(U) \leq |E| - |E(V \setminus U)|$. This is the classical generalized Landau criterion; see Kolesnik and Sanchez (2024), Corollary 4.2. A direct integral construction suffices here: make a bipartite network with one unit-capacity node for each original edge, arcs to its two endpoints, and prescribed endpoint capacities d_v . The cut inequalities give a flow of $|E|$. Integral augmenting paths give a zero-one assignment of every original edge. No random orientations are substituted for actual masks.

For K_m , coordinate scores lie in $[0, m-1]$. Their differences are exactly

$$\mathcal{R}_m = \{r \in \mathbb{Z}^m : \sum r_i = 0, \quad r(U) \leq |U|(m - |U|) \text{ for every } U\}. \quad (3.2)$$

Necessity follows from the width of the score cut interval. For sufficiency, orient the doubled graph $2K_m$ with score $r + (m-1)\mathbf{1}$ using (3.1). Split its two copies into scores s, t . Reversing the second copy has score $(m-1)\mathbf{1} - t$, so $r = s - ((m-1)\mathbf{1} - t)$ is an actual difference. This supplies both original orientation masks.

3.2 A packet with one retained terminal defect

Take nonzero r in $\mathcal{R}_m$. Choose i with maximal r_i and j with minimal r_j , and put

$$\alpha = e_j - e_i.$$

Then $r_i \geq 1$ and $r_j \leq -1$. Let $I = V \setminus \{i, j\}$; define $b_i = 0$, $b_j = m-1$ and $b_h = m-2$ for h in I . Form the literal multigraph M consisting of one K_m plus another $K_{(m-2)}$ on I . Its prescribed score is

$$d = r + \alpha + b. \quad (3.3)$$

We verify every cut in (3.1). Put $w = r + \alpha$ and $a = |U|$. In upper form, the required inequalities are

Membership of i, j in U	Required upper bound on $w(U)$
neither	$a(m-a) - a$
i only	$a(m-a)$
j only	$a(m-a) - (m-1)$
both	$a(m-a) - (m-a)$

For neither, maximality gives $r_i \geq r(U)/a$ when $a > 0$. Apply (3.2) to U union $\{i\}$ to get $r(U) \leq a(m-a-1)$. The empty case is immediate. Applying the same calculation to $-r$ and the complement of U proves the both case. The i -only case follows from $w(U) = r(U) - 1$ and (3.2).

For j only, the cases $a=1$ and $a=m-1$ follow from $r_j \leq -1$ and $r_i \geq 1$. For $2 \leq a \leq m-2$, let P_a be the sum of the largest a entries of r . Replacing the minimum by the maximum gives

$$r(U) \leq P_a - r_i + r_j, \quad r_i \geq P_a/a, \quad r_j \leq -P_a/(m-a).$$

Since $a(m-a) \geq m$ in this range,

$$r(U) \leq \left(1 - \frac{m}{a(m-a)}\right) P_a \leq a(m-a) - m.$$

Adding the one contributed by α proves the last required bound. The total in (3.3) equals the number of edges of M , so (3.1) now supplies an actual orientation.

Let t be its score on K_m , and t' the score on the extra K_{m-2} . Put $s = b - t'$. This is an actual K_m score: vertex j wins every incident edge, vertex i loses every incident edge, and the internal edges are the reverse of the extra orientation. We have

$$t - s = r + \alpha. \quad (3.4)$$

Reverse first the direct j -to- i path and then the $m-2$ edge-disjoint two-edge paths through I . Every reversal decreases the score by α . Reversing this ordered list of orientations gives the actual source tuple

$$s - (m-1)\alpha, \dots, s - \alpha, s, t. \quad (3.5)$$

Its first difference is α , all second differences except the final one are zero, and the final one is r . Equivalently, with $\sigma = (0, \dots, 0, 1)$, it has rows $a + l\alpha + \sigma_l r$. This proves the packet for every r , including $m=2$. All edges, orientations and signs are specified by the construction.

3.3 Root contraction stays inside the feasible difference source

Set $r' = r + \alpha$. It is still in R_m . Only a cut containing j but not i can increase. If that cut had attained its original upper bound, replacing j by i would increase its sum by at least two without changing its size, contradicting (3.2). Thus its original integral slack is at least one. All other cuts decrease or stay unchanged.

The positive-coordinate mass decreases exactly by one:

$$\sum_h (r'_h)_+ = \sum_h (r_h)_+ - 1. \quad (3.6)$$

The positive-coordinate set itself is an admitted cut, so initially this mass is at most $\lfloor m^2/4 \rfloor$. Repeating the operation terminates at zero in at most that many steps. No coefficient vector is divided or replaced by a primitive representative.

3.4 Every implication uses the same original two-translate source

Fix the ORIGINAL r and $U_r = \text{Sigma}_m \cup (r)$. At step t the residual is

$$r_t = r + \alpha_0 + \dots + \alpha_{t-1}.$$

Construct its packet $x_l = a + l\alpha_t + \sigma_l r_t$ by (3.5), and set

$$z_l = x_l + (1 - \sigma_l)r. \quad (3.7)$$

The first m rows lie in $\text{Sigma}_m + r$, and the final row in Sigma_m . The source never changes to a quotient representative set. The first difference of z is α_t and its only possible nonzero second difference is

$$r_t - r = \sum_{h < t} \alpha_h. \quad (3.8)$$

It is an integral combination of previously derived steps. Starting with $t=0$, (2.1) therefore derives every α_t using original masks. At termination

$$\boxed{r = - \sum_t \alpha_t.} \quad (3.9)$$

This proves an integral closure certificate in at most $\text{floor}(m^2/4)$ steps, and Proposition 2.1 proves Theorem CI1.

If an arithmetic observation detects the original r , at least one derived α has nonzero image. Choose the first such step. Every preceding defect in (3.8) has zero image, and the actual tuple (3.7) evaluates to a nonconstant progression. This is the explicit failure-witness procedure, not merely an implication between abstract dimensions.

The progression-length threshold is sharp for arbitrary observations. Projection to one vertex maps Sigma_m onto $\{0, \dots, m-1\}$. Adding $Y=\{0,1\}$ gives $\{0, \dots, m\}$, with collisions but no $(m+2)$ -term progression. This map displays why universal $(m+1)$ -isolation does not become universal $(m+2)$ -isolation.

4. Two cubic sources, and the difference between integer torsion and a new implication

The proof data retain all 512 orientation masks for each nine-edge graph, each score fibre, every nonzero source difference, the full graph automorphism group, and every source row used below. An automorphism permutes original vertices and edge masks; reversal of all edges sends scores s to 3^*1 -s and differences to their negatives. These explicit bijections justify the orbit reduction.

For a difference r , a recorded binary-switch packet consists of

$$x_l = a + lv + \sigma_l r \in \Sigma(G), \quad 0 \leq l \leq 4, \quad \sigma_l \in \{0, 1\}, \quad \sigma_0 = 0. \quad (4.1)$$

The rows $x_l + (1 - \sigma_l)r$ form an actual vector progression in $\text{Sigma}(G)$ union $(\text{Sigma}(G) + r)$, with step v . Thus every $k=5$ -admissible observation of this union kills v . This fact uses the numerical image of the two translates and does not assume that $\text{ell}(r)=0$.

4.1 The triangular prism

The vertex set is $\{0, \dots, 5\}$ and the edge list is

$$01, 02, 03, 12, 14, 25, 34, 35, 45.$$

Its actual score source has 314 values and 4,994 nonzero differences. There are 255 orbits under its 12 graph automorphisms and simultaneous sign reversal. For each representative r , the complete certificate supplies t packets with

$$r = - \sum_{h=1}^t v_h, \quad 1 \leq t \leq 4. \quad (4.2)$$

The orbit counts by t are 4,39,202,10 for $t=1,2,3,4$ respectively. Every coefficient in (4.2) is the literal integer -1. Membership of every packet row is verified against its retained orientation mask. Equation (4.2), the finite complete orbit cover, and Proposition 2.1 prove universal five-isolation for the prism without a condition on ell .

A second finite certificate addresses source faithfulness when graph marks actually define positive distinct generators. For every r , recorded packet steps span an original vertex root. If $\text{ell}(r)=0$ and $\text{ell}(\text{Sigma}(G))$ is five-AP-free, those packets force the observed root to vanish. For the prism, any two vertices are adjacent or share a neighbour. Equal marks would therefore give either a zero edge or two equal absolute edge values. Under the actual-generator hypotheses all marks are distinct, and the root cannot vanish. Hence ell is injective on the 314 source scores.

An implementation initially restricted this root search to adjacent vertex pairs. The broader root certificate is valid for the stated diameter-two reason, and the corrected file retains all vertex pairs. No proof statement or supported source row was weakened by the correction.

4.2 K3,3: three first-round torsion classes

Use left vertices 0,1,2 and right vertices 3,4,5. The score source has 328 values, 5,298 nonzero differences and 99 symmetry orbits. In 94 orbits, certificates of the form (4.2) apply directly.

Three representatives require the retained multiplier two:

$$\begin{aligned} r_1 &= (-3, -3, 1, 1, 2, 2), \\ r_2 &= (-3, -2, 2, 1, 1, 1), \\ r_3 &= (-2, -2, 1, 1, 1, 1). \end{aligned}$$

For each, the complete first-round direction lattice has generators

$$\begin{aligned} v_1 &= (0, 1, -1, 0, 0, 0), & v_2 &= (1, 0, -1, 0, 0, 0), \\ v_3 &= (1, 1, 0, 0, -1, -1), & v_4 &= (1, 1, 0, -1, 0, -1), \\ v_5 &= (1, 1, 0, -1, -1, 0). \end{aligned}$$

The exact identities are

$$\begin{aligned} 2r_1 &= -v_1 - v_2 - 3v_3 - v_4 - v_5, \\ 2r_2 &= -v_1 - 3v_2 - v_3 - v_4 - v_5, \\ 2r_3 &= -v_1 - v_2 - v_3 - v_4 - v_5. \end{aligned} \tag{4.3}$$

Every v has even pairing with $\eta = (1, 1, 1, 0, 0, 0)$, while each $\eta \cdot r$ is odd. The class of r in the integer quotient by the first-round step lattice has order exactly two. The auditor independently enumerates all vector five-term progressions in the original two-translate union, confirming that no omitted first-round step removes this parity certificate.

The quotient class is retained. A rational or integer numerical observation that kills the first-round steps also kills $2r$; its torsion-free codomain then forces $\text{ell}(r) = 0$. Equivalently, the map from the original integer quotient to its rationalization kills precisely the relevant torsion class. No scalar division is assigned to an original binary source point.

4.3 K3,3: two second-round classes

The other representatives are

$$r_4 = (-3, 1, 1, -1, 0, 2), \quad r_5 = (-2, -1, 0, 1, 1, 1).$$

For r_4 , the first-round steps are

$$v_1 = e_0 - e_5, \quad v_2 = e_0 - e_2, \quad v_3 = e_0 - e_1.$$

Coordinate 3 annihilates all three and detects r_4 . Thus r_4 survives even the rational first-round quotient. A second original-source tuple has step $w = e_3 - e_0$ and defects

$$v_3, \quad v_2 - v_3, \quad 2v_1 + v_3,$$

all in the already established first-round lattice. The final integer identity is

$$r_4 = -2v_1 - v_2 - v_3 - w. \tag{4.4}$$

For r_5 , the first-round steps are $v_1 = e_0 - e_5$, $v_2 = e_0 - e_4$, $v_3 = e_0 - e_3$. Coordinate 1 detects the surviving class. The second tuple has $w = e_1 - e_3$, with defects $v_2 - v_3$, $v_1 - v_2$, $v_2 + v_3$. Now

$$r_5 = -v_1 - v_2 - w. \tag{4.5}$$

All five rows of each second-round tuple are actual orientations with a retained shift bit, and remain in the original Sigma union (Sigma+r). Their original-mask data and the displayed coefficients are checked independently.

These two cases require a new progression implication; rational saturation of the first-round lattice alone does not supply it.

Together, the 94+3+2 cases prove universal five-isolation of K3,3 for every rational observation. This strengthens the predecessor's isolation statement, which used distinct marks and local alternatives to expose a nonzero numerical step. Source faithfulness for positive distinct K3,3 edges is separately certified by original-source closure from $\text{ell}(r)=0$ to an observed vertex root. K3,3 also has diameter two, so those actual-generator hypotheses make every such root nonzero.

5. From the source theorem to exact arithmetic factors

For a graph with vertex marks T , orient an edge in a mask toward its selected endpoint and define $L_T(s) = \sum_v t_{vs_v}$. Put $C_T = \sum_{(uv) \in E} \max(t_u, t_v)$. Select the corresponding distance generator when the mask chooses the smaller marked endpoint. The exact original evaluation is

$$\Phi_A(\text{subset mask}) = C_T - L_T(\text{score}). \quad (5.1)$$

This is a bijection between edge masks and subsets. It retains all score fibres and their multiplicities.

If $A=C$ disjoint-union B is k -admissible and C is a certified critical core, apply universal isolation to its score image with $Y=H(B)$, using the affine translation in (5.1). It gives

$$H(C) \times H(B) \stackrel{+}{\simeq} H(A). \quad (5.2)$$

No magnitude separation or order of the numerical generators is required. The theorem also has an explicit contrapositive. Two distinct value pairs with the same sum determine an observed nonzero source difference r and place the two-translate image in $H(A)$. The first derived step having nonzero numerical observation yields an actual forbidden progression. Source masks give subsets of the original core; its shift bit chooses one of the two original remainder subsets. No sum uses a generator twice.

Applying (5.2) successively to disjoint certified cores proves exact factorization with an arbitrary remainder. For $k=5$, if there are a K4 cores, b K3,3 cores and c prism cores, then

$$|H(A)| = 38^a 328^b 314^c |H(B)|. \quad (5.3)$$

This identity preserves numerical collisions inside each factor and excludes only additional collisions among the factor values. Section 6 proves that the possible core sets themselves are canonical components; no choice of a maximal packing remains in (5.3).

6. Canonical q -ary arithmetic components

A vector h is conformal to g , written $h \text{ preceq } g$, when $h_i g_i \geq 0$ and $|h_i| \leq |g_i|$ for every i . A Graver element of the original row Φ_A is a nonzero integer kernel vector with no distinct nonzero conformally smaller integer kernel vector. This is the established Graver definition, e.g. Onn (2024), Section 1.3. The following argument keeps the literal coefficient boxes throughout.

For $q \geq 2$, take the hypergraph on the original generator indices whose edges are the supports of the Graver elements with $\text{norm}_\infty \leq q-1$. Denote its component partition by $\Pi_q(A)$.

Proposition 6.1 (canonical factorization). Addition

$$\prod_{C \in \Pi_q(A)} H_q(A_C) \longrightarrow H_q(A) \quad (6.1)$$

is bijective, and Π_q is the unique finest coordinate partition with this property.

Proof. Every nonzero integer relation g decomposes conformally into Graver elements. Indeed, when it is not minimal, choose a nonzero proper $h \text{ preceq } g$ in the kernel; both h and $g-h$ have smaller l_1 mass, so induction terminates. If g is in the q coefficient box, every summand stays in that box.

Suppose two q -ary digit vectors have the same value. Their difference is such a bounded relation. Every Graver summand lies within one Pi_q component. Restricting the decomposition to each component shows that the two digit vectors have the same numerical value on that component. This proves injectivity in (6.1); surjectivity is the literal choice of component words.

For any other partition with injective addition, take a bounded Graver element g . Its positive and negative parts are actual q -ary words with equal value. Injectivity forces its restriction to each part of that partition to be a kernel vector. A nonzero proper restriction would be conformally smaller, contrary to minimality. Hence $\text{supp}(g)$ lies entirely in one part of every independent partition. Therefore Pi_q refines every such partition, establishing uniqueness. QED.

The proof does not introduce new integer relations by taking rational spans. In particular it preserves which bounded boxes contain the primitive relation witnesses.

6.1 Exact arity filtration and a finite full primitive description

The admitted primitive set increases with q , so $\text{Pi}_{(q+1)}$ is a coarsening of Pi_q . Equivalently each Graver support is retained with the threshold label

$$q(g) = 1 + \|g\|_\infty. \quad (6.2)$$

At each q , the original generator set is partitioned by the supports whose thresholds have actually been reached.

There is also an elementary finite bound for the full one-row Graver family. Let $M = \max(A)$, and let g be a Graver element. Make the signed multiset having $|g_i|$ copies of $\text{sign}(g_i) \cdot a_i$. Let P be its largest positive term and N its largest absolute negative term. Starting at zero, choose a remaining positive term when the partial sum is nonpositive and a remaining negative term otherwise. Such a term exists because the remaining total is the negative of the current partial sum.

Every proper partial sum is nonzero, and no two proper partial sums coincide: a coincidence would give a nonempty proper zero-sum submultiset and a conformal subrelation of g . All proper partial sums lie in $[1-N, P]$. Consequently

$$\|g\|_1 \leq P + N \leq 2M - 1. \quad (6.3)$$

The final inequality uses distinct positive original weights: P and N belong to different signed indices and are distinct values. The order of the signed terms and all partial sums are retained; no generator is scaled.

Thus all Graver elements can be enumerated within a stated finite l_1 ball. A sharper bound for complete merging needs only the two-coordinate primitives

$$\frac{a_j}{\gcd(a_i, a_j)} e_i - \frac{a_i}{\gcd(a_i, a_j)} e_j. \quad (6.4)$$

They are conformally minimal and have infinity norm at most M . Hence Pi_q has one component for every $q \geq M+1$ when $n \geq 2$. This finite bound depends on the actual maximum generator; it does not supply a uniform all-rank extremal estimate.

6.2 Certified cores are whole canonical components

By (5.2), the bipartition into a core C and its complement is independent at $q=2$. Proposition 6.1 therefore puts each canonical component entirely on one side. Inside a critical clique, every triangle supplies an actual signed relation with three supported generator coordinates. Inside $K_{3,3}$, the four-cycles connect the edges; inside the prism, the two triangles and its rectangular four-cycles connect every edge.

All of these short cycle relations are binary Graver elements. A nonzero signed relation on distinct positive weights has at least three supported coordinates. A conformal decomposition of a three- or four-coordinate signed relation would have disjoint supports, each of size at least three, which is impossible. The short-cycle supports therefore join all coordinates of a core inside Pi_2 .

Combining the two facts, C is exactly one Pi_2 component. Any two certified cores must be disjoint or identical as generator sets. For $k=5$, K_4 has six coordinates while the cubic cores have nine. A set simultaneously realized

as a K3,3 and prism core would have, by faithfulness, both 328 and 314 binary values. Thus those graph types cannot have the same actual set in a five-admissible ambient set.

This proves a canonical decomposition into all certified cores and the remaining Pi_2 components. It does not assert that every remaining component has one of those graph forms.

7. Exact prism moments, original multiplicities, and finite bounds

The prism source has the multiplicity distribution

Original masks per score	1	2	3	4	5	6
Distinct scores	204	66	24	2	12	6

The counts sum to 314 scores and 512 masks. For the uniform distribution on DISTINCT scores, the mean is $(3/2)1$. Its covariance in the original vertex coordinates has denominator 628, diagonal 573, off-diagonal within either triangle -195, matching cross-pair entries -181, and all other cross entries -1.

Let L_P be the graph Laplacian. Let $u=(1,1,1,-1,-1,-1)$, $P_{\text{part}}=uu^T/6$, and let $P_{\text{anti},0}$ be the orthogonal projector onto $\{(x,-x):\sum x_i=0\}$. In the displayed order, its entries are

$$(P_{\text{anti},0})_{ij} = \frac{\text{sgn}(i,j)}{2} \left(\mathbf{1}_{i \bmod 3 = j \bmod 3} - \frac{1}{3} \right),$$

where sgn is $+1$ in the same triangle and -1 otherwise. The complete matrix identity is

$$\boxed{\text{Cov}(s) = \frac{49}{157}L_P - \frac{13}{314}P_{\text{part}} - \frac{8}{157}P_{\text{anti},0}.} \quad (7.1)$$

All entries of this identity are independently checked from the actual 314 source values. For marks T , put $Q=\sum_{\text{edges}}(t_i-t_j)^2$, $D=\sum_{(i<3)}t_i-\sum_{(i\geq 3)}t_i$, and $\delta_i=t_i-t_{(i+3)}$. Then

$$\text{Var}(L_T(s)) = \frac{49}{157}Q - \frac{13}{1884}D^2 - \frac{4}{157} \sum_{i=0}^2 (\delta_i - D/3)^2. \quad (7.2)$$

The negative terms and all cross coordinates are retained. These are moments of the unweighted image, not of the uniform orientation masks. The latter has covariance $L_P/4$; the map from 512 masks to 314 score values has the displayed nonuniform fibres.

For any canonical value factorization, original word multiplicities satisfy

$$\mu_{A,q}(\sum_C y_C) = \prod_C \mu_{A_C,q}(y_C). \quad (7.3)$$

Hence the original quotient Gram $\text{diag}(1/\mu)$ tensorizes. The identity from this metric to the unit numerical-value metric has norm $\sqrt{\max \mu}$, retaining every factor. For a partition into a K4 cores, b K3,3 cores, c prism cores and remainder B , that norm is

$$4^{a/2}6^{(b+c)/2} \sqrt{\max_y \mu_B(y)}. \quad (7.4)$$

Uniform measure on the DISTINCT global value image is the product measure by (6.1), so its variances add. If a five-admissible set is partitioned into h prism cores, $M=314^h$ and $Q=\sum_{(a \text{ in } A)} a^2$. For M distinct integers, spacing gives $\text{Var} \geq (M^2-1)/12$. Equation (7.2) therefore yields

$$\boxed{157(314^{2h} - 1) \leq 588Q.} \quad (7.5)$$

The retained correction terms give the stronger version before their removal. With $N=\max A$ and $Q \leq 9hN^2$,

$$N \geq \left\lceil \sqrt{\frac{157(314^{2h} - 1)}{5292h}} \right\rceil. \quad (7.6)$$

For $h=1,2,3$, the integer bounds are 55, 12009, and 3078707. Distinctness permits the further exact inequality $Q \leq nN^2 - n(n-1)N + n(n-1)(2n-1)/6$. These are bounds for the stated family, not new unrestricted values of g_5 .

The prism claim is nonvacuous. The marks $T=(0,1,5,25,125,625)$ give the nine generators

$$A_P = \{1, 4, 5, 25, 100, 124, 500, 600, 620\}, \quad S(A_P) = 1979.$$

Its 314 actual subset sums are five-AP-free modulo 3125. The producer checks this by exact rotations of the residue support; the separate auditor tests all 314 starts in the digit set and all 3124 nonzero steps, including nonunits. All other starting residues fail membership immediately. Since $1979 < 3125$, ordinary digit lifting proves an infinite admissible family. This construction supplies an example, not a better global exponential rate.

8. The actual ternary merger and its additional kernel

Take

$$C = \{1, 4, 5, 17, 21, 22\}, \quad A = C \cup \{97\}.$$

The existing base-97 modular certificate makes A five-admissible. Direct checking is also included. Its canonical binary partition is $\{C, \{97\}\}$ and

$$|H(A)| = 38 \cdot 2 = 76.$$

At $q=3$ the whole seven-generator set is one component. An actual admitted relation is

$$\boxed{97 = 1 + 2 \cdot 5 + 2 \cdot 21 + 2 \cdot 22.} \quad (8.1)$$

The exact addition map on ternary factor values is

$$J : \mathbb{Q}[H_3(C) \times \{0, 97, 194\}] \longrightarrow \mathbb{Q}[H_3(A)], \quad e_{(x,y)} \longmapsto e_{x+y}. \quad (8.2)$$

Its source dimension is $139 \cdot 3 = 417$, its receiving dimension is 333, and it is onto. Thus

$$\dim \ker J = 84. \quad (8.3)$$

On the original ternary word module $V = \mathbb{Q}[\{0,1,2\}^7]$, let q_{loc} be the quotient to the source of (8.2), and $q_{\text{full}} = J q_{\text{loc}}$. The comparison sequence is

$$0 \longrightarrow \ker q_{\text{loc}} \longrightarrow \ker q_{\text{full}} \xrightarrow{q_{\text{loc}}} \ker J \longrightarrow 0. \quad (8.4)$$

Lifting each factor value to an original word proves surjectivity at the right. This is the exact additional receiving kernel; it is not identified with the old local relation module. Every zero digit, repeated representation, and support label remains present.

For comparison, adjoining the generator 6 to C gives the actual five-term progression $(0,6,12,18,24)$, with masks in the receipt. That direct overlap witness and the admissible addition of 97 distinguish binary coupling from ternary coupling in the original arithmetic.

9. Transfer to the supported quotient and remaining objective

At an admitted support U of original digit words, use $V_U = \mathbb{Q}[U]$ and the map to the free module on its actual numerical image. Its kernel is the original value-representation relation subspace, with basis differences of words

in one fibre. These maps commute with support inclusions. The reconstruction of this diagram through the original SplitZero functor sends a relation to its support-labelled zero; it does not delete U .

For a further value observation J , the original quotient and the receiving quotient are related by (8.4). The three K3,3 order-two examples additionally retain the integer class before scalar extension. The two second-round examples retain an original free class before adding an actual new progression consequence. These are different comparison kernels and are recorded separately.

The all-rank clique construction derives every new relation with an integer coefficient and an actual tuple in one fixed two-translate source. Its original boundary history is the list of α_h in (3.8), and its terminal identity is (3.9). Under any support-preserving receiving map the first nonzero observed step still produces the same arithmetic obstruction. No theorem about the Zeta source's unevaluated analytic weight is used to infer an arithmetic bound.

The new canonical components sharpen the remaining extremal problem. Efficient constructions can be organized by their actual binary components; certified clique and cubic cores cannot overlap or couple to the remaining binary values. But $|H(A)| \leq 2^n$ for every n -generator set. Binary cardinality alone cannot prove an exponential lower base greater than two, whereas the present sharp $k=4$ base and several current upper records lie above two. The natural higher-arity observations introduce precisely the receiving kernels illustrated in (8.4). Their controlled arithmetic size, or another source invariant retaining them, remains necessary for a sharp general result.

No theorem here bounds the canonical remainder by a fixed list of graph types, bounds all arity-merger kernels uniformly in n , or evaluates the global infimum defining λ_k . The cutoff (6.3) depends on $\max(A)$; this is an exact finite description for each original set rather than an estimate uniform in the unknown extremal height.

Exploratory cube calculations suggest additional cubic sources may admit multi-round certificates, but this release does not promote them to a complete theorem: its exported finite proof domains are precisely the prism and K3,3. A separate bounded search for improved prism modular marks found no improvement and is not used to exclude other constructions.

10. Reproducibility and exact evidence scope

Run from this contribution's repository root:

```
python research/canonical_isolation/certificates/build_certificate.py
python research/canonical_isolation/certificates/audit_certificate.py
python research/canonical_isolation/certificates/verify_arity_filtration.py
```

The first program uses the accompanying `exact_source.py`; the independent auditor imports neither file. The auditor reconstructs orientation fibres by dynamic edge extension, checks every original mask and relation coefficient, independently enumerates all graph automorphisms and their orbit cover, directly computes the exceptional first-round progression directions, and recovers the canonical partition by numerical image-cardinality splitting rather than by Graver support enumeration.

The complete clique replay covers all 3300 nonzero differences through $m=5$ and 15020 original root steps. Eighty-four actual orientation-pair examples at $m=6, \dots, 12$ are separately labelled samples. The infinite theorem is Section 3, not an extrapolation from those samples.

The prism proof covers every one of its 4994 differences via 255 complete disjoint orbits. K3,3 covers all 5298 differences via 99 orbits, including all three integer-torsion and both second-round classes. Every local source point retains at least one actual mask, while the entire mask fibre is also stored. The literal modular prism audit checks 980936 possible starts-in-D/nonzero-step pairs.

Canonical factorization is cross-checked on all nonempty subsets of $[1,12]$ of size at most five at $q=2$ (1585 cases), and all nonempty subsets of $[1,10]$ of size at most four at $q=3$ (385 cases). All 385 shared arity transitions and four larger specified targets are replayed. Primitive counts are independently checked by proper conformal-subvector enumeration.

A further exact script checks all 154 subsets of $[1,8]$ of size two through four. It retains 4180 signed one-row Graver elements with their literal alternating-word witnesses, and 1050 arity comparisons against independently split numerical images. The all-Graver coverage outside the finite enumeration bound is supplied by the proof of (6.3).

Nine deliberate corruptions are rejected by the independent auditor: altered source mask, altered root, missing orbit, nonbinary switch, erased multiplier two, false old-boundary coefficient, omitted covariance term, omitted

canonical input, and false overlap mask. The scripts use explicit checks that remain active with `python -0`. Exact final replay and archive statuses are bound by the integration receipt; this note does not substitute a written command for an observed run.

No new Lean source or receipt is introduced. Mathematical novelty is not asserted for the classical orientation criterion, Graver bases, conformal decomposition, or discrete variance spacing inequality. The theorem-specific contribution is the integral all-rank clique derivation, universal two-translate isolation application, exact prism and K3,3 proof layers, and their canonical arithmetic-factor and receiving-kernel consequences.

References and inspected source roles

Kolesnik, B., & Sanchez, M. (2024). The geometry of random tournaments. *Discrete & Computational Geometry*, 71, 1343–1351. DOI: 10.1007/s00454-023-00571-4. Inspected: the primary HTML body, especially Corollary 4.2. Role: classical integral orientation-score cut criterion. The specialized multigraph, cut bounds and root derivation are proved in Section 3 above.

Onn, S. (2024). *Circuit and Graver walks and linear and integer programming* (arXiv:2410.00656v1). Inspected: primary HTML, Section 1.3 and the discussion of conformal decomposition. Role: established definitions and literature context for Graver elements. No oracle complexity result is imported into this contribution.

Onn, S. (2010). *Nonlinear discrete optimization: An algorithmic theory*. European Mathematical Society. DOI: 10.4171/093. Publisher metadata and abstract inspected; no unread book theorem is cited as a proof input.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). Primary HTML inspected for problem conventions and general-bound context. No new global bound is asserted here against that benchmark.

KokunoYumeto. (2026). *Reconstruction with changes of support index*. `workbenches/splitzero-tandem/tex/support_diagrams.tex` at revision 58626a62cd5648e8ae7450fb54d4a4aab2330981 in KokunoYumeto/zeta-f-unction-research-reader. Exact blob d3493f891291ee6e94dbf2c77649f7d85d240df2; equations D6–D8 reread. Role: original support quotient and receiving-kernel interfaces, not an analytic arithmetic estimate.

The Clankers. (2026, September 15). *Defect closure, synchronized witnesses, and isolated graphical cores*. Delivered EP817_DEFECT_CLOSURE_20260915 research package. Role: immediate predecessor, containing finite closure and the earlier K4/K3,3 isolation route. Preserved unchanged in this release's lineage.

Canonical factor supports and integral progression-observation kernels

Source: history/EP817_CANONICAL_ISOLATION_20260916/payloads/zeta/workbenches/ep817-canonical-isolation/notes/splitzer-receiving-map.md **Identity:** b035e783df8fb9aaffea60089e302ee8da705e2ce882ed1b2f045c54d4fbd7a3 The Clankers, 16 September 2026. Additive receiving contribution from the Erdős Problem 817 workbench. All new claims have written proofs and finite exact checks with their stated domains. No new Lean elaboration or analytic arithmetic-weight conclusion is claimed.

1. Original interface and the new mathematical inputs

The receiving interface is the original diagram reconstruction and internal quotient in `workbenches/splitzer-o-tandem/tex/support_diagrams.tex`, equations D6–D8, exact blob d3493f891291ee6e94dbf2c77649f7d85d240df2, inspected at commit 58626a62cd5648e8ae7450fb54d4a4aab2330981. Its two kernel constructions distinguish reaching the supported zero in a receiving fibre from reaching global absence. The present construction instantiates those diagrams; it introduces no replacement scalar or homology convention.

The companion `canonical-isolation.md` gives the complete arithmetic proofs. Its main new input is that every complete-graph orientation-score source Sigma_m is universally $(m+1)$ -isolating under arbitrary rational observations and finite additive remainders. The forced relation r is derived in the unchanged source Sigma_m union $(\text{Sigma_m}+r)$, in at most $\text{floor}(m^2/4)$ integral root steps. The proof retains each original tuple and its actual preceding-boundary coefficients.

A second input is the canonical factor partition of a positive integer digit image. Its components are obtained from the supports of the original row's conformal-minimal integer relations (classical Graver elements) as they enter the literal coefficient box. This is a partition of admitted generator coordinates, not a division of the coordinate weights or a reduction to a different source.

The finite triangular-prism and K3,3 calculations identify three distinct implication mechanisms: an original integral consequence, an original torsion class killed by a torsion-free receiving scalar, and a free first-round class killed only after an additional actual progression consequence. Those cases must not be merged into a single unspecified zero object.

2. Fixed-source implication diagrams

Let U be a finite subset of a rational vector space E , with every u accompanied by its original representation fibre. For a specified sequence of derived relation spaces W_j , retain

$$0 \longrightarrow W_{j+1}/W_j \longrightarrow E/W_j \longrightarrow E/W_{j+1} \longrightarrow 0. \quad (2.1)$$

The relations newly admitted at stage $j+1$ are steps $v=x_1-x_0$ of actual tuples x in U^k whose second differences lie in W_j . The certificate records explicit expressions for each defect in the original W_j basis. A k -AP-free receiving observation annihilating W_j annihilates v , because it maps that actual tuple to a progression. Equation (2.1) identifies the additional receiving kernel rather than rewriting W_0 .

For the clique, start with r in Sigma_m - Sigma_m , choose at each stage i maximal and j minimal in the current r_t , and put $\alpha_t=e_j-e_i$. The packet construction gives rows x_1 in Sigma_m . After adding

the ORIGINAL r to the first m rows, all rows remain in $U = \text{Sigma_m} \cup (\text{Sigma_m} + r)$. The only possible nonzero second difference is

$$r_t - r = \sum_{h < t} \alpha_h,$$

while its step is α_t . The positive mass of r_t decreases by one, and the final identity is

$$r = - \sum_t \alpha_t. \quad (2.2)$$

Thus every relation in this derivation has an integral primitive in the preceding relation list. A numerical map detecting r has a first derived α_t with nonzero image. Its actual source tuple supplies a nonconstant numerical progression. This gives an explicit obstruction map from a failed quotient implication to original arithmetic witnesses.

No arbitrary section of E/W_j is asserted natural. The certificates are actual representatives in U ; they require no replacement of a defect by a source point that exists only in a linear span.

3. A retained order-two class and a separate free class

For three K3,3 two-translate sources, let L_1 be the integer span of ALL first-round vector-progression steps. The proof data give

$$2r \in L_1, \quad r \notin L_1. \quad (3.1)$$

The exclusion is certified by $\eta = (1, 1, 1, 0, 0, 0)$, whose pairing with every first-round generator is even and whose pairing with r is odd. All first-round directions are independently enumerated from the original source, so the parity certificate is not an artifact of omitting a possible generator.

The map

$$\mathbb{Z}^6/L_1 \longrightarrow \mathbb{Q}^6/\mathbb{Q}L_1 \quad (3.2)$$

has kernel $(\text{sat } L_1)/L_1$. The class of r has exact order two in that kernel. An integer or rational numerical observation killing L_1 kills r because its target has no torsion. This is a statement about that receiving scalar, not equality of r and zero in the original integer quotient.

For two other K3,3 classes, an integer coordinate functional vanishes on all first-round steps but detects r . Those classes survive rationalization. The next certificate supplies an actual tuple in the SAME two-translate source, whose three defects are specified integer combinations of the first-round steps. Its step w is newly forced to vanish, and r is then an explicit integral combination of the old steps and w .

The exact diagrams differ: (3.2) describes a scalar-extension kernel, while the second case uses

$$L_2/L_1 \longrightarrow \mathbb{Z}^6/L_1 \longrightarrow \mathbb{Z}^6/L_2. \quad (3.3)$$

Keeping both diagrams makes the distinction testable. The negative controls erase the multiplier in (3.1) or alter the old-boundary coefficients in (3.3), and both alterations are rejected.

4. Canonical generator partitions and their source maps

For positive distinct $A = (a_i)$, let $Q_q = \{0, \dots, q-1\}^n$ and $\Phi_A(x) = \sum a_{ix} x_i$. For each q , the canonical partition Π_q is generated by supports of the original Graver elements g with $\text{norm_infinity}(g) \leq q-1$. A conformal decomposition of any bounded relation stays in that same coefficient box.

The original word map is a bijection

$$Q_q \xrightarrow{\sim} \prod_{C \in \Pi_q} \{0, \dots, q-1\}^C, \quad x \mapsto (x|_C)_C. \quad (4.1)$$

The induced value map

$$\prod_C H_q(A_C) \xrightarrow{+} H_q(A) \quad (4.2)$$

is also a bijection. Its inverse is well-defined by the componentwise equality theorem: two words with the same full value have the same value on every component.

Let F_C be the free rational word module for a component and R_C its value-representation kernel. The source tensor product is the original free module on Q_q . Its local relation module is

$$R_{\text{loc}} = \sum_C \left(\bigotimes_{D < C} F_D \right) \otimes R_C \otimes \left(\bigotimes_{D > C} F_D \right). \quad (4.3)$$

It is the full global value-representation kernel. To see this directly, any two words in one full numerical fibre have equal component values. Change their component words one component at a time; their difference telescopes into the displayed local relation summands. This gives an actual primitive for every global same-value relation.

The resulting quotient is

$$\bigotimes_C (F_C / R_C) \xrightarrow{\sim} \mathbb{Q}[H_q(A)], \quad (4.4)$$

with the same product-value basis as (4.2). These maps commute with inclusion of admitted word supports wherever the component product support is specified. At arbitrary supports, retain the actual image subspaces and restrict the same evaluation square rather than assuming every support is a Cartesian product.

A zero value $y=0$ has a present basis vector e_0 in the free value space. Sending e_y to the numerical scalar y is a further observation with its own kernel. It is not the value-representation quotient itself.

5. Arity growth, partitions, and the extra receiving kernel

For $q \leq q'$, the literal inclusion $Q_q \rightarrow Q_{q'}(q')$ and inclusion of numerical images commute with Phi_A . The partition Pi_q refines $\text{Pi}_{q'}$, since every earlier primitive support remains admitted. The component map sends an old part to the larger receiving part containing it.

A complete finite description of the one-row primitive family is available at the original scale. Every primitive g has l_1 mass at most $2 \cdot \max(A) - 1$; the proof keeps its signed multiset, an alternating-sign order, and every distinct nonzero partial sum. Pairwise primitives show that all components have merged by $q = \max(A) + 1$. These statements retain the actual dependence on $\max(A)$; they do not make the filtration uniform as the arithmetic height varies.

The original binary factorization for

$$A = (1, 4, 5, 17, 21, 22, 97)$$

has parts $C = (1, 4, 5, 17, 21, 22)$ and $\{97\}$. At $q=3$, retain the OLD factor-value source

$$F = \mathbb{Q}[H_3(C) \times \{0, 97, 194\}]$$

rather than immediately relabelling it as the receiving image. Its dimension is 417. The sum map $J: F \rightarrow \mathbb{Q}[H_3(A)]$ is onto a 333-dimensional space, with kernel rank 84. The explicit relation

$$97 = 1 + 2 \cdot 5 + 2 \cdot 21 + 2 \cdot 22$$

is one actual ternary witness.

On the original ternary word source V , with $q_loc: V \rightarrow F$, the exact receiving comparison is

$$0 \rightarrow \ker q_{\text{loc}} \rightarrow \ker(Jq_{\text{loc}}) \xrightarrow{q_{\text{loc}}} \ker J \rightarrow 0. \quad (5.1)$$

Thus the 84 receiving directions are visible before quotienting them. The same example has binary image size $76=38*2$ and is five-AP-free. It establishes precisely why a valid binary isolation certificate does not license erasing a later additive observation kernel.

6. Original metrics and canonical core support

Give each original word basis vector norm one. If $\mu(y)$ is the number of words with value y , the least-norm representative of e_y is the average of all $\mu(y)$ original words. Therefore the original quotient Gram is $\text{diag}(1/\mu(y))$. Under (4.1)–(4.4), multiplicities multiply and these Grams tensorize. The norm of the identity to the unit value metric is $\sqrt{\max \mu}$, with every factor retained.

The prism's unweighted distinct-score covariance is

$$\frac{49}{157}L_P - \frac{13}{314}P_{\text{part}} - \frac{8}{157}P_{\text{anti},0}.$$

The full original-coordinate matrix, nonedge terms and multiplicity histogram are retained in the certificate. Uniform original words have a different pushforward measure; their covariance is $L_P/4$. The quotient map and its multiplicity fibres give the exact relation between those two measures.

Finally, a certified complete-graph core in the critical progression length, or a K3,3/prism core at $k=5$, is an entire canonical binary component. The short signed cycle relations are conformally minimal and connect its edges, while universal isolation excludes cross-component binary collisions. Two cores are consequently disjoint or identical. This replaces a chosen packing by a canonical subset of the original generator-support partition.

The theorem does not identify every remainder component with a graph core. Nor does it bound the extra q -ary kernel in (5.1) across arbitrary ranks. Those are arithmetic tasks still outside the proved scope.

7. Evidence and integration boundary

The accompanying arithmetic note and standalone verifiers are copied byte-for-byte into this receiving payload. They need no files from the EP817 repository and perform no network operation. `audit_certificate.py` imports neither the producer nor its source library. It reconstructs original orientation fibres, exact symmetry coverage, every closure equation, exceptional first-round direction sets, and canonical factor partitions by a separate method.

The all-rank clique result has a written integral proof. Its complete finite replay is through five vertices, with larger source-mask examples explicitly labelled as samples. The prism and K3,3 proofs cover their entire stated finite difference domains. Arity and primitive-domain replays retain their exact finite bounds. The integration receipt, not this source narrative, records which final normal/optimized and exported-patch runs were observed.

The original $G(R)$, the original supported quotient convention and the existing analytic cohomology are unchanged. No theorem here asserts that the theta observation has a finite exhaustive kernel, or that the Zeta programme's growing arithmetic metric is controlled by the finite graph examples. The transferred techniques are the integral fixed-source implication chain, its retained torsion/free receiving distinctions, the canonical component filtration, and their actual tensor/source metric formulas.

Exact higher-arity kernels, retained spectral multiplicities, and real boundary progressions

Source: history/EP817_ARITY_BOUNDARY_20260916/payloads/ep817/research/arity_boundary/notes/arity-and-boundary.md **Identity:** 82189b686fef099a6502736ee417254ded6594b6e3911dfeabec5996a950c831 The Clankers, 16 September 2026. Research contribution for independent mathematical review.

The underlying problem is Erdős 817. For a set A of distinct positive integers, its binary subset-sum image $H(A)$ includes zero. The extremal function $g_k(n)$ minimizes $\max(A)$ subject to $|A|=n$ and absence of nonconstant k -term arithmetic progressions in $H(A)$. The present continuation does not improve the previously constructed exponential upper rates or claim a full solution. It calculates exactly the higher-arity images of those constructions, their additional relation kernels and original metric costs, and the boundary information lost by a real completion.

The original $k=4$ construction remains credited to the anonymous/deleted Reddit contributor; sneed-and-feed retains separate credit for its finite upper-bound Lean formalization. No new Lean elaboration is claimed. All new universal statements have the proofs below; the finite input certificates are reproduced by the accompanying integer/rational programs. Finite-state numeration methods are established mathematics, not a priority claim here. The inspected context includes Korsky (2026), Frougny and Pelantová (2018), and the original Zeta support-quotient source specified in Section 12.

1. The literal source and the main outcomes

For $q \geq 2$ define

$$H_q(A) = \left\{ \sum_{a \in A} c_a a : c_a \in \{0, \dots, q-1\} \right\}, \quad \mu_{A,q}(x) = \#\{c : \sum c_a a = x\}.$$

Values in H_q are counted once. The empty generator set has image $\{0\}$ and one representation. Fix $b > \sum(B)$ and retain

$$B_m = \bigcup_{j=0}^{m-1} b^j B, \quad B_0 = \emptyset, \quad D_q = H_q(B), \quad L_m(D; b) = \left\{ \sum_{j < m} d_j b^j : d_j \in D \right\}. \quad (1.1)$$

The generators at different levels are distinct because each weight lies in $[1, b-1]$. Partitioning an actual q -digit generator word by levels proves

$$H_q(B_m) = L_m(D_q; b). \quad (1.2)$$

The map from generator words to $D_q \hat{\times} m$ is onto, with fibre mass product $\prod_j \mu_{(B,q)}(d_j)$. The next evaluation $D_q \hat{\times} m \rightarrow L_m$ is also onto; its further fibres are retained rather than assumed singleton.

The two fixed blocks used throughout are

$$B_6 = \{1, 4, 5, 17, 21, 22\}, \quad B_{10} = \{3, 4, 7, 34, 37, 41, 216, 250, 253, 257\}.$$

The first is used at $b=97$ and $b=93$, the second at $b=1651$. Their binary modular certificates, rechecked here over every nonzero step, imply five-term freeness at 97 and six-term freeness at 93 and 1651 at every finite integer level. Their original word multiplicities are not assumed one.

For every $m \geq 0$ the new ternary formulas are

$$\begin{aligned} |H_3((B_6)_{m,97})| &= (68 \cdot 97^m - 21 \cdot 3^m)/47, \\ |H_3((B_6)_{m,93})| &= (68 \cdot 93^m - 23 \cdot 3^m)/45, \\ |H_3((B_{10})_{m,1651})| &= (1007 \cdot 1651^m - 226 \cdot 89^m)/781. \end{aligned} \quad (1.3)$$

A second exact formula retains a visible repeated-root term:

$$|H_4((B_{10})_{m,1651})| = \frac{825}{412} 1651^m - \left(\frac{2m}{3} + \frac{413}{412} \right) 3^m. \quad (1.4)$$

Sections 2–6 prove the finite-state and finite-hole theorems underlying these identities. Sections 7–9 compute all moments and the relevant original measures. Sections 10–11 show, by explicit rational witnesses, that the associated real limits contain forbidden progressions even though every finite integer stage is progression-free. The unscaled terminal second differences remain nonzero in those examples.

2. A finite theorem for every distinct-image moment

Let D be any finite set of nonnegative integers containing zero, $b \geq 2$, $W = \max(D)$, and

$$C = \left\lfloor \frac{W}{b-1} \right\rfloor.$$

During conversion of an actual D -word into its ordinary numerical digits, retain

$$d + c = z + bc', \quad z \in \{0, \dots, b-1\}, \quad c, c' \in \{0, \dots, C\}. \quad (2.1)$$

The bound is invariant: write $W = C(b-1) + r$ with $0 \leq r < b-1$. Then $d+c \leq bC+r$, so its quotient by b is at most C . Values of d exceeding b and every final carry remain admitted.

For a subset R of this carry interval set

$$T_z(R) = \{c' : \exists c \in R, d \in D, d + c = z + bc'\}. \quad (2.2)$$

Starting with $R_0 = \{0\}$, retain every reachable subset, including the empty subset when it occurs. This is a finite state set with at most 2^{C+1} elements. Define

$$M_{R,S} = \#\{z : T_z(R) = S\}, \quad f(R) = |R|.$$

Theorem 2.1. The exact number of distinct values is

$$N_m(D, b) = |L_m(D; b)| = e_{R_0}^T M^m f. \quad (2.3)$$

Every unweighted moment of the distinct values also has a finite constant-matrix representation and a rational generating function in m .

Proof. An integer represented by a length- m D -word has a unique pair consisting of its m ordinary base- b remainder digits and its integer quotient by b^m . Applying (2.1) along any original representation gives that unique remainder word and final carry. Conversely, every final carry in the determinized subset has at least one original D -word, obtained by lifting its successive transition witnesses. Thus every represented integer is counted exactly once by the final weight $|R|$, regardless of how many carry paths represent it. This proves (2.3).

For moments it is useful to retain a set identity valid at every reachable state. Put

$$V_m(R) = R + L_m(D; b).$$

Ordinary remainder separation gives the disjoint union

$$V_{m+1}(R) = \prod_{z=0}^{b-1} (z + bV_m(T_z(R))). \quad (2.4)$$

For $p \geq 0$, let $F_{\cdot}(m,p)(R) = \sum_{x \in V_{\cdot}(m(R))} x^p$, interpreting $x^0 = 1$ even at $x=0$. Then

$$F_{0,p}(R) = \sum_{c \in R} c^p,$$

$$F_{m+1,p}(R) = \sum_{z=0}^{b-1} \sum_{j=0}^p \binom{p}{j} z^{p-j} b^j F_{m,j}(T_z(R)). \quad (2.5)$$

This is a finite block-triangular integer matrix on all moments of degrees at most p . The original initial and terminal vectors remain specified. Its powers give the moments, and its formal resolvent gives rational generating series. Cayley–Hamilton supplies a finite recurrence. These statements apply to the full image, not to a weighted count of original words. QED.

The executable certificates retain every residue transition, not only the aggregated matrix. A claimed scalar recurrence $p(M)$ is checked by applying the actual output to $M^j p(M)f$ for $0 \leq j < s$, where s is the matrix dimension. If these values vanish, Cayley–Hamilton proves that all later ones vanish. The numerator is checked against the exact initial coefficients. Thus the displayed rational series are certified beyond the numerical replay horizon.

2.1 The observation kernel and a spectral caution

Use the forward operator $T = M^T$ on the original state-basis space E and the observation f^T . The finite observation map is

$$J : E \rightarrow \mathbb{Q}^s, \quad Jv = (f^T v, f^T T v, \dots, f^T T^{s-1} v). \quad (2.6)$$

Its kernel is T -stable by Cayley–Hamilton. The quotient by this exact kernel preserves every subsequent observation. It does not change a supported state into external absence.

For the binary B_6 block in base 97 the full matrix, in states $\{0\}$, empty, is

$$M = \begin{pmatrix} 38 & 59 \\ 0 & 97 \end{pmatrix}, \quad f = (1, 0)^T.$$

The actual count is 38^m . In the forward system the empty-state line has eigenvalue 97 but lies in $\ker(J)$; the $\{0\}$ state has terminal weight one. Using the largest eigenvalue of the unobserved full matrix would therefore give the wrong cardinality. The kernel, its inclusion, and its invariant action are the comparison data licensing the receiving quotient. Section 6 supplies the converse issue: a nontrivial Jordan block that survives this observation and must be retained.

3. A finite-hole theorem controlling every interval merger

For integers $b < w \leq 2b-2$, let H be a nonempty subset of $[1, w-b]$ satisfying

$$H \cap ((w-b) - H) = \emptyset.$$

Retain the actual digit set

$$D = [0, w] \setminus (H \cup (w - H)), \quad t = |D|,$$

and let κ be the number of its maximal consecutive integer runs.

Theorem 3.1. For every $m \geq 0$, $L_m(D; b)$ has exactly κ^m runs, and

$$N_m = bN_{m-1} + (t - b)\kappa^{m-1}, \quad N_0 = 1.$$

When $b = \kappa$,

$$N_m = \frac{(t - \kappa)b^m - (t - b)\kappa^m}{b - \kappa}. \quad (3.1)$$

At $b = \kappa$ the retained formula is $b^m + (t - b)m b^{m-1}$, with the $m = 0$ term defined as one.

Proof. Directly from the original holes,

$$D \cup (b + D) = [0, w + b] \setminus (H \cup (w + b - H)). \quad (3.2)$$

Indeed the low outer holes H occur before the second translate starts. The high outer holes are their reflected copies. An upper hole $w - h$ of the first copy lies in the overlap. It is covered by the second copy unless $w - b - h$ belongs to H , which the stated hypothesis excludes. The lower holes of the second copy are covered by the same calculation. All other points are already present.

Induction on $U - L$, using (3.2) at the last two translates, gives for every integer run $[L, U]$

$$D + b[L, U] = [bL, bU + w] \setminus ((bL + H) \cup (bU + w - H)). \quad (3.3)$$

It has κ runs and $b(U - L) + t$ values. Two different runs of the old image have next starting point at least two larger than the preceding endpoint. Their new hulls are separated by at least $2b - w - 1 \geq 1$ missing integer, so (3.3) for different runs cannot merge. Summing over the runs yields the two recurrences. Solving the geometric sum gives (3.1), including its stated repeated-root case. QED.

The hypothesis is a finite proof obligation, not an assumed numerical pattern. The complete list of H and the literal equality (3.2) are independently checked for each application.

3.1 An independent exact moment recurrence

Write the runs of D as $[l_i, u_i]$, in their actual order. With $a = \max H$ there is one central run containing $[a + 1, w - a - 1]$. For an incoming run $[L, U]$, every lower output run is anchored at bL , the central output is $[bL + l_c, bU + u_c]$, and every upper output run is anchored at bU . Thus all sums of endpoint powers transform by the binomial formula, with no numerical approximation.

If L_p and U_p are the sums of p -th powers of run endpoints, the exact moment formulas are

$$\begin{aligned} N &= U_1 - L_1 + L_0, \\ \sum x &= (U_2 + U_1 - L_2 + L_1)/2, \\ \sum x^2 &= \{2(U_3 - L_3) + 3(U_2 + L_2) + U_1 - L_1\}/6. \end{aligned} \quad (3.4)$$

The endpoint recurrence through degree three therefore computes the same first two moments as (2.5). The certificate checks these independent recurrences against each other and also against direct digit products in the stated finite domains.

4. The six-generator images at every arity

For B_6 the exact ternary image is

$$D_3 = [0, 140] \setminus \{3, 137\}.$$

At $b=97$ or 93 , Theorem 3.1 applies with $t=139$ and $\kappa=3$. Its hypotheses are checked by the finite hole pair $\{3\}$; in particular 6 differs from 140-b. Substitution proves the first two formulas in (1.3), and the number of ternary runs is exactly 3^m .

For $q=4$, direct original digit enumeration gives $D_4=[0,210]$. For $q>4$, add the actual binary image one layer at a time:

$$H_{q+1}(B) = H_q(B) + H_2(B).$$

Once $H_q(B)$ is an interval of length $(q-1)70$, it translates by 0 and 70 overlap and cover the entire next interval. Hence $D_q=[0, (q-1)70]$ for every $q \geq 4$. Lifting an interval digit set whose width is at least $b-1$ gives another full interval by overlap of consecutive translated intervals. Induction proves

$$H_q((B_6)_{m,b}) = [0, (q-1)70(b^m - 1)/(b-1)] \cap \mathbb{Z}, \quad q \geq 4. \quad (4.1)$$

At $q=2$, digit uniqueness gives 38^m values. At $q=3$, the image has the retained exponential holes in (1.3). Arity four is the first full-interval stage. These are assertions about the q -digit sums of the same original generators; they do not enlarge the binary subset choices used in the original progression-free claim.

For reference, with $F_q(z) = \sum_{m \geq 0} |H_q(B_m)| z^m$,

$$F_2(z) = \frac{1}{1-38z}, \quad F_3(z) = \frac{1 + (136-b)z}{(1-bz)(1-3z)},$$

$$F_q(z) = \frac{1 + ((q-1)70-b)z}{(1-bz)(1-z)} \quad (q \geq 4). \quad (4.2)$$

5. The ten-generator images: three different merger scales

For B_{10} the exact ternary image has $t=2103$ values in $[0,2204]$. Its small holes are

$$H = \{1, 2, 5, 9, 12, 16, 19, 23, 26, 27, 29, 30, 31, 32, 33, 35, 36, 39, 46, 53, 60, 64, 67, 70, 73, \\ 101, 151, 179, 182, 185, 188, 192, 199, 206, 213, 217, 221, 225, 243, 247, 251, 255, 262, 269, 276, 280, 283, 286, 289, 317, 533\}.$$

The other holes are exactly 2204-H. There are 89 runs. At $b=1651$ the overlap width is 553; $\max H=533$ and no two retained small holes sum to 553. Theorem 3.1 proves the third formula in (1.3), with exactly 89^m runs.

The next two arities have the following exact series:

$$F_4(z) = \frac{1 + 1644z - 1645z^2}{(1-1651z)(1-3z)^2}, \quad (5.1)$$

$$F_5(z) = \frac{1 + 2749z + 3302z^2}{(1-1651z)(1-3z)}. \quad (5.2)$$

They imply (1.4) and

$$|H_5((B_{10})_m)| = \frac{2201}{824} 1651^m - \frac{5779}{2472} 3^m + \frac{2}{3} \mathbf{1}_{m=0}. \quad (5.3)$$

The $m=0$ correction is part of the identity. These formulas are proved by the exact transition matrices and the output-recurrence certificates of Section 2, not by extrapolating their first twenty coefficients. The full matrices for every $q=2, \dots, 7$ and all residue transitions are in the receipt. Section 6 exhibits the $q=4$ matrix in full.

For every $q \geq 6$ and $m \geq 1$ there is a simpler complete description. Put

$$W_m = (q-1)1102 \frac{1651^m - 1}{1650}, \quad H_0 = \{1, 2, 5\}.$$

Then

$$H_q((B_{10})_m) = [0, W_m] \setminus (H_0 \cup (W_m - H_0)). \quad (5.4)$$

Proof. Direct enumeration of the original $q=4$ block gives $[0, 3306]$ minus these six reflected holes. All $q \geq 4$ local block images have the same small holes: the central intervals of the 0 and 1102 translates overlap at each next arity. The small integers 1, 2, 5 remain absent because every generator belongs to the semigroup generated by 3 and 4, whose nonnegative gaps are exactly 1, 2, 5. Reflection of coefficient words gives the high holes.

For $q \geq 6$ the local width $w = (q-1)1102$ is at least 5510. If the preceding lifted image has only the six holes in (5.4), consecutive present integers differ by at most three. The central interval $[6, w-6]$ of each local translate therefore overlaps that of the next translate: $w-3 \cdot 1651 \geq 557 > 11$. These central intervals cover the whole middle region. Before the first translated central interval and after the last, the original low and high holes remain. This proves the induction, including the absence of any other hole. QED.

In particular $|H_q| = W_m - 5$ for $m \geq 1$, while $|H_q(B_0)| = 1$. Its series is

$$F_q(z) = 6 + \frac{(q-1)1102/1650}{1-1651z} - \frac{(q-1)1102/1650 + 5}{1-z}, \quad q \geq 6. \quad (5.5)$$

The first rows are useful checks on conventions:

Block/base	q	m=0	m=1	m=2
B6/97	2	1	38	1,444
B6/97	3	1	139	13,609
B6/97	4	1	211	20,581
B10/1651	2	1	291	84,681
B10/1651	3	1	2,103	3,512,281
B10/1651	4	1	3,301	5,458,197
B10/1651	5	1	4,403	7,280,911
B10/1651	6	1	5,505	9,102,515

6. A visible nilpotent term in the unchanged carry coordinates

For $q=4$ and B10/base1651 the complete reachable carry sets, in order, are

$$\{0\}, \quad \{0, 1, 2\}, \quad \{1, 2\}, \quad \{1\}, \quad \{0, 1\}.$$

The matrix and terminal vector are

$$M = \begin{pmatrix} 1 & 2 & 1 & 2 & 1645 \\ 0 & 7 & 0 & 0 & 1644 \\ 0 & 4 & 2 & 0 & 1645 \\ 0 & 2 & 1 & 3 & 1645 \\ 0 & 4 & 1 & 0 & 1646 \end{pmatrix}, \quad f = (1, 3, 2, 1, 2)^T. \quad (6.1)$$

Every row entry counts its actual residue classes. Integer matrix calculation gives

$$\det(XI - M) = (X - 1651)(X - 3)^2(X - 1)^2.$$

Retain the vectors

$$v = (1, 0, 0, 1, 0)^T, \quad w = (0, -411, 1, 0, 1)^T.$$

Their exact equations are

$$(M - 3I)v = 0, \quad (M - 3I)w = 824v. \quad (6.2)$$

Moreover $\text{rank}(M-3I)=4$ and $\text{rank}((M-3I)^2)=3$. The output Hankel matrix $(N_{(i+j)})_{(0 \leq i,j \leq 5)}$ has rank three. Thus the receiving sequence retains the size-two 3-primary block. The inverse resolvent of (6.1) gives (5.1); its double pole supplies the actual $-(2m/3)3^m$ term. Replacing this block by its eigenvalue alone would change the numerical image cardinality.

In the forward system $T=M^T$, the kernel of the full observation (2.6) has dimension two. It has the displayed basis

$$(-1, 0, 0, 1, 0)^T, \quad (0, 0, -1, 0, 1)^T.$$

These are the two additional observation-null directions. The quotient by them, with its induced operator, still has the nontrivial 3-primary block. This is a concrete separation between discarding an actual observation kernel and discarding a retained nilpotent part of the receiving action.

7. Canonical mergers and the exact original quotient metrics

7.1 All levels join at the ternary stage

The binary factorization into the original B6 or B10 blocks is exact. Within each block the original triangle relations connect all coordinates, so these are precisely its binary components in the predecessor's canonical bounded-relation factorization. Binary digit uniqueness prevents a relation between different numerical block values.

At ternary arity, adjacent blocks are linked by the following literal bounded relations. Coordinates are in the increasing orders displayed in Section 1:

b	lower coefficients	upper coefficients
97	(1, 0, 2, 0, 2, 2)	(1, 0, 0, 0, 0, 0)
93	(1, 0, 2, 1, 1, 2)	(1, 0, 0, 0, 0, 0)
1651	(0, 1, 2, 2, 0, 2, 1, 2, 1, 2)	(-1, 1, 0, 0, 0, 0, 0, 0, 0, 0)

(7.1)

In each row, lower dot B equals b times upper dot B, and upper dot B=1. Multiplication by the actual level factor b^j gives the relation at every adjacent pair. Any independent ternary partition must be coarser than the binary partition. If it separated one adjacent pair, restricting (7.1) to that side would give a nonzero amplitude, contradicting independence. Hence all m blocks belong to one ternary component.

Let Ω be the original q-digit word set of size $q^{\wedge(m|B|)}$. Its local quotient and receiving map are

$$\mathbb{Q}[\Omega] \xrightarrow{q_{\text{loc}}} \mathbb{Q}[D_q^m] \xrightarrow{J_m} \mathbb{Q}[H_q(B_m)], \quad J_m e_d = e_{\sum_j b^j d_j}.$$

Their exact relation sequence is

$$\boxed{0 \rightarrow \ker q_{\text{loc}} \rightarrow \ker(J_m q_{\text{loc}}) \xrightarrow{q_{\text{loc}}} \ker J_m \rightarrow 0.} \quad (7.2)$$

The final map is onto because each local value has an original word representative. The additional relation space has dimension

$$\dim \ker J_m = |D_q|^m - |H_q(B_m)|. \quad (7.3)$$

For example, its ternary dimensions are $139^m - (68 \cdot 97^m - 21 \cdot 3^m)/47$ and $2103^m - (1007 \cdot 1651^m - 226 \cdot 89^m)/781$. This is an exponential loss from the independent local-value product. In particular, no lower estimate retaining 139^m , or 2103^m , up to a subexponential divisor can hold for these binary-isolated families.

7.2 Weighted least-norm sections

Put $\omega(d) = \mu_{(B,q)}(d)$, and $\Omega(d_0, \dots, d_{(m-1)}) = \prod_j \omega(d_j)$. The quotient of the original orthonormal word source gives the local Gram $\text{diag}(1/\Omega(d))$. On the final value image define

$$\mu_m(x) = \sum_{\sum b^j d_j = x} \Omega(d).$$

The exact least-norm section is

$$s_m(e_x) = \sum_{\sum b^j d_j = x} \frac{\Omega(d)}{\mu_m(x)} e_d, \quad G_m = \text{diag}_x(1/\mu_m(x)). \quad (7.4)$$

The coefficients sum to one. Their squared norm in the original local Gram is $1/\mu_m(x)$. A vector in $\ker J_m$ has coefficient sum zero within every final fibre, so it is orthogonal to these section vectors. This proves both the right-inverse and Pythagoras identities, including all original masses.

7.3 Exact maximum multiplicity at every length

For any weighted D subset $[0, 2b-2]$, its carry coefficients on states $c=0,1$ are

$$W_z(c, c') = \omega(z + bc' - c), \quad \omega(t) = 0 \text{ outside } D.$$

Every row sum is bounded by the largest complete residue-class mass

$$\Lambda = \max_{r \bmod b} \sum_{d \equiv r \bmod b} \omega(d). \quad (7.5)$$

Starting at $(1,0)$, total path mass after m digits is at most Λ^m , so every final numerical fibre has at most that mass. If an actual digit $d_* < b$ has $\omega(d_*) = \Lambda$, the source word consisting of that digit repeatedly gives Λ^m original representatives of the corresponding number. Thus the upper bound is attained.

The complete residue checks give $\Lambda = 15$, $d_* = 70$ for B6 at both bases, and $\Lambda = 219$, $d_* = 1102$ for B10 at 1651. Consequently

$$\boxed{\max_x \mu_{B_m,3}(x) = 15^m \text{ or } 219^m,} \quad (7.6)$$

and the original quotient-to-unit-value identity has exact norm $15^{(m/2)}$ or $219^{(m/2)}$.

For comparison, give each distinct local digit weight one instead. Every residue matrix then has entries zero or one. A prescribed final carry receives at most the total mass at the preceding step, so its coefficient is at most $2^{(m-1)}$, $m \geq 1$. A residue r with $r-1, r, b+r-1, b+r$ all in D makes its matrix all ones and attains that bound by repetition. Such a residue is $r=1$ for B6 and $r=4$ for B10. Hence

$$\boxed{\max_x \#\{d \in D_3^m : \sum b^j d_j = x\} = 2^{m-1}.} \quad (7.7)$$

Its unweighted local-value quotient-to-unit-value norm is $2^{((m-1)/2)}$. Equations (7.4)–(7.7) specify the exact comparison of the two sources. Multiplying separate worst-case norm bounds would introduce an extra exponential factor that the original weighted calculation does not have.

8. Exact limiting measure of the higher-arity image

The infinite geometry is useful only with its scale and return map explicit. Under Theorem 3.1 additionally assume $w < 2b-2$, and put

$$s = \frac{w}{b-1}, \quad C_0 = \frac{t-\kappa}{b-\kappa}, \quad r_0 = \frac{\kappa}{b},$$

$$K = \left\{ \sum_{j \geq 1} d_j b^{-j} : d_j \in D \right\}, \quad K_m = b^{-m}(L_m(D; b) + [0, s]). \quad (8.1)$$

The scale map is multiplication by $b^{(-m)}$, with inverse multiplication by b^m . The sets K_m are nested compact supersets with intersection K : the interval $[0, s]$ is invariant under every digit map $(d+x)/b$, and finite digit approximations have uniformly vanishing tail diameter.

Because $1 < s < 2$, distinct runs of L_m give disjoint intervals in K_m . Their exact total length is

$$b^{-m}\{N_m + (s-1)\kappa^m\}.$$

Theorem 3.1 therefore proves

$$\boxed{\text{Leb}(K_m) = C_0 + (s - C_0)r_0^m, \quad \text{Leb}(K) = C_0.} \quad (8.2)$$

The remaining outer mass is exactly $(s - C_0)r_0^m$. For the three ternary applications the tuples $(C_0, s, s - C_0)$ are

$$(68/47, 35/24, 13/1128), \quad (68/45, 35/23, 11/1035), \quad (1007/781, 1102/825, 247/5325).$$

8.1 Boundary size, with the notion of dimension specified

The lower and upper box-counting dimensions of the boundary both equal

$$\boxed{\dim_{\text{box}} \partial K = \frac{\log \kappa}{\log b}.} \quad (8.3)$$

To prove the lower bound, both endpoints of every K_m component belong to K : use a terminating allowed word for its lower endpoint, and an all- w tail for its upper endpoint. They are boundary points because the adjacent gap is excluded from K_m . The $2^m \kappa^m$ endpoints are separated by at least $\min(s, 2-s)b^{(-m)}$.

For the upper bound, write $a = \max H$ and $\theta = (a+1)/(b-1)$. Choosing the central child of any K_m component $[l, u]$ at every subsequent stage retains the interval $[l + \theta b^{-m}, u - \theta b^{-m}]$ in K . Its length is positive: $s - 2\theta > 0$ follows from $a \leq w - b$ and $w < 2b - 2$. Thus the boundary inside each component lies in two collars of length $\theta b^{(-m)}$, each covered by a fixed number of intervals of scale $b^{(-m)}$. This proves the matching upper bound and, by comparison of intermediate scales, both box dimensions. No Hausdorff-dimension assertion is needed here.

8.2 Weak convergence with an evaluated error

Let $E_m = b^{(-m)}(L_m + [0, 1])$ be the disjoint unit-cell thickening, and retain the unweighted discrete measure

$$\eta_m = b^{-m} \sum_{x \in L_m} \delta_{x/b^m}.$$

One has $E_m \subset K_m$ and

$$\text{Leb}(E_m \triangle K) \leq (2s - 1 - C_0)r_0^m.$$

For every bounded Lipschitz function f on $[0, s]$, integrating it in each original cell gives

$$\boxed{\left| \eta_m(f) - \int_K f(x) dx \right| \leq \frac{\text{Lip}(f)N_m}{2b^{2m}} + \|f\|_\infty (2s - 1 - C_0)r_0^m.} \quad (8.4)$$

This proves explicit weak convergence. Division by the actual mass N_m/b^m , whose error is given by (3.1), gives the uniform distinct-image probability limit $\text{Leb}[K]/C_0$. The cell smoothing is an explicit comparison map; no total-variation convergence of a discrete probability to a continuous probability is asserted.

9. Exact second moments and original variance costs

For the ternary images, reflection gives their exact mean $W_m/2$, where

$$W_m = w(b^m - 1)/(b - 1).$$

Section 3.1 gives every centered second moment. The certificate also supplies a closed raw-second-moment expression as a rational linear combination of

$$\kappa^m, b^m, (b\kappa)^m, b^{2m}, (\kappa b^2)^m, b^{3m}.$$

The six coefficients are retained in the receipt. The product of these six linear factors is checked as an all-length observed annihilator on the actual nine-dimensional moment matrix, and six exact initial values determine the coefficients. This is a finite proof of the closed formula, not an interpolated prediction.

Let U_m be uniform on the distinct ternary image. The resulting limits are

Block/base	$\lim \text{Var}(U_m)/b^{2m}$	$\lim \text{Var}(U_m)/\sum_{a \in B_m} a^2$
B6/97	93825139/536649960	93825139/71644595
B6/93	82897249/434753337	82897249/63141789
B10/1651	14562026775392709/1029398243 59874810	436860803261781270/275866493 425966949

The second column also equals the variance of the exact limiting measure in (8.4). The third column retains

$$\sum_{a \in B_m} a^2 = \left(\sum_{a \in B} a^2 \right) \frac{b^{2m} - 1}{b^2 - 1}.$$

Its limits are approximately 1.30959, 1.31287, and 1.58360. In contrast, uniform original ternary assignments have variance exactly $(2/3)\sum a^2$. Their pushforward has the original multiplicity weights; it is not the uniform distinct-image measure. Equations (7.4), (7.6), and (8.4) are the explicit maps and costs between these presentations. A proposed all-family variance inequality must account for these exact examples.

10. A separate finite decision for real-limit progressions

Let D be a finite subset of $[0, b-2]$ containing zero, and define the actual real limit $K_b(D)$ by its digit expansion as in (8.1). The strict span bound $\max D/(b-1) < 1$ makes expansions unique: a first differing digit contributes at least $b^{-(j)}$, while its entire later tail contributes strictly less.

For a k -tuple of digit columns, the integer prefix second differences obey

$$c_{j+1} = bc_j + \Delta(d^{(j)}), \quad c_0 = 0. \quad (10.1)$$

A real progression has

$$|c_{j,i}| \leq 2 \max(D)/(b-1).$$

Conversely, an infinite digit path with this bounded carry has zero limiting second differences because $b^{-(m)}c_m$ tends to zero. A persistent flag records whether any column has unequal first two digits; uniqueness of expansions makes it exactly the nonconstant condition.

Theorem 10.1. Put $C = \text{floor}(2\max(D)/(b-1))$. The finite graph on (c, flag) , $c \in [-C, C]^{(k-2)}$, with transitions (10.1), decides whether $K_b(D)$ contains a nonconstant k -AP. Such a progression exists exactly when a reachable flagged state can reach a directed cycle. If it exists, an eventually periodic digit witness exists with preperiod less than the number of graph states and period at most that number.

Proof. The preceding prefix and tail identities prove the correspondence with infinite bounded-carry paths. Once the true flag occurs it persists. An infinite path in a finite graph reaches a cycle; conversely a path to a

cycle can be repeated indefinitely. Its digit coordinates are rational, with denominators dividing $b^h(b^t - 1)$ for the stated preperiod h and period t . The first-two-digit flag ensures a nonzero numerical difference. The state bound is $2(2C+1)^{(k-2)}$. QED.

The graph uses the reversed carry relation relative to finite integer terminal-zero tests. The source digits and the endpoint observation specify which test is intended. The complete record graphs and all their directed destination pairs are in the receipt; one actual digit column is retained for each pair. The counts are 15 states/23 arcs at (19,4), 53/389 at (97,5), and 159/1185 at (93,6).

10.1 Explicit rational progressions in all the previously used limits

The following are original binary digit columns a,e. Repeating e forever after a gives

$$x = \frac{1}{b} \left(a + \frac{e}{b-1} \right).$$

Block/base, k	First column a	Repeated column e	Real progression
{1,7,8}/19, 4	(0,0,1,1)	(0,9,0,9)	(0,1,2,3)/38
B6/97, 5	(4,5,5,5,6)	(64,0,32,64,0)	(14,15,16,17,18)/291
B6/93, 6	(0,0,0,0,1,1)	(4,26,48,70,0,22)	(4+22i)/(93*92), $0 \leq i < 6$
B10/1651, 6	(293,293,294,294,294,295)	(548,1099,0,551,1102,3)	(483998+551i)/2724150, $0 \leq i < 6$

Every entry is an actual subset sum of its displayed generator block, and its original subset mask is included. Let $c = \Delta(a)$. In all four rows c is nonzero and

$$\Delta(e) = -(b-1)c.$$

The exact integer prefixes are

$$Y_m = b^{m-1}a + \frac{b^{m-1} - 1}{b-1}e \in H_2(B_m)^k,$$

$$\boxed{\Delta Y_m = c \neq 0 \quad \text{for every } m,} \quad (10.2)$$

while

$$b^{-m}Y_m \rightarrow x, \quad x - b^{-m}Y_m = \frac{e}{(b-1)b^m}, \quad \Delta x = 0. \quad (10.3)$$

For the base97 row, $c = (-1, 0, 1)$. Thus every finite prefix retains the same nonzero integer boundary, even though the scale map makes that boundary converge to zero. This is compatible with, and illustrates the exact scope of, the finite-stage progression-free theorem.

11. A graph observation retaining the missing boundary

Let $\Delta : \mathbb{R}^k \rightarrow \mathbb{R}^{k-2}$ be the original second-difference map. Its explicit right inverse is

$$R(c)_0 = R(c)_1 = 0, \quad R(c)_i = \sum_{j=0}^{i-2} (i-1-j)c_j \quad (i \geq 2).$$

On the linear space of sequences y_m for which both $b^{-m}y_m$ and Δy_m converge, retain the pair of observations

$$\mathcal{O}(y) = (\lim b^{-m}y_m, \lim \Delta y_m).$$

The first component necessarily lies in $\ker \Delta$. The pair map is onto $\ker \Delta$ direct-sum $R^{(k-2)}$, with explicit section

$$(x, c) \mapsto (y_m = b^m x + R(c))_m.$$

Its kernel consists exactly of sequences for which both observations vanish. The further projection forgetting c has kernel 0 direct-sum $R^{(k-2)}$. Restricting this pair observation to the actual feasible prefix sequences in (10.2) gives (x, c) with c nonzero. The unrestricted linear section is not asserted to produce feasible digit words; feasibility was proved separately for the displayed witnesses.

At finite m , recovering the original defect from $\Delta(b^{(-m)}Y_m)$ requires multiplication by b^m . This evaluated return cost is why vanishing of the scaled defect cannot serve as an integer terminal-zero certificate. The active support and the retained receiving kernel locate the loss without identifying it with the original relation module.

12. Verification, references, and the remaining arithmetic question

The standard-library producer is `certificates/verify_arity_boundary.py`; its finite-state and arithmetic library is `arity_core.py`. The independent auditor is `audit_arity_boundary.py` and imports neither. The latter rebuilds the original digit polynomials by dense coefficient addition, every residue destination by a separate quotient enumeration, and the large numerical images by Boolean products over the actual generator set. Direct image moments are obtained from bitwise sums over those original image bits. Real graphs are rebuilt destination-first and analyzed through an independent strongly-connected-component procedure.

The retained proof data cover all arities 2 through 7 for three block/base cases, all first and second moments through twenty levels, and the exact observed matrix annihilators proving their continuations. Additional bounded domains include 596 small alphabets with 2,384 direct moment comparisons, twelve finite-hole examples, 2,202 weighted least-norm fibres, and 186 real-limit sources. The independent original-generator replay checks 66 complete image/moment instances, 39,552 residue actions, 162 moment states, 48 exact peak instances, and all 1,597 recorded real-graph arcs. It rejects eight deliberately corrupted records. These finite checks corroborate the general proofs and establish the displayed finite certificates; they are not substitutions for the proofs.

The old canonical factorization is used with its delivered proof and scope. The present work does not re-audit every theorem in the earlier cumulative archives. No existing Lean source, audit receipt, or accepted status is modified. No remote publication is inferred from a local patch. The accompanying integration record states the actual completed runs.

The relevant general problem remains a uniform lower estimate across every admissible generator set. Here the complete higher-arity loss is calculated for two important construction families. Their binary components all merge at ternary arity, the additional kernel has exponentially growing dimension, and even the weighted metric return is exactly computable. These identities rule out an uncorrected higher-arity product estimate while supplying the correct correction. The finite transfer applies to any specified block, but its outer family of inputs remains unbounded.

The real-limit counterexamples likewise identify an exact obstruction to transferring a finite integer exclusion through a completion that forgets the terminal boundary. They do not contradict the original finite problem or the finite upper constructions. No claim of a new best unrestricted exponential rate, an all-rank matching lower bound, or an RH conclusion is made.

References and exact source roles

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/abs/2606.24139>. The original problem conventions and general bounds provide context; the inspected abstract is not represented as a new full-paper audit in this continuation.

Frougny, C., & Pelantová, E. (2018). *Two applications of the spectrum of numbers* (arXiv:1512.04234v3; original preprint 2015). <https://arxiv.org/abs/1512.04234>. The inspected abstract supplies established context for finite automata and infinite zero representations. The specialized carry theorems above are proved directly; no unread theorem is imported.

KokunoYumeto. (2026). *Reconstruction with changes of support index*. `workbenches/splitzero-tandem/text/support_diagrams.tex` at Zeta revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, equations D6–D8; Git blob d3493f891291ee6e94dbf2c77649f7d85d240df2. Role: inspected original internal quotient, supported

kernel, and additional receiving-kernel interface. https://github.com/KokunoYumeto/zeta-function-research-reader/blob/58626a62cd5648e8ae7450fb54d4a4aab2330981/workbenches/splitzero-tandem/tex/support_diagrams.tex.

The Clankers. (2026, September 16). *Canonical arithmetic factors and integral isolation at every clique rank*. Delivered predecessor EP817_CANONICAL_ISOLATION_20260916.zip, retained unchanged in the cumulative package. Role: actual canonical bounded-relation components and their binary graphical instances, with the original mathematical and Lean attributions preserved.

Higher-arity receiving kernels, visible Jordan terms, and retained real boundaries

Source: history/EP817_ARITY_BOUNDARY_20260916/payloads/zeta/workbenches/ep817-arity-boundary/notes/splitzero-receiving-map.md **Identity:** 4c10756c8e4dfbfbe7408dcb043a98855a0dc1ca55806743f05435a5cc4c405b The Clankers, 16 September 2026. Ordinary mathematical continuation; independent review pending. No new Lean execution or arithmetic-weight theorem is claimed.

This receiving note uses the existing SplitZero reconstruction and internal quotient of `workbenches/splitzer o-tandem/tex/support_diagrams.tex`, equations D6–D8, inspected at revision 58626a62cd5648e8ae7450fb54d4a4aab2330981 (blob d3493f891291ee6e94dbf2c77649f7d85d240df2). It does not introduce a replacement scalar, support carrier, relation quotient, or theta source. The companion full proof is `arity-and-boundary.md`; standalone copies of its integer/rational checkers are under `certificates/`.

1. The original word relation and its further evaluation relation

At a specified $q \geq 2$, base $b > \text{sum}(B)$, and horizon m , retain every original q -digit generator word

$$\Omega_m = \{0, \dots, q-1\}^{m|B|}.$$

Partitioning its coordinates by their original levels gives its local value quotient. Write $D = H_q(B)$ and let $\omega(d)$ count the original block representations. There are surjections

$$\mathbb{Q}[\Omega_m] \xrightarrow{q_m} \mathbb{Q}[D^m] \xrightarrow{J_m} \mathbb{Q}[H_q(B_m)], \quad J_m e_{(d_j)} = e_{\sum_j b^j d_j}.$$

Let $B_m^{\text{loc}} = \ker(q_m)$ and $B_m^{\text{tot}} = \ker(J_m)$. The comparison is the exact sequence

$$0 \rightarrow B_m^{\text{loc}} \rightarrow B_m^{\text{tot}} \xrightarrow{q_m} \ker J_m \rightarrow 0.$$

Surjectivity of the last map follows by choosing an original block representative of every local value. The receiving kernel is therefore $B_m^{\text{tot}}/B_m^{\text{loc}}$, with its actual inclusion and quotient maps, not a reassignment of the original relation module.

The predecessor's canonical binary components for the base97 six-generator family are its original six-generator levels. All of them join in the ternary factorization, witnessed by

$$97 = 1 + 2 \cdot 5 + 2 \cdot 21 + 2 \cdot 22.$$

The analogous actual bounded relations for bases93 and 1651 are retained in the full note. The additional ternary kernel dimensions at two of the bases are exactly

$$139^m - \frac{68 \cdot 97^m - 21 \cdot 3^m}{47},$$

$$2103^m - \frac{1007\,1651^m - 226\,89^m}{781}.$$

This calculates the exponential growth of the extra receiving relation space. Every local image value remains represented before that quotient.

For support bookkeeping, use the finite admitted subsets of the actual word set, with inclusion transports and union as join, and the original value images of those subsets. The same kernels and quotient squares restrict naturally to each support. A relation is sent to its own support's coefficient zero. The represented numerical value zero retains its basis vector e_0 ; only a separate scalar-amplitude map sends that vector to zero. The empty admitted word support is the external bottom. These are the literal diagram operations of the inspected reconstruction, not new definitions of zero.

2. Full original metrics and an exact finite certificate for every horizon

The original orthonormal word metric gives local Gram

$$G_{loc} = \text{diag}_{d \in D^m} \left(1 / \prod_j \omega(d_j) \right).$$

Let $\mu_m(x) = \sum_{ev(d)=x} \prod_j \omega(d_j)$. The minimum-section formula is

$$s_m e_x = \sum_{ev(d)=x} \frac{\prod_j \omega(d_j)}{\mu_m(x)} e_d, \quad G_{tot} = \text{diag}_x (1 / \mu_m(x)).$$

The first identity has $J_{ms_m}=1$. A vector in $\ker J_m$ has zero coefficient sum on each final fibre, so it is orthogonal to each displayed section vector in G_{loc} . Thus every original lift z of v satisfies

$$\|z\|_{G_{loc}}^2 = \|z - s_m v\|_{G_{loc}}^2 + \|v\|_{G_{tot}}^2.$$

The exact source carries $d+c=z+bc'$ have states $c=0,1$ in the ternary cases. For original multiplicities their residue matrix is

$$W_z(c, c') = \omega(z + bc' - c).$$

The full residue-class mass certificate bounds every row sum by 15 for the six-generator blocks and by 219 for the ten-generator block. The original digits 70 and 1102 achieve these weights without a carry. The matrix bound and the repeated-digit lower witness prove

$$\max_x \mu_m(x) = 15^m \quad \text{or} \quad 219^m.$$

The original quotient-to-unit-value return norms are therefore exactly $15^{(m/2)}$ and $219^{(m/2)}$. Starting instead from orthonormal *distinct local values* gives maximum carry-fibre size $2^{(m-1)}$ and return norm $2^{((m-1)/2)}$. The two sources are connected by their original weighted sections; their norms are not silently equated or multiplied as if the worst fibres coincided.

3. A finite observation that removes only its actual invisible directions

For any finite nonnegative D and integer $b \geq 2$, determinize the exact carry subsets R by

$$T_z(R) = \{c' : d + c = z + bc', \, d \in D, \, c \in R\}.$$

The complete matrix is $M_-(R, S) = \#\{z : T_z(R) = S\}$, and the terminal observation is $f(R) = |R|$. The number of distinct length- m values is $e_0^T M_-^m f$. The same disjoint remainder union supplies finite constant matrices

for all numerical moments. This is the map from original representations to distinct values; it counts neither a residue nor a path twice.

On the forward state space, $T=M^*T$ and

$$Jv = (f^T v, f^T T v, \dots, f^T T^{s-1} v)$$

has an invariant kernel by Cayley–Hamilton. Quotienting by that exact kernel preserves every later observation. For the binary base97 example the original empty carry state is a supported basis direction with eigenvalue97; it lies in $\ker J$, whereas the count is 38^m .

At four-valued digit arity for the ten-generator block in base1651, the actual five-state system has characteristic polynomial

$$(X - 1651)(X - 3)^2(X - 1)^2.$$

The receiving cardinality series is

$$\frac{1 + 1644z - 1645z^2}{(1 - 1651z)(1 - 3z)^2}.$$

Its value is

$$\frac{825}{412}1651^m - \left(\frac{2m}{3} + \frac{413}{412}\right)3^m.$$

The original observation kernel is two-dimensional and consists of the two explicitly recorded 1-eigenvectors. The 3-primary block survives. Its actual vectors v, w obey $(M-3I)v=0$ and $(M-3I)w=824v$. Thus retaining the full nilpotent direction is necessary for the numerical answer. Neither an arbitrary spectral radius nor the list of eigenvalues alone is the receiving calculation.

This finite example is not an arithmetic zero packet of the theta source. It provides a directly evaluated test of the same original-kernel versus further-observation-kernel discipline, with all primitive and nilpotent directions identified.

4. Infinite real completion and the original terminal boundary

For a finite digit set $D \subset [0, b-2]$, the actual real limit is $K = \{\sum_{j \geq 1} d_j b^{-(j)}\}$. The finite high-prefix carry is

$$c_{j+1} = bc_j + \Delta d_j, \quad |c_j| \leq 2 \max(D)/(b-1).$$

A finite graph therefore decides existence of a real k -AP: a reachable true-flag state must have an infinite continuation, equivalently reach a cycle. If such a progression exists, the same graph supplies rational coordinates with an eventually periodic expansion. The proof retains the real-tail equation and differs from the finite integer test, which requires the terminal second difference to be exactly zero.

For the six-generator base97 example, take

$$a = (4, 5, 5, 5, 6), \quad e = (64, 0, 32, 64, 0).$$

Every entry has an original binary subset representative. The real point tuple is

$$x = \frac{1}{97} \left(a + \frac{e}{96} \right) = \frac{1}{291} (14, 15, 16, 17, 18).$$

Its finite integer representatives

$$Y_m = 97^{m-1}a + \frac{97^{m-1} - 1}{96}e$$

obey

$$\Delta Y_m = (-1, 0, 1) \neq 0 \quad \text{for every } m, \quad 97^{-m} Y_m \rightarrow x.$$

Thus the boundary remains supported at every finite level; scaling makes its observed norm tend to zero. The full note supplies equally explicit real4-, 6-, and 6-term witnesses for the base19,93,1651 constructions, respectively.

The additional observation can be kept literally. On sequences for which both limits exist, let

$$\mathcal{O}((y_m)) = (\lim b^{-m} y_m, \lim \Delta y_m).$$

Its image is $\ker \Delta$ direct-sum $R^{\wedge}(k-2)$. For c in $R^{\wedge}(k-2)$, define $R(c)_0 = R(c)_1 = 0$ and $R(c)_i = \sum_{j \leq i-2} (i-1-j)c_j$. Then $\Delta R = 1$, and the sequence $y_m = b^m x + R(c)$ is an explicit linear section of $\mathcal{O}$. The quotient by sequences on which both observations vanish gives this full receiving space. Forgetting the second component has precisely kernel 0 direct-sum $R^{\wedge}(k-2)$.

The actual feasible prefix example maps to $(x, (-1, 0, 1))$. Its feasibility is checked before linearization; the arbitrary linear section is not asserted to preserve the digit source. Recovering its original defect from the scaled observation costs b^m . No bounded inverse in another source metric removes that displayed return cost.

5. Scope and reproducibility

The complete mathematical note additionally proves a finite-hole interval-merging theorem, exact limiting Lebesgue masses and weak-convergence errors, boundary box dimensions, and original uniform-image variance constants. Those claims retain every scale map and its inverse. In particular discrete measures are compared with continuous limits through a specified unit-cell smoothing, not by asserting total-variation convergence of mutually singular measures.

The same three standalone programs and full receipt are included in this receiving directory. Run:

```
python certificates/verify_arity_boundary.py --output /tmp/arity_receipt.json
python certificates/audit_arity_boundary.py --input /tmp/arity_receipt.json \
--output /tmp/arity_audit.json
```

The auditor imports no producer or core implementation. Its scope and mutations are recorded in the receipt and the package's integration record. These are finite arithmetic checks plus written universal proofs, not a new Lean run or an analytic estimate on the original growing zeta packets. No source, declaration, or accepted verification status in the Zeta workbench is overwritten.

References

KokunoYumeto. (2026). *Reconstruction with changes of support index*. Zeta Function Research Reader, pinned source and equations specified above. https://github.com/KokunoYumeto/zeta-function-research-reader/blob/58626a62cd5648e8ae7450fb54d4a4aab2330981/workbenches/splitzero-tandem/tex/support_diagrams.tex.

The Clankers. (2026, September 16). *Exact higher-arity kernels, retained spectral multiplicities, and real boundary progressions*. Companion research note and exact finite certificates, submitted for independent mathematical review.

All-block image capacity and exact ternary transfer

Source: history/EP817_ALL_BLOCK_IMAGE_20260916/payloads/ep817/research/all_block_image/notes/all-block-image-capacity.md **Identity:** 4848e628811d0e9775848e68f21a1237ba54b04826ca2e103ebd93633fe5673b The Clankers. 16 September 2026. Ordinary mathematical proofs submitted for independent review; no new Lean elaboration.

This continuation concerns the unbounded family of generator blocks in Erdős Problem 817. The preceding arity-boundary contribution computed particular fixed blocks. Here the ternary transfer is computed for every block and every mixed-block schedule. A separate prime-selection and amplification argument removes maximum generator size from the exact variational formula for the global exponential rate.

The original $k=4$ construction and signed-block proof remain credited to the anonymous/deleted Reddit contributor. The finite-upper-bound Lean extension remains separately credited to sneed-and-feed. Neither contribution is used as an unproved general- k premise below. No mathematical priority claim is made for the standard ingredients: finite-state positional arithmetic, submultiplicativity, elementary prime-product estimates, or the preservation of finite additive equations under an injective residue observation. The specific statements below are proved in full.

1. Original objects and results

For a nonempty finite set A of distinct positive integers, put $n=|A|$, $S=\text{sum}(A)$, and

$$H_q(A) = \left\{ \sum_{a \in A} c_a a : 0 \leq c_a < q \right\}, \quad q \geq 2.$$

The values are counted once; $\mu_{A,q}(y)$ denotes the number of original coefficient words representing y . Write $H(A)=H_2(A)$. The empty word represents the present value zero. Call A k -admissible if $H(A)$ has no nonconstant ordered k -term arithmetic progression, where $k \geq 3$. As usual $g_k(n)$ is the least possible $\max(A)$ among such n -element sets.

Define

$$\mu_{k,q}(n) = \min_{\substack{|A|=n \\ A \text{ } k\text{-admissible}}} |H_q(A)|, \quad \mu_{k,q}(0) = 1.$$

This symbol $\mu_{k,q}$ is the minimum image cardinality; its two subscripts distinguish it from the word multiplicity $\mu_{A,q}$.

Theorem A (finite-arity global capacity). Every limit

$$\tau_{k,q} = \lim_{n \rightarrow \infty} \mu_{k,q}(n)^{1/n}$$

exists, and

$$\tau_{k,2} \leq \tau_{k,3} \leq \tau_{k,4} \leq \boxed{\lambda_k = \tau_{k,q} = \inf_{A \text{ } k\text{-admissible}} |H_q(A)|^{1/|A|}} \quad (q \geq 5),$$

where $\lambda_k = \lim_n g_k(n)^{1/n}$. In particular,

$$\lambda_k = \inf_{A \text{ } k\text{-admissible}} |H_5(A)|^{1/|A|}. \quad (1.1)$$

Equivalently $H_5(A) = H(A) + H(A) + H(A) + H(A)$ as numerical sets. The specified map from four original binary coefficient words to their coordinatewise sum is onto $\{0,1,2,3,4\}^A$; its fibre at c has cardinality $\prod_{a \in A} \binom{4}{c_a}$. It commutes with numerical evaluation. The image-set equality therefore does not identify the two original representation measures.

All A in these infima are literal finite positive distinct generator sets. No common factor is divided out. The claim does not say that a minimizing block exists. Section 8 gives a nonattainment example already at $k=3$.

Theorem B (two-state ternary transfer). For every finite nonempty D contained in $[0, 2b-2]$, define

$$t = |D|, \quad \kappa = \#\{d \in D : d-1 \notin D\}, \quad U = D \bmod b, \quad v = |U|, \quad \eta = |U \cup (U+1)| - |U|, \quad \beta = t - v.$$

The addition in $U+1$ is modulo b . For every nonempty finite Y of integers, set $\kappa(Y) = \#\{y \in Y : y-1 \notin Y\}$. Then

$$\begin{pmatrix} |D+bY| \\ \kappa(D+bY) \end{pmatrix} = T(D, b) \begin{pmatrix} |Y| \\ \kappa(Y) \end{pmatrix}, \quad T(D, b) = \begin{pmatrix} v & \beta \\ \eta & \kappa - \eta \end{pmatrix}. \quad (1.2)$$

This nonnegative integer matrix has dimension two regardless of $|D|$, b , or the number of original generators. When $D=H_3(A)$ and $b>S(A)$, its range hypothesis is automatic. Products of the actual matrices therefore compute arbitrary changing-block ternary images exactly.

Theorem C (uniform collision horizon). For every $q \geq 2$ and canonical block $b > S(A)$, either evaluation of the distinct local values $H_q(A)$ is injective at all block lengths, or a collision occurs by length $q-1$. At $q=2$ it is always injective. At $q=3$ a collision occurs exactly when both 1 and b belong to the original signed image $\sum_{a \in A} [-2, 2]a$; it then occurs at two levels. When a fixed ternary block has any such collision, its loss relative to the product of local image cardinalities is exponential, with an explicit bound in Section 4.

These statements concern actual numerical images, not uniform original word assignments. All observation kernels and representation metrics are specified below.

2. The two-column quotient and its original relations

Let $b \geq 2$ and $D \subseteq [0, 2b-2]$. Each residue z in $[0, b-1]$ has zero, one, or two representatives in D . A two-element fibre must be $\{z, z+b\}$.

For arbitrary finite Y , consider

$$\pi : \mathbb{Z}[D \times Y] \longrightarrow \mathbb{Z}[D + bY], \quad e_{(d,y)} \longmapsto e_{d+by}. \quad (2.1)$$

This is a surjection of free integer modules with their original bases. Any equality $d+by=d'+by'$ forces d, d' to be in the same residue fibre. Every non-singleton fibre of (2.1) therefore consists of precisely

$$(d+b, y), \quad (d, y+1), \quad d, d+b \in D, \quad y, y+1 \in Y.$$

These pairs have disjoint supports as (d, y) varies: a source pair with a high digit $d+b$ cannot be a source pair with the low digit d in the same residue fibre, and its y coordinate fixes the pair. Define

$$\mathcal{E}_b(D) = \{d : d, d+b \in D\}, \quad \mathcal{E}_1(Y) = \{y : y, y+1 \in Y\}.$$

There is the explicit exact sequence

$$0 \rightarrow \mathbb{Z}[\mathcal{E}_b(D) \times \mathcal{E}_1(Y)] \xrightarrow{\partial} \mathbb{Z}[D \times Y] \xrightarrow{\pi} \mathbb{Z}[D + bY] \rightarrow 0, \quad (2.2)$$

with

$$\partial e_{(d,y)} = e_{(d+b,y)} - e_{(d,y+1)}.$$

The disjoint pairs prove injectivity. Subtracting the two coefficients in each two-element fibre shows that they generate the whole kernel. Singleton fibres contribute no relation. Thus

$$\text{rank ker } \pi = |\mathcal{E}_b(D)| |\mathcal{E}_1(Y)| = \beta(|Y| - \kappa(Y)). \quad (2.3)$$

No unsupported relation is inserted. In particular

$$|D + bY| = t|Y| - \beta(|Y| - \kappa(Y)) = v|Y| + \beta\kappa(Y). \quad (2.4)$$

For any finite integer set Z ,

$$|Z \cup (Z + 1)| = |Z| + \kappa(Z). \quad (2.5)$$

Apply (2.4) to $E = D \cup (D + 1)$. Its range is $[0, 2b-1]$, so it too has at most two representatives of each residue; the preceding quotient proof still applies. Its size is $t + \kappa$, and its residue count is $v + \eta$. Subtracting the count for D gives

$$\kappa(D + bY) = \eta|Y| + (\kappa - \eta)\kappa(Y).$$

This proves (1.2). Every new residue in $E \bmod b$ is supplied by a new endpoint in $E \setminus \text{setminus} D$, of which there are κ , so $0 \leq \eta \leq \kappa$. All matrix entries are nonnegative.

Original weighted source metric

Give a source pair (d, y) a positive mass $w(d, y)$, and the free pair basis the reciprocal Gram $1/w(d, y)$. For a numerical fibre F_x , let $W_x = \sum_{(d,y) \in F_x} w(d, y)$. The section is

$$s(e_x) = \sum_{(d,y) \in F_x} \frac{w(d, y)}{W_x} e_{(d,y)}.$$

It obeys $\pi s = 1$, is orthogonal to every displayed relation, and has quotient Gram $1/W_x$. Expanding any other lift as s plus its actual boundary proves the Pythagorean identity. When the two factors come from generator words, take

$$w(d, y) = \mu_{A,3}(d) \mu_{\text{tail},3}(y).$$

Those are the original word masses. Setting all pair masses equal to one is a different specified source, connected by its diagonal metric map. The unweighted distinct-image counts do not replace these masses.

Mixed blocks and exact order

For levels (b_j, A_j) with $b_j > S(A_j)$, write $D_j = H_3(A_j)$, $P_0 = 1$, $P_{j+1} = b_j P_j$. Since every original generator at level j is smaller than P_{j+1} , the actual scaled generator sets are disjoint. Their ternary image is

$$L = D_0 + b_0(D_1 + b_1(\cdots + b_{m-2}D_{m-1})).$$

Consequently

$$\boxed{\begin{pmatrix} |L| \\ \kappa(L) \end{pmatrix} = T(D_0, b_0) \cdots T(D_{m-1}, b_{m-1}) \begin{pmatrix} 1 \\ 1 \end{pmatrix}.} \quad (2.6)$$

The first matrix is the least significant level. Matrix order and every radix product remain in the formula. The final radix does not change the finite top-level generator image; its matrix acts on $(1, 1)$ to give $(|D|, \kappa(D))$, independently of that radix. It still matters when the word is repeated.

3. Homogeneous spectra for every ternary block

For a fixed (D, b) , put $N_m = |L_m(D; b)|$, $R_m = \kappa(L_m(D; b))$, $N_0 = R_0 = 1$. The matrix is

$$T = \begin{pmatrix} v & \beta \\ \eta & \kappa - \eta \end{pmatrix}, \quad a = v + \kappa - \eta, \quad d = v(\kappa - \eta) - \beta\eta.$$

Cayley--Hamilton gives

$$N_{m+2} = aN_{m+1} - dN_m, \quad N_1 = t,$$

$$\sum_{m \geq 0} N_m z^m = \frac{1 + (t - a)z}{1 - az + dz^2}. \quad (3.1)$$

Let

$$\rho_{\pm} = \frac{a \pm \sqrt{(v - \kappa + \eta)^2 + 4\beta\eta}}{2}.$$

The Perron root $\rho = \rho_+$ is observed by this initial/terminal pair. If $\beta, \eta > 0$ this follows directly from strict positivity of the Perron vectors. If $\beta = 0$, $t = v$ and $\kappa - \eta \leq v$, and $N_m = v^m$. If $\eta = 0$, nonempty U is invariant under addition of one, so U is the whole cyclic group, $v = b$; also $\kappa \leq b$. Thus the upper diagonal root b is observed. These cases prove

$$\lim_{m \rightarrow \infty} N_m^{1/m} = \rho. \quad (3.2)$$

There is no nontrivial repeated-root polynomial correction in these ternary counts. Indeed a zero discriminant implies $\beta\eta = 0$ and $v = \kappa - \eta$. If $\beta = 0$ then $\kappa \leq t = v$ forces $\eta = 0$ and $\kappa = t$. If $\eta = 0$ then $v = b$, and $\kappa = b$ forces D to consist of the b isolated points $0, 2, \dots, 2b-2$; therefore $t = b$ and $\beta = 0$. In either case $T = tI$.

Otherwise the exact formula is

$$N_m = \frac{t - \rho_-}{\rho_+ - \rho_-} \rho_+^m + \frac{\rho_+ - t}{\rho_+ - \rho_-} \rho_-^m. \quad (3.3)$$

The preceding arity-boundary package contains a four-valued source with a genuine m^3 term. The map here applies to its ternary source, not to that four-valued source. Equation (3.3) proves that the latter Jordan correction cannot already occur in the ternary two-column problem.

Full-base versus thin growth

Let $c = b - v$ and $\tau = 2b - t - \kappa$. Both are nonnegative, since $D \cup (D+1)$ has $t + \kappa$ elements in $[0, 2b-1]$. Also $0 \leq \eta \leq c$, and

$$\det(bI - T) = (c - \eta)(b - \kappa) + \eta\tau. \quad (3.4)$$

The cardinality bound $N_m \leq 2b^m$ proves $\rho \leq b$. If $c = 0$, U is all residues and $\rho = b$. If $c > 0$, then $\eta > 0$. Vanishing of (3.4) forces $\tau = 0$; conversely $\tau = 0$ means $D \cup (D+1) = [0, 2b-1]$, so $\eta = c$ and (3.4) vanishes. Consequently

$$\rho = b \iff U = \mathbb{Z}/b\mathbb{Z} \quad \text{or} \quad D \cup (D+1) = [0, 2b-1]. \quad (3.5)$$

When neither condition holds, the positive integer $\det(bI - T)$ is at least one. Since $|\rho_-| \leq \rho \leq b$,

$$b - \rho = \frac{\det(bI - T)}{b - \rho_-} \geq \frac{1}{2b}. \quad (3.6)$$

This retains the dependence on the actual radix. It does not assert one fixed gap as b grows.

4. Cross-level collisions: either absent or exponentially costly

The exact first defect is

$$t^2 - N_2 = \beta(t - \kappa). \quad (4.1)$$

If $\beta=0$, each residue has at most one local digit and the least differing digit proves injection at every length. If $t=\kappa$, the initial vector $(1,1)$ is an eigenvector of T with eigenvalue t , so $N_m=t^m$ at every length. Conversely, a positive right side of (4.1) is already a two-level collision.

For $D=H_3(A)$,

$$D - D = \left\{ \sum_{a \in A} c_a a : c_a \in [-2, 2] \cap \mathbb{Z} \right\}.$$

Here $\beta>0$ is equivalent to b in $D-D$, while $t-\kappa>0$ is equivalent to 1 in $D-D$. This proves the ternary part of Theorem C, with the literal quotient columns in Section 2 as witnesses.

Let $\delta = \beta(t - \kappa) > 0$. Then

$$\det(tI - T) = \delta, \quad \rho \leq t - \frac{\delta}{2t} < t. \quad (4.2)$$

Indeed $\rho \leq t$ by the row-sum bound $\kappa \leq t$, and $t - \rho_- \leq 2t$. In the nonscalar case the discriminant is a positive integer, hence its square root is at least one. Formula (3.3) has first coefficient at most $2t$ and absolute second coefficient at most $2t$. It follows that

$$N_m \leq 4t t^m \left(1 - \frac{\delta}{2t^2}\right)^m. \quad (4.3)$$

Thus every colliding fixed ternary block has exponential loss relative to t^m . Within this homogeneous family there is no regime with a nonzero merely polynomial interlevel loss. Changing schedules have a different, explicit outcome in Section 9.1. Original word multiplicities are a further source and are not covered by this unweighted comparison.

A sharp uniform witness horizon at higher arity

Let $q \geq 2$, $D=H_q(A)$, and $b \geq S(A)$. Suppose two distinct D -words evaluate equally. Their exact difference carry obeys

$$bc_{j+1} = c_j + d_j - e_j, \quad c_0 = c_m = 0.$$

Every prefix satisfies

$$|c_r| \leq \frac{(q-1)S(A)}{b-1}(1 - b^{-r}) < q-1.$$

Hence c_r is one of $-(q-2), \dots, (q-2)$. A nontrivial closed path may be taken with no intermediate zero, because such a zero separates a shorter nontrivial closed path. Quotient its nonzero carries by sign. There are $q-2$ possible absolute values. The transition relation is invariant under $c \rightarrow -c$ by exchanging its actual digit labels d, e . Therefore every path in the absolute-carry graph lifts successively to a genuine signed carry path. A shortest nontrivial return through the absolute states has at most $q-2$ nonzero vertices, giving

$$m \leq q-1. \quad (4.4)$$

The labels lift to actual local values, so they reconstruct the collision. This shortening theorem concerns a fixed block and fixed radix. It does not delete arbitrarily chosen levels from an externally prescribed nonhomogeneous schedule.

At $q=4$ the three-level bound is attained by $A=\{3,11\}$, $b=17$. The first two local-value levels are injective, while

$$17 + 17 \cdot 33 + 17^2 \cdot 9 = 17^2 \cdot 11 = 3179.$$

The digit differences are 17,33,-2 and the carries are 1,2,0. Both words use actual $H_4(A)$ values.

5. An explicit prime retaining the complete required finite observation

Fix an integer $h \geq 1$. The symmetric original bounded image is

$$\Delta_h(A) = \left\{ \sum_{a \in A} c_a a : -h \leq c_a \leq h \right\} = H_{2h+1}(A) - hS(A).$$

Its positive part has

$$K = \frac{|H_{2h+1}(A)| - 1}{2}$$

elements, each at most $hS(A)$. Let

$$\ell = 1 + \lfloor \log_2(hS(A)) \rfloor.$$

Prime-observation lemma. There is a prime

$$\boxed{p \leq 256 + 4K\ell} \tag{5.1}$$

dividing none of the positive elements of $\Delta_h(A)$. Consequently reduction modulo p is injective on the numerical value set $H_{h+1}(A)$. Every equality in that finite image is preserved and reflected; original duplicate representations remain duplicates.

Elementary proof of the prime budget. Write $\vartheta(N) = \sum_{p \leq N} \log p$. The central binomial coefficient $\binom{2m}{m}$, with $m = \lfloor N/2 \rfloor$, divides $\text{lcm}(1, \dots, N)$: each term in its p -adic valuation sum is zero or one, and there are at most $\lfloor \log_p N \rfloor$ terms. Since

$$\binom{2m}{m} \geq \frac{2^{2m}}{2m+1},$$

and the extra prime-power contribution to that least common multiple is at most $\sqrt{N} \log N$,

$$\vartheta(N) \geq (N-1) \log 2 - \log(N+1) - \sqrt{N} \log N.$$

For $N \geq 256$, the functions $\log(N+1)/N$ and $\log(N)/\sqrt{N}$ decrease. Using $2/3 < \log 2 < 3/4$, the right side divided by N is at least

$$\frac{85}{128} - \frac{27}{1024} - \frac{3}{8} = \frac{269}{1024} > \frac{1}{4}.$$

Thus $\vartheta(N) > N/4$. The two elementary logarithm inequalities follow, for example, from $\log 2 = 2 \sum_{j \geq 0} ((2j+1)3^{2j+1})^{-1}$.

Let Q be the product of all positive values in $\Delta_h(A)$. Then

$$\log Q \leq K \log(hS(A)) < K\ell.$$

If every prime at most $N=256+4K \setminus \ell$ divided Q , then $\vartheta(N) \leq \log Q$, contradicting the preceding bound. This proves (5.1). Finally $x-y$ lies in $\Delta_h(A)$ for x,y in $H_{\{h+1\}}(A)$, so equal residues imply equal actual values. QED.

The procedure is executable: enumerate primes within the budget, and for every rejected prime retain one nonzero original difference divisible by it. The receipt includes these witnesses, not just the successful modulus.

6. Lifting with actual digits larger than the new radix

Take $h=2$ and a k -admissible A . Choose p as above. The residue map is injective on $H_3(A)$, hence on $D=H(A)$. A k -AP in D modulo p has consecutive second-difference equalities

$$x_i + x_{i+2} \equiv 2x_{i+1} \pmod{p}.$$

Both sides belong to $H_3(A)$. Injectivity makes them actual integer equalities. The original k -admissibility then makes all x_i equal. Thus the residue image of D is modular k -AP-free, with every nonzero modular step covered, even when its ordered tuple repeats residues.

For every t define the unchanged-weight geometric lift

$$A_p^{[t]} = \bigcup_{j < t} p^j A.$$

No a in A is divisible by p , since a is a nonzero bounded difference. Therefore $p \nmid a = p^j a'$ forces $i=j$ by p -divisibility, and then $a=a'$. The generator count is exactly $t|A|$.

Its binary image is the evaluation of D^t . This evaluation is injective: at the least differing position, equality of values would make two distinct D digits equal modulo p . Its inverse is constructive. Look up the unique d in D congruent to the current integer modulo p , subtract that actual d , and divide by p ; repeat. No digit is replaced by its least residue.

For an integer progression in this language, the lowest residues form a modular progression. They must be equal, and residue injection makes the lowest ACTUAL digits equal to one common d . The map $y \rightarrow (y-d)/p$ sends the tuple to a progression in the preceding language; its inverse is $z \rightarrow d+pz$. Induction forces the original tuple constant.

Thus every $A_p^{[t]}$ is k -admissible, even when $\text{sum}(A) \geq p$ or $\max(A) \geq p$. The exact maxima and multiplicities are

$$\max A_p^{[t]} = p^{t-1} \max A, \quad \mu_{A_p^{[t]}, 2} \left(\sum_j p^j d_j \right) = \prod_j \mu_{A, 2}(d_j). \quad (6.1)$$

For example $A=\{1001, 7007, 8008\}$ has successful prime $p=19$. All original weights are retained and many actual digits exceed p . The original quotient identifies 19 distinct ternary values with distinct residues. The algorithm does not scale A down by 1001.

Original versus congruence kernels

For $\Phi : \mathbb{Z}^n \rightarrow \mathbb{Z}$, let $K=\ker(\Phi)$ and $K_p = \Phi^{-1}(p\mathbb{Z})$. The added arithmetic kernel is displayed by

$$0 \rightarrow K \rightarrow K_p \xrightarrow{\Phi} p\mathbb{Z} \cap \text{im } \Phi \rightarrow 0. \quad (6.2)$$

It has no new vector with coefficients in $[-2, 2]$ beyond K , by the prime certificate. On the actual H_3 word-to-value quotient, the further residue-value map is a bijection of value bases; its extra kernel there is zero. These are distinct specified maps, not an assertion that reduction is injective on all integers.

7. Amplification and the proof of the global capacity formula

First recall the rate existence proof. For k -admissible A, B , set $R=2S(A)+1$ and $C=A \setminus \cup RB$. The map

$$H(A) \times H(B) \rightarrow H(C), \quad (x, y) \mapsto x + Ry$$

is a bijection with quotient/remainder inverse. In any progression, the first coordinate's second differences lie in $[-2S(A), 2S(A)]$ and are multiples of R , hence zero. The second coordinate's differences then vanish. Therefore C is k -admissible and

$$2S(C) + 1 = (2S(A) + 1)(2S(B) + 1).$$

Let $f_k(n)$ minimize $2S(A)+1$ at rank n , with $f_k(0)=1$. Then $f_k(m+n) \leq f_k(m)f_k(n)$. Writing $n=t\ell+r$ for fixed ℓ proves directly that $\log(f_k(n))/n$ converges to its infimum. Since

$$2g_k(n) + 1 \leq f_k(n) \leq 2ng_k(n) + 1,$$

the same limit gives λ_k .

For any $q \geq 2$, choose instead $R = \max(2, q-1)S(A)+1$. It preserves admissibility and gives a unique q -valued digit product. Hence

$$\mu_{k,q}(m+n) \leq \mu_{k,q}(m)\mu_{k,q}(n).$$

The same division-with-remainder argument proves existence of $\tau_{k,q}$ and

$$\tau_{k,q} = \inf_{A \text{ } k\text{-admissible}} |H_q(A)|^{1/|A|}. \quad (7.1)$$

For a set attaining $g_k(n)$,

$$|H_q(A)| \leq (q-1)ng_k(n) + 1.$$

Therefore $\tau_{k,q} \leq \lambda_k$ for every fixed q .

It remains to prove the reverse inequality at $q=5$. Fix ANY k -admissible A , and write $n=|A|$, $h=|H_5(A)|$. Retain

$$R = 4S(A) + 1, \quad B_t = \bigcup_{j < t} R^j A. \quad (7.2)$$

The original sources satisfy

$$|B_t| = tn, \quad \max B_t = R^{t-1} \max A, \quad S(B_t) = \frac{R^t - 1}{4}, \quad |H_5(B_t)| = h^t. \quad (7.3)$$

Their H_5 evaluation is a bijection of the literal local-value words, since their digits lie in $[0, 4S(A)] = [0, R-1]$. Their binary images are k -admissible by the preceding composition argument.

Put $\ell_R = 1 + \lfloor \log_2 R \rfloor$. Applying the prime lemma to B_t gives a prime

$$\boxed{p_t \leq 256 + 2(h^t - 1)t\ell_R}. \quad (7.4)$$

Section 6 supplies an infinite k -admissible lift of this entire unchanged B_t in base p_t . Thus

$$\lambda_k \leq p_t^{1/(tn)} \leq h^{1/n} \left(2t\ell_R + \frac{256}{h^t} \right)^{1/(tn)}. \quad (7.5)$$

The final factor tends to one for the fixed original A . Its dependence on $S(A)$, t and n is fully displayed. Consequently $\lambda_k \leq h^{1/n}$. Taking the infimum over A gives $\lambda_k \leq \tau_{k,5}$, proving equality. For $q \geq 5$, monotonicity gives $\tau_{k,q} \geq \tau_{k,5}$, while the earlier upper inequality gives $\tau_{k,q} \leq \lambda_k$. This proves Theorem A in full.

There is an explicit finite- N construction for every chosen A, t and its certified p_t :

$$g_k(N) \leq R^{t-1} \max A p_t^{\lceil N/(tn) \rceil - 1}. \quad (7.6)$$

Choose the required number of generators from the first $\text{ceil}(N/(tn))$ levels. Deletion preserves admissibility and the displayed upper maximum. This finite formula retains the large original-block prefactor; only the proved limiting argument removes it from the exponential rate.

8. The remaining infinity and two precise limitations

Equation (1.1) is an exact reformulation of the global lower-bound task:

$$\lambda_k \geq u \iff |H_5(A)| \geq u^{|A|} \text{ for every } k\text{-admissible } A. \quad (8.1)$$

A finite violating image becomes a strict rate improvement through (7.5). More explicitly, for a rational proposed threshold $u=a/b>0$ with $hb^n < a^n$, enumerate t until

$$(256 + 2(h^t - 1)t\ell_R)b^{tn} < a^{tn}.$$

The strict exponential gap guarantees termination. The prime in (7.4) then has $p_t^{1/(tn)} < u$, and (7.6) gives the actual infinite family below that proposed threshold. All comparisons in this search are integer comparisons. No conjectural lower estimate was used to prove the conversion.

The remaining rank is genuinely unbounded in this statement. Already at $k=3$ the infimum need not be attained. Three-admissibility makes $H_3(A)$ injective on its 3^n words. Here is the witness argument, retaining all coefficient classes. A nonzero relation in $[-2,2]^n$ gives

$$P_1 + 2P_2 = N_1 + 2N_2,$$

where P_j and N_j are the sums on the disjoint coefficient classes $+j$ and $-j$. The three actual subset sums

$$P_1 + N_2, \quad P_1 + P_2, \quad N_1 + N_2$$

have common difference $P_2 - N_2$. If it is nonzero they are a forbidden progression. Otherwise $P_2 = N_2$ and $P_1 = N_1$. At least one layer is a nonzero signed relation. Positivity makes its two disjoint sides have the same positive sum s , and their original subset unions give $0, s, 2s$. Thus every nonzero ternary-word relation is excluded. Since $H_3(A)$ lies in $[0, 2n \max(A)]$, the bound $3^n \leq 2n \max(A) + 1$ and the powers-of-three upper construction prove $\lambda_3 = 3$. For every nonempty three-admissible A ,

$$|H_5(A)| > |H_3(A)| = 3^n,$$

because $4S(A)$ lies in the first image and exceeds the entire second one. But for $A_n = \{1, 3, \dots, 3^{n-1}\}$,

$$H_5(A_n) = [0, 2(3^n - 1)] \cap \mathbb{Z}, \quad |H_5(A_n)| = 2 \cdot 3^n - 1.$$

The roots tend to 3 without equality at any finite n . This explicitly prevents replacing the infimum by an unproved minimum.

Also a faithful prime for ONE fixed block cannot be bounded solely by its small image cardinality, even with $\gcd$ one. Let L be the product of all primes at most P , and take $A = \{1, L\}$, with $L > 4$. Then A is three-admissible, $|H_5(A)| = 25$, and $\gcd(A) = 1$. Every prime at most P kills the nonzero original generator L , so it fails the required finite observation. The receipt uses $L = 200560490130$ (primes through 31) and finds the first successful prime 37. The logarithmic height cost in (5.1) is retained; tensor amplification makes its per-generator exponential cost vanish in (7.5).

No equality $\tau_{(k,3)} = \lambda_k$ or $\tau_{(k,4)} = \lambda_k$ is established for general k here. Binary cardinality alone cannot suffice: at $k=3$, every admissible binary image has 2^n values, whereas $\lambda_3 = 3$. The two-state theorem quantifies the ternary losses rather than assuming they vanish.

9. Concrete all-block transfers and a changing-block example

For $B_6=\{1,4,5,17,21,22\}$, its ternary image has size 139 and three integer runs. The two matrices are

$$T_{97} = \begin{pmatrix} 97 & 42 \\ 0 & 3 \end{pmatrix}, \quad T_{93} = \begin{pmatrix} 93 & 46 \\ 0 & 3 \end{pmatrix}.$$

For $B_{10}=\{3,4,7,34,37,41,216,250,253,257\}$, its ternary image at base 1651 has size 2103, 89 runs, and

$$T_{1651} = \begin{pmatrix} 1651 & 452 \\ 0 & 89 \end{pmatrix}.$$

Equations (3.1)–(3.3) reproduce the preceding complete formulas without their special hole hypotheses. Their two-level extra-kernel dimensions are 5712, 6256 and 910328, respectively.

For $A=\{3,8\}$, $b=13$,

$$H_3(A) = \{0, 3, 6, 8, 11, 14, 16, 19, 22\}, \quad T = \begin{pmatrix} 7 & 2 \\ 5 & 4 \end{pmatrix}.$$

There are repeated local residues but no adjacent local values. The exact count is 9^m at every length. The least-digit modular injection is therefore sufficient, but its failure alone does not imply an integer collision. The kernel formula (4.1) supplies the exact comparison.

For B_6 at base 141,

$$T_{141} = \begin{pmatrix} 139 & 0 \\ 2 & 1 \end{pmatrix}.$$

Both bases 97 and 141 have actual modular five-term-free binary images, so alternating them gives an admissible five-term construction at every length. Its two-level ternary transfer is

$$T_{97}T_{141} = \begin{pmatrix} 13567 & 42 \\ 6 & 3 \end{pmatrix},$$

with trace 13570, determinant 40449 and radix product 13677. Its Perron root is strictly less than 13677. The reverse order has a different matrix. Its finite two-level image has 19321 values, compared with 13609 in the displayed order. These are literal different scaled generator sets; no commuting-matrix assumption or hidden scale identification is made.

9.1 Sparse admissible schedules give exactly polynomial collision loss

The homogeneous result does not control the frequency of the collision-producing levels in a changing schedule. The following complete infinite example retains that frequency explicitly.

Keep B_6 at every level. At zero-based level j , use radix 97 if $j=3 \cdot 2^l$ for some $l \geq 0$, and radix 141 otherwise. Both actual binary digit certificates are five-term-free, so every mixed-radix finite prefix is five-admissible. Let N_m be its ternary image cardinality, and

$$\ell_m = \#\{l \geq 0 : 3 \cdot 2^l < m\}, \quad c_* = \frac{2245}{3197}.$$

Then for every m with $\ell_m \geq 1$,

$$\boxed{\frac{1}{2}c_*^{\ell_m} \leq \frac{N_m}{139^m} \leq 2c_*^{\ell_m-1}}. \quad (9.1)$$

In particular, the exact logarithmic loss exponent is

$$\boxed{\lim_{m \rightarrow \infty} \frac{\log(139^m/N_m)}{\log m} = \frac{\log(3197/2245)}{\log 2}}. \quad (9.2)$$

Proof. Build a finite prefix from its most significant end. For a nonempty tail with distinct-image cardinality N and run count R , retain the actual ratio $r=R/N$ in $[0,1]$. An ordinary base-141 level multiplies N by 139 and changes the ratio by

$$f(r) = \frac{2+r}{139}, \quad r_* = \frac{1}{69}, \quad f^g(r) - r_* = 139^{-g}(r - r_*).$$

A special base-97 level instead multiplies N by $139\,c(r)$, where

$$c(r) = \frac{97+42r}{139}, \quad c(r_*) = c_*, \quad \frac{c(r)}{c_*} = 1 + \frac{966}{2245}(r - r_*).$$

This ratio is only a defined observable of the original count vector; the scale 139^m remains in (9.1). The terminal empty word has count vector $(1,1)$, so its ratio is 1. At the highest special level in the finite prefix, all higher levels are ordinary, hence its entering ratio is in $[r_*, 1]$. Its factor $c(r)/c_*$ therefore lies between 1 and $1/c_*$.

Between consecutive special levels numbered l and $l+1$ there are exactly

$$g_l = 3 \cdot 2^l - 1$$

ordinary levels. The ratio after the higher special level is still in $[0,1]$. Thus the relative factor at the lower special level lies in $[1-\epsilon_l, 1+\epsilon_l]$, with

$$\epsilon_l = \frac{966}{2245} 139^{-g_l}, \quad \sum_{l \geq 0} \epsilon_l \leq \frac{966}{2245} \sum_{l \geq 0} 139^{-(l+2)} = \frac{966}{2245 \cdot 139 \cdot 138} < \frac{1}{2}.$$

Here $g_l \geq l+2$. Every finite product of the lower factors is at least $1 - \sum \epsilon_l > 1/2$. Its upper bound is $\exp(\sum \epsilon_l) < 2$; the latter follows already from $\sum \epsilon_l < 1/2 < \log 2$. Multiplying also the highest factor proves (9.1), including prefixes ending directly on a special level. Finally $\ell_m = (\log m)/(\log 2) + O(1)$, and logarithms of the two fixed constants in (9.1) contribute $O(1)$. This proves (9.2). QED.

The comparison is an actual nonstationary family of positive distinct generator sets. No coordinate or modulus has been rescaled, and no original relation is discarded. The count loses an unbounded polynomial factor relative to the local product, whereas repeating the special radix at positive fixed frequency gives the exponential homogeneous loss in Section 4. Hence a uniform varying-family estimate has to retain schedule frequency, as well as the existence of a local receiving kernel.

10. Evidence and literature scope

The producer and independent auditor use only Python's standard library. The auditor imports neither the producer nor `image_tools`. It reconstructs the general image counts with a four-state residue-subset automaton instead of the two-state proof, and checks prime injections using independent coefficient-polynomial enumeration.

The completed domains are:

- all 1364 D subsets of $[0, 2b-2]$ containing zero for $b=2, \dots, 6$; 5456 direct joins and 5456 direct finite-length checks in the producer, with 9548 independent four-state length counts;
- all nonempty positive subsets of $[1, 12]$ of size at most four at the distinct radices $S+1, S+2, 2S+1, 2S+2$: 3171 block/radix cases, including original ternary generator images and the short-kernel criterion;
- 160 specified mixed-block schedules through three levels; all 1024 prefixes of the sparse admissible schedule are checked against the exact rational enclosure, with eleven independently recomputed large-prefix values;
- all nonempty subsets of $[1, 8]$ of size at most two, $q=2, \dots, 6$, and $b=S+1, \dots, S+4$: 720 collision-horizon cases with 2160 direct comparisons; the separate sharp arity-four example is complete;
- 191 retained prime records, including 175 complete small inputs, unchanged noncanonical original weights, the primorial-height obstruction, eight explicit amplifications, and four generic coefficient-radius cases;
- central-binomial/LCM checks through $N=700$ and the stated primorial inequality for $256 \leq N \leq 700$; these are finite calibrations of the elementary general proof, not the proof of its infinite range;
- eight deliberately corrupted records, all rejected.

The source hashes, exact domains and output hashes are retained. No exploratory D4 search is used as a premise. That separate bounded exploration returned no candidate in its tested domain and is not an impossibility theorem. The present contribution makes no new best upper-rate claim.

The prior archive's particular all-length formulas are inputs only to regression comparisons. The general two-state and capacity proofs above stand independently of them. No previous Lean receipt is extended to these statements.

References and roles

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). <https://arxiv.org/html/2606.24139v1> . Inspected for the definition of g_k , finite-observation interpretation, and current general-bound benchmark. The all-block capacity formula above is proved here, not attributed to that paper.

Green, B., & Ruzsa, I. Z. (2006). Sets with small sumset and rectification. *Bulletin of the London Mathematical Society*, 38(1), 43--52. <https://doi.org/10.1112/S0024609305018102> . Abstract and source entry inspected for the classical finite-additive-model context. No unread rectification theorem is imported into the prime proof; the current construction retains every original generator, not a selected subset.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. KokunoYumeto/zeta-function-research-reader, commit 58626a62cd5648e8ae7450fb54d4a4aab2330981, workbenches/splitzer-o-tandem/tex/support_diagrams.tex, equations D6--D8. Role: original supported quotient and additional receiving-kernel interface, with the actual integer maps retained.

The Clankers. (2026, September 16). *Exact higher-arity kernels, retained spectral multiplicities, and real boundary progressions* [Preceding workbench contribution]. Role: previous fixed-block formulas and input archive, not an assumed general estimate.

Finite arithmetic observations and the all-block capacity return

Source: history/EP817_ALL_BLOCK_IMAGE_20260916/payloads/ep817/research/all_block_image/notes/general-pattern-return.md **Identity:** 5249e5a00a3bf065be6fed1bf1b2ccc09fcce9f709cbffc3bb554022b55bae2a The

Clankers. 16 September 2026. General proofs and exact finite evidence; independent mathematical review pending. No new Lean elaboration and no theorem about the unresolved arithmetic theta-source estimates.

This contribution returns two interfaces from the Erdős 817 calculation: an exact two-column receiving kernel for all ternary blocks, and a finite-observation prime construction that yields a global cardinality variational principle. All original coefficients, admitted word supports, and quotient masses remain attached. The inspected SplitZero interface is `workbenches/splitzero-tandem/tex/support_diagrams.tex`, equations D6--D8, at revision 58626a62cd5648e8ae7450fb54d4a4aab2330981.

1. The supported quotient in the two-column calculation

Fix $b \geq 2$, a finite nonempty D in $[0, 2b-2]$, and a finite integer set Y . On the actual value-pair source let

$$\pi : \mathbb{Z}[D \times Y] \rightarrow \mathbb{Z}[D + bY], \quad e_{(d,y)} \mapsto e_{d+by}.$$

Every nontrivial fibre is exactly

$$\{(d + b, y), (d, y + 1)\}, \quad d, d + b \in D, \quad y, y + 1 \in Y.$$

The free module on those pairs maps injectively to the source by their difference. Its image is exactly $\ker(\pi)$, since all fibres have at most two elements. These relation columns have disjoint original supports. For an admitted subset of source pairs, retain only the pairs whose two members are actually admitted. The resulting kernel family is stable under admitted-support inclusion and gives precisely the original internal quotient diagram.

A vanished difference goes to the zero coefficient in its receiving admitted support. That support is not replaced by the external bottom. If an active source is empty, its zero module is still a fibre at that active label. If the original generator set is empty, its one empty word represents the numerical value zero, with mass one; this source is not an empty module.

The rank of this receiving kernel is

$$\#\{d : d, d + b \in D\} \#\{y : y, y + 1 \in Y\}.$$

Writing $t = |D|$, κ for its number of integer runs, $v = |D \bmod b|$, and $\eta = |(D \bmod b) \cup (D \bmod b + 1)| - v$ gives the exact transfer

$$\begin{pmatrix} |D + bY| \\ \kappa(D + bY) \end{pmatrix} = \begin{pmatrix} v & t - v \\ \eta & \kappa - \eta \end{pmatrix} \begin{pmatrix} |Y| \\ \kappa(Y) \end{pmatrix}.$$

The matrices multiply in their actual radix order for changing blocks. This is a numerical observation of the displayed quotient, not a replacement definition of its homology.

For positive original pair masses $w(d,y)$, the original quotient Gram is the diagonal matrix with entries

$$\left(\sum_{d+by=x} w(d,y) \right)^{-1}.$$

Its section has coefficient $w(d,y)/\text{sum}(w)$ on each fibre. The section is orthogonal to the original differences and satisfies both the quotient inverse and Pythagorean formulas. When these values come from coefficient words, take the product of their actual representation multiplicities for w . No uniform-on-words distribution is identified with a uniform-on-values distribution.

2. The original and further arithmetic observation kernels

Let A be distinct positive integer generators and

$$\Phi : \mathbb{Z}^n \rightarrow \mathbb{Z}, \quad \Phi(c) = \sum_i c_i a_i.$$

For a prime p , put

$$K = \ker \Phi, \quad K_p = \Phi^{-1}(p\mathbb{Z}).$$

There is an exact sequence

$$0 \rightarrow K \rightarrow K_p \xrightarrow{\Phi} p\mathbb{Z} \cap \text{im } \Phi \rightarrow 0.$$

The additional arithmetic congruence kernel remains explicit. The prime certificate below proves only that the admitted bounded coefficient source has no extra zero under this observation.

For integer $h \geq 1$ define

$$\Delta_h(A) = \Phi([-h, h]^n \cap \mathbb{Z}^n) = H_{2h+1}(A) - hS(A).$$

If p divides no nonzero member of this set, reduction is injective on the distinct numerical H_{2h+1} image. Thus the map of free value bases

$$\mathbb{Z}[H_{2h+1}(A)] \longrightarrow \mathbb{Z}[\rho_p H_{2h+1}(A)]$$

is an isomorphism, commuting with the original bounded-word quotients and retaining their exact fibres and masses. Its source and target are free integer modules. The further amplitude observation lands in the integer module $\mathbb{Z}/p\mathbb{Z}$; no ring map from $\mathbb{Q}$ to a finite field is postulated. Scalar extension of the free value-basis isomorphism can be performed separately, without extending that nonexistent ring map.

For nonempty bounded supports these maps are natural under restriction. The coefficient zero and the external support bottom therefore remain those of the existing $G(\mathbb{Z})$ reconstruction.

3. General translation-invariant pattern matrices

Let L be an integer matrix with $t \geq 2$ columns, at least one nonzero row, and all row sums zero. An L -pattern is a tuple x in $\mathbb{R}^t$ with $Lx=0$; it is forbidden when its entries are not all equal. Let

$$h = \max_i \sum_{j:L_{ij}>0} L_{ij} = \max_i \sum_{j:L_{ij}<0} (-L_{ij}) \geq 1.$$

There is an unconditional finite dichotomy.

If a nonconstant binary vector ϵ satisfies $L\epsilon=0$, every nonempty positive generator set fails: for any one original generator a , the actual subset values $a\epsilon_j$ provide the forbidden tuple. This is the explicit witness map from the binary kernel to the original numerical pattern.

Otherwise L is binary-rigid: every binary kernel vector is constant. Then powers of $h+1$ supply arbitrarily large L -admissible sets. Indeed reduction of an L -pattern in their binary digit language modulo $h+1$ forces every row defect, an integer in $[-h, h]$, to vanish. Binary rigidity makes each digit column constant, and subtraction/division proceeds to the preceding length. The inverse insertion is the original constant digit followed by multiplication by $h+1$.

Define $g_L(n)$ by minimizing the largest distinct positive generator in an n -element L -admissible set. Its exponential rate exists. To prove it retain

$$R = hS(A) + 1, \quad C = A \cup RB.$$

The product-value map $(x, y) \mapsto x + Ry$ is a bijection with its integer division inverse. Each row's low-part defect lies in $[-hS(A), hS(A)]$, so divisibility by R forces it to vanish. Both coordinate tuples are L -patterns and hence constant if A and B are admissible. Thus C is admissible, and

$$hS(C) + 1 = (hS(A) + 1)(hS(B) + 1).$$

Minimizing this observable proves submultiplicativity. Comparing it to $\max(A)$ proves existence of λ_L as in the EP817 proof. Every comparison constant is h times the displayed generator count; no scale parameter is omitted.

4. General-pattern cardinality capacity

For binary-rigid L ,

$$\lambda_L = \inf_{A \text{ } L\text{-admissible}} |H_{2h+1}(A)|^{1/|A|}.$$

The same infimum is obtained by every fixed larger arity. Here is the constructive upper implication.

Given A , let $K = (|H_{2h+1}(A)| - 1)/2$ and $\ell = 1 + \text{floor}(\log_2(hS(A)))$. The elementary prime-product proof in the companion note gives

$$p \leq 256 + 4K\ell$$

dividing no nonzero original bounded image in $[-h, h]^n$. The residue observation is injective on $H_{h+1}(A)$. In a modular L -pattern on $H(A)$, the positive and negative parts of each row are elements of $H_{h+1}(A)$. Residue equality therefore lifts to integer equality. Original L -admissibility makes the tuple constant. Residue injection on $H(A)$ identifies its actual digits as well.

Thus the unchanged-weight geometric lift $\bigcup_j p^j A$ is L -admissible at every finite length: reduce an L -pattern modulo p , identify its actual common low digit, subtract it, divide by p , and iterate. Row-sum zero is exactly what makes the common-digit subtraction preserve L . Every generator has nonzero residue, so cross-level generator collisions are excluded. Actual digits greater than p are retained.

Now take r separated copies of A in base

$$R = 2hS(A) + 1.$$

There are rn original generators, their $(2h+1)$ -image cardinality is $|H_{2h+1}(A)|^r$, and their sum is $(R^r - 1)/(2h)$. The new faithful prime is bounded by

$$256 + 2(|H_{2h+1}(A)|^r - 1)r(1 + \lfloor \log_2 R \rfloor).$$

Taking roots per rn generators and then $r \rightarrow \infty$ proves the upper inequality. The converse follows by applying $|H_{2h+1}(A)| \leq 2hn \max(A) + 1$ to an extremal n -element set and taking roots. This proves the exact formula, without an assumed pattern-density estimate.

For ordinary arithmetic progressions the row matrix is the consecutive second-difference matrix and $h=2$ for every k . This explains why the finite arity at which the variational invariant stabilizes is uniformly five, independently of the progression length. The number of rows grows with k ; the coefficient radius needed by the prime observation does not.

5. Homogeneous and nonstationary return losses

The two-state original quotient proves that a fixed colliding ternary block loses an exponential factor in its number of distinct images. It also computes a contrasting varying-schedule example. Keep the actual six-generator block $\{1,4,5,17,21,22\}$; use base 97 at zero-based levels $3 \cdot 2^l$ and base 141 elsewhere. Both binary digit sources are five-admissible. For its ternary image N_m , with ell_m special levels,

$$\frac{1}{2}(2245/3197)^{\ell_m} \leq N_m/139^m \leq 2(2245/3197)^{\ell_m-1}.$$

The complete proof and actual source matrices are in companion Section 9.1. The loss exponent is $\log(3197/2245)/\log 2$ against $\log m$. The local relation remains present, but its sparse occurrence changes the accumulated arithmetic return. A uniform estimate over schedules cannot be inferred merely from the fact that one supported receiving kernel is nonzero. The exact frequency and matrix order supply the comparison.

6. Limits, provenance, and checking scope

This construction controls the two-column image transfer uniformly over the full block family, and replaces the global height/radix optimization by an exact finite-image cardinality infimum. The generator count in that infimum remains unbounded. The proof does not establish the optimal value for general L or general k .

The one-block prime choice has a real height cost: $\{1, L\}$ with L a large primorial has fixed $|H_5|=25$, while every prime factor of L fails the observation. The amplification construction retains that logarithmic cost before proving that its exponential per-generator contribution tends to zero. It does not set the cost to a constant.

There is also a genuine nonattainment phenomenon at $k=3$: $|H_5(A)| > 3^{|A|}$ for every nonempty admissible A , but powers of three have $|H_5|=2 \cdot 3^{n-1}$. Thus the capacity equals 3 without a minimizing finite block. Finite-state control of a specified block is not being used as a proof that the all-block infimum is attained.

The standalone finite evidence files are copied byte-for-byte from the EP817 contribution. The auditor imports neither its producer nor its helper library. The generic prime lemma is checked at several coefficient radii; the universal matrix-pattern theorem rests on the proof in Section 4, not an exhaustive enumeration of every integer pattern matrix. No theta-source primitive, analytic quotient, nilpotent spectral packet, or arithmetic moment is claimed to have been evaluated by these finite checks.

The original $k=4$ construction and mathematical proof remain credited to the anonymous/deleted Reddit contributor. sneed-and-feed retains the distinct finite-upper-bound Lean credit. Existing SplitZero source and verification pins are unchanged.

Source roles

KokunoYumeto. (2026). *Reconstruction with changes of support index*. `zeta-function-research-reader`, revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, `support_diagrams.tex`, D6--D8. Actual reconstructed quotient and additional observation-kernel interface.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). Original extremal definitions and benchmark; no imported all-block lower estimate.

The Clankers. (2026, September 16). *All-block image capacity and exact ternary transfer*. Companion proof, including the fully elementary prime-budget calculation and actual mixed-block matrices.

Universal image observations, bounded-fibre resolutions, and spectral capacity

Source: `history/EP817_SEVEN_OBSERVABLE_20260916/payloads/ep817/research/seven_observable/notes/seven-observable-capacity.md` **Identity:** `6aaae799ab5dabb576d00f2144c8d75be8b782052f0330fe150481f1d0a58eff` The Clankers. 16 September 2026.

Evidence status. This contribution contains ordinary mathematical proofs and exact, explicitly bounded computational replays. It is submitted for independent mathematical review. No new Lean elaboration is claimed. The original anonymous/deleted Reddit contributor retains credit for the workbench's four-term construction and signed-block argument; **sneed-and-feed** retains separate credit for the finite-upper-bound Lean formalization. None of those formal certificates is extended by this note. No general priority claim is made for finite-state positional arithmetic, inclusion-exclusion, augmented simplex contractions, or finite-dimensional spectral theory.

The immediate predecessor is *All-block image capacity and exact ternary transfer*, in `EP817_ALL_BLOCK_IMAGE_20260916.zip`. That complete archive is retained unchanged with this delivery. The current results extend its two-dimensional ternary transfer to every coefficient arity, determine the sharp universal linear count dimension at arity five, and give a dimension-free multiplicative error for each homogeneous block. The resulting spectral variational formula concerns the full family of blocks; its remaining infimum is not evaluated here.

1. Original arithmetic objects and principal results

Let A be a finite set of distinct positive integers. Put $S(A)=\text{sum}(A)$, $n=|A|$, and

$$H_q(A) = \left\{ \sum_{a \in A} c_a a : 0 \leq c_a < q \right\}, \quad q \geq 2.$$

The numerical values are counted once. Their original word multiplicities are denoted by $\mu_{-}(A,q)(y)$. An admissible generator set for progression length $k \geq 3$ is one whose binary image $H_2(A)$ has no nonconstant ordered k -term arithmetic progression. Let $g_k(n)$ be the least possible largest generator, and let $\lambda_k = \lim_n g_k(n)^{1/n}$. The existence and the finite-image characterization needed here are proved again in Section 10.

For a canonical block (A,b) , meaning $b > S(A)$, retain

$$A_b^{[m]} = \bigcup_{j=0}^{m-1} b^j A, \quad L_m(D;b) = \left\{ \sum_{j < m} b^j d_j : d_j \in D \right\}, \quad D = H_q(A).$$

All $m \cdot n$ generators of $A_b^{[m]}$ are distinct. Their q -ary image is literally $L_m(D;b)$, including all word collisions. Its maximum possible digit is $(q-1)S(A)$, which may exceed b . The actual digit values are never replaced by their residues.

Theorem 1 (universal finite observation). For every fixed q , every canonical block and every changing-block schedule, all distinct-image counts are computed by a nonnegative integer matrix on $2^{(q-2)}$ specified translated-image observations. On the actual centrally symmetric images H_q , reflection gives dimension

$$d_q = 1 + \frac{1}{2} \sum_{d=1}^{q-2} \left(2^{d-1} + 2^{\lceil (d-1)/2 \rceil} \right). \quad (1.1)$$

In particular $d_2=1$, $d_3=2$, $d_4=4$, and $d_5=7$. This count dimension does not depend on the generator count, radix, number of levels, or forbidden progression length. The raw count-and-moment carrier through moment order p has dimension $(p+1)2^{(q-2)}$.

Theorem 2 (sharp fifth-arity linear dimension). Seven observations suffice and are necessary for a universal linear tail-state count realization, even on the finite subfamily of five-admissible concatenations displayed in Section 4. The lower bound is an actual 7 by 7 numerical-image matrix of determinant 1,228,800. It excludes a smaller universal linear representation of those concatenation counts, not arbitrary nonlinear encodings or restricted subclasses.

Theorem 3 (uniform block-length control). Let D be finite, nonempty, contain 0, and lie in $[0, (q-1)(b-1)]$. Write $W = \max D$ and

$$C_D = \max \left\{ 1, \left\lceil \frac{W}{b-1} \right\rceil \right\} \leq q-1.$$

Then $\rho = \lim_m |L_m(D; b)|^{1/m}$ exists and

$$\rho^m \leq |L_m(D; b)| \leq C_D \rho^m \quad (m \geq 0). \quad (1.2)$$

The ρ is the spectral radius of the complete applicable count matrix, rather than an eigenvalue selected after dropping an observation kernel. Its peripheral eigenvalues are semisimple. For $q=5$, the logarithmic per-generator error is at most $\log(4)/(mn)$.

Theorem 4 (seven-matrix global capacity). Let M_k be the class of nonempty canonical pairs (b, A) whose actual binary image, reduced modulo b , has no ordered nonzero-step k -AP. Composite moduli and repeated points in a modular progression are included. For every $k \geq 3$,

$$\lambda_k = \inf_{(b, A) \in \mathcal{M}_k} \rho(M_5(A, b))^{1/|A|}. \quad (1.3)$$

Here $M_5(A, b)$ is the specified seven-dimensional matrix, not a new arithmetic source. Its integer entries and the admissible pairs still form an unbounded family. Theorem 4 supplies neither an all-rank lower certificate nor a convergence modulus for this outer infimum.

2. Complete translated-image carrier

Fix $q \geq 2$. Set $C=q-1$ and $Q=q-2$, and define the raw shape set

$$\mathcal{R}_q = \{R \subseteq \{0, \dots, Q\} : 0 \in R\}.$$

For a nonempty finite integer set Y , retain the actual observations

$$F_R(Y) = |Y + R|, \quad R \in \mathcal{R}_q. \quad (2.1)$$

There are exactly 2^Q of them. They count numerical unions, not sums of word multiplicities.

Take a nonempty D subset $[0, C(b-1)]$ and $b \geq 2$. For a specified R , put $E_R = D + R$ as a numerical set. Its largest element is at most $Cb-1$. For every residue z in $[0, b-1]$ actually present in E_R , define

$$Q_{R,z} = \{(e-z)/b : e \in E_R, e \equiv z \pmod{b}\}, \quad h_{R,z} = \min Q_{R,z}, \quad R_{R,z} = Q_{R,z} - h_{R,z}. \quad (2.2)$$

The original quotient coordinates belong to $[0, Q]$, and $R_{R,z}$ belongs to R_q . Both the shift h and the entire original quotient set are retained. An absent residue contributes no term. The present singleton quotient $\{0\}$ is a different object and contributes the shape $\{0\}$.

The residue decomposition is the disjoint union

$$(D + bY) + R = \coprod_z (z + bh_{R,z} + b(Y + R_{R,z})). \quad (2.3)$$

For each z the displayed affine map has the exact inverse $(x - z - bh)/b$ on its numerical image. This proves (2.3), not just its cardinality.

Define

$$M_q(D, b)_{R,T} = \#\{z : R_{R,z} = T\}. \quad (2.4)$$

The matrix is nonnegative integral, and every row sum is at most b . Equation (2.3) proves

$$\boxed{\mathbf{F}(D + bY) = M_q(D, b)\mathbf{F}(Y).} \quad (2.5)$$

The initial vector for $Y = \{0\}$ is $v_R = |R|$. Repetition gives

$$|L_m(D; b)| = e_{\{0\}}^T M_q(D, b)^m v. \quad (2.6)$$

For changing levels (A_j, b_j) , the q -ary image has the original form

$$D_0 + b_0(D_1 + b_1(\cdots + b_{m-2}D_{m-1})), \quad D_j = H_q(A_j).$$

Consequently the product is $M_0 M_1 \dots M_{(m-1)} v$, with the least significant level on the left. The last radix does not alter that finite generator image, but it remains relevant if another level is appended or the word is repeated. No commutation of different matrices is assumed.

2.1 Every raw moment closes in the same shapes

For $p \geq 0$ put

$$F_{R,p}(Y) = \sum_{x \in Y+R} x^p.$$

Apply the binomial formula to the literal affine maps in (2.3). With $a_{-}(R,z) = z + bh_{-}(R,z)$,

$$\boxed{F_{R,p}(D + bY) = \sum_z \sum_{j=0}^p \binom{p}{j} a_{R,z}^{p-j} b^j F_{R_{R,z},j}(Y).} \quad (2.7)$$

The initial moments are $\sum_{(r \in R)} r^p$. Thus all raw moments through order p are computed by a block triangular matrix of size $(p+1)2^Q$. Its j -th diagonal block is $b^j M_q(D, b)$. Homogeneous generating functions are rational by the finite resolvent formula. Neither their raw moments nor their measures are obtained by declaring reflected tails equivalent; (2.7) uses the unreflected labels and original offsets.

3. Exact reflection transport and seven observations

For a shape R define $R^\vee = \max(R) - R$. If Y is centrally symmetric, write $c_Y = \min Y + \max Y$, so $c_Y - Y = Y$. The map

$$x \mapsto c_Y + \max R - x \quad (3.1)$$

is an inverse-to-itself affine bijection between $Y+R$ and $Y+R^\vee$. It proves equality of the two cardinalities and records how odd moments would change.

Every $H_q(A)$ is centrally symmetric under the original coefficient-complement involution $c_a \rightarrow (q-1)-c_a$. The symmetry constant is $(q-1)S(A)$. It also preserves every original representation multiplicity by a bijection of words.

Choose the lexicographically smaller shape from each reflection pair. For centrally symmetric D , folding the target shapes in (2.4) produces a nonnegative integer matrix on these representatives. Let E be the raw-coordinate embedding that duplicates the reflected coordinate. Then

$$M_q^{\text{raw}} E = E M_q^{\text{sym}}. \quad (3.2)$$

To see this directly, reflect $D+R$ about $c_{D+\max R}$. A residue z is sent to the unique residue of $c_{D+\max R}-z$. Its quotient set is reflected, with its actual minimum shift adjusted by the corresponding integer quotient. The original sets and affine maps in (2.3) are paired bijectively. Folded row coefficients agree. This proves (3.2) without imposing a relation on nonsymmetric tails.

A shape of span $d \geq 1$ contains both endpoints and an arbitrary subset of its $d-1$ interior positions. There are 2^{d-1} such shapes and $2^{\lceil (d-1)/2 \rceil}$ reflection-fixed shapes. Counting reflection orbits, plus the singleton, proves (1.1).

At $q=5$ the exact ordered basis is

$$\{0\}, \{0,1\}, \{0,2\}, \{0,1,2\}, \{0,3\}, \{0,1,3\}, \{0,1,2,3\}. \quad (3.3)$$

The omitted raw shape $\{0,2,3\}$ is related to $\{0,1,3\}$ by (3.1), not by an unrestricted equality of source sets. For $Y=\{0,1,3\}$, the difference of their union counts is -1 . This actual nonsymmetric example is retained as a regression.

4. Sharpness of the linear carrier

4.1 The raw observation basis is integral and complete

For a shape R put

$$C_R(Y) = \#\{t : t + R \subseteq Y\}.$$

Inclusion-exclusion on the actual translated copies $Y+r$ gives

$$F_R(Y) = \sum_{\emptyset \neq T \subseteq R} (-1)^{|T|+1} C_{\max T - T}(Y). \quad (4.1)$$

Order the shapes by cardinality. The full-subset term is signed reflection of R ; every proper-subset term has smaller cardinality. Thus the integer change-of-observation matrix U is unimodular. Reflection permutes its largest-cardinality blocks, with diagonal coefficients ± 1 .

Evaluate C_R on the finite set $Y=P$ for every shape P . A pattern of larger cardinality cannot fit into P ; patterns of equal cardinality fit exactly when $R=P$, since both contain zero. The evaluation matrix is unit triangular. It follows that the raw union observations are linearly independent and have a unimodular complete evaluation matrix. This proves sharp dimension 2^Q for this universal translated-union observation space.

On symmetric sets, incidence observations obey $C_R = C_{(R^{\vee})}$. The transformation (4.1) commutes with reflection and descends to an invertible rational transformation on reflection orbits. These remaining functions are independent: test them on

$$Y_P = P \cup (L - P), \quad L > 3Q.$$

Since $P \subseteq [0, Q]$, the inter-component gap is at least $L - 2Q > Q$. No pattern of width at most Q crosses the gap. Its incidence count is $C_R(P) + C_{(R^{\vee})}(P)$. The orbit evaluation matrix is triangular in cardinality, with diagonal one for a two-shape orbit and two for a reflection-fixed shape. Its determinant is nonzero. This proves the generic symmetric dimension d_q .

4.2 An integral seven-coordinate source packet

The seven symmetric sets

$$\{0\}, \{0,1\}, \{0,2\}, \{0,1,2\}, \{0,3\}, \{0,1,2,3\}, \{0,1,3,4\}$$

give the following matrix of observations (3.3):

$$E_7 = \begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 & 4 \\ 2 & 3 & 4 & 4 & 4 & 5 & 5 \\ 2 & 4 & 3 & 5 & 4 & 5 & 6 \\ 3 & 4 & 5 & 5 & 6 & 6 & 6 \\ 2 & 4 & 4 & 6 & 3 & 5 & 7 \\ 4 & 5 & 6 & 6 & 7 & 7 & 7 \\ 4 & 6 & 7 & 7 & 6 & 8 & 8 \end{pmatrix}, \quad \det E_7 = -1. \quad (4.2)$$

Thus the seven-coordinate observation has full integral image on this specified packet. Adding $Y_8 = \{0,2,3,5\}$ produces the original nonzero free-object relation

$$e_{Y_8} + e_{\{0,1\}} - e_{\{0,2\}} - e_{\{0,1,3,4\}}. \quad (4.3)$$

Its seven observations vanish. The eight-object source has rank eight; its observation quotient has rank seven and kernel rank one. This additional kernel is not confused with the original generator-word relations.

4.3 Seven states are necessary even for actual admissible concatenations

The sharper linear-realization lower bound uses only actual fifth-arity images of nonempty five-admissible sets. Take tail blocks, in this order,

$$(4), (1), (2), (3), (1,2), (1,6), (2,9). \quad (4.4)$$

Their binary images have at most four values, hence are five-admissible. Their seven-observation matrix E_A has determinant -10,240.

Take the prefix pairs

$$((1),3), ((1),4), ((2),3), ((3),4), ((1,2),5), ((1,3),5), ((1,6),8). \quad (4.5)$$

Every binary image in (4.5) is modular-five-free, with every nonzero step tested, including steps of small additive order. The rows of the actual count carrier that observe the final cardinality are

$$P = \begin{pmatrix} 1 & 2 & 0 & 0 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 2 \\ 0 & 2 & 0 & 1 & 0 & 2 & 3 \end{pmatrix}, \quad \det P = -120.$$

For a prefix (A_i, b_i) and a tail C_j , the actual combined generator set is A_i union $b_i C_j$. Its generators are distinct because $b_i > S(A_i)$. Reduction modulo b_i forces the lowest binary digits of any five-term progression to be constant; after subtraction and division, the tail admissibility forces the progression to be constant. Thus every one of the 49 concatenations is five-admissible.

Their numerical count matrix is

$$H_{ij} = |H_5(A_i \cup b_i C_j)| = P E_A^T, \quad \boxed{\det H = 1,228,800 \neq 0.} \quad (4.6)$$

Any representation of these counts by a d -dimensional tail vector and a prefix-dependent linear functional factors H through dimension d . Hence $d \geq 7$. The result applies a fortiori to a universal linear transfer realization on all

five-admissible concatenations, and also to the higher-k classes containing them. No minimality claim is made for every specialized k=3 or k=4 subfamily or for nonlinear encodings.

5. A bounded-fibre resolution at every original radix cut

Let X, Y be finite nonempty integer sets, with X contained in $[0, C(P-1)]$ and $P \geq 2$. The original map is

$$\pi : X \times Y \rightarrow X + PY, \quad (x, y) \mapsto x + Py. \quad (5.1)$$

If two pairs have the same value, their X coordinates differ by an integer multiple of P . The displayed interval admits at most C values in one residue. Therefore every fibre $I_{z \equiv \pi^{-1}(-1)(z)}$ satisfies

$$1 \leq |I_z| \leq C. \quad (5.2)$$

For canonical q -ary mixed-radix prefixes, $C=q-1$ always applies. Indeed $\max X$ is at most $(q-1)\sum_j P_j(b_j-1) = (q-1)(P-1)$. Thus

$$\boxed{\frac{|X||Y|}{q-1} \leq |X + PY| \leq |X||Y|.} \quad (5.3)$$

For a homogeneous D with width W and a cut after m digits, the sharper bound C_D in Theorem 3 follows from

$$\max L_m = \frac{W(b^m - 1)}{b - 1}.$$

The number of points in any residue modulo b^m is at most $1 + \text{floor}((W/(b-1))(1-b^{-m}))$, which is at most $\text{ceil}(W/(b-1))$ when $W > 0$. $W=0$ has singleton fibres. The $q-1$ bound is sharp in the unrestricted count theorem: at $q=5$, take the low generators $\{1, 2, 4\}$, cut $P=8$, and tail generators $\{1\}$; the value 24 has the four distinct local-value pairs $(0, 3), (8, 2), (16, 1), (24, 0)$. That example is not used as a claim of five-admissibility.

5.1 The complete integral resolution, not just its rank

For each numerical fibre I_z , retain every nonempty subset J of its original pair labels. Let C_j have basis (z, J) with $|J|=j+1$, for $j \geq 0$; let $C_{-1} = Z[X+PY]$. The augmentation sends every pair over z to e_z . The other differentials delete one ordered vertex with alternating signs. Thus $C_j = 0$ for $j > C$, and

$$\cdots \rightarrow C_2 \rightarrow C_1 \rightarrow C_0 \rightarrow C_{-1} \rightarrow 0 \quad (5.4)$$

is the direct sum of the actual augmented simplex complexes of the fibres.

Choose the least original pair v_z in each fibre, and let h insert it into a face, with its ordering sign; h kills a face already containing v_z . At degree -1 put $h(e_z) = e_{\{v_z\}}$. Cancellation of the alternating faces proves on each basis vector

$$\boxed{\partial h + h \partial = I, \quad \partial^2 = 0.} \quad (5.5)$$

All coefficients are integral. This proves exactness and retains the higher relations between pairwise collision relations. At arity five, fibres can have four vertices, so terms through C_3 can be necessary. Keeping only pair differences without their face relations is not the full complex.

With unit face bases and the prescribed coefficient Hilbert norm, each h_j is a signed partial isometry and has norm at most one. For $j \geq 0$, each differential column has at most $j+1$ entries, each row at most $C-j$ entries; the matrix row/column estimate gives

$$\|\partial_j\| \leq \sqrt{(j+1)(C-j)}. \quad (5.6)$$

For augmentation this reads $\sqrt{C}$. These bounds persist for countably many fibres and any coefficient Hilbert space: extend from finite supports by continuity. Equation (5.5) remains valid on the Hilbert direct sums, so the

completed complex is still contracted. Its constants depend on the coefficient arity, not the number of values, source rank, or radix horizon.

For support inclusions the face and value maps are natural. The chosen least-vertex section need not be natural. Its exact difference on a retained value is

$$e_{v_{\text{new}}} - e_{v_{\text{old}}} = \partial[v_{\text{old}}, v_{\text{new}}], \quad (5.7)$$

with the orientation sign retained. This is an original boundary, not a declared absent action.

5.2 Original representation masses and quotient metrics

Give a pair (x, y) its specified positive original mass $w(x, y)$, with source Gram $1/w(x, y)$. For actual generator images the mass is $\mu_{\text{left}}(x)\mu_{\text{right}}(y)$. On a numerical fibre set $W_z = \sum (I_z)w$. Its least-norm section is

$$s(e_z) = \sum_{(x, y) \in I_z} \frac{w(x, y)}{W_z} e_{(x, y)}, \quad \pi s = I, \quad G_{\text{quotient}}(z) = 1/W_z. \quad (5.8)$$

Pairing with every difference of two vertices gives zero, and expansion proves the full Pythagorean quotient identity. These actual masses can be much larger than C . In the separately specified unit distinct-pair metric, $W_z = |I_z| \leq C$ and the return to unit numerical-value mass has norm at most $\sqrt{C}$. Equation (5.8), rather than that unit-pair bound, governs the original word metric. The unit-face contraction estimate (5.6) is not silently asserted in a different weighted face metric.

For an explicit unbounded return cost, take the original generators $\{1, 3, \dots, 3^{(m-1)}\}$ and $q=5$. There are 5^m original coefficient words and exactly $2 \cdot 3^{m-1}$ numerical values. Thus the largest original mass satisfies

$$\mu_{\max} \geq \frac{5^m}{2 \cdot 3^m - 1}, \quad \|\text{id} : (G_{\text{quotient}}) \rightarrow (I_{\text{values}})\| = \sqrt{\mu_{\max}} \geq \sqrt{\frac{5^m}{2 \cdot 3^m - 1}}. \quad (5.9)$$

The finite images are intervals by their actual base-three recursion. The numerical bound follows from averaging the original masses. It grows exponentially even though each distinct-value cut has at most two representatives and its unit-face resolution has the uniform contraction. The source measures, not a failed inverse identity, explain this difference.

6. Uniform spectral estimates for the whole homogeneous source

Let $N_m = |L_m(D; b)|$. The actual concatenation map in Section 5 gives

$$N_m N_n / C_D \leq N_{m+n} \leq N_m N_n. \quad (6.1)$$

Subadditivity of $\log N_m$ proves existence of $\rho = \inf_m N_m^{1/m}$. For a direct proof, fix m and write a large index as $tm+r$. The upper inequality controls $\log N_{(tm+r)}/(tm+r)$ by $\log N_m/m$ and finitely many remainder terms. The lower defining infimum gives the opposite bound. Repeating the lower inequality in (6.1) and taking t -th roots also gives $\rho^m \geq N_m/C_D$. Consequently

$$\boxed{\rho^m \leq N_m \leq C_D \rho^m.}$$

This is Theorem 3. In particular

$$\left(\frac{N_m}{C_D}\right)^{1/(mn)} \leq \rho^{1/n} \leq N_m^{1/(mn)}, \quad 0 \leq \frac{\log N_m}{mn} - \frac{\log \rho}{n} \leq \frac{\log C_D}{mn}. \quad (6.2)$$

No diagonalization or condition number is needed to evaluate these endpoints. Matrix powering uses fixed dimension at fixed arity; integer bit lengths still depend on the actual input and m . The error estimate is not a bound on that input size or on the rank at which the outer extremal infimum is approached.

6.1 Every matrix spectral component is actually observed

On the count-coordinate space define the explicit auxiliary norm

$$\|x\|_v = \max_R \frac{|x_R|}{|R|}, \quad v_R = |R|.$$

This is a norm on observation coordinates, not the original word or quotient metric. For a nonnegative matrix P its induced norm is $\max_R (Pv)_R / v_R$. For every original word product of our carriers,

$$(Pv)_R = |L + R| \leq |R||L|,$$

and the singleton coordinate gives equality. Hence

$$\boxed{\|P\|_v = |L|.} \quad (6.3)$$

In particular, the spectral radius of M is exactly ρ . This follows either from the finite-dimensional spectral-radius formula for matrix powers or from the elementary Jordan decomposition and (6.3). Thus one does not accidentally use an unobserved spectral radius of a larger automaton.

Equation (6.2) bounds $\|M^m\|_v / \rho^m$ by C_D . A Jordan block of length greater than one on an eigenvalue of modulus ρ would make this ratio grow polynomially. Therefore every peripheral eigenvalue is semisimple. Interior Jordan blocks remain possible and are retained in Section 8. Every fifth-arity ρ is an algebraic integer of degree at most seven; sharpness of the state dimension does not assert sharp algebraic degree for a single block.

6.2 Moment and variance growth retain the physical scale

Suppose $W = \max D > 0$. For $p \geq 1$ the moment at m levels has the bounds

$$W^p (\rho b^p)^{m-1} \leq \sum_{x \in L_m} x^p \leq C_D \left(\frac{W}{b-1} \right)^p (\rho b^p)^m. \quad (6.4)$$

The lower bound uses the actual subset $W b^{m-1} + L_{m-1}$; the upper uses $\max L_m < W b^m / (b-1)$. Thus its exponential growth is ρb^p .

For $m \geq 2$, use the low cylinder with the top two digits zero and the high cylinder with both top digits W . Each contains N_{m-2} values. Every high value exceeds every low value by at least

$$\frac{W}{b-1} (1 - 2/b^2) b^m.$$

The variance identity for a uniform finite numerical set is the sum of squared unordered pair differences divided by its cardinality squared. These two cylinders and (6.2) give

$$\boxed{\frac{W^2 (1 - 2/b^2)^2}{(b-1)^2 C_D^2 \rho^4} b^{2m} \leq \text{Var}(\text{Unif}(L_m)) \leq \frac{W^2}{4(b-1)^2} b^{2m}.} \quad (6.5)$$

The upper bound follows from the interval support; its exact finite width $W(b^m - 1)/(b-1)$ can be substituted. The constants in (6.5) retain W and ρ . These uniform distinct-value statements do not replace original assignment-weighted moments.

6.3 Arbitrary schedules still need a limit hypothesis or a limiting inferior

Uniform cut control does not force every changing schedule to have a limiting rate. At stage j use $j!$ successive copies of $A = \{1\}$, with base 3 for odd j and base 5 for even j . Both actual binary digit sets are modular-three-free. Therefore every finite prefix is three-admissible.

For arity five every image is the interval

$$[0, 4 \sum_{i < m} P_i], \quad N_m = 1 + 4 \sum_{i < m} P_i.$$

The interval assertion follows by induction: the lower interval reaches at least P_{j-1} because each previous radix is at most 5, so the next translates overlap or adjoin. Also $\sum P_i$ is bounded above and below by fixed multiples of P_m , since the radices lie in $\{3, 5\}$. Thus $\log N_m - \log P_m$ stays bounded.

At the end of stage j the last $j!$ positions dominate all preceding stages, whose total divided by $j!$ tends to zero. The average log radix therefore approaches $\log 3$ along odd stage endpoints and $\log 5$ along even endpoints. Every intermediate average lies between them. Hence

$$\boxed{\liminf N_m^{1/m} = 3, \quad \limsup N_m^{1/m} = 5.} \quad (6.6)$$

The growing generators and scale product are exactly the original ones. The six finite stage endpoints in the receipt corroborate the explicit formula; the argument proves the infinite oscillation.

7. The full spectral variational formula

Let M_k be the canonical modular-safe pairs specified in Theorem 4. For each pair, its binary digit language is admissible at all lengths: reduce a progression modulo b , identify its common lowest actual binary digit, subtract it, divide by b , and induct. The digit values lie below b , so that lookup is the actual canonical inverse. Composite b causes no difficulty because the finite certificate tests every nonzero modular step.

Every prefix $A_b^{[m]}$ is therefore k -admissible. The all-block fifth-image theorem in Section 10 gives

$$\lambda_k \leq |H_5(A_b^{[m]})|^{1/(m|A|)}.$$

Taking m to infinity and using Section 6 proves

$$\lambda_k \leq \rho(M_5(A, b))^{1/|A|}. \quad (7.1)$$

Conversely, for any finite k -admissible A choose $b=4S(A)+1$. Its actual binary image is modular-safe because all integer second differences have absolute value at most $2S(A) < b$. The fifth-arity digits are below b , so all digit words are distinct and

$$|H_5(A_b^{[m]})| = |H_5(A)|^m.$$

Thus $\rho = |H_5(A)|$, and the all-block infimum gives the reverse inequality in (1.3). This proves Theorem 4. Enlarging the class to all canonical pairs whose binary languages are safe by the full carry criterion gives the same infimum, by the same two inclusions.

At $k=3$ this spectral infimum is attained by $A=\{1\}$, $b=3$, with

$$|H_5(A_b^{[m]})| = 2 \cdot 3^m - 1, \quad \rho = 3 = \lambda_3.$$

In contrast, no finite nonempty three-admissible A attains $|H_5(A)|^{1/|A|}=3$: its ternary assignments are distinct, so $|H_3(A)|=3^n$, and its fifth-arity set includes a strictly larger value. The two conclusions concern different explicitly connected infima, with the word-length limit retained in the spectral one. The standard three-term injectivity statement is proved in the cited primary literature and can also be recovered by expressing a short relation as a binary second difference and using positivity to handle an initially constant numerical tuple.

8. Exact examples and retained interior spectral data

All following matrices use (3.3). Their count series are checked by their exact characteristic identities and sufficient initial coefficients, with original generator images independently recomputed.

8.1 A genuinely quartic observed rate

For $A=\{3,10\}$, $b=19$, the binary residue set has four elements. Since 19 is prime, a nonzero modular step has order 19, so it cannot produce a five-term progression entirely inside those four residues. The original construction is five-admissible at every length.

The fifth-arity matrix is

$$\begin{pmatrix} 9 & 8 & 0 & 0 & 0 & 0 & 0 \\ 2 & 9 & 2 & 6 & 0 & 0 & 0 \\ 2 & 6 & 1 & 10 & 0 & 0 & 0 \\ 0 & 2 & 0 & 17 & 0 & 0 & 0 \\ 6 & 12 & 0 & 0 & 0 & 0 & 0 \\ 1 & 7 & 1 & 10 & 0 & 0 & 0 \\ 0 & 1 & 0 & 18 & 0 & 0 & 0 \end{pmatrix}.$$

Its characteristic polynomial is $x^7 f(x)$, where

$$f(x) = x^4 - 36x^3 + 382x^2 - 1116x + 65.$$

There is one root in each interval $(0,1), (4,5), (12,13), (18,19)$, as verified by the endpoint signs; degree four gives exactly those four real roots. The largest is ρ . Modulo three, f becomes $x^4 + x^2 + 2$. It has no linear root and is divisible by none of the three monic irreducible quadratics x^2+1 , x^2+x+2 , x^2+2x+2 . Therefore f is irreducible over $\mathbb{Q}$. The observed rate is genuinely quartic.

Its complete count series is

$$\sum_{m \geq 0} N_m z^m = \frac{1 - 11z + 43z^2 + 87z^3}{1 - 36z + 382z^2 - 1116z^3 + 65z^4}. \quad (8.1)$$

Here $C_D=3$. Theorem 3 gives $\rho^m \leq N_m \leq 3\rho^m$ with $18 < \rho < 19$. This example shows a real strict spectral improvement on its own radix cost, but it is not a competitive global $k=5$ record.

8.2 A surviving interior Jordan chain

For $A=\{7,10\}$, $b=19$, again every binary level is modular-five-free. The seven-dimensional matrix has characteristic polynomial

$$(x-19)(x-13)^2(x-5)(x-2)(x-1)(x+3).$$

In the original coordinates set

$$v = (2, 0, 0, 0, 1, 0, 0)^T, \quad w = (-227, 198, 132, 0, 0, 120, 0)^T.$$

Then

$$(M - 13I)v = 0, \quad (M - 13I)w = 462v.$$

The observed count series is

$$\sum_{m \geq 0} N_m z^m = \frac{1 - 25z + 243z^2 - 795z^3}{(1-5z)(1-13z)^2(1-19z)}. \quad (8.2)$$

The numerator is nonzero at $z=1/13$, so the double pole and its m^{13} contribution survive. This does not contradict peripheral semisimplicity: $13 < \rho = 19$, and N_m remains between 19^m and $4 \cdot 19^m$.

For comparison, $A=\{6,9\}$, $b=17$ has characteristic polynomial $(x-17)^3(x-3)^3(x+1)$, while the squarefree polynomial $(x-17)(x-3)(x+1)$ annihilates the actual matrix. Its repeated top eigenvalue is semisimple, and

$$N_m = (8 \cdot 17^m - 3^m)/7.$$

The original common factor three in the generators is not divided out.

8.3 The existing record sources

For $B_6 = \{1, 4, 5, 17, 21, 22\}$, its fifth-arity image is the interval $[0, 280]$. Its matrices at bases 97 and 93 have $\rho_3 = \rho_5$ equal to the respective radix. For $B_{10} = \{3, 4, 7, 34, 37, 41, 216, 250, 253, 257\}$, at base 1651 the ternary and fifth-arity spectral radii are both 1651. The same equality is checked for the original block $\{1, 7, 8\}$ at base 19.

The full-base root is certified by the two exact facts $\det(bI - M) = 0$ and every row $\text{sum} \leq b$. The latter bounds the spectral radius above; the former supplies the eigenvalue. Both arity matrices are constructed from the actual numerical images. These particular formulas were available in the preceding packages; they are regression inputs and applications of the new uniform framework, not relabelled new record constructions.

9. Faithful re-encoding: exact scope and a fixed-block barrier

For a canonical block (A, b) , let $C_m = A_b^{[m]}$. Let r_m be the least integer modulus ≥ 2 on which the actual numerical image $H_3(C_m)$ reduces injectively; let p_m be the least prime with that property. Both exist. Write ρ_3 and ρ_5 for the two actual image spectral radii and $L_b = \lceil \log_2 b \rceil$. Section 10's elementary prime estimate gives

$$\rho_3^m \leq |H_3(C_m)| \leq r_m \leq p_m \leq 256 + 2(|H_5(C_m)| - 1)(mL_b + 2) \leq 256 + 8\rho_5^m(mL_b + 2). \quad (9.1)$$

The height bound uses $S(C_m) \leq b^{m-1}$, hence $1 + \lfloor \log_2(2S(C_m)) \rfloor \leq mL_b + 2$. No generator is changed to a least residue.

Faithfulness on H_3 preserves the binary second-difference equations. If C_m is k -admissible, its binary residues are modular- k -free. Each actual generator is nonzero modulo the modulus, so the repeated scaled families are distinct even when the actual generator exceeds the modulus: a cross-level equality would make one of the original generators zero modulo it. Lookup of the unique actual binary digit of a residue, subtraction of that digit, and division give the inverse at every level.

When $\rho_3 = \rho_5 = \rho$, (9.1) proves

$$\lim_m r_m^{1/m} = \lim_m p_m^{1/m} = \rho. \quad (9.2)$$

Every faithful modulus sequence has lower per-generator exponential rate at least $\rho^{1/|A|}$, and prime choices attain it asymptotically. For all four record pairs in Section 8.3, $\rho = b$. Therefore faithful H_3 residue re-encodings of their full repeated blocks cannot improve their existing exponential rates. This conclusion covers every such modulus and every horizon, not just a searched prime range.

The class is deliberately stated. It maps into the larger class of modular- k -free binary observations by the second-difference argument. The inclusion can be strict: B_6 has 139 distinct ternary values, more than 97, although its binary image is modular-five-free at 97. Thus (9.2) is not a lower bound against every possible observation or a different block family.

For a strict spectral example, take $A = \{1, 9\}$, $b = 27$. The ternary and fifth-arity radii are respectively 9 and 17. The actual fifth-arity matrix satisfies $M^2 = 17M$, and

$$|H_5(C_m)| = 25 \cdot 17^{m-1} \quad (m \geq 1).$$

Faithful prime choices at $m = 1, 2, 3$ are respectively 13, 151, 1091. The unchanged generator lists, every preceding rejected prime with a nonzero original bounded-difference witness, and the exact digit inverses are in the receipt. These re-encodings preserve three-term exclusion; their asymptotic upper cost is at most $\sqrt{17}$, strictly below the original $\sqrt{27}$. The known global three-term rate is 3, so this is a checked mechanism rather than a new numerical record.

10. The inherited capacity identity, proved with its height cost

This section supplies the finite-image interface used in Theorem 4. It also specifies precisely what is imported from the immediately preceding note.

10.1 Existence of the maximum-generator rate

For admissible A, B set $R=2S(A)+1$ and $C=A$ union RB . The map

$$H_2(A) \times H_2(B) \rightarrow H_2(C), \quad (x, y) \mapsto x + Ry$$

is a bijection with the original division inverse. A progression pulls back to progressions in both factors: each lowest second difference is a multiple of R in $[-2S(A), 2S(A)]$ and hence zero. A nonconstant target progression has a nonconstant factor. The original sum cost obeys

$$2S(C) + 1 = (2S(A) + 1)(2S(B) + 1).$$

Minimizing that cost on n generators gives a submultiplicative sequence $F_k(n)$. Splitting a large n into blocks of a fixed length proves that its n th-root limit is $\inf_n F_k(n)^{1/n}$. The exact inequalities

$$2g_k(n) + 1 \leq F_k(n) \leq 2ng_k(n) + 1$$

give the same limit for g_k . Powers of three supply admissible sets at every n . Thus

$$\lambda_k = \inf_{A \text{ } k\text{-admissible}} (2S(A) + 1)^{1/|A|}. \quad (10.1)$$

10.2 Elementary prime avoidance

Let $h \geq 1$ and A be a fixed positive generator set. The nonzero bounded differences in $\sum_a [-h, h]a$ occur in opposite pairs. Let K be their number of positive values and put $\ell = 1 + \text{floor}(\log_2(hS(A)))$. Then there exists a prime

$$p \leq 256 + 4K\ell \quad (10.2)$$

dividing none of those positive differences.

Here is a complete prime-product bound. For integer $N \geq 256$ put $m = \text{floor}(N/2)$. The central binomial coefficient $\text{binom}(2m, m)$ divides $\text{lcm}(1, \dots, N)$: at each prime, each term in its factorial valuation difference is zero or one, and there are at most $\text{floor}(\log_p N)$ such terms. Also $\text{binom}(2m, m) \geq 2^{2m}/(2m+1)$. Higher prime powers contribute at most $\sqrt{N} \log N$ to the logarithm of the least common multiple. Consequently, with $\theta(N) = \sum_{p \leq N} \log p$,

$$\theta(N) \geq (N-1) \log 2 - \log(N+1) - \sqrt{N} \log N > N/4.$$

For the last bound, use $2/3 < \log 2 < 3/4$, monotonic decrease of $\log(N+1)/N$ and $\log N/\sqrt{N}$ in this range, and $\log 257 < \log 512 = 9 \log 2$. The resulting lower coefficient is $85/128 - 27/1024 - 3/8 = 269/1024 > 1/4$. The elementary inequalities for $\log 2$ follow, for example, by convexity and the chord bound for $1/x$ on $[1, 2]$.

The product Q of the K positive bounded differences has $\log Q \leq K \log(hS(A)) < K\ell$. If every prime up to $256 + 4K\ell$ divided Q , the lower bound just proved would give $\log Q > 64 + K\ell$. This contradiction proves (10.2).

For $h=2$, the positive difference count is $K = (|H_5(A)| - 1)/2$. The successful prime makes reduction injective on the distinct values of $H_3(A)$. Every modular binary k -AP then has equal adjacent ternary values $x_i + x_{i+2}$ and $2x_{i+1}$; injectivity converts their equations to integer equations. Original admissibility makes the tuple constant. The actual binary digit lift is safe at every length by the lookup/subtract/divide map, even if original weights exceed p .

10.3 Amplification keeps and then removes the exponential height contribution

Fix admissible A with n generators and put $h_5 = |H_5(A)|$, $R = 4S(A) + 1$. The original block

$$B_t = \bigcup_{j < t} R^j A$$

has tn distinct generators, sum $(R^t - 1)/4$, is k -admissible, and has exactly h_5^t fifth-arity values. The prime lemma gives

$$p_t \leq 256 + 2(h_5^t - 1)t\ell_R, \quad \ell_R = 1 + \lfloor \log_2 R \rfloor.$$

Its unchanged-weight prime lift proves $\lambda_k \leq p^{-t/(tn)}$. Therefore

$$\lambda_k \leq h_5^{1/n} (2t\ell_R + 256h_5^{-t})^{1/(tn)}.$$

The final factor tends to one for that specified A . Taking the infimum proves $\lambda_k \leq \inf_A |H_5(A)|^{1/|A|}$.

For the reverse inequality, minimizing $|H_5|$ at rank n gives a submultiplicative sequence under sufficiently separated composition, hence its root limit equals the infimum over all ranks. On a max-optimal admissible set, $|H_5(A)| \leq 4n g_k(n) + 1$. The root limit is therefore at most λ_k . This proves

$$\lambda_k = \inf_{A \text{ } k\text{-admissible}} |H_5(A)|^{1/|A|}.$$

The finite construction retains its large prefactor $R^{(t-1)\max A}$. No height-independent finite modulus bound is asserted for one unamplified block. Neither this proof nor the seven-state reformulation establishes equality with arity three or four in general.

11. Observation quotients and the SplitZero receiving interface

The seven counts are not a linear map on individual generator-word amplitudes: union cardinality is nonlinear on such a presentation. The correct linear source for the count observation is the free module on the specified finite numerical-set objects.

For a finite packet P of centrally symmetric nonempty integer sets, let $V_P = Z[P]$ and

$$J_P e_Y = (|Y + R|)_{R \in \mathcal{R}_q/\vee}.$$

Its quotient is the actual image lattice $J_P(V_P)$, not automatically all of $Z^{(d_q)}$. Formula (4.2) supplies a particular packet where it is all of Z^7 . Formula (4.3) supplies a nonzero original source element in an additional packet's kernel.

A block acts on the set-object basis by $e_Y \rightarrow e_{(D+bY)}$. Its exact observation square is

$$J_{T(P)} \mathcal{T}_{D,b} = M_q^{\text{sym}}(D, b) J_P.$$

Thus observation kernels are transported to observation kernels. Packet inclusions give the original natural diagrams and their internal quotients. In the SplitZero reconstruction, a relation maps to its own packet-labelled zero; the empty support remains the external bottom. The source interface is the one in `support_diagrams.tex`, (D6)–(D8), at the pinned revision in the references.

The simplex resolution of Section 5 is a different, explicitly related presentation: it resolves actual pair-to-value fibres, and (5.8) retains their original masses. The set-object observation is not asserted to identify those entire value modules with seven coordinates. Its additional kernel must remain in any comparison. The change from union observations to incidence observations in (4.1) is unimodular; it preserves the observation kernel, with the transported Gram $U^{(-T)} G U^{(-1)}$ if a metric was specified. It is not an isometry by declaration.

The contribution therefore transfers exact support quotients, original boundary primitives, a uniform bounded-fibre contraction, and explicit observation dynamics. It does not claim a finite replacement for the original infinite theta quotient or a Zeta arithmetic weight estimate.

12. Verification and remaining general question

The dependency-free producer and the independently implemented auditor accompany this note. The latter imports neither the producer nor its carrier library. It reconstructs raw matrices by interpolation on literal finite numerical joins, whereas the producer constructs them from exact residue fibres. It recomputes selected original large-generator moments by bitwise bit-image arithmetic, uses Gaussian rational elimination instead of the producer's fraction-free determinants, and contracts each simplex to the last rather than the first vertex.

The exact completed domains are recorded in the receipts:

- every D subset $[0, (q-1)(b-1)]$ containing zero, for $q=2, \dots, 5$ and $b=2, 3, 4$: 5,050 alphabets, 15,150 literal joins, 45,450 raw moment-vector checks, and 384 symmetric-source intertwining checks;
- all nonempty positive subsets of $[1, 8]$ of size at most three, arities two through six, and distinct radices $S+1, S+2, 2S+1$: 1,375 original two-level generator comparisons;
- 140 specified changing schedules through four levels, with 210 macro-cut inequalities and exact observation-norm comparisons;
- raw and symmetric dimension witnesses through arity seven, the complete seven-coordinate integral packet, all 49 actual five-admissible Hankel concatenations, and the additional observation-kernel example;
- all augmented-simplex basis identities for fibre sizes one through six: 126 identities, plus 2,828 actual numerical cut fibres and their original weighted sections;
- 18 specified original block/arity matrices, 36 direct one-/two-level images, 108 first and second moment entries including counts, exact characteristic identities, and count vectors through length ten;
- three unchanged-weight prime re-encodings with every preceding rejected prime, six factorial-schedule stage endpoints, eight original-word metric-return examples, and the stated invalid-input and nonsymmetric-observation regressions.

The independent auditor reproduces all 5,050 raw matrices by its alternative construction and checks each on an additional tail; it also replays the complete 1,375-generator domain, the 18 selected cases, the prime data, the integral minimality and Hankel certificates, and the 2,828 cut-fibre counts. Eight deliberately corrupted records are rejected. Its full independent scope is stated in its receipt; it does not claim an independent replay of every one of the producer's 45,450 general raw-moment comparisons.

The ordinary theorems quantify over every finite block and every stated infinite homogeneous length. The finite tests certify their concrete inputs and check the formulas on the displayed domains; they are not substituted for the proofs.

What remains is an inequality over all admissible arithmetic blocks, or a construction violating a proposed one. The rank does not enter the size of the fifth-arity count carrier or its uniform length error, but it remains in the generator reward and in the allowed integer entries. There is no proved cutoff on the rank needed to approach the global infimum. The faithful-compression barrier in Section 9 also concerns its precisely specified class, not all possible modular observations. No new best numerical bound or complete solution of Erdős Problem 817 is asserted.

References and exact source roles

The Clankers. (2026, September 16). *All-block image capacity and exact ternary transfer* [Preceding research note]. EP817_ALL_BLOCK_IMAGE_20260916.zip, `research/all_block_image/notes/all-block-image-capacity.md`. Role: immediate predecessor and provenance for the finite-image capacity/prime interface, re-proved in Section 10; its archive is retained unchanged.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1) [Preprint]. <https://arxiv.org/html/2606.24139v1>. Inspected: definitions, three-term injectivity, and general exponential benchmarks. Role: surrounding problem and prior three-term result; its June version is not used as a September status report about the separate historical prefactor question.

Frougny, C., & Pelantová, E. (2018). *Two applications of the spectrum of numbers* (arXiv:1512.04234v3; original preprint 2015) [Preprint]. <https://arxiv.org/abs/1512.04234>. Source entry inspected for classical finite-state positional-arithmetic context; no uninspected theorem is an input to the carrier or sharpness proofs above.

Green, B., & Ruzsa, I. Z. (2006). Sets with small sumset and rectification. *Bulletin of the London Mathematical Society*, 38(1), 43–52. <https://arxiv.org/abs/math/0403338>. Source entry inspected for classical finite-additive-model context. The prime lemma here is the explicit elementary proof in Section 10 and retains every original generator.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. KokunoYumeto/zeta-function-research-reader, revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, workbenches/splitzer-o-tandem/tex/support_diagrams.tex, equations D6–D8; Git blob d3493f891291ee6e94dbf2c77649f7d85d240df2. Role: original support reconstruction, internal quotient, and additional receiving-kernel distinction. It supplies no unstated analytic or arithmetic estimate to this contribution.

Seven image observations and the original supported quotient

Source: history/EP817_SEVEN_OBSERVABLE_20260916/payloads/zeta/workbenches/ep817-seven-observable/notes/splitzero-receiving-map.md **Identity:** b064cbc73a4582fa2bb7aaf45aac4357eea1a0c62eccaf307ca3af16a8f23305 The Clankers. 16 September 2026. Additive receiving note for the SplitZero workbench. The complete arithmetic and operator proofs are in the accompanying `seven-observable-capacity.md`. These are ordinary proofs and exact finite replays, not new Lean certificates or an arithmetic weight theorem for the theta source.

1. Two sources, with their actual comparison maps

The finite count carrier has seven coordinates at coefficient arity five. Those coordinates do not replace the free module on generator words or the free module on distinct numerical values. The latter can have arbitrarily large rank. The new finite source is an observation quotient of a free module on *finite numerical-set objects*.

For a nonempty finite centrally symmetric integer set Y and a shape R , define

$$F_R(Y) = |Y + R|.$$

The seven shapes are $\{0\}$, $\{0,1\}$, $\{0,2\}$, $\{0,1,2\}$, $\{0,3\}$, $\{0,1,3\}$, and $\{0,1,2,3\}$. A missing residue is not one of those shapes. The omitted eighth raw shape $\{0,2,3\}$ has the same observation only on centrally symmetric Y , via the actual bijection

$$x \mapsto (\min Y + \max Y) + 3 - x.$$

On $Y=\{0,1,3\}$ the two raw observations differ by -1. This unpaired value is retained before imposing the symmetry hypothesis.

Let P be a finite packet of actual symmetric numerical sets, $V_P = \mathbb{Z}[P]$, and

$$J_P : V_P \rightarrow \mathbb{Z}^7, \quad e_Y \mapsto (F_R(Y))_R.$$

Its target image is the literal lattice $J_P(V_P)$. Packet inclusions transport source bases by inclusion and image lattices by their actual inclusion. Thus

$$0 \rightarrow \ker J_P \rightarrow V_P \rightarrow J_P(V_P) \rightarrow 0$$

is a natural diagram of original internal quotients. A seven-set packet in the full note has determinant -1 and hence image $\mathbb{Z}^7$. An eight-set packet has kernel rank one, generated by

$$e_{\{0,2,3,5\}} + e_{\{0,1\}} - e_{\{0,2\}} - e_{\{0,1,3,4\}}.$$

This is a nonzero element at its original packet support. Its receiving coefficient is zero; it is not external absence. For a symmetric digit set D and base b , the block map on set-object bases is

$$\mathcal{T}_{D,b} : e_Y \mapsto e_{D+bY}.$$

The residue construction in the main proof gives a nonnegative integer matrix $M(D,b)$ and the exact square

$$J_{\mathcal{T}(P)} \mathcal{T}_{D,b} = M(D,b) J_P.$$

Consequently $\ker J_P$ is taken into $\ker J_{\mathcal{T}(P)}$. The induced map on the internal quotient is the restriction of the actual matrix to the image lattice. None of this assumes that arbitrary word amplitudes have a linear union-cardinality map.

The original SplitZero reconstruction applies to this packet-indexed diagram. Support join is packet union and bottom is the empty packet. Addition transports coefficients to the join. A supported zero remains $(P,0)$; the external scalar zero sends it to bottom. These are the same operations and quotient conventions as the inspected `support_diagrams.tex`, equations D6--D8. No replacement scalar or cohomology definition is introduced.

2. The original pair-to-value quotient and its complete resolution

For a canonical radix cut retain the accumulated scale P_0 and the actual q -ary numerical images X and Y . The map is

$$\pi : X \times Y \rightarrow X + P_0 Y, \quad (x, y) \mapsto x + P_0 y.$$

Every X lies in $[0, (q-1)(P_0-1)]$, so each numerical fibre has at most $q-1$ original pairs. This proves the uniform count inequality, but the full original relations are also retained.

For each value z let I_z be its fibre. Define C_j with basis (z, J) , where J is a $(j+1)$ -element subset of I_z , and let $C_{-1} = Z[X + P_0 Y]$. Alternating deletion of a pair gives the boundary; augmentation sends each pair over z to e_z . This is the actual augmented simplex resolution of every fibre.

Selecting the least original pair gives the explicit cone homotopy. On the unchanged unit face bases,

$$\partial h + h \partial = I, \quad \partial^2 = 0, \quad \|h\| \leq 1, \quad \|\partial_j\| \leq \sqrt{(j+1)(q-1-j)}.$$

These constants are independent of block rank and horizon. For countably many fibres, Hilbert direct sum and the same bounded maps preserve the identities. The augmented receiving space and each support label remain in the completion.

At fifth arity, the original fibre may have four vertices. Its triangular and tetrahedral relations are therefore part of the resolution. Keeping the pair-difference generators while dropping their higher relations would change the complex.

The chosen section can change under support extension. Its difference is the specified old/new pair boundary

$$e_{v_{\text{new}}} - e_{v_{\text{old}}} = \partial[v_{\text{old}}, v_{\text{new}}].$$

The face maps and quotients are natural without declaring this arbitrary section itself natural.

This resolution is distinct from the set-object count observation in Section 1. The former retains each numerical value and its original pair representations. The latter observes seven aggregate counts of entire numerical sets. The receiving-kernel distinction in D7--D8 identifies where the extra information is removed; it does not identify the two sources.

3. Preserve the specified source measures

For original generator-word masses, the pair (x, y) has mass

$$w(x, y) = \mu_{\text{left}}(x) \mu_{\text{right}}(y).$$

Write $W_z = \sum_{(x, y) \in I_z} w(x, y)$. In the original reciprocal-mass Gram, the minimum section is

$$s(e_z) = \sum_{(x, y) \in I_z} \frac{w(x, y)}{W_z} e_{(x, y)}, \quad G_{\text{quotient}}(z) = 1/W_z.$$

It is orthogonal to every pair-difference boundary and satisfies the exact Pythagorean identity. A unit distinct-pair source instead has $W_z = |I_z| \leq q-1$ and return norm at most $\sqrt{q-1}$. The original word mass W_z may be arbitrarily large; the unit-pair estimate is not silently applied to it. For the three-admissible generators $\{1, 3, \dots, 3^{m-1}\}$ at $q=5$, the original 5^m words have only $2 \cdot 3^m - 1$ numerical values. Therefore the original quotient-to-unit-value return norm is at least $\sqrt{5^m / (2 \cdot 3^m - 1)}$, exponentially growing, while every numerical cut has at most two pair representatives. Both observations and their metrics remain attached to their own sources.

The aggregate coordinate norm used for spectral control is another explicitly specified norm:

$$\|v\|_{\text{obs}} = \max_R |v_R|/|R|.$$

Its role is only to measure the seven count observations. For the positive transfer of an actual word,

$$\|M_0 \cdots M_{m-1}\|_{\text{obs}} = |L_m|.$$

The equality follows from $|L+R| \leq |R||L|$ and equality in the singleton coordinate. It is not an isometry claim for the original word or quotient metric.

For a repeated block with digit width W and base b , set $C = \max(1, \text{ceil}(W/(b-1)))$. The exact bounded-fibre quotient gives

$$\rho^m \leq |L_m| \leq C\rho^m, \quad C \leq q-1.$$

This controls every homogeneous horizon with a constant independent of source rank. In particular the complete count matrix's Perron root is observed, and its peripheral eigenvalues are semisimple. Interior nilpotents survive: the main note gives an actual fifth-arity matrix with eigenvalue 13, vectors v, w satisfying $(M-13)v=0$ and $(M-13)w=462v$, and a surviving double pole in the numerical count series. The coefficient and its receiving contribution are both retained.

4. Observation coordinates, integral changes, and kernel identity

For each shape R the incidence observation is

$$C_R(Y) = \#\{t : t + R \subseteq Y\}.$$

Inclusion-exclusion gives the integral change

$$F_R(Y) = \sum_{\emptyset \neq T \subseteq R} (-1)^{|T|+1} C_{\max T - T}(Y).$$

The transformation is unimodular, so the two original observation maps have the same kernel and isomorphic image lattices. If a Gram G has been specified in one observation basis, it is transported as $U^*(-T)G U^*(-1)$, not reassigned to the coordinate identity. Reflection then takes a separate quotient only on the symmetric source; the nonsymmetric defect remains an explicit counterexample to imposing it globally.

The seven-coordinate realization is sharp as a linear count model. A 7 by 7 matrix of actual five-admissible prefix/tail concatenation counts has determinant 1,228,800. Therefore a tail representation with linear prefix readouts has dimension at least seven even on that finite arithmetic subfamily. This is a rank theorem about this stated observation problem, not a claim that every aspect of arithmetic cohomology has dimension seven.

5. Return to the arithmetic capacity and the faithful-prime kernel

For each $k \geq 3$ the main proof gives the exact formula

$$\lambda_k = \inf_{(b,A) \in \mathcal{M}_k} \rho(M_5(A,b))^{1/|A|}.$$

Every pair has distinct positive actual generators, $S(A) < b$, and a tested modular k -AP-free binary image. The reverse map sends an arbitrary integer admissible A to $b = 4S(A) + 1$; the forward map repeats its unchanged block. The required prime-selection and image-capacity proof is included with all height constants, rather than importing an unevaluated source comparison.

For a fixed source C , let $\Phi: Z^C \rightarrow Z$ be its original weight map, $K = \ker \Phi$, and $K_p = \Phi^{-1}(pZ)$. The additional congruence observation has the exact sequence

$$0 \rightarrow K \rightarrow K_p \xrightarrow{\Phi} p\mathbb{Z} \cap \text{im } \Phi \rightarrow 0.$$

A prime faithful on the numerical $H_3(C)$ introduces no additional zero on that specified bounded image. It does not have zero kernel on all of Z . No ring map from $\mathbb{Q}$ to a finite field is used.

The observed ternary and fifth-arity spectral radii give lower and upper bounds on every such faithful re-encoding. When they agree, the optimal asymptotic cost of this entire residue-faithful class is exactly that common radius. The earlier record blocks have equal radii at their stated bases, so this type of faithful repetition/compression cannot lower those bases' exponential costs. Other modular observations may allow ternary collisions while still excluding the required binary progression; that larger class is related by the explicit implication, not identified with the faithful one.

6. Scope of the transfer

The new infinite control consists of the bounded-fibre Hilbert contraction, the rank-independent count matrix, and the uniform homogeneous spectral error. The free set-object observation kernel, the original pair-to-value kernel, the word masses, and the congruence kernel are all specified separately.

What remains unbounded is the admissible family of arithmetic blocks and their generator reward. Neither the seven-dimensional count carrier nor the contraction provides an all-rank inequality on that family by itself. The theta-function source, its original quotient, its finite packet kernel, and its analytic metric comparisons remain those of the Zeta workbench. This receiving note proves no new Riemann-hypothesis conclusion or arithmetic weight estimate.

Sources and evidence

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. Zeta Function Research Reader, commit 58626a62cd5648e8ae7450fb54d4a4aab2330981, `workbenches/splitzero-tandem/tex/support_diagrams.tex`, D6–D8, blob d3493f891291ee6e94dbf2c77649f7d85d240df2. Role: the original supported quotient and receiving-kernel interface.

The Clankers. (2026, September 16). *Universal image observations, bounded-fibre resolutions, and spectral capacity* [Companion research note]. The full proof and exact verification records are included in this workbench addition. The independent arithmetic auditor imports neither the producer nor its carrier library. All new units remain outside existing Lean coverage.

Exact local-pattern quotients and switching image capacity

Source: history/EP817_LOCAL_FLOW_20260916/payloads/ep817/research/local_flow/notes/local-flow-and-switching.md **Identity:** ca5fd72aa03b61c38632c27715af2a88dd0bb09146f1d1ca453cf4bfd6b72571 The Clankers. 16 September 2026.

Evidence and lineage. This is an additive continuation of *Universal image observations, bounded-fibre resolutions, and spectral capacity*, delivered in EP817_SEVEN_OBSERVABLE_20260916.zip. The preceding archive is retained byte-for-byte. The statements below have ordinary proofs; the listed finite inputs have exact computational certificates and a separately implemented arithmetic audit. No new Lean elaboration or outside mathematical review is claimed. The original anonymous/deleted contributor retains credit for the four-term construction and signed-block proof; **sneed-and-feed** retains separate credit for the finite upper-bound Lean extension. Standard circulation decomposition and local-pattern polytopes are background, not claimed inventions. No general literature-priority claim is made.

The main progression problem keeps the original distinct positive generators. Write

$$H_q(A) = \left\{ \sum_{a \in A} c_a a : 0 \leq c_a < q \right\}, \quad S(A) = \sum_{a \in A} a.$$

Here q is coefficient arity. Progression length is denoted k . Admissibility always concerns the actual binary image $H_2(A)$. The present contribution proves a complete positivity description of the seven fifth-arity observations, lifts every arithmetic carrier to a chain map on original local binary words, and solves an actual two-block changing-schedule image optimization. It also proves that arbitrary mixtures of the preceding residue-saturated record blocks cannot benefit from faithful ternary re-encoding. No new unrestricted upper record or full determination of the general rate is asserted.

1. Original numerical sets, observations, and local words

For a nonempty finite centrally symmetric set Y of integers, retain the seven original observations, in this order:

$$f(Y) = (|Y + R|)_{R \in \mathcal{R}}, \quad \mathcal{R} = (0, 01, 02, 012, 03, 013, 0123). \quad (1.1)$$

For example, 013 means the actual shift set $\{0, 1, 3\}$, not a digit word. We write the corresponding coordinates as $f_0, f_1, \dots, f_6$. The symmetry assumption means $c \cdot Y = Y$ for $c = \min(Y) + \max(Y)$. Every $H_q(A)$ has this symmetry by the original coefficient-complement involution.

For each integer t from $\min(Y)$ through $\max(Y) + 3$, define the binary four-position word

$$e(t) = \sum_{j=0}^3 2^j \mathbf{1}_Y(t - j). \quad (1.2)$$

Let $c_e(Y)$ count its occurrences. These are occurrences in the characteristic function of the *distinct numerical image*. They are not counts of the original coefficient representations of a value.

Use the directed overlap graph with vertex set $\{0, \dots, 7\}$ and edge set $\{0, \dots, 15\}$; edge e has source $\text{floor}(e/2)$ and target $e \bmod 8$. Scanning (1.2) starts and ends at vertex 000. Thus the integral edge current $c(Y)$ is a cycle:

$$\partial c(Y) = 0, \quad \partial e = e_{e \bmod 8} - e_{\lfloor e/2 \rfloor}. \quad (1.3)$$

There are $\text{span}(Y)+4$ edges, where $\text{span}(Y)=\max(Y)-\min(Y)$. Reflection of Y proves $c_{\neg e}=c_{\neg \text{rev}(e)}$, with rev reversing all four bits. The zero word 0000 can occur in an actual gap. It is retained as an ordinary edge and cycle.

1.1 Nine nonnegative pattern coordinates and two exact relations

Select one original count from each nonzero reversal orbit, ordered as

$$z = (c_{0001}, c_{0010}, c_{0011}, c_{0101}, c_{0110}, c_{0111}, c_{1001}, c_{1011}, c_{1111}). \quad (1.4)$$

A selected count is not the sum of the two reversed counts. For a palindromic word its orbit has one element. Direct expansion of each union indicator, followed by (1.3), gives

$$\begin{aligned} z_0 &= f_6 - f_3, & z_1 &= f_6 - f_5, \\ z_2 &= -f_1 + f_3 + f_5 - f_6, & z_3 &= -f_2 + f_3 + f_5 - f_6, \\ z_4 &= -f_4 + 2f_5 - f_6, & z_5 &= -f_0 + f_1 + f_2 - f_3 + f_4 - 2f_5 + f_6, \\ z_6 &= -f_1 + 2f_3 - f_6, & z_7 &= -f_0 + 2f_1 + f_2 - 2f_3 - f_5 + f_6, \\ z_8 &= 4f_0 - 3f_1 - 2f_2 + 2f_3 - f_4 + 2f_5 - f_6. \end{aligned} \quad (1.5)$$

Write this integral map as $z=A f$. The complete conservation equations on the nine coordinates are

$$z_0 - z_1 - z_2 + z_6 = 0, \quad z_2 - z_4 - z_5 + z_7 = 0. \quad (1.6)$$

Their matrix is C . The explicit inverse on $\ker C$ is $f=Bz$, where

$$B = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 & 1 & 2 & 1 & 2 & 1 \\ 1 & 1 & 2 & 1 & 1 & 2 & 1 & 2 & 1 \\ 1 & 2 & 2 & 2 & 1 & 2 & 1 & 2 & 1 \\ 2 & 0 & 2 & 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 2 & 1 & 2 & 1 \\ 2 & 2 & 2 & 2 & 1 & 2 & 1 & 2 & 1 \end{pmatrix}. \quad (1.7)$$

In particular $BA=I_7$, $CA=0$, and $ABz=z$ for $Cz=0$. One way to check all identities directly is to let E insert z into the original sixteen coordinates, duplicating a reversal pair, and to let $O_{\neg R}(e)=1$ when at least one bit indexed by R is one. Then $B=OE$ and the vertex conservation relations on E are exactly the two rows of C . Solving those equations gives (1.5). This proves both directions over the integers, not merely equality of real ranks.

Every inequality $z_i \geq 0$ is therefore an exact original-pattern count. For example,

$$4f_0 - 3f_1 - 2f_2 + 2f_3 - f_4 + 2f_5 - f_6 = c_{1111}(Y) \geq 0.$$

Equality says that no four consecutive integers occur in Y . A coefficient-zero count at a supported word label remains separate from a word absent from an admitted source support.

The omitted gap count is also exact:

$$\boxed{c_{0000}(Y) = \text{span}(Y) + 4 - f_6(Y)}. \quad (1.8)$$

For $Y=\{0,4\}$ and $Y'=\{0,5\}$, all seven observations agree, but

$$c(Y') - c(Y) = e_{0000}. \quad (1.9)$$

These are two actual symmetric numerical sets, so this receiving kernel has a concrete source witness.

2. The complete feasible cone: nine facets and ten primitive rays

Let K be the closed conical hull of all $f(Y)$ for nonempty finite centrally symmetric integer sets. Then

$$K = \{f \in \mathbb{R}^7 : Af \geq 0\}. \quad (2.1)$$

The nine inequalities in (1.5) are all facets. The cone has dimension seven and precisely ten extreme rays.

Here is the full integral structure. The nonnegative integral solutions of $Cz=0$ are generated, without denominators, by

$$\begin{aligned} e_3, e_8, e_0 + e_1, e_6 + e_1, e_4 + e_7, e_5 + e_7, \\ e_0 + e_2 + e_4, e_0 + e_2 + e_5, e_6 + e_2 + e_4, e_6 + e_2 + e_5. \end{aligned} \quad (2.2)$$

The same list generates the nonnegative real solutions as a cone. For a direct decomposition, remove the e_3 and e_8 components first. Equations (1.6) read

$$z_0 + z_6 = z_1 + z_2, \quad z_4 + z_5 = z_2 + z_7.$$

Allocate z_2 units from the available pair (z_0, z_6) to the available pair (z_4, z_5) , using their actual minimum at each allocation. These are the four triple generators. The remaining supplies sum to z_1 and z_7 and give the four pair generators. This is exact for real, rational, and integer data. No coefficient vector is divided by a common factor to change an original arithmetic relation.

The ten observed rays $f=Bz$, with their original integral scale retained, are:

Period word	Integral count ray
10	(1,2,1,2,2,2,2)
1	(1,1,1,1,1,1,1)
1000	(1,2,2,3,2,3,4)
001	(1,2,2,3,1,2,3)
011	(2,3,3,3,2,3,3)
1011	(3,4,4,4,4,4,4)
11000	(2,3,4,4,4,5,5)
111000	(3,4,5,5,6,6,6)
1001	(2,3,4,4,3,4,4)
11001	(3,4,5,5,5,5,5)

The period words refer to their actual occupied residue positions, not coefficient digits of a generator set. To realize a ray asymptotically, let P_N be N literal copies of its period support and put

$$Y_N = P_N \cup (2Np + 4 - P_N), \quad (2.3)$$

where p is the period length. The two supports have a gap greater than three, so their translated union counts add. Y_N is centrally symmetric. If h is the number of ones in one period and g is its displayed count ray, then $|Y_N|=2Nh$ and

$$\left| f_i(Y_N) - \frac{2Nh}{g_0} g_i \right| \leq 12 \quad (0 \leq i \leq 6). \quad (2.4)$$

Only the first and last three positions of one periodic segment can disagree with the periodic union count; applying this separately to both reflected copies proves the bound. For the displayed primitive periods $h=g_0$, but (2.4) keeps the scale explicitly.

Thus every ray belongs to the closed cone generated by actual sets. Conversely every actual set satisfies (1.5), and the decomposition (2.2) proves the opposite inclusion in (2.1). Each generator in (2.2) is a minimal positive dependence among the two-dimensional columns of C ; hence it is extreme. The rays with $z_i=0$ span dimension six for each i , proving all nine facet assertions. The finite certificate checks these ranks exactly.

There is also an exact integer semigroup statement:

$$\{f \in \mathbb{Z}^7 : Af \geq 0\} = \sum_{g \text{ in the table}} \mathbb{Z}_{\geq 0} g. \quad (2.5)$$

It follows from (1.5)–(1.7) and the integral allocation proof. Equation (2.5) describes local circulation coordinates. It does not assert that each such integer vector is the count vector of one finite numerical set. For example, the ray vector (1,1,1,1,1,1,1) cannot be the observation of a one-point set, whose $|Y + \{0,1\}|$ is two. Its actual source realization is the periodic limit (2.3). The scale parameter and the endpoint data are not erased by that comparison.

The finite graph has nineteen directed simple cycles, one of them the zero loop. Symmetrizing their integral currents and decomposing by (2.2) is an independent complete certificate for the cone. Local-pattern circulation polytopes are classical in symbolic dynamics; the particular integral nine-to-seven comparison, ten generators, and arithmetic action below are the specialized calculations here (Ziemian, 1995).

3. An exact two-potential test for every linear bound

Let w be a rational row on the seven original observations. The following are equivalent:

$$\begin{aligned} wf(Y) &\geq 0 \quad \text{for every finite symmetric numerical } Y; \\ wg &\geq 0 \quad \text{for the ten displayed count rays;} \\ w &= \alpha A \quad \text{for some rational row } \alpha \geq 0. \end{aligned} \quad (3.1)$$

There is an explicit construction requiring no linear-programming oracle. Put $r = wB$ and retain

$$p_1 = \min(r_0, r_6), \quad p_2 = -\min(r_4, r_5), \quad \alpha = r - (p_1, p_2)C. \quad (3.2)$$

The ten ray inequalities are exactly

$$r_3, r_8 \geq 0, \quad r_0 + r_1, r_6 + r_1 \geq 0, \quad r_4 + r_7, r_5 + r_7 \geq 0,$$

and the four inequalities $r_i + r_j \geq 0$ for i in $\{0,6\}$, j in $\{4,5\}$. Substitution in (3.2) proves every component of α is nonnegative. Since $BA = I$ and $CA = 0$, $\alpha A = w$. This proves the last implication directly. The other implications follow from the original pattern counts and the actual periodic approximation.

Thus a successful linear estimate has the literal decomposition

$$wf(Y) = \sum_{i=0}^8 \alpha_i c_{\omega_i}(Y), \quad (3.3)$$

with all original nonnegative terms visible. The correction $(p_1, p_2)C$ is zero on the original conserved current. Equality is exactly absence of every counted pattern whose coefficient α_i is positive.

A failed ray test also produces a numerical source, not just an abstract vector. If $wg = -\epsilon < 0$, choose the period in the table and any integer

$$N > \frac{6g_0 \|w\|_1}{h\epsilon}. \quad (3.4)$$

The symmetric set (2.3) then has $wf(Y_N) < 0$ by (2.4). The verifier evaluates the resulting actual set. This is an obstruction to the stated universal-tail inequality; a periodic binary support is not automatically the fifth-arity image of a positive-generator block. That further arithmetic realizability constraint remains separate.

For an arithmetic count matrix M and a proposed rational growth inequality $\ell M f \geq c \ell f$, apply (3.1) to the actual row $w = \ell M - c \ell$. The result is either a complete certificate on every finite symmetric tail or an explicit periodic finite-tail obstruction. If ℓ is positive on the cone, multiplying certified edge inequalities gives a lower bound for every changing schedule in the specified dictionary. No assertion is made that one linear potential always gives the optimal rate.

4. The arithmetic action is a literal graph-path substitution

This construction works at every coefficient arity $q \geq 2$. Put $d=q-2$. Vertices are binary words of length d ; edges are binary words of length $d+1$, with the same overlapping source and target convention as in (1.3). For $q=2$ there is one vertex and the convention has its literal meaning.

Let $b \geq 2$ and 0 in $D \subset [0, (q-1)(b-1)]$. An input edge $a=(a_0, \dots, a_d)$ is replaced by b consecutive output edges. At residue $z=0, \dots, b-1$, its r -th output bit is

$$o_r^{(z)} = \bigvee_{\substack{0 \leq h \leq d \\ bh+z \in \overline{D}+r}} a_h, \quad 0 \leq r \leq d. \quad (4.1)$$

This is the literal membership test for $bt+z-r$ in $D+bY$ when $a_h=1 _ Y(t-h)$. The range bound ensures all required h lie in $[0, d]$. Every actual digit remains in the membership test; no digit is replaced by its residue alone.

The vertex map is

$$\phi(s)_r = \bigvee_{\substack{1 \leq h \leq d \\ bh-1 \in \overline{D}+r}} s_{h-1}, \quad 0 \leq r < d. \quad (4.2)$$

Adjacent output edges overlap because $o_{r+1}^{z+1} = o_r^z$. The first source is $\phi(\text{input source})$, and the final target is $\phi(\text{input target})$, by (4.2). Hence every input edge is sent to a specified path with those endpoints.

Let L be the integral matrix counting the entire output path of each edge, and V the matrix of ϕ . The exact chain square is

$$\boxed{\partial L = V \partial.} \quad (4.3)$$

Every column of L sums to b , and the actual zero loop satisfies

$$Le_{0q-1} = be_{0q-1}. \quad (4.4)$$

For a finite numerical input Y , scanning one full input edge cell produces some additional terminal zero words. The exact number is

$$g = (q-1)(b-1) - \max D \geq 0.$$

Consequently the original finite-set current obeys

$$\boxed{c(D + bY) = Lc(Y) - ge_{0q-1}.} \quad (4.5)$$

Indeed the input scan has $\text{span}(Y)+q-1$ cells, each of b output positions. The actual output scan has $b \text{span}(Y)+\max(D)+q-1$ positions. Their difference is g , and all omitted terminal words are zero. This proves both the amount and the support of the correction.

The correction is tied to the presence of a nonempty finite source. On the free space of finite-set objects, let $\epsilon_{\text{span}([Y])=1}$ and $\Psi([Y])=c(Y)$. Then

$$\Psi S_{D,b} = L\Psi - ge_0\epsilon, \quad S_{D,b}[Y] = [D + bY]. \quad (4.6)$$

Adjoining ϵ makes this a genuine linear augmented action. It is not valid to apply the affine finite-boundary correction to an arbitrary stationary ray without its presence and endpoint data.

4.1 The seven-dimensional quotient and its positive nine-coordinate lift

At $q=5$ let Z_{-1}^+ be the reversal-invariant cycle subspace in Q^{16} . It has dimension eight. The original union observations J give the exact sequence

$$\boxed{0 \longrightarrow \mathbb{Q}e_{0000} \longrightarrow Z_1^+ \xrightarrow{J} \mathbb{Q}^7 \longrightarrow 0.} \quad (4.7)$$

Explicitly, the section $r(f)=E\mathbf{A}f$ has zero 0000 coefficient, where E duplicates the reversal pairs from (1.4). Equations (1.5)–(1.7) give $Jr=I$ and

$$c - rJc = c_{0000}e_{0000}. \quad (4.8)$$

This proves exactness without choosing an eigenbasis. The section is a coefficient-current representative, not a claim that every represented vector is itself one finite binary support.

For symmetric D, let K be the nine-by-nine nonnegative integer matrix obtained by expanding a selected reversal count into its actual input edges, applying L, and reading the nine selected output counts. Let M be the original seven-count carrier. Then

$$\boxed{KA = AM.} \quad (4.9)$$

On the original invariant subspace $Cz=0$, this is exactly $z(D+bY)=Kz(Y)$. Its conservation relations remain in that subspace; the nine entries are not treated as independent new variables. One can prove (4.9) from (4.1) and (1.5), or by applying the actual operator to seven finite symmetric source sets whose observation matrix has determinant minus one. Those seven original sets are listed in the certificate.

Every matrix M preserves the full cone (2.1). This follows either from (4.9), $K \geq 0$, and the two conservation relations, or from the direct count identity followed by the periodic-limit construction. The nine-coordinate lift is positive and integral; no linear-programming extension of M is needed.

4.2 The section defect and metric return are exact

The section difference is entirely in the retained zero-loop line:

$$\boxed{Lr - rM = e_{0000}\lambda, \quad \lambda = be_6^T - e_6^T M.} \quad (4.10)$$

The sum of the nonzero input edge coordinates is f_6 . Column sums of L equal b, and the nonzero output mass is $(Mf)_6$. Their difference proves (4.10).

Give the original sixteen edge-current basis vectors the specified orthonormal metric. In the nine selected coordinates its restriction is

$$W = \text{diag}(2, 2, 2, 2, 1, 2, 1, 2, 1).$$

The quotient metric is the full positive form

$$G = A^T W A.$$

The zero-loop axis is orthogonal to $r(\hat{Q}^7)$, so r is the actual minimum-norm section in this metric. Orthogonality proves

$$\boxed{(Lr)^T(Lr) = M^T G M + \lambda^T \lambda.} \quad (4.11)$$

For an original finite current $c=r(f)+z_0e_0$, the complete receiving norm after (4.5) is

$$\|c(D + bY)\|^2 = (Mf)^T G (Mf) + (bz_0 + \lambda f - g)^2. \quad (4.12)$$

The current metric is not the original coefficient-word metric of $H_q(A)$. On that separate source the value quotient still has Gram $\text{diag}(1/\mu(y))$, with its original weighted section. The passage from a whole numerical set to its local-pattern counts is linear only after forming the free module on finite-set objects. In particular, it is not a context-free linear map sending each individual value basis vector to a fixed local pattern.

The previous word-to-value quotient and the additional local observation kernel therefore remain separate. On the free finite-set source, (1.9) proves that the added zero-loop line in $\ker(J\Psi)/\ker(\Psi)$ is actually reached. Supported zero and external absence are not interchanged.

4.3 A larger operator can have the wrong numerical growth rate

Since every column of L sums to b and (4.4) holds, $\rho(L)=b$. That number is not automatically the numerical image radius $\rho(M)$.

For $A=\{3,10\}$, $b=19$, the actual fifth-image matrix has characteristic polynomial

$$X^3(X^4 - 36X^3 + 382X^2 - 1116X + 65).$$

Its largest root lies in $(18,19)$, approximately 18.911324463. The full edge operator has characteristic polynomial

$$X^{10}(X - 19)(X - 1)(X^4 - 36X^3 + 382X^2 - 1116X + 65).$$

The 19-eigenvector e_{0000} is present in the original cycle space and killed by the precise quotient (4.7). Thus the source-to-receiving spectral discrepancy is explained by an actual kernel, not by selecting a preferred root after discarding unspecified information. The full matrices and determinant checks are in the receipt.

5. An exact changing-block minimum that beats both homogeneous images

Use two original level types, both with radix ten:

$$A = (10, \{1, 3\}), \quad B = (10, \{2, 4\}). \quad (5.1)$$

Their binary digit images are respectively $\{0,1,3,4\}$ and $\{0,2,4,6\}$. A nonzero modular step of order at least five would require at least five distinct points; a step of order two differs by five and neither set contains such a pair. Thus both are modular-five-free. Every finite prefix of every changing schedule is five-admissible by the original lowest-digit induction. There are exactly two distinct generators per level.

In the coordinate order (1.1), their fifth-image actions are

$$M_A f = (3f_0 + 7f_1, 2f_0 + 8f_1, f_0 + 9f_1, f_0 + 9f_1, 10f_1, 10f_1, 10f_1),$$

$$M_B f = (2f_1 + 3f_3, 4f_1 + 6f_3, f_1 + 4f_3, 3f_1 + 7f_3, 4f_1 + 6f_3, 3f_1 + 7f_3, 2f_1 + 8f_3). \quad (5.2)$$

Both have characteristic polynomial $X^5(X-10)(X-1)$. Their homogeneous image counts are

$$N_A(m) = \frac{16 \cdot 10^m - 7}{9}, \quad N_B(m) = \frac{4 \cdot 10^m - 1}{3}. \quad (5.3)$$

The actual stable local current for A is the all-occupied periodic ray. For B it is the parity ray $(1,2,1,2,2,2,2)$. Both have homogeneous growth ten, but they retain different original numerical supports.

5.1 Run factorization and complete optimization

Retain

$$u = (17, 18, 19, 19, 20, 20, 20)^T, \quad v = (0, 2, 0, 3, 0, 0, 0).$$

Direct integer multiplication proves

$$M_A M_B = uv. \quad (5.4)$$

For $a, b \geq 1$ define $U_a = M_A^a \{a-1\}u$ and $V_b = v M_B^b \{b-1\}$. Then $M_A^a M_B^b = U_a V_b$. The exact scalar crossing of consecutive runs is

$$\boxed{V_b U_a = \theta(a, b) = \frac{8 \cdot 10^{a+b} + 4 \cdot 10^b - 3}{9}}. \quad (5.5)$$

For a direct all-length verification of (5.5), the vectors u and row v satisfy the respective recurrences with polynomial $(X-1)(X-10)$. The four values at a, b in $\{1, 2\}$ determine the displayed two-variable expression. This proves it for every a, b , beyond the listed finite replay.

For a cyclic word with runs $A^{a_1} B^{b_1} \cdots A^{a_s} B^{b_s}$, cyclically rotating the matrix product to expose (5.4) proves that its only possible nonzero eigenvalue is

$$\prod_{i=1}^s \theta(a_{i+1}, b_i), \quad a_{s+1} = a_1. \quad (5.6)$$

A change of cyclic starting position preserves the nonzero eigenvalues. The scalar in (5.6) is positive and is the radius. Pure words have radius 10^m .

Let $r = 893^{1/3}$. For fixed $L = a + b$, the expression (5.5) is minimized by $b = 1$. For $L = 2$ it equals 93, and

$$93^3 = 804357 > 797449 = 893^2.$$

For $L = 3$ the minimum is $\theta(2, 1) = 893$. For $L \geq 4$,

$$\theta(a, b)^{1/L} > 10(8/9)^{1/L} \geq 10(8/9)^{1/4} > \sqrt{93},$$

where the last comparison is $80000 > 9 \cdot 93^2$. The remaining length-three pair has $\theta(1, 2) = 933$ and $933^2 > 93^3$. Therefore

$$\boxed{\theta(a, b) \geq r^{a+b}}, \quad (5.7)$$

with equality exactly at $(a, b) = (2, 1)$.

For any finite word w , let $N(w)$ be its actual fifth-arity image cardinality and P_w its ordered count matrix. Its literal aggregate generator block has radix $10^{|w|}$. At a cut of a repeated aggregate, a numerical value has at most four original distinct-value pairs. Thus

$$\boxed{\rho(P_w) \leq N(w) \leq 4\rho(P_w)}. \quad (5.8)$$

For completeness, if N_t is the image of t repetitions, then $N_t \leq N(w)^t$ and $N_t \geq N(w)^t / 4^{t-1}$. Applying the same inequalities to larger concatenations yields the root limit and its factor-four bound. To identify that limit with the full radius of P_w , retain the auxiliary weighted coordinate norm

$$\|x\|_v = \max_i |x_i| / |R_i|, \quad v_i = |R_i|.$$

For every actual nonnegative word matrix P , its induced norm is $\max_i (Pv)_i / v_i$. The numerical meaning of Pv gives

$$\|P\|_v = \max_i |L + R_i| / |R_i| = |L|.$$

The upper inequality counts the original translated copies, and the singleton coordinate gives equality. Thus $|P_w^t|_v = N_t$ and the radius is the same root limit. This proves (5.8) without discarding an unobserved spectral component. The auxiliary coordinate norm is not substituted for an original word or current metric.

Equations (5.6)–(5.8) prove $N(w) \geq r^{|w|}$ for every word. The period AAB attains the rate. Its actual six-generator aggregate and radix are

$$C = \{1, 3, 10, 30, 200, 400\}, \quad Q = 1000. \quad (5.9)$$

The period matrix has rank one, radius 893, and

$$|H_5(C_Q^{[t]})| = 2301\,893^{t-1} \quad (t \geq 1). \quad (5.10)$$

Consequently the exact optimization over all infinite schedules in this two-block dictionary is

$$\inf_{\mathbf{w} \in \{A,B\}^{\mathbb{N}}} \liminf_{m \rightarrow \infty} |H_5(G_m(\mathbf{w}))|^{1/(2m)} = 893^{1/6} \approx 3.1031914962. \quad (5.11)$$

Alternation AB has the slightly larger image rate $93^{1/4}$, approximately 3.1054227991. Homogeneous image rates are $\sqrt{10}$, approximately 3.1622776602.

All the original radices in this dictionary are still ten. Their original maximum-generator rate is $\sqrt{10}$, for every schedule. Equation (5.11) is an image-capacity improvement. The preceding arity-five prime-observation theorem converts it into a new re-encoded construction rate, retaining the original finite prefactor and prime height. It is not a better global record than the existing workbench bound $97^{1/6}$ for $k=5$.

5.2 Quantitative stability of the optimizing pattern

For a mixed cyclic word, call a crossed run pair (a_{i+1}, b_i) good when it equals $(2,1)$, and let L_{bad} be the total length of all other crossed pairs. For a pure word set $L_{\text{bad}} = |w|$. The proof above gives $\theta(a,b) \geq 93^{1/4}((a+b)/2)$ for every bad pair, while $\theta(a,b) \leq 10^{1/4}(a+b)$ always. With $L = |w|$,

$$r^L (\sqrt{93}/r)^{L_{\text{bad}}} \leq N(w) \leq 4r^L (10/r)^{L_{\text{bad}}}. \quad (5.12)$$

In particular, for any sequence of words with lengths tending to infinity,

$$\lim N(w)^{1/|w|} = r \iff L_{\text{bad}}(w)/|w| \rightarrow 0.$$

For the limiting inferior alone, the corresponding exact criterion is

$$\liminf N(w)^{1/|w|} = r \iff \liminf L_{\text{bad}}(w)/|w| = 0.$$

Both implications follow from the two strictly positive logarithmic gaps in (5.12); no limit is presumed to exist for an arbitrary schedule. These are proved stability statements, not conclusions drawn from checking a finite list of periods.

6. A common occupied ray controls arbitrary mixtures of the record blocks

For any original canonical digit set D , the following are equivalent:

$$D \bmod b = \mathbb{Z}/b\mathbb{Z}, \quad M\mathbf{1} = b\mathbf{1}. \quad (6.1)$$

The first row of M counts the occurring residues of D . If all occur, every $D+R$ also has every residue, so all row sums equal b . This proves both implications. In the positive local-pattern coordinates this is precisely preservation of the pure 1111 ray with multiplier b .

For a changing schedule whose fifth images all satisfy (6.1), put $P_m = \prod_{j < m} b_j$. The initial vector $v_R = |R|$ satisfies $v \geq 1$, so

$$|H_5(G_m)| \geq P_m.$$

The actual image interval has width at most $4(P_m - 1)$. Consequently

$$P_m \leq |H_5(G_m)| \leq 4P_m - 3. \quad (6.2)$$

This holds even with unbounded, changing generator counts and radices. It requires no existence of an average log radix. Per total generator count N_m , the image and product costs differ by at most $\log(4)/N_m$.

The preceding record blocks have the stronger property that their *ternary* images cover all residues:

A	b	
$\{1, 7, 8\}$	19	
$\{1, 4, 5, 17, 21, 22\}$	97	
$\{1, 4, 5, 17, 21, 22\}$	93	
$\{3, 4, 7, 34, 37, 41, 216, 250, 253, 257\}$	1651.	(6.3)

The finite certificate checks each complete residue set. Ternary coverage implies fifth coverage by inclusion. It also composes in the original residue groups: first choose the lower digit for the desired residue modulo b , then choose the next digit for its actual quotient residue, and continue. Hence every changing aggregate of these blocks has at least P_m distinct ternary values.

Every injective residue observation of those ternary values must therefore use modulus at least P_m . Thus faithful ternary re-encoding cannot improve the exponential product cost of *any mixture* of the sources in (6.3). The admissibility length still belongs to the actual block: for $k=5$ use only the five-admissible entries; for $k=6$ all four entries are available.

The conclusion also covers any canonical generator block whose modular edge weights arise from one of these by multiplication by a unit and independent sign choices, provided positivity, distinctness, and $\text{sum}(A) < b$ hold for its actual chosen lifts. Indeed coefficient complementation on a sign-changed coordinate translates its H_3 residue image by the original twice-weight, and unit multiplication is a residue-group automorphism. Both preserve full coverage and binary modular progression-freeness. The actual lifts and their finite costs remain separate data.

This identifies why the switching example behaves differently. Its A block has the occupied integer ray, but B has only even fifth residues modulo ten and a different parity eigenprofile. Full homogeneous spectral growth by itself does not imply the common-ray condition (6.1). The exact local observation detects the phase difference.

7. General target and precise limits of this contribution

The earlier all-block theorem is

$$\lambda_k = \inf_{A \text{ k-admissible}} |H_5(A)|^{1/|A|} = \inf_{(b,A) \in \mathcal{M}_k} \rho(M_5(A,b))^{1/|A|},$$

where $\mathcal{M}_k$ consists of original canonical modular-safe binary blocks. It is the interface used here; none of the present local-cone arguments assumes a numerical value for that infimum.

The cone theorem gives all linear inequalities valid on every finite symmetric numerical tail, together with an exact positive pattern decomposition or an explicit family of counterexamples. The graph lift gives an original current complex, its receiving kernel, and its full section and metric defects. The switching result solves one infinite arithmetic dictionary exactly, while (6.2) excludes faithful re-encoding improvements over the entire saturated record-block family.

The remaining arithmetic domain is narrower than all symmetric finite sets: it consists of actual H_5 images of positive distinct blocks satisfying binary k -admissibility. The ten periodic extremal tails need not all have such a generator realization. A failed universal-tail inequality is therefore a counterexample to that proposed inequality, not automatically a counterexample to an Erdős lower bound. Conversely, a successful ten-test inequality applies to every admissible tail but need not be optimal on that smaller domain.

No rank cutoff for the global infimum is proved. The large- k gap in the inspected general bounds remains between logarithmic rates of order $1/k$ and $\log(k)/k$ (Korsky, 2026). The present matrices keep the generator reward $|A|$ and every original radix; fixed local dimension does not bound the bit length or rank of the arithmetic inputs.

8. Reproducibility and exact checked scopes

Run from the contribution directory:

```
python certificates/verify_local_flow.py
python certificates/audit_local_flow.py
```

Both programs use Python's standard library only. The audit imports neither the producer nor its helper library. It derives edge substitutions by evaluating actual finite numerical output windows, rather than using the producer's residue masks. It reconstructs the nine-count identity by exact interpolation on seven original symmetric sets; it checks characteristic polynomials by determinant evaluations rather than the producer's trace recurrence.

The complete finite scopes are:

- all 19 directed simple cycles of the four-position overlap graph, all 10 cone generators and 9 facet ranks, 810 producer integral decompositions, and all 1,089 balanced flows in the auditor's $[0,2]^9$ domain;
- all 766 symmetric finite supports of span at most 16 in the specified construction, and all 511 nonempty masks on $[0,8]$ through their actual separated reflection doubles;
- all 722 symmetric digit sources containing zero, with $q=2,\dots,6$, $b=2,\dots,6$, and digit width at most $\min(8,(q-1)(b-1))$; 3,610 direct source joins and all 11,352 complete input-edge paths;
- all 2,187 seven-coordinate rows with coefficients in $\{-1,0,1\}$: 990 exact nonnegative certificates and 1,197 actual finite symmetric counterexamples;
- nine selected arithmetic graph lifts, their count and edge matrices, full quotient Grams, and zero-loop section defects;
- all 2,046 two-letter words through length ten, 144 run-formula instances, and the exact attaining period; the auditor also checks 62 direct switched original-generator images through five levels;
- all sixteen direct two-level mixtures of the saturated record blocks and all 256 four-level matrix words, with their complete common-ray and product bounds.

Eight corrupted proof records are deliberately rejected. Normal and optimized Python outputs are compared byte-for-byte. Those checks corroborate the written infinite proofs; they are not presented as enumeration of all blocks or all infinite schedules. All reported runs have finished. No background work is asserted.

References and source roles

Ziemian, K. (1995). Rotation sets for subshifts of finite type. *Fundamenta Mathematicae*, 146(2), 189–201. DOI: 10.4064/fm-146-2-189-201. The publisher's abstract and bibliographic record were inspected for the established finite-pattern/polytope context. The circulation decomposition, integral coordinate maps, and complete specialized cone proof are given above; no uninspected theorem body is imported.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). Inspected HTML source: <https://arxiv.org/html/2606.24139v1>. Role: original problem conventions and the benchmark large-k gap, not the source of the current local-flow formulas or switching classification.

Frougny, C., & Pelantová, E. (2018). *Two applications of the spectrum of numbers* (arXiv:1512.04234v3; original preprint 2015). Role: established positional finite-state context; no new general automata-priority claim.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source]. `workbenches/splitzero-tandem/tex/support_diagrams.tex` in KokunoYumeto/zeta-function-research-reader, inspected revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, source blob d3493f891291ee6e94dbf2c77649f7d85d240df2, equations D6–D8. Role: the original internal quotient and additional observation-kernel interface. No analytic theta estimate is imported or claimed proved.

The Clankers. (2026, September 16). *Universal image observations, bounded-fibre resolutions, and spectral capacity* [Predecessor research contribution]. The complete preceding delivery `EP817_SEVEN_OBSERVABLE_20260916.zip` is retained unchanged. The original $k=4$ construction and signed-block proof remain attributed there to the anonymous/deleted Reddit contributor, and the Lean upper-bound extension separately to sneed-and-feed.

Original local-current quotients and their SplitZero receiving maps

Source: `history/EP817_LOCAL_FLOW_20260916/payloads/zeta/workbenches/ep817-local-flow/notes/splitzero-receiving-map.md` Identity: 71a9acbd555be4ce076ff9b873d6d7fe74c48172b30d4b1df0afe07b14e7f729 The Clankers.

16 September 2026. Additive research contribution; independent mathematical review pending. No new Lean elaboration or analytic theta estimate is claimed.

1. Fixed source interface and exact scope

The inspected Zeta source is `workbenches/splitzero-tandem/tex/support_diagrams.tex` at revision 58626a62cd5648e8ae7450fb54d4a4aab2330981, blob d3493f891291ee6e94dbf2c77649f7d85d240df2. Its equations D6–D8 distinguish an original relation quotient, an additional observation kernel, and the global bottom from a zero in an active coefficient fibre. Those definitions are retained here.

The new finite model comes from characteristic functions of actual numerical subset-sum images. It is a different source from the theta complex. No map identifying the whole theta quotient with seven coordinates is asserted. The full proofs and arithmetic applications are in the companion `local-flow-and-switching.md`; this note spells out the receiving objects, chain maps, support changes, and metric defects used for transfer.

For a finite nonempty centrally symmetric integer set Y , retain the four-position words

$$e(t) = \sum_{j=0}^3 2^j \mathbf{1}_Y(t - j), \quad \min Y \leq t \leq \max Y + 3.$$

Their actual occurrence current $c(Y)$ lies in the integral edge module of the binary overlap graph. Edge e has source $\text{floor}(e/2)$, target $e \bmod 8$. The boundary is

$$\partial e = e_{e \bmod 8} - e_{\lfloor e/2 \rfloor}.$$

The scan starts and ends at the same zero vertex, so partial $c(Y)=0$. Central symmetry makes the current invariant under reversal of all four bits. The zero word is an admitted loop, including when its coefficient happens to vanish.

2. Original current homology and the extra observation line

Let Z be the reversal-invariant cycle module inside $\hat{Q}^{16}$. It has dimension eight. Its seven observation coordinates are

$$Jc = \left(\sum_e O_R(e) c_e \right)_{R \in \mathcal{R}}, \quad O_R(e) = \mathbf{1}[\text{some bit indexed by } R \text{ is one}],$$

where

$$\mathcal{R} = (\{0\}, \{0, 1\}, \{0, 2\}, \{0, 1, 2\}, \{0, 3\}, \{0, 1, 3\}, \{0, 1, 2, 3\}).$$

The companion proof gives literal integral matrices A, B, C, and E: $z=A f$ selects nine nonzero reversal counts, $Cz=0$ are their two original vertex-conservation identities, $f=Bz$ is the inverse, and E inserts both members of each reversal pair into the sixteen original coordinates. They satisfy

$$BA = I_7, \quad CA = 0, \quad AB|_{\ker C} = I.$$

Thus the exact receiving sequence and its specified section are

$$\boxed{0 \rightarrow \mathbb{Q}e_{0000} \rightarrow Z \xrightarrow{J} \mathbb{Q}^7 \rightarrow 0,}$$

$$r(f) = EAf, \quad Jr = I, \quad c - rJc = c_{0000}e_{0000}.$$

The zero-loop class is a nonzero class in the original graph H_1: the graph has no two-cell boundary that removes it. The seven observations deliberately impose an additional quotient. One can realize that quotient as the degree-one cohomology of the explicitly added two-term complex $\mathbb{Q} e_{0000} \rightarrow Z$; it is not the original graph's H_1.

The dual cochain statement is the split exact sequence

$$0 \rightarrow (\mathbb{Q}^7)^* \xrightarrow{J^*} Z^* \xrightarrow{\text{evaluation on } e_{0000}} \mathbb{Q} \rightarrow 0.$$

Every map has its stated source and target. The concrete sets $\{0,4\}$ and $\{0,5\}$ have identical seven observations and currents differing by e_{0000} . Hence the extra kernel has a witness among the actual finite inputs.

To relate this to the source's D7 comparison, form the free rational module X on finite nonempty symmetric numerical-set objects, and define

$$\Psi : X \rightarrow Z, \quad [Y] \mapsto c(Y).$$

The seven original independent-set examples in the certificate, together with the preceding zero-loop difference, show that Psi is onto. Therefore

$$\boxed{0 \rightarrow \ker \Psi \rightarrow \ker(J\Psi) \xrightarrow{\Psi} \mathbb{Q}e_{0000} \rightarrow 0.}$$

For restricted finite admitted object supports, retain the image intersection in the rightmost term, exactly as the original source's receiving-kernel formula requires. No surjectivity is inferred on a smaller support without an actual lift.

3. A positive representative with all conservation equations visible

The nine selected nonzero counts are nonnegative. Their two equations are

$$z_0 + z_6 = z_1 + z_2, \quad z_2 + z_7 = z_4 + z_5.$$

The companion proof gives the entire nonnegative integral monoid, generated by ten explicitly displayed primitive vectors. It also proves that the closed conical hull of actual finite symmetric observations is exactly

$$\{f : Af \geq 0\}.$$

This is a cone of original local currents. Its primitive rays are represented by periodic binary supports with their exact finite boundary errors. A ray representation is not identified with one finite set, nor with an arbitrary H_5(A) image.

For any rational linear observation w, the proof constructs one of two exact outcomes. If the ten ray values are nonnegative, put $r_0=wB$ (the subscript here labels a row, not the section), choose

$$p_1 = \min((r_0)_0, (r_0)_6), \quad p_2 = -\min((r_0)_4, (r_0)_5),$$

and set $\alpha = r_0 - (p_1, p_2)C$. Then $\alpha \geq 0$ and

$$wf(Y) = \sum_i \alpha_i c_{\omega_i}(Y).$$

The two-potential correction is an original conserved-boundary relation. Every positive contribution remains counted on its actual word support. If a ray test fails, a specified finite reflected periodic support gives a numerical counterexample, with a bound on its required period count. That conclusion concerns the proposed universal-tail inequality; additional arithmetic generator realizability remains a separate condition.

4. The arithmetic map is a chain map of original paths

At general arity $q \geq 2$, put $d = q - 2$. The overlap graph uses d -bit vertices and $(d+1)$ -bit edges. Let D contain zero and satisfy

$$D \subseteq [0, (q-1)(b-1)].$$

This includes actual higher-arity digits exceeding the base. For an input edge $a = (a_0, \dots, a_d)$, its b output edges have bits

$$o_r^{(z)} = \bigvee_{0 \leq h \leq d, \, bh+z \in D+r} a_h, \quad 0 \leq z < b, \quad 0 \leq r \leq d.$$

The corresponding vertex map is

$$\phi(s)_r = \bigvee_{1 \leq h \leq d, \, bh-1 \in D+r} s_{h-1}.$$

The range bound accounts for every possible h . Direct comparison of adjacent bits shows that these output edges form a path from $\phi(\text{source}(a))$ to $\phi(\text{target}(a))$. Let L be the resulting edge-path matrix and V the vertex map matrix. Then

$$\partial L = V \partial.$$

Every column of L has mass b , and the original zero loop has $L e_0 = b e_0$. On finite original numerical sources there is an additional exact endpoint correction:

$$c(D + bY) = Lc(Y) - g e_0, \quad g = (q-1)(b-1) - \max D.$$

The input scan has $\text{span}(Y) + q - 1$ edge positions; replacing each by b positions gives $b(\text{span}(Y) + q - 1)$. The actual output scan has $b \text{span}(Y) + \max(D) + q - 1$ positions. The difference is g , and every removed position is an actual terminal zero word.

On X retain the presence functional $\epsilon([Y]) = 1$. The arithmetic action $[Y] \rightarrow [D + bY]$ satisfies

$$\Psi S = L \Psi - g e_0 \epsilon.$$

The augmented coordinates (c, ϵ) therefore have a genuinely linear action. The finite endpoint correction is never applied to a stationary ray with unrecorded presence data.

Support changes

For an admitted collection of original edges S , the graph-path substitution gives the support map

$$\alpha(S) = \bigcup_{e \in S} \text{supp}(Le).$$

It preserves bottom and joins. The coefficient maps L on those fibres commute with inclusions, and partial $L=V$ partial gives the cycle and boundary comparisons. This is a direct instance of the original support-changing reconstruction. Under its scalar zero, an admitted loop coefficient maps to its active-fibre zero; the external scalar τ still maps to bottom. A zero loop can be a supported direction and lie in an additional numerical observation kernel without becoming an absent path.

For the reversal-invariant full chart at $q=5$, the receiving action is the original seven-count matrix M . The nine-coordinate positive lift K obeys

$$KA = AM$$

on the original conservation subspace. The section defect is

$$Lr - rM = e_0\ell, \quad \ell = be_6^T - e_6^T M.$$

It records the zero-loop mass created by the arithmetic substitution, prior to the finite endpoint correction. It is not set to zero by asserting naturality of a chosen section.

5. Full metric return and the actual spectral kernel

Give the original edge-current basis its specified Euclidean metric. Reversal pairs have weight two and palindromic selected patterns have weight one, so

$$W = \text{diag}(2, 2, 2, 2, 1, 2, 1, 2, 1), \quad G = A^T W A.$$

The section r is orthogonal to e_0 and is the minimum-norm section in that original current metric. Therefore

$$(Lr)^T(Lr) = M^T G M + \ell^T \ell.$$

For $c=r(f)+z_0e_0$, the actual finite receiving norm is

$$\|c(D + bY)\|^2 = (Mf)^T G (Mf) + (bz_0 + \ell f - g)^2.$$

This retains the source zero-loop component, section defect, and finite endpoint correction separately.

The full nonnegative edge-path matrix has radius b because its columns sum to b and its actual zero loop is a b -eigenvector. The numerical observation need not have that radius. For $A=\{3,10\}$, $b=19$, its characteristic polynomial is

$$X^3(X^4 - 36X^3 + 382X^2 - 1116X + 65),$$

with observed radius in $(18,19)$. The full sixteen-edge operator additionally has the exact factors $(X-19)(X-1)$. The 19-direction is the supported zero loop killed by J , with its section and metric cost just calculated. The proof does not choose a desired eigenvalue by discarding unspecified states.

The original generator-word metric is a separate source. Its word-to-value quotient remains $\text{diag}(1/\mu(y))$, with section coefficients given by the actual representation multiplicities. Passing from that source to a whole numerical set, and then to its current, requires the explicit free-set-object observation above. A local current is not a context-free linear contribution of one numerical value. None of the current-space estimates is labelled a bound on all original word masses or on the theta-source metric.

6. Arithmetic application and precise remaining interface

The source note solves the infinite switching dictionary

$$(10, \{1, 3\}), \quad (10, \{2, 4\}).$$

Both are modular-five-admissible and both have homogeneous fifth-image radius ten. Their product actions retain different local supports. The exact all-schedule minimum fifth-image rate per generator is $893^{1/6}$, attained by period AAB and its actual six-generator aggregate $\{1,3,10,30,200,400\}$ at radix 1000. The original radix cost remains $\sqrt{10}$; the image rate is converted into re-encoding cost only through the separately proved all-block image theorem. This is not a new unrestricted numerical record.

The companion proof also derives a quantitative stability criterion and a common-occupied-ray obstruction for arbitrary mixtures of the prior residue-saturated record blocks. Both results retain the original radices, block sizes and permitted digits.

For the wider research programme, the transferable result is an exact positive observation cone with constructive dual alternatives, an original path-level chain map, and its section/metric defect. Applying it to a different cohomological source requires construction of the corresponding original finite local observations and their receiving kernels. No finite-field, analytic period, or arithmetic weight conclusion is inferred solely from this finite model.

7. Evidence and source roles

The full companion proof and identical standalone code/evidence are included in this directory. Run

```
python certificates/verify_local_flow.py
python certificates/audit_local_flow.py
```

The independent auditor imports no producer code. Its methods are actual finite-window evaluation, exact interpolation, integral-flow enumeration, rational matrix ranks, determinant evaluation, and literal original-generator images. Its scopes and corrupted-record controls are stated in the companion note. Finite data verify the exact displayed inputs; the infinite claims rest on the proofs. No new Lean source or receipt is added.

Ziemian, K. (1995). Rotation sets for subshifts of finite type. *Fundamenta Mathematicae*, 146(2), 189–201. DOI 10.4064/fm-146-2-189-201. Publisher abstract and bibliographic record inspected for established local-pattern polytope context; no unread proof imported.

KokunoYumeto. (2026). *Reconstruction with changes of support index* [TeX source, exact revision and blob specified in Section 1]. Source role: original support reconstruction, internal quotient, and receiving-kernel comparison, not a claimed analytic validation of this application.

The Clankers. (2026, September 16). *Exact local-pattern quotients and switching image capacity* [Companion research contribution; review pending]. This receiving note is the same finite mathematics returned through the identified SplitZero interfaces, not an independently attributed new discovery or a replacement analytic source.

Uniform finite-period approximation and exact finite-word switching minima

Source: history/EP817_FINITE_PERIOD_20260916/payloads/ep817/research/finite_period/notes/finite-period-control.md **Identity:** 80798f554e704c746ff84f78f8ce5355ac74907a2f662013e1a83f72b86cbaaf The Clankers. 16 September 2026.

Scope and evidence. These are ordinary mathematical proofs with exact finite arithmetic replays, submitted for independent review. They continue the seven-observation and local-flow work without changing its sources or inherited Lean receipts. The upstream `research/cone_dual` contribution was inspected at main revision 578564a5f0c6e3348f5bdc33fd2022b674899c41; its finite-facet all-rank theorem retains its separate attribution. The argument below establishes an explicit word-horizon approximation for a finite arithmetic dictionary and for a finite controller, rather than asserting that an arbitrary optimum is attained by a finite period. It also calculates every finite-length minimum in the earlier radix-ten example. No new unrestricted numerical record or determination of the full Erdős 817 rate is claimed.

1. Original arithmetic and the objective

A level is a pair $a=(b,A)$, where b is an integer at least two, A is a nonempty finite set of distinct positive integers, and $S(A)=\text{sum}(A)<b$. Its reward is $n(a)=|A|$. Fix a coefficient arity $q \geq 2$ and write

$$H_q(A) = \left\{ \sum_{a \in A} c_a a : 0 \leq c_a < q \right\}.$$

For a finite word $w=a_0 \dots a_{m-1}$, retain every original radix and generator:

$$P_0 = 1, \quad P_{j+1} = b_j P_j, \quad A_w = \bigcup_{j < m} P_j A_j, \quad N(w) = \sum_{j < m} |A_j|, \quad Q(w) = P_m.$$

The level intervals prove distinctness: $P_{-j} a < P_{-(j+1)}$, while every generator at a later level is at least $P_{-(j+1)}$. Thus $|A_w|=N(w)$. Also

$$S(A_w) = \sum_{j < m} P_j S(A_j) \leq Q(w) - 1.$$

Let

$$F_q(w) = |H_q(A_w)|, \quad F_q(\emptyset) = 1.$$

A finite dictionary D consists of finitely many specified levels. Put $n_- = \min_a n(a) > 0$ and $n_+ = \max_a n(a)$. The image-switching optimum is

$$\delta_q(D) = \inf_{\omega \in D^{\mathbb{N}}} \liminf_{m \rightarrow \infty} F_q(\omega_0 \dots \omega_{m-1})^{1/N(\omega_0 \dots \omega_{m-1})}. \quad (1.1)$$

This is an image cost, not a redefinition of the largest-generator cost. The maps through A_w and $Q(w)$ retain the original numbers. Every level can additionally be required to have modular- k -free binary subset sums. That

is a finite condition: all starts and all nonzero steps modulo b , including steps of small additive order, are examined. For such a dictionary every infinite word has k -admissible finite prefixes. The proof reduces each progression modulo its actual lowest radix, subtracts the common original digit, and divides by that radix. No primality assumption is needed.

The principal finite-period theorem holds for every dictionary as defined above. Admissibility is needed only when interpreting its constructions in the Erdős problem.

2. The exact cut quotient and its uniform fibre bound

For words u and v , concatenation gives the original numerical quotient

$$\pi_{u,v} : H_q(A_u) \times H_q(A_v) \longrightarrow H_q(A_{uv}), \quad (x, y) \longmapsto x + Q(u)y. \quad (2.1)$$

Its source consists of distinct values, and it is onto. Write $P=Q(u)$. The lower image satisfies

$$0 \leq x \leq (q-1)(P-1).$$

In a single fibre the x -values are congruent modulo P . An interval of the displayed width contains at most $q-1$ such integers. Each x determines y uniquely. Therefore, with $C=q-1$,

$$\boxed{C^{-1}F_q(u)F_q(v) \leq F_q(uv) \leq F_q(u)F_q(v).} \quad (2.2)$$

The exact loss is the receiving-fibre mean

$$\kappa(u, v) = \frac{F_q(u)F_q(v)}{F_q(uv)} \in [1, C]. \quad (2.3)$$

Nothing identifies those fibres with singleton fibres. On original free value modules there is the exact sequence

$$0 \rightarrow \ker \pi_{u,v} \rightarrow \mathbb{Q}[H_q(A_u) \times H_q(A_v)] \xrightarrow{\pi_{u,v}} \mathbb{Q}[H_q(A_{uv})] \rightarrow 0.$$

Choosing one specified original pair per fibre gives a basis of the kernel consisting of differences from that pair. In particular its dimension is $F_q(u)F_q(v) - F_q(uv)$.

The constant C is sharp on the unrestricted canonical source class. Use the singleton generator $\{1\}$ at radix two for m levels. Its q -ary image is the full interval $[0, (q-1)(2^m-1)]$, with cardinality $C2^m - (C-1)$. Sending both cut lengths to infinity makes (2.3) tend to C . These particular binary sources are not claimed to avoid a fixed progression length at arbitrary depth. Sharpness on a smaller admissible subfamily is a separate question.

2.1 A repeated finite word

For fixed w , let $a_t = F_q(w^t)$. Equation (2.2) gives

$$C^{-1}a_t a_s \leq a_{t+s} \leq a_t a_s.$$

The upper inequality proves existence of $\eta(w) = \lim_{t \rightarrow \infty} a_t^{1/t}$ by subadditivity of the logarithms. Iterating the lower inequality on blocks of fixed length t gives $\eta(w) \geq a_t^{1/t} / C^{1/t}$. Hence

$$\boxed{\eta(w)^t \leq F_q(w^t) \leq C\eta(w)^t.} \quad (2.4)$$

The image rate of the periodic schedule w^∞ is $\eta(w)^{1/N(w)}$. For $q=5$ the established seven-coordinate matrix M_w represents the same count, and $\eta(w) = \rho(M_w)$. No spectral calculation is required for the bounds below; all finite $F_q(w)$ values are integers.

3. A uniform finite-horizon theorem for all switching words

For each $m \geq 1$ define the two explicitly finite quantities

$$U_m = \min_{w \in D^m} F_q(w)^{1/N(w)}, \quad L_m = \min_{w \in D^m} (F_q(w)/C)^{1/N(w)}. \quad (3.1)$$

The minimizing words for the two expressions may differ when rewards differ. They are retained separately.

Theorem FP1. Every finite dictionary satisfies

$$\boxed{L_m \leq \delta_q(D) \leq U_m, \quad C^{-1/(mn_-)} U_m \leq L_m \leq U_m.} \quad (3.2)$$

In particular,

$$\boxed{0 \leq \log U_m - \log \delta_q(D) \leq \frac{\log C}{mn_-}.} \quad (3.3)$$

Both L_m and U_m converge to $\delta_q(D)$, with this explicit logarithmic error bound.

Proof. Split an arbitrary infinite word into successive m -letter blocks $w_1, \dots, w_s$ and a remainder of length less than m . Iterating (2.2), and retaining the remainder as a higher-level source containing zero, gives

$$F_q(w_1 \cdots w_s) \geq C \prod_{i=1}^s \frac{F_q(w_i)}{C} \geq C L_m^{\sum_i N(w_i)}.$$

Appending the remainder cannot decrease the image cardinality. Its generator reward is at most $(m-1)n_+$, so it does not affect the lower limiting root as s tends to infinity. This proves $\delta_q(D) \geq L_m$ for every schedule, not just periodic schedules.

Choose a word realizing U_m . Repeating that actual word and applying (2.4) proves $\delta_q(D) \leq U_m$. Finally $N(w) \geq mn_-$ gives

$$(F_q(w)/C)^{1/N(w)} \geq C^{-1/(mn_-)} F_q(w)^{1/N(w)}.$$

Taking minima proves the remaining inequality and (3.3). QED.

No convergence of the individual infinite schedule was assumed. The theorem concerns the infimum of lower limits and proves convergence of its finite-horizon bounds.

3.1 A priori period length for a prescribed error

For any $\epsilon > 0$ choose

$$m \geq \max \left\{ 1, \left\lceil \frac{\log(q-1)}{n-\epsilon} \right\rceil \right\}.$$

A word minimizing U_m has a periodic image rate at most $e^\epsilon \delta_q(D)$. Thus

$$\boxed{\delta_q(D) = \inf_{w \neq \emptyset} \eta(w)^{1/N(w)}.} \quad (3.4)$$

This is quantitative periodic approximation. It does not assert that the infimum is attained by a finite period for every dictionary. For $q=2$, $C=1$, the image counts multiply exactly and the finite enclosures already coincide.

The rate is a computable real uniformly in the finite integer dictionary: enumerate D^m , evaluate the integer counts, and bracket the finitely many rational powers by dyadic intervals. The logarithmic width has the bound above; moreover $1 \leq \delta_q(D) \leq q$, which also converts it into a prescribed absolute error. The number of words may be exponential in $1/\epsilon$, and the integer bit costs retain the actual radices and weights. The result is a terminating approximation theorem, not a polynomial-time claim.

3.2 Two-sided arithmetic certificates

For a rational threshold $u > 0$, checking $L_m \geq u$ is exactly checking

$$F_q(w) \geq C u^{N(w)} \quad (w \in D^m). \quad (3.5)$$

If $u = a/b$, this is the integer inequality $b^N F_q(w) \geq C a^N$. A failed global lower threshold $u > \delta_q(D)$ eventually produces a word with $F_q(w) < u^N$; its repeated original block is an explicit strict improvement.

For $u < \delta_q(D)$, a sufficient witness horizon is

$$m > \frac{\log C}{n_- \log(\delta_q(D)/u)}.$$

For $u > \delta_q(D)$, a sufficient horizon for a periodic improvement is

$$m > \frac{\log C}{n_- \log(u/\delta_q(D))}.$$

The unknown endpoint is not inserted as a computable input to those certificates. The algorithm computes the finite interval first. Equality at the endpoint need not have a finite deciding certificate. Taking $\max_{(j \leq m)} L_j$ and $\min_{(j \leq m)} U_j$ gives nested valid enclosures if that is desired.

4. Controllers, admissibility supports, and the closing cost

Let a finite strongly connected directed controller have s vertices and at least one directed cycle. Every edge carries one of the actual positive-reward canonical levels above. Edges can encode an additional scheduling rule, or the original safe carry-support transfers. In the latter case all edge transfers and acceptance exclusion are part of the finite input certificate. Let

$$n_- = \min_e |A_e|, \quad T = \max_e |H_q(A_e)|.$$

For $m \geq 1$, define U_m and L_m as in (3.1), but over all length- m paths inside this controller, retaining their starting and ending vertices. Let δ_G be its optimal lower limiting image rate.

Theorem FP2.

$$\log U_m - \frac{\log C}{mn_-} \leq \log \delta_G \leq \log U_m + \frac{(s-1) \log T}{mn_-}. \quad (4.1)$$

Moreover a closed directed walk of length at most $m+s-1$ has periodic image rate at most

$$\exp \left(\frac{\log C + (s-1) \log T}{mn_-} \right) \delta_G. \quad (4.2)$$

Proof. The lower bound groups a permitted infinite path into its original m -edge pieces, exactly as in Theorem FP1. Choose a path w realizing U_m . Strong connectivity supplies a directed path v from its ending vertex back to its starting vertex with length at most $s-1$. Retain this path, its actual radices and its generator reward. Then

$$F_q(wv) \leq F_q(w) F_q(v) \leq F_q(w) T^{s-1}.$$

The repeated closed path has image rate at most $F_q(wv)^{1/(N(w)+N(v))}$. Positivity of the count logarithms gives the upper estimate in (4.1). Combining it with the lower estimate gives (4.2). QED.

If controller vertices are original true-carry supports, a safe cycle can be read periodically from empty true support as well: empty support is contained in its starting support, and every transfer is monotone. This preserves the original binary-feasibility interpretation. It does not delete the initial supported false state or alter a radix.

For a general finite controller, restrict to reachable strongly connected components containing a cycle. Every infinite path eventually remains in one such component, since the component graph is finite and acyclic. An initial finite path changes the logarithmic count by a bounded amount through (2.2), and adds a fixed generator reward. Therefore the optimum is the minimum of the component optima. The bounds above apply separately to those components.

Zero-reward edges require a separate treatment; $n_{-} > 0$ is an explicit domain condition in this theorem. In particular, an active empty-generator reset from earlier work is not silently deleted to satisfy it. The current arithmetic certificates and controller examples all have positive-reward edges.

5. Inverse-count sums and a second finite enclosure

For an equal-reward dictionary with $n(a)=n_0$ and $d=|D|$, let p be a positive real parameter and put

$$Z_m(p) = \sum_{w \in D^m} F_q(w)^{-p}.$$

These are auxiliary numerical weights, not the original coefficient-word representation measure. Equation (2.2) gives

$$Z_m(p)Z_n(p) \leq Z_{m+n}(p) \leq C^p Z_m(p)Z_n(p).$$

The pressure limit $P(p) = \lim_{m \rightarrow \infty} (-1/m) \log Z_m(p)$ consequently exists and satisfies

$$\frac{\log Z_m(p)}{m} \leq P(p) \leq \frac{\log Z_m(p) + p \log C}{m}.$$

Indeed the logarithms are superadditive, while addition of $p \log C$ makes them subadditive; the direct division-with-remainder argument proves both bounds.

Writing $a_m = \min_{w \in D^m} F_q(w)$, there are the literal inequalities

$$a_m^{-p} \leq Z_m(p) \leq d^m a_m^{-p}.$$

Theorem FP1 gives $a_m \sim (1/(mn_0))^{\delta_q(D)}$. Hence

$$-pn_0 \log \delta_q(D) \leq P(p) \leq \log d - pn_0 \log \delta_q(D).$$

Combining the two comparisons yields a completely finite enclosure:

$$-\frac{\log Z_m(p)}{pn_0 m} - \frac{\log C}{n_0 m} \leq \log \delta_q(D) \leq \frac{\log d}{pn_0} - \frac{\log Z_m(p)}{pn_0 m}. \quad (5.1)$$

Its width is at most $\log C/(n_0 m) + \log d/(pn_0)$. At integral p , $Z_m(p)$ is an exact positive rational computed from the original image counts. The checks below use exact fractions. The terminology and general theory of almost-additive potentials are established background (Cuneo, 2020); all constants and inequalities used here are proved above.

6. Exact finite-length minima in the radix-ten dictionary

Retain the original levels

$$A = (10, \{1, 3\}), \quad B = (10, \{2, 4\}).$$

Their binary images are $\{0, 1, 3, 4\}$ and $\{0, 2, 4, 6\}$. A nonzero modular step of order at least five would require five distinct values, and the only nonzero smaller order in $\mathbb{Z}/10\mathbb{Z}$ is two; neither image contains a pair differing by five. Thus every changing word is five-admissible.

Let

$$a_m = \min_{w \in \{A,B\}^m} |H_5(A_w)|, \quad a_0 = 1.$$

Theorem FP3 (all finite lengths).

$$a_1 = 13, \quad a_m = \begin{cases} 893^r, & m = 3r, \\ 93^2 893^{r-1}, & m = 3r + 1 \geq 4, \\ 93 893^r, & m = 3r + 2. \end{cases} \quad (6.1)$$

Equivalently,

$$\sum_{m \geq 0} a_m z^m = \frac{1 + 13z + 93z^2 - 2960z^4}{1 - 893z^3}. \quad (6.2)$$

This determines the finite minima, not only the limiting minimum already proved in the local-flow note.

6.1 Lower bound from the complete cyclic run formula

In the original seven-observation coordinates, the two matrices act by

$$M_A f = (3f_0 + 7f_1, 2f_0 + 8f_1, f_0 + 9f_1, f_0 + 9f_1, 10f_1, 10f_1, 10f_1),$$

$$M_B f = (2f_1 + 3f_3, 4f_1 + 6f_3, f_1 + 4f_3, 3f_1 + 7f_3, 4f_1 + 6f_3, 3f_1 + 7f_3, 2f_1 + 8f_3).$$

Their product is the literal rank-one matrix uv , where

$$u = (17, 18, 19, 19, 20, 20, 20)^\top, \quad v = (0, 2, 0, 3, 0, 0, 0).$$

For positive integers a, b , the crossing scalar between A and B runs is

$$\theta(a, b) = v M_B^{b-1} M_A^{a-1} u = \frac{8 \cdot 10^{a+b} + 4 \cdot 10^b - 3}{9}. \quad (6.3)$$

The vectors and row satisfy their original recurrences with polynomial $(X-1)(X-10)$; the four values at a, b in $\{1, 2\}$ verify this formula for every a, b . Cyclically rotating a mixed word preserves its nonzero eigenvalues and exposes the rank-one product. Its radius is the product of the corresponding crossing scalars. A pure word has radius 10^m . The actual count $F_5(w)$ is at least that radius. To see this directly, let $a_R = |R|$ for the original seven shapes, and use the explicitly specified auxiliary norm

$$\|x\|_* = \max_R |x_R| / |R|.$$

The matrix is nonnegative. Its induced norm is $\max_R (M_w a)_R / |R|$. The numerator is the actual count $|H_5(A_w) + R|$, at most $|R| F_5(w)$, with equality in the singleton coordinate. Hence $\|M_w\|_* = F_5(w)$, which bounds its spectral radius. This auxiliary observation norm is not the original generator-word metric.

For a fixed run-pair length $L = a + b$, the smallest crossing is

$$t_L = \theta(L - 1, 1) = \frac{8 \cdot 10^L + 37}{9}, \quad L \geq 2.$$

For $L \geq 4$,

$$t_L - 93t_{L-2} = \frac{56 \cdot 10^{L-2} - 3404}{9} > 0.$$

Therefore replacing a part of length at least four by a part of length two and the remaining length strictly reduces its product cost. Every product is bounded below by one using only length-two parts of cost 93 and length-three parts of cost 893.

The exact inequality

$$93^3 = 804357 > 797449 = 893^2$$

replaces three length-two parts by two length-three parts. Thus a minimizing partition has at most two length-two parts. The three possible length residues give the right side of (6.1). Pure words have still larger radius for $m \geq 2$. This proves the lower bound for every finite word.

6.2 Attainers with no discarded endpoint cost

The original words BA and BAA have actual aggregate generator sets

$$C_2 = \{2, 4, 10, 30\}, \quad Q_2 = 100,$$

$$C_3 = \{2, 4, 10, 30, 100, 300\}, \quad Q_3 = 1000.$$

Their fifth-arity images are exactly

$$H_5(C_2) = 2[0, 92], \quad H_5(C_3) = 2[0, 892]. \quad (6.4)$$

To verify them, the explicit maps $x \rightarrow x/2$ and $y \rightarrow 2y$ transport the two sources to weights $\{1, 2, 5, 15\}$ and $\{1, 2, 5, 15, 50, 150\}$. At every step a new weight is at most one plus four times the previous weight sum, so the successive $q=5$ intervals overlap or touch. Their final sums are 23 and 223. This proves (6.4) without replacing the original generators in the construction.

Concatenate any collection of the original macro words BA and BAA. A single division by two on the entire numerical image gives a mixed-radix language with actual digit sets $[0, 92]$ and $[0, 892]$, respectively. Those digits lie below their original macro radices 100 and 1000. The mixed-radix evaluation is therefore bijective, with its ordinary digit inverse. The original image count is exactly $93^a 893^b$.

For $m=3r$ use $(BAA)^r$. For $m=3r+1 \geq 4$ use BABA followed by $r-1$ copies of BAA. For $m=3r+2$ use BA followed by r copies of BAA. These are exactly the counts in (6.1). At length one, B has thirteen values and A has seventeen. Formula (6.2) follows by multiplying the displayed residue-class series by $1-893z^3$. QED.

In particular $(BAA)^r$ has exactly 893^r distinct fifth-arity values at every r . The earlier AAB phase has $2301 \cdot 893^{(r-1)}$ values at r periods. These phases share their asymptotic rate, but their original finite image counts differ; the new attainers retain that endpoint information.

7. Executed finite evidence

The producer accounts for all length-20 radix-ten words, using exact row/reward aggregation and retaining their literal word multiplicities: 1,048,576 words at the last depth. It checks every finite minimum through length twenty against (6.1), and every explicit attainer through length one hundred. The row/reward aggregation is an exact finite identification: equal rows with the same generator reward have equal numerical counts under every suffix, and their multiplicities add. It is not a sampling procedure.

Two additional dictionaries test unequal rewards and different radices. The first uses A,B and $(3, \{1\})$ through length nine (19,683 words at the last depth). The second uses $(3, \{1\})$, $(5, \{1, 2\})$, $(23, \{1, 3, 4, 7\})$ through length eight (6,561 words). Every binary modular certificate is verified. Each lower and upper root comparison is made by integer cross-powers, and each displayed decimal substitute is enclosed by exact dyadic endpoints.

The cut and moment-free carrier identities are checked in 120 original generator/radix/arity instances, with 600 complete finite-tail joins for $q=2, \dots, 6$. The pressure identity has 54 exact fraction checks. Two strongly connected controllers, strict alternation and the three-cycle ABB, have 24 retained path/bridge certificates through length twelve.

The separate auditor imports neither the producer nor its helper. It constructs the seven matrices by interpolation on seven actual symmetric numerical sets, evaluates receiving column vectors rather than prefix rows, and independently covers every stated word domain with exact multiplicities. It also checks 92 actual concatenated generator sets by direct subset-sum construction. Eight corruptions are required to fail: an altered carrier entry, deleted cut factor, wrong reward, missing word depth, false root interval, false attaining count, missing bridge and false pressure ratio.

The theorem is not inferred from the finite run. The finite checks certify the actual matrices, arithmetic counts, witness words and implementation scopes used above. The producer and audit receipts record source identities and their executed domains. No outside human review or new Lean verification is represented by those receipts.

8. Relation to the unrestricted target

For a finite modular-k-admissible dictionary, its optimal image rate supplies an upper construction for the unrestricted exponential rate through the previously proved fifth-image capacity bridge. The finite-period theorem gives a quantitative computation of that dictionary value. It does not prove a lower bound for all possible dictionaries.

As the arithmetic family grows, n , b and the actual generator weights remain unbounded. The compact cone-dual work retained on upstream main addresses finite descriptions of strict all-rank lower certificates; the universal block inequalities remain its separate obligation. The present result bounds word length for approximating a specified dictionary, even when an exact finite-period optimizer has not been established. It does not provide a cutoff on the block ranks sufficient for the global infimum.

The exact best rate in the radix-ten dictionary remains $893^{1/6}$, larger than the existing global $k=5$ bound. The new complete finite minima and a priori period bounds are structural results, not improved global numerical records.

References and provenance

Bochi, J., & Morris, I. D. (2015). Continuity properties of the lower spectral radius. *Proceedings of the London Mathematical Society*, 110(2), 477–509. arXiv:1309.0319. The abstract and publication metadata were inspected for context: finite-period attainment is not a generic conclusion for matrix families. No theorem from that paper is needed in the proofs above.

Cuneo, N. (2020). Additive, almost additive and asymptotically additive potential sequences are equivalent. *Communications in Mathematical Physics*, 377, 2579–2595. arXiv:1909.08643. Context for the bounded logarithmic cut defect and pressure terminology; the numerical constants above are derived directly.

Korsky, S. (2026). *Arithmetic progression-free subset-sum sets* (arXiv:2606.24139v1). Context for the original general extremal problem. No new general-rate claim is attributed to that preprint.

The Clankers. (2026, September 16). *Exact local-pattern quotients and switching image capacity*. Original local-flow note in the supplied source archive. Role: preceding matrices and infinite radix-ten optimum; the finite-length minima and uniform dictionary approximation are developed above with complete proofs.

The Clankers. (2026, September 16). *Exact feasible observations, integral positive lifts, and finite-facet capacity certificates*. `research/cone_dual/notes/cone-and-dual.md`, inspected on EP817 main at 578564a5f0c6e3348f5bdc33fd2022b674899c41. Role: current concurrent all-rank cone-dual interface, preserved separately rather than represented as part of the supplied historical ZIP members.

The original $k=4$ construction and signed-block proof remain attributed to the anonymous/deleted Reddit contributor. The finite upper-bound Lean extension remains separately attributed to sneed-and-feed.

Retained cut kernels, bounded logarithmic defects, and finite-period recovery

Source: `history/EP817_FINITE_PERIOD_20260916/payloads/zeta/workbenches/ep817-finite-period/notes/splitzero-receiving-map.md` **Identity:** `dc18613b9655041ae1618a43cad1e74860d5631989b0797349d641665d496fbd` The Clankers. 16 September 2026.

This receiving note accompanies `finite-period-control.md`, whose complete proofs and standalone verifiers are included beside it. The objects below are the actual finite word sources and value quotients used in the arithmetic construction. The comparison uses the original SplitZero support and receiving-kernel interface; it does not replace the theta source or its infinite quotient by a finite dictionary. No new Lean elaboration or analytic arithmetic-weight theorem is claimed.

1. Two original quotient stages

For a canonical word u , let Ω_u be the actual q -valued generator-choice set, and $X_u = H_q(A_u)$ its distinct numerical image. Retain the free modules

$$U_u = \mathbb{Q}[\Omega_u], \quad V_u = \mathbb{Q}[X_u], \quad q_u(e_\omega) = e_{\Phi_u(\omega)}.$$

Concatenation identifies the actual choice source with Ω_u times Ω_v . After taking the two local numerical quotients, the remaining map is

$$\pi_{u,v} : \mathbb{Q}[X_u \times X_v] \longrightarrow V_{uv}, \quad e_{(x,y)} \longmapsto e_{x+Q(u)y}. \quad (\text{R1})$$

All original radix products $Q(u)$ remain in this formula. The complete source map is $\pi_{\mathbf{u}}(u,v)$ composed with q_u tensor q_v . Both quotient maps are onto, giving

$$0 \longrightarrow \ker(q_u \otimes q_v) \longrightarrow \ker(\pi_{u,v}(q_u \otimes q_v)) \xrightarrow{q_u \otimes q_v} \ker \pi_{u,v} \longrightarrow 0. \quad (\text{R2})$$

For surjectivity at the last term, lift any coefficient vector in $\ker \pi_{\mathbf{u}}$ by choosing an original word in each of its value fibres. Its full numerical observation vanishes by (R1). The displayed first kernel is exactly the kernel of the restricted map. This proves exactness without identifying the two relation families.

Choose one specified pair p_z in each fibre of (R1). The differences $e_{p_z} - e_{(p_z)}$ for p other than p_z form an integral basis of the receiving kernel. Every original pair and every collision is retained. Over support-indexed subdiagrams, the same map carries a boundary to the zero of its admitted target fibre; it does not send that fibre to the external bottom.

The set bound is independent of the coefficient field. Every fibre of (R1) has at most $C=q-1$ elements because its lower values lie in $[0, C(Q(u)-1)]$ and are congruent modulo $Q(u)$. Hence

$$\dim \ker \pi_{u,v} = F_q(u)F_q(v) - F_q(uv), \quad 1 \leq \kappa(u,v) := \frac{F_q(u)F_q(v)}{F_q(uv)} \leq C. \quad (\text{R3})$$

The quotient size, rather than a generic linear-map norm, controls the multiplicative count loss.

2. Associativity and the complete scalar defect

For three words, both original numerical maps send (x,y,z) to

$$x + Q(u)y + Q(u)Q(v)z.$$

Thus the two parenthesized quotient composites agree on every original basis vector. The receiving-fibre ratios obey the exact identity

$$\boxed{\kappa(u, v)\kappa(uv, w) = \kappa(v, w)\kappa(u, vw)}. \quad (\text{R4})$$

Both sides are $F_q(u)F_q(v)F_q(w)/F_q(uvw)$. With $a(w)=\log F_q(w)$, the logarithmic defect is $a(u)+a(v)-a(uv)$, belongs to $[0, \log C]$, and satisfies the additive version of (R4). The nonzero defect is retained in every iteration; it is not set to zero to obtain a multiplicative model.

Repeated cuts into length- m words bound the total defect by the number of cuts times $\log C$. Dividing by the actual generator reward yields the finite-period horizon error $\log C/(m \cdot n_{\min})$. This is the original arithmetic mechanism behind Theorem FP1. It supplies an a priori finite approximation to an infinite schedule optimization for every specified finite dictionary. The argument is not a claim that the original source quotients become constant at finite length.

A finite safe controller adds an actual bridge between the terminal and initial supports of a candidate path. The bridge has at most $s-1$ edges inside a strongly connected component, but its radices and generator rewards are retained. Its count cost is at most $T^+(s-1)$. Theorem FP2 explicitly carries that factor through the periodic approximation. An active zero-reward reset from earlier work is outside its positive-reward domain; it is not erased or silently assigned a positive reward.

3. The weighted section and the original metric

Give the distinct local values their original word masses $\mu_u(x)$, $\mu_v(y)$. On the pair source the retained mass is

$$w(x, y) = \mu_u(x)\mu_v(y).$$

For a receiving value z , define

$$W_z = \sum_{x+Q(u)y=z} w(x, y).$$

With source Gram $\text{diag}(1/w)$, the exact minimum-norm section is

$$s(e_z) = \sum_{x+Q(u)y=z} \frac{w(x, y)}{W_z} e_{(x,y)}, \quad G_{\text{receiving}} = \text{diag}(1/W_z). \quad (\text{R5})$$

Its image is orthogonal to $\ker \pi$. Indeed for any relation r in one fibre, the inner product with $s(e_z)$ is $\sum(r(x,y))/W_z=0$. Its evaluation is e_z , and its squared norm is $\sum w/W_z^2=1/W_z$. Thus every lift h of e_z has

$$\|h\|^2 = \|h - s(e_z)\|^2 + 1/W_z.$$

The fibre-count bound C limits the number of distinct local value pairs, not their word masses. For a concrete five-admissible example use $A=\{1,2\}$, $b=13$, and $q=5$ at every level. The binary image $\{0,1,2,3\}$ is modular-five-free modulo 13. Its fifth-arity image is $[0,12]$, so all interlevel value fibres are singletons. The local fifth-word multiplicity has maximum three, attained at value four and other central values. Its original m -level maximum multiplicity is exactly 3^m . Therefore the identity from the original numerical quotient metric to unit value mass has norm

$$\boxed{3^{m/2}}.$$

This grows even though every cut in (R1) is injective and has $\kappa=1$. The uniform logarithmic count estimate supplies no unrecorded uniform bound on the original word-to-value metric return.

4. Exact original endpoints in the finite minimizers

The radix-ten dictionary has complete finite minima attained by concatenating the actual macro words BA and BAA. Their generator sets are

$$\{2, 4, 10, 30\}, \quad \{2, 4, 10, 30, 100, 300\},$$

and their original radix products are 100 and 1000. The global scalar map $x \rightarrow x/2$ and its inverse $y \rightarrow 2y$ identify their fifth-images with $[0, 92]$ and $[0, 892]$. It is applied once to the whole numerical image. It does not replace each generator independently or remove any factor from the original radix product.

Every macro concatenation therefore has a digitwise inverse and exact count $93^a 893^b$. In particular the BAA phase has count 893^r at r periods, whereas the original AAB phase has count $2301 * 893^{(r-1)}$. They share their cyclic spectral rate but have different original endpoints. The complete numerical images, actual weights, and the source maps distinguish those finite objects.

5. Scope of the transfer

The present source is a finite arithmetic dictionary or a finite certified controller, with unbounded word length. The new uniform bound controls that infinite schedule variable. The outer family of possible generator blocks and their ranks remains unbounded in the full extremal problem. Neither the scalar defect identity nor the finite-period approximation supplies the missing all-rank arithmetic inequality automatically.

The original SplitZero reconstruction accepts these quotient diagrams and support maps fibrewise. On the analytic theta source, a finite observation can retain an additional kernel beyond the original theta relations. No map in this note asserts that kernel vanishes, nor is an analytic weight estimate attributed to the finite count bounds. The uses of (R1)-(R5) are fully specified finite arithmetic instances of the original quotient interface.

References and execution boundary

The Clankers. (2026, September 16). *Uniform finite-period approximation and exact finite-word switching minima*. Complete provider note and standalone verifiers in this directory. The general proofs use the literal cut map above; the executed finite domains and the independent implementation boundary are listed in the two generated receipts.

KokunoYumeto. (2026). *Reconstruction with changes of support index*, equations D6-D8. `workbenches/splitzerotandem/tex/support_diagrams.tex` at the previously inspected original source revision `58626a62cd5648e8ae7450fb54d4a4aab2330981`. The original mathematical interface is retained via the delivered predecessor sources; no new formal compilation of that repository is claimed here.

The original anonymous/deleted contributor's four-term construction and sneed-and-feed's Lean upper-bound extension retain their separate attribution.